



Decentralized Crypto World Bank

A Blockchain-Based Banking Platform
with AI-Enhanced Security and Assistance

by

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A pre-thesis 1 report submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science

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Declaration

It is hereby declared that

1. The report submitted is our own original work while completing degree at BRAC University.
2. The report does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The report does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Ethics Statement

This project operates exclusively on public blockchain test networks using test tokens with no real monetary value. No real financial transactions, personal banking data, or user financial records are involved at any stage of development or evaluation. All transaction data used for Artificial Intelligence and Machine Learning (AI/ML) model training and evaluation is either synthetically generated or sourced from publicly available anonymized datasets. The platform prototype does not process Know Your Customer (KYC) or Anti-Money Laundering (AML) data in its current scope, and all wallet addresses used during testing are disposable testnet addresses. We acknowledge the ethical considerations inherent in AI-assisted financial decision-making and have incorporated explainability mechanisms (SHapley Additive exPlanations, or SHAP) to ensure that automated risk assessments remain transparent and auditable by human reviewers. Future phases plan to incorporate a privacy-preserving ZKP-based identity compliance layer in which users prove KYC status without exposing personal data on-chain; the ethical implications of this design will be evaluated in the final thesis.

Abstract

Global development finance relies on multilayered institutional structures to distribute capital across borders and communities, yet these systems often struggle with fragmented transparency, procedural friction, and uneven access to credit. Although decentralized finance has demonstrated that programmable infrastructures can automate lending and enhance auditability, existing models largely employ flat architectures that do not reflect institutional hierarchies or structured governance. As a result, there remains limited exploration of how tiered financial systems might be represented within decentralized environments while preserving oversight and adaptability. Here we present *Crypto World Bank*, a formally specified, planning-phase research prototype that specifies a blockchain-based institutional finance platform coordinating capital allocation across a four-tier hierarchical governance structure on EVM-compatible Layer 2 infrastructure. The primary contribution is a smart-contract hierarchy with oracle-mediated risk analytics (Random Forest, Isolation Forest, SHAP via Chainlink Functions); an optional conversational agent (MCP tool server) provides an alternate UI path while core banking remains form- and wallet-driven. The architecture spans six functional domains—deposit mobilization, credit allocation, payment settlement, risk intermediation, liquidity management, and ancillary financial services. Smart contracts, frontend, and oracle integration are specified in this pre-thesis; implementation and test-net deployment are planned across four phases in the final thesis stage. This hybrid design illustrates how institutional finance and decentralized systems may converge, contributing an open-source, hierarchically governed crypto financial services architecture as a foundation for further research.

Keywords: Blockchain, Decentralized Finance, Institutional DeFi, Hierarchical Lending, Smart Contracts, Oracle-Mediated Risk Analytics, Explainable AI, Group Lending, Multi-Entity Co-Lending, Deposit Mobilization, Reserve Management, Financial Sustainability

Dedication

Dedicated to our families for their unwavering support throughout our academic journey.

Acknowledgment

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List of Formulas

Utilization Rate (U): $U = \frac{L}{L+B}$ (lending utilization; L is lent amount, B is available balance/liquidity; drives the kinked rate model in Formulas 18–19)

Kinked Borrowing Rate (Formula 18): $r_b(U) = r_0 + \frac{U}{U^*} \cdot r_1$, $U \leq U^*$ (gentle slope below optimal utilization U^* ; base rate r_0 , below-kink slope r_1 ; enforced on-chain in retail and interbank pools; Section 3.9)

Kinked Borrowing Rate (Formula 19): $r_b(U) = r_0 + r_1 + \frac{U-U^*}{1-U^*} \cdot r_2$, $U > U^*$ (steep slope above U^* to incentivise repayment and new deposits; jump multiplier r_2 ; on-chain parameter guards apply before updates)

Collateral Ratio (CR): $CR = \frac{C}{D}$ (collateral-to-debt ratio)

Loan-to-Value (LTV): $LTV = \frac{\text{Loan Amount}}{\text{Collateral Value}}$

APR (simple): $APR = r_{\text{period}} \times N$ (annualized simple rate over N periods)

Compound Interest (Savings): $A = P \left(1 + \frac{r}{n}\right)^{nt}$ (deposit growth; P principal, r annual rate, n compounding frequency)

EMI (Installment Payment): $EMI = \frac{P \cdot r \cdot (1+r)^n}{(1+r)^n - 1}$ (installment for principal P , periodic rate r , n installments)

Health Factor (HF): $HF = \frac{C \times LT}{D}$ (liquidation trigger for *over-collateralized retail loans only*; if $HF < 1.0$ liquidation is triggered, where LT is liquidation threshold and C is oracle-priced collateral; Section 3.10)

Group Health Factor (HF_{group}): $HF_{\text{group}} = \frac{\text{total_group_collateral} \times LT}{\text{total_group_outstanding_debt}}$ (pool-level liquidation score for `GroupLendingPool`; shared shortfall across members)

Institutional Health Factor (HF_{inst}): $HF_{\text{inst}} = \frac{\text{on_chain_reserve} - \text{min_reserve}}{\text{outstanding_borrowing}}$ (Tier 1–3 reserve-surplus analogue to collateral; parent-tier surplus above mandatory minimum reserve)

Reserve Ratio (RR): $RR = \frac{\text{Reserves}}{\text{Total Deposits}}$ (solvency constraint enforced at each tier; governs the maximum downward capital that may be allocated, *not* a money-creation parameter)

Platform Solvency Ratio (PSR): $PSR = \frac{\text{TotalReserveBalance}}{\text{TotalOutstandingLoans}}$ (on-chain analogue of capital adequacy, measuring whether the system’s reserve cover exceeds outstanding loan obligations; replaces the Basel III CAR formula, which is undefined for this platform because USDC assets have no regulatory risk-weight classification under Basel III and “Tier 1 Capital” in the Basel sense cannot be computed from an on-chain USDC balance alone)

Credit Velocity (CV): $CV = \frac{\text{Capital Recycled per Period}}{\text{Total Reserve}}$ (rate at which reserve capital flows through the lending hierarchy and returns as repayment; replaces the inapplicable fractional-reserve money multiplier, which presupposes new money creation and does not apply to fixed-supply USDC)

Interbank Rate Cap: $r_{\text{IB}}(U) \leq r_{\text{downward}}(U) - \delta$ (same-tier `InterBankLendingPool` borrowing rate capped below parent-tier downward rate; δ is governance-set anti-arbitrage)

spread; Section 3.14.1)

Upward Deposit Rate Cap: $r_{\text{up}} < r_{\text{down}} - \delta$ (asymmetric tier spread for UpwardDepositFacility; upward yield stays below downward lending rate; Section 3.14.2)

FX Conversion: $\text{Amount}_B = \text{Amount}_A \times \left(\frac{P_A}{P_B}\right)$ (oracle-priced conversion between assets A and B ; Chainlink AggregatorV3Interface feeds)

Treasury Swap (oracle-priced): $\text{Amount}_{\text{to}} = \text{Amount}_{\text{from}} \times \frac{P_{\text{from}}}{P_{\text{to}}} \times (1 - \text{spread})$ (atomic cross-asset swap in TreasurySwap; P from Chainlink feeds; treasury spread 5–10 bps; Section 3.14.5)

Oracle Commit–Reveal Hash: $h = \text{keccak256}(s \parallel \text{nonce})$ (prototype-phase ML score commitment to LoanController; contract verifies $h = \text{keccak256}(s \parallel \text{nonce})$ before storing score s ; immutable on-chain audit trail; Section 3.3.2)

Group Loan Individual Share: $\text{Share}_i = \frac{\text{LoanTotal}}{N_{\text{members}}}$

On-Chain Credit Score (conceptual): $CS = \sum_i w_i \cdot \text{feature}_i$ (weighted features learned off-chain and stored as a score on-chain)

Isolation Forest Anomaly Score: $s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$ (anomaly scoring for instance x ; where $h(x)$ is the path length of instance x from the root of an isolation tree and $c(n)$ is the average path length for n samples)

Random Forest Fraud Probability: $P(\text{fraud} \mid x) = \frac{1}{T} \sum_{t=1}^T f_t(x)$ (ensemble average over T trees; where $f_t(x)$ is the binary fraud prediction of the t -th decision tree)

Platform Net Interest Allocation: $\text{NetInterest} = \text{InterestCollected} - \text{DepositorYield} - \text{InsuranceAllocation}$

SHAP (Shapley value): $\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$

List of Abbreviations

AA: Account Abstraction (ERC-4337)

ALM: Asset-Liability Management

AI: Artificial Intelligence

AI/ML: Artificial Intelligence / Machine Learning

AML: Anti-Money Laundering

APR: Annual Percentage Rate

API: Application Programming Interface

BRAC: Bangladesh Rural Advancement Committee

CAR: Capital Adequacy Ratio (Basel III; not directly applicable to this platform — see Platform Solvency Ratio, PSR, in List of Formulas)

CBDC: Central Bank Digital Currency

CCIP: Cross-Chain Interoperability Protocol (Chainlink)

CCS: ACM Computing Classification System

CR: Collateral Ratio

CVL: Certora Verification Language

DeFi: Decentralized Finance

DID: Decentralized Identifier (W3C)

DON: Decentralised Oracle Network (Chainlink)

EIP: Ethereum Improvement Proposal

EIP-7702: Ethereum Improvement Proposal 7702 — Session Key standard enabling scoped, time-bound wallet authorisations for an autonomous agent

EMI: Equated Monthly Installment

ERC: Ethereum Request for Comments (token / interface standard)

EVM: Ethereum Virtual Machine

FL: Federated Learning

FATF: Financial Action Task Force

FX: Foreign Exchange

GNN: Graph Neural Network

HF: Health Factor

KYC: Know Your Customer

LLM: Large Language Model

L2: Layer 2

LTV: Loan-to-Value Ratio

MFI: Microfinance Institution

MiCA: Markets in Crypto-Assets Regulation (EU)

ML: Machine Learning

MCP: Model Context Protocol — tool-calling interface used by the AI agent to interact with CWB banking operations

NIM: Net Interest Margin

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

PoS: Proof of Stake

PoR: Proof of Reserve (Chainlink on-chain reserve verification)

PSR: Platform Solvency Ratio (on-chain capital adequacy analogue; see List of Formulas)

QRC: QR code

RBAC: Role-Based Access Control

RLS: Row-Level Security (PostgreSQL policy feature enforcing append-only access on audit tables)

RR: Reserve Ratio

SAR: Suspicious Activity Report

SBT: Soulbound Token (non-transferable ERC standard)

SHAP: SHapley Additive exPlanations

SOFR: Secured Overnight Financing Rate

SSE: Server-Sent Events

SSI: Self-Sovereign Identity

SoK: Systematization of Knowledge

SWC: Smart Contract Weakness Classification

TVL: Total Value Locked

U: Utilization

UUPS: Universal Upgradeable Proxy Standard (ERC-1822)

VC: Verifiable Credential (W3C)

XAI: Explainable Artificial Intelligence

ZKP: Zero-Knowledge Proof

zkAML: Zero-Knowledge Anti-Money-Laundering proof

zkEVM: Zero-Knowledge Ethereum Virtual Machine — a ZK rollup that inherits Ethereum L1 security via cryptographic validity proofs

zk-SNARK: Zero-Knowledge Succinct Non-Interactive Argument of Knowledge

Chapter 1

Introduction

The Crypto World Bank is a Final Year research-based project prototype demonstrating the feasibility of a blockchain-based institutional finance platform on programmable blockchain infrastructure. Where existing decentralized finance protocols address isolated financial functions—Aave handles collateralized lending, Uniswap handles exchange, MakerDAO issues stablecoins—the Crypto World Bank formally specifies a hierarchically governed crypto financial services platform spanning the full spectrum of banking functions. These functions include deposit mobilization, credit allocation, installment-based repayment, interbank liquidity management, foreign exchange facilitation, solidarity group lending for retail borrowers, syndicated co-lending and co-funding by groups of institutional entities, reserve enforcement, and risk-based governance—coordinated across four institutional tiers that mirror the structural logic of real-world development finance. The Tier 1 World Bank Reserve and core lending hierarchy are formally specified in Solidity; the multi-entity contract suite (IBLP, SyndicatedLoan, UpwardDepositFacility, TreasurySwap, TranchedReader, NettingEngine) is fully specified in Chapter 3 and scheduled across Phases II–III.

The system is designed to serve three categories of participants who are systematically underserved by existing DeFi infrastructure—not as end consumers of a blockchain application, but as institutional actors who intermediate between wholesale capital and underserved borrowers. **First**, national development banks and regional financial institutions in emerging markets that depend on correspondent banking for cross-border capital allocation face the \$42-per-transaction settlement cost and 2–5 day delays documented by BIS [25]—not because they lack technology, but because they rely on pre-funded nostro/vostro accounts in opaque bilateral relationships. **Second**, local microfinance institutions (MFIs) and cooperative banks—institutions like those in Bangladesh’s regulated MFI sector, where female-to-male account parity reached 49%/49% in 2025 [R44]—possess formal licensing and KYC infrastructure but lack programmable, auditable credit facilities and reserve management tools. **Third**, the semi-formal clients of those MFIs—borrowers with some digital identity or mobile account access but without structured installment-based credit—are reachable through MFI-sponsored onboarding without requiring the platform to resolve fiat on-ramp infrastructure in the prototype scope. Development-finance literature cites large unbanked and MSME financing gaps [14, 20] as evidence of structural intermediation failure; the CWB contribution targets the institutional layer above end users. Evidence from CBDC deployments targeting financial inclusion—including Nigeria’s eNaira, which saw 98.5% wallet inactivity one year post-launch [161]—confirms that primary barriers are infrastructural and trust-based, not technical, reinforcing the institutional-intermediary design focus. Design artifacts, specifications, and supporting material for this project are available in our GitHub repository.

Blockchain functions here as a coordination technology, providing visibility of shared states, audit trails that are difficult to alter, and programmable enforcement of rules among in-

stitutions that may have competing interests. Governance design, security controls, and regulatory considerations are incorporated into the platform according to a staged implementation roadmap that includes academic validation and regulatory sandbox pilots. Analogous hybrid institutional structures in blockchain-based finance include the World Bank's FundsChain programme, which tracks fund disbursement across projects on a permissioned ledger, and JPMorgan Kinexys, which settles multi-billion-dollar institutional payments daily on a private EVM-compatible chain—demonstrating that open ledger infrastructure and institutional hierarchy are not mutually exclusive.

Figure 1.1 maps the whole-system architecture at a glance: four institutional tiers, the application stack, oracle-mediated analytics, and cross-cutting compliance controls. Detailed layer and component specifications follow in Chapter 3.2.

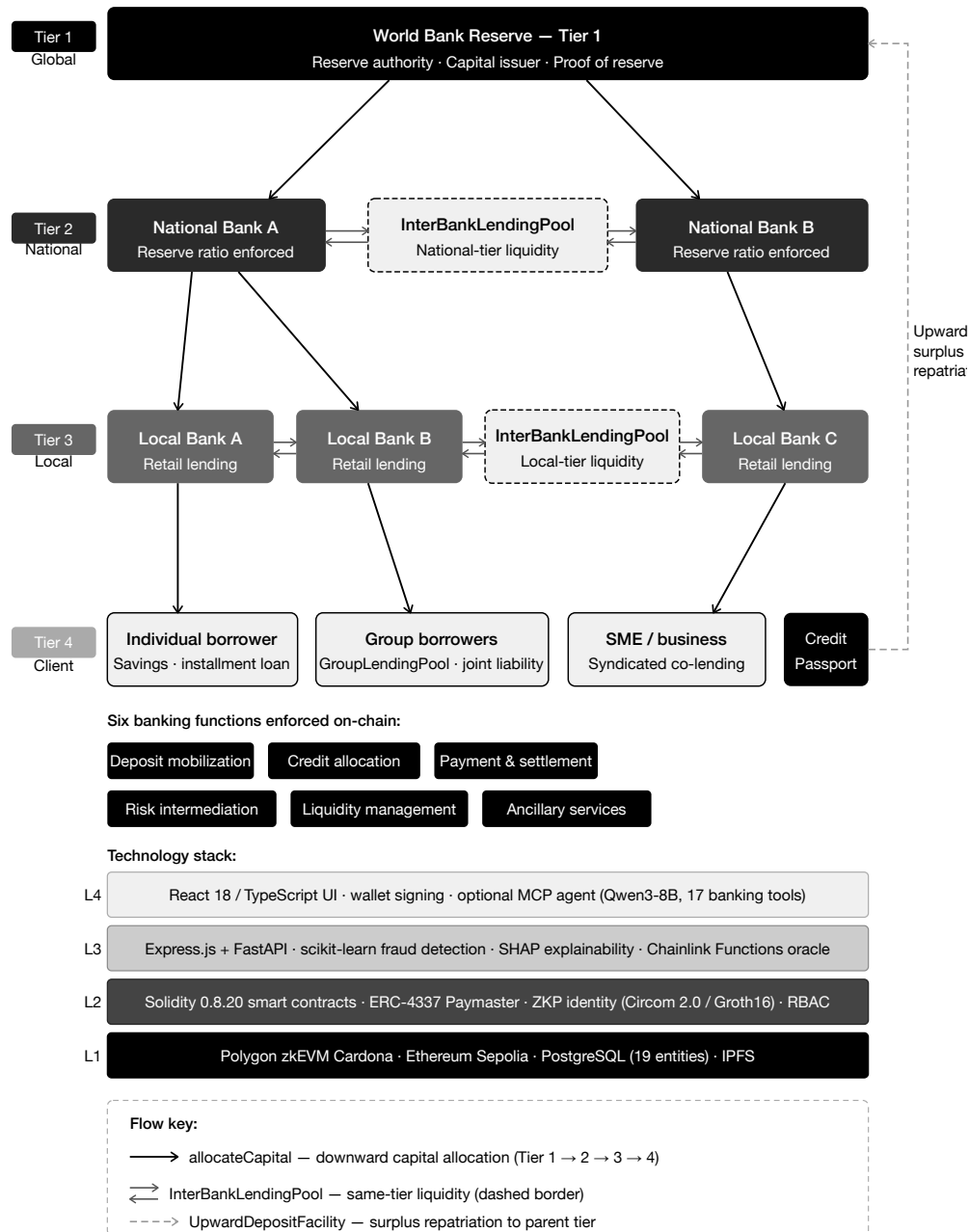


Figure 1.1: Whole-system overview: four-tier capital hierarchy (top) and four-layer technology stack (bottom). Reserve ratios, RBAC, oracle-mediated ML, compliance controls, and modular contracts are specified in Chapter 3.2.

1.1 Background

Understanding why such a system is necessary requires examining the structural failures of the institutions it is designed to complement.

Capital is distributed to local borrowers through layered institutional arrangements in which supranational development institutions such as the World Bank and IMF, national financial intermediaries, and local lenders collaborate to channel development finance to end borrowers. Although this model allows risk sharing and penetrating local markets, it

is linked to great operational issues. In cross-border correspondent banking, financial institutions are obligated to have pre-funded nostro and vostro accounts at correspondent banks in every currency corridor, which maroon huge amounts of capital in idle balances unable to be invested or lent out [24]. Crossborder transactions are regularly settled in two to five business days with an average cost of about \$42 per transaction by using the correspondent banking network [25]. The remittance market, which is valued at about \$860 billion annually, loses between \$48–56 billion per annum to transfer charges, with the average cost to transfer in the world being 6.49% as of 2024, more than the United Nations Sustainable Development Goal limit of 3% [26]. The World Bank Global Findex estimates 1.4 billion unbanked adults [14] and the IFC estimates a \$4.5 trillion annual MSME financing gap [20]—structural indicators of intermediation failure that the institutional actors above are positioned to address, not evidence that unbanked populations can directly access a prototype requiring wallets, KYC, and stablecoin literacy.

The DeFi industry has already shown that lending, collateral management, and interest calculation can be automated on a smart contract platform and that all transactions are transparent, on-chain. The largest DeFi lending protocol, Aave v3, has amassed \$26.3 billion in total value locked across ten blockchain networks [27] (accessed March 2026), and the DeFi lending market has more than \$55 billion in TVL [13] (accessed March 2026). Nevertheless, these protocols utilize flat, pool-based designs where all the lenders feed into one pool and all borrowers withdraw off the same source—irrespective of institutional status, creditworthiness, and geography. There is no current DeFi scheme that reflects the multi-tier institutional structure, inter-tier capital movement, and the client-via-Local-Bank lending path with bank-level syndication for large exposures that is typical of real-world development finance.

What smart contracts and blockchain actually provide. A **smart contract** is a program deployed to a blockchain’s state machine—on the Ethereum Virtual Machine (EVM), this means a deterministic function executed by every validating node simultaneously. Its key properties distinguish it sharply from conventional software: (1) **Governance-bounded execution**: once deployed, contract logic cannot be modified by any single party unilaterally. Modifications require a recorded, multi-signature governance vote subject to a minimum 48-hour timelock delay between proposal and execution (Section 3.18). This prevents flash-loan governance attacks—the attack class responsible for the \$182M Beanstalk exploit (April 2022), in which an attacker acquired a supermajority of governance tokens in a single block and immediately executed a malicious proposal [158]. The CWB governance specification requires: (a) a snapshot of voting power at the proposal creation block, not at execution time; (b) a 60% quorum floor for contract upgrade proposals; (c) a 48-hour TimelockController delay before any upgrade becomes executable; and (d) a 2-of-3 emergency pause multisig that can halt operations but cannot bypass the upgrade timelock. (2) **Atomic settlement**: a multi-step operation—such as releasing a loan, recording a repayment, and updating a reserve ratio—either completes entirely or reverts entirely; there is no half-settled state. (3) **Governance-bounded trustless enforcement**: the contract enforces its rules against *any* caller, including the institution that deployed it, within the bounds of the governance process specified in Section 3.18; no administrator can unilaterally override a reserve-ratio check or bypass a confirmation gate outside that process. (4) **Transparent audit**: every function call, parameter, and state change is permanently recorded on the chain and queryable by any participant.

Blockchain itself adds the coordination layer: it is a replicated, cryptographically linked log that produces *shared state* among participants who may not trust each other. In a correspondent banking network, each institution maintains its own private ledger and reconciles with counterparties through batch messaging and manual reconciliation—the source of the 2–5 day settlement delays and the \$42 per-transaction cost [25]. A shared blockchain ledger replaces this bilateral reconciliation with a single authoritative record: when a Local Bank disburses a loan, the World Bank Reserve’s reserve ratio is updated in the same transaction, with no inter-institution messaging required. This is not merely a database with replication; it is a *coordination mechanism* that removes the need for trusted intermediaries between institutions with potentially conflicting interests.

Decentralized Finance (DeFi) is the application of smart contracts to financial services, eliminating the need for a licensed intermediary to hold assets in custody, process payments, or enforce loan terms. Aave and Compound demonstrate that the core banking function of deposit-funded lending can operate without a bank: depositors supply assets to a pool contract, borrowers draw from the pool, and interest accrues automatically per a utilization-curve formula. The Crypto World Bank extends this pattern in three directions not explored by any existing DeFi protocol. First, it introduces *institutional hierarchy*: rather than a single flat pool, capital flows through a four-tier structure (World Bank → National Bank → Local Bank → Client) in which each tier’s reserve ratio, governance rights, and borrowing limits are distinct and enforced on-chain. Second, it introduces *multi-entity operations*: groups of entities—both retail borrowers and institutional banks—can co-fund and co-lend through programmable solidarity group loans, syndicated loans, tranching pools, and the interbank lending pool, none of which exist in any current DeFi protocol. Third, it connects DeFi’s programmability to the real-world institutional framework of development finance—a combination that the academic literature has identified as structurally absent from the existing DeFi landscape [1].

Optional conversational interface. Primary interaction is through the React UI and wallet signing. Optionally, a locally hosted conversational agent (Qwen3-8B + MCP tool server, 17 banking tools) can answer policy questions and execute banking operations on behalf of retail clients, subject to a mandatory human confirmation gate and EIP-7702 scoped session keys. Every write operation available through the agent is equally available through the web interface; the agent does not gate core banking. Application-layer enforcement (confirmation audit hook, HTTP 403 on missing confirmation) is specified in Chapter 4.

1.2 Rationale of the Study

The combination of three forces, namely: (1) the long-standing inefficiencies in traditional development finance, which restrict transparency and settlement speed; (2) the absence of multi-tier banking functions in existing DeFi designs, particularly deposit mobilization and solidarity group lending alongside hierarchical lending; (3) the possibility of integrating blockchain auditability with lightweight analytics and monitoring support to enhance operational control, drives this study. With an architecture that bridges these areas, we aim to show a plausible path toward transparent, programmable, and institutionally significant banking services for hierarchical finance participants—institutional tiers and retail clients alike.

1.3 Problem Statement

The international system of the development finance works based on a stratified system in which funds are directed down by large supranational organizations such as the World Bank and IMF to national institutions, which transmit them to regional and local banks, and they are ultimately delivered to individual borrowers. This intermediary chain brings into being a number of significant inefficiencies:

1. **Lack of transparency.** Financial records at each tier of the intermediation chain are not openly accessible to participants below that tier. Lower-level institutions and individual borrowers cannot independently verify how capital is being managed, what reserve levels are maintained, or on what basis lending decisions are made at higher tiers. Reserve adequacy is predominantly self-reported and audited at most quarterly, providing no mechanism for real-time verification of institutional financial health.
2. **Sluggish settlements and unproductive capital.** Cross-border money transmission requires multiple intermediaries, compliance verification steps, and correspondent bank confirmations that routinely delay settlement by two to five business days. Banks must also maintain pre-funded accounts at partner institutions in every currency corridor, locking up capital that would otherwise be available for lending [24]. An average transaction via the correspondent banking system costs approximately \$42, a burden that falls disproportionately on small-value transactions and participants in developing economies [25].
3. **Inconsistent risk evaluation.** Fraud screening and credit assessment rely heavily on individual human judgment, introducing bias, inconsistency, and limited scalability. Because no unified and verifiable credit history exists across institutional tiers, borrowers must be re-evaluated from scratch at each stage of the intermediation chain, adding time and cost without improving accuracy.
4. **Obstacles to access and institutional trust.** Establishing inter-institutional trust through legal agreements, compliance processes, and third-party audits is both expensive and time-consuming, placing smaller institutions and borrowers in developing countries at a particular disadvantage. According to the World Inequality Report 2026, the global monetary system continues to redirect resources from developing to wealthier economies, with developing-country investors receiving returns approximately three percentage points lower than those in developed economies [28]—a structural gap that further impedes access to international credit markets.
5. **Absence of programmable reserve-management tools.** National and local MFIs lack real-time, auditable reserve-ratio enforcement and programmable capital-allocation rules. Reserve adequacy is predominantly self-reported and audited at most quarterly, providing no mechanism for wholesale lenders or regulators to verify institutional financial health between audit cycles.
6. **Inaccessibility of programmable group credit mechanisms.** Many borrowers lack individual collateral but could access credit through solidarity group structures demonstrated in microfinance literature (BRAC, Grameen Bank, ASA). No existing DeFi protocol models programmable mutual liability or on-chain group repayment enforcement with MFI-appropriate stress sizing.

At the same time, the DeFi ecosystem has shown that the lending, collateral management, and interest calculation can be completely automated through smart contract platforms

with complete transparency. The TVL of Aave v3 is \$26.3 billion and active borrows are \$17.7 billion [27]; the TVL of Compound v3 is \$1.4 billion [29]; and the combined DAI and USDS stablecoin issuance of MakerDAO (renamed Sky Protocol) is approximately \$10.5 billion with projected 2026 revenue of \$611 million [30]. Despite this scale, these protocols exhibit fundamental limitations in the context of institutional development finance:

- They employ **flat, peer-to-peer, or over-collateralized** lending models. While Aave v3 implements risk-parameterized sub-markets (Isolation Mode, Efficiency Mode) and Compound v3 deploys separate Comet markets with independent risk parameters, none of these protocols model institutional hierarchy, cross-tier capital allocation, or differentiated governance access by institutional role. The CWB's four-tier structure is distinct from Aave's sub-markets: it models the *institutional relationships* between a reserve authority, national lenders, local lenders, and retail clients—not merely risk-differentiated asset buckets within a single lending pool.
- They are not connected with **AI-based risk analytics** and **explainable decision support** systems. DeFi lending decisions are determined solely using collateral ratio and utilization curve and no behavioral fraud detection, anomaly detection or explainable credit judgement is present.
- They do not model **role based, tiered governance** of the development finance institutions. It is a form of leadership, in which a majority of the large token holders will wield the power of decision-making, without the institutional role distinction (regulator, wholesale lender, retail lender, retail client) of hierarchical finance.
- The credit-based and emerging-market lending has also been introduced with other new institutional DeFi projects such as Maple Finance (\$2.6–3.8 billion TVL, undercollateralized institutional lending) [31] and Goldfinch (\$680 million loan originations in 18+ developing countries) [32], however, are single-level projects with no hierarchical capital flows and no interbank lending facilities.

1.4 Objectives

The objectives of this project are:

1. To **design and formally specify** a four-tier lending architecture (World Bank → National Bank → Local Bank → Client) on an EVM-compatible blockchain that maintains institutional hierarchy and provides shared access to the ledger. Smart contract interfaces and the lending request/approval workflow are specified in this pre-thesis; full implementation and testnet deployment are planned for the final thesis phase (Phases I–IV).
2. To specify and justify an extended banking product suite—including deposit mobilization, savings products, and solidarity group lending—that can be integrated on top of the hierarchical lending foundation.
3. To investigate a lightweight off-chain analytics support layer, including fraud detection, anomaly detection, and explainable review support, to supplement human credit governance; and to specify an optional autonomous AI agent system using a Model Context Protocol (MCP) tool server with 17 banking tools (9 read tools and 8 write tools), enabling clients to execute banking operations through a conversational interface as an alternative path to the standard React UI.
4. To specify transparent and programmable lending processes with configurable borrowing limits, installment payments, and role-based access control implemented

through smart contracts.

5. To plan testnet deployment on Polygon zkEVM Cardona (with Ethereum Sepolia retained only as a Certora formal-verification target in Phase IV) and evaluate hierarchical controls, transparency properties, and operational workflows in the final thesis phase.
6. To document the design of governance mechanisms, security controls, regulatory considerations, and a controlled rollout pathway toward academic validation and potential regulatory sandbox piloting.

1.5 Research Questions

The research questions used in the study are as follows:

1. **RQ1:** Does a hierarchical blockchain-based lending architecture reflect more faithfully the real-world capital flow mechanism of development finance than present decentralized lending?
2. **RQ2:** Can on-chain transparency, programmable reserve controls and role-based governance system make settlements less opaque and less frictional across institutional tiers?
3. **RQ3:** Does a lightweight off-chain analytics layer offer practical and auditable support to monitor fraud and lending in such a system?
4. **RQ4:** Can the proposed architecture be deployed on current public test networks such that contracts exhibit the role-based access control, capital-flow, and reserve-invariant properties required for institutional banking use cases? *Scope note: RQ4 evaluates prototype-level technical viability on testnets only; it does not constitute a claim of production readiness, regulatory compliance, or real-asset deployment.*
5. **RQ5:** Can deposit-funded lending and solidarity group lending be represented as programmable, auditable mechanisms on-chain while preserving practical feasibility for retail users?

1.6 Research Contribution

This work makes four original and precisely scoped research contributions to the field of decentralized finance and blockchain-based development banking:

Contribution 1 — Four-Tier Hierarchical DeFi Architecture. No prior published work combines (a) a four-tier smart-contract lending hierarchy (World Bank Reserve → National Bank → Local Bank → Client) with reserve-ratio enforcement at every tier, (b) oracle-mediated explainable ML credit governance, and (c) programmable mutual-liability group lending in a single open architecture. Partial counterexamples exist on individual axes: Maple Finance (pool delegate → borrower, three-tier underwriting), Goldfinch (senior pool → backers → borrowers, two-tier tranching), Morpho V2 (modular curator layer, institutional-scale fixed-term lending), and Aave Arc (permissioned pool with KYC whitelist—reported at only $\approx \$50$ k TVL in February 2026 despite permissioned architecture [R21]). None models multilateral development-finance capital flows with enforced reserve ratios at all four tiers. We extend flat DeFi lending [1] and modular institutional credit [43] to incorporate this hierarchy, addressing the gap identified by Werner et al. [1] and Aspris and Svec (2025) [118], who formalise risk curation as a modular curator layer distinct from flat pool-based lending.

Implementation scope: The World Bank Reserve (Tier 1) contract and the lending request/approval workflow are formally specified in Solidity interface form (Appendices B and D). National Bank (Tier 2) and Local Bank (Tier 3) contracts are likewise specified. Full cross-tier fund-transfer automation and testnet deployment are scoped to Phases I–II of the final thesis phase; Appendix C records the planned deployment manifest template. The architectural contribution—the four-tier design pattern and its formal specification—holds at the pre-thesis stage as a design contribution independent of deployment completeness.

Contribution 2 — On-Chain Solidarity Group Lending. To our knowledge, no prior published academic work specifies a programmable mutual-liability group lending mechanism in Solidity/EVM that encodes group formation consent, installment splitting, and over-indebtedness controls into smart contract logic. Goldfinch’s Backers/Senior Pool model provides partial structural precedent but does not implement per-member mutual liability (Section 3.17.3). The economic foundation is Jaffer’s solidarity-lending mechanics [119]; Sherratt (2016) documents that mutual liability in analogue MFIs can become coercive—a risk the on-chain design mitigates by eliminating human collection pressure but at the cost of losing beneficial social monitoring [120]. **Scope honesty:** the `GroupLendingPool` implements *programmable joint liability with cryptographic enforcement* (closer to cosigning than to BRAC/Grameen solidarity groups); BRAC and Grameen provide motivational context, not a direct technical analogy. Without weekly meetings and field-officer social pressure, on-chain group default rates are modeled conservatively at 15–25% during early deployment; a 20% stress scenario sizes the `InsuranceFund` at 5% of outstanding group principal, with per-member collateral absorbing residual shortfalls—a reduced-collateral, not zero-collateral, product.

Implementation scope: Contribution 2 is a specification contribution: no prior academic work formally specifies the algorithmic structure of solidarity group lending as a smart contract interface. The formal specification is given in Section 3.17.3. A minimal proof-of-concept deployment is scoped to Phase II of the final thesis phase; full lifecycle implementation with mutual liability enforcement and over-indebtedness control is likewise Phase II.

Contribution 3 — Oracle-Mediated AI/ML Integration with Explainability. We propose an architectural pattern for integrating off-chain Random Forest fraud detection, Isolation Forest anomaly detection, and SHAP- plus deviation-report explainability (Section 4.3.2) into an on-chain lending decision workflow via a trusted oracle relay, providing a blueprint for auditable, AI-assisted credit governance in DeFi. The design addresses the oracle problem—the challenge of securely bridging on-chain logic with off-chain data—and specifies three concrete integration options (centralized relay, Chainlink Functions, and commit-reveal scheme) with explicit trust trade-offs. The centralized relay is a Phase I–II development tool only; Chainlink Functions is the production oracle from Phase III onward. Oracle calls are priced at approximately \$0.10–\$1.00 per invocation; at a \$25,000 loan this is negligible, but at a \$500 micro-loan a single \$1.00 oracle call can exceed 0.2% of annual interest—establishing a practical minimum ticket size of approximately \$500 for oracle-mediated ML scoring. Explainability outputs are audience-specific: plain-language decline reasons for retail clients, an Authority Brief dashboard for bank approvers, and a full inference audit trail in `LOAN_RISK_ASSESSMENT` (FK to `MODEL_REGISTRY`) for compliance reviewers; system-wide AML alerts append to `SECURITY_EVENT_LOG`.

Implementation scope: The Random Forest fraud detector is trained and benchmarked on the BCCC-DeFiFraudTrans-2025 dataset [R47] (1,026,867 annotated DeFi transactions, 79

features), with cross-validation against the Elliptic Bitcoin Dataset [R48]. Datasets, training steps, feature mapping, and pre-submission deliverables are documented in Section 4.3.1; explainability is formalised in Section 4.3.2. Oracle wiring to the LocalBank contract via Chainlink Functions is scoped to Phase III; the architectural specification, ML workload plan, benchmark, and explainability design together constitute the Contribution 3 claim at pre-thesis stage.

Extension — Optional autonomous agent architecture. Contribution 3 is optionally extended by a conversational agent add-on (Qwen3-8B + MCP, human confirmation gate, EIP-7702 session keys). The six-step pipeline, Authority Brief for approvers, and `agent_action_log` audit trail are specified in Chapter 4; implementation is a Phase III–IV Could deliverable and does not gate core blockchain functionality.

Contribution 4 — Compliance-Aware ZKP / zkAML Identity Pathway. We design a two-circuit privacy-preserving compliance architecture combining a zk-SNARK KYC verifier (Groth16 proofs via Circom 2.0 + snarkjs) with a zkAML circuit [R19] that proves a wallet has not transacted with sanctioned addresses and remains within velocity limits, without revealing the underlying transaction graph. Both circuits compose with a W3C Decentralized Identifier (DID) and Verifiable Credential (VC) layer [R20] anchoring user identity off-chain. On-chain Soulbound Tokens (SBTs) for credit passports create a complementary privacy risk: repayment history becomes permanently public on-chain. This risk is mitigated by composing off-chain VC storage with ZKP selective disclosure (Contribution 4), so only eligibility proofs—not raw credit attributes—appear on-chain. This positions the platform’s compliance gateway against the broader Self-Sovereign Identity (SSI) framework and is grounded in the Piper et al. (2025) TU Berlin permissioning paper [R8].

Implementation scope: A Groth16 KYC age-range circuit is specified using Circom 2.0 with an auto-generated Solidity verifier planned for Polygon zkEVM Cardona deployment in Phase I (manifest template in Appendix C). The circuit proves knowledge of an age satisfying KYC eligibility without revealing the age on-chain, using a Poseidon hash commitment for proof freshness. The wallet-velocity AML circuit [R19] (IACR ePrint 2025/465, benchmarked at 55 TPS / 226 ms) requires a significantly larger circuit and is scoped to Future Work.

Positioning against institutional DeFi. The Sygnum Bank *Institutional DeFi in 2025* report (February 2026) documents a specific gap: protocols work technically, but pension-fund and endowment allocators do not participate until *legal enforceability*—not technology—is resolved [R21]. The Crypto World Bank is a **planning-phase research prototype** targeting development-finance and MFI use cases (national development banks, local MFIs), not large-cap pension-fund DeFi; those populations operate under different mandates and risk tolerances. Combined with the mBridge / Agora positioning in Section 1.10.3, this reframes the contribution from “an alternative to traditional banking” to “a composable lending-layer specification compatible with emerging CBDC and tokenized-deposit settlement rails”—a substantially more defensible and publishable claim.

The remainder of this paper is organized as follows. Chapter 2 surveys the academic and industry literature on DeFi lending, hierarchical financial architecture, institutional DeFi, AI-assisted credit assessment, and development-finance economics, identifying the research gaps that motivate this work and presenting the synthesis under a PRISMA-style

methodological frame. Chapter 3 presents the system architecture and design, including the smart contract hierarchy, banking product specifications, data model (51 normalized entities), identity and onboarding flows, the formal actor taxonomy and permission matrix, the Chainlink oracle stack, and the compliance and governance framework. Chapter 4 describes the development methodology, the Minimum Viable Thesis checklist (Section 4.1), blockchain-first implementation phase plan, AI/ML datasets and training workload (Section 4.3.1), explainability methodology (Section 4.3.2), optional agent add-on (Section 4.4; harness detail in Appendix B), formal-verification strategy, and technology stack justification. Chapter 5 evaluates the technical, economic, regulatory, and sustainability feasibility of the proposed system, including the revenue stream taxonomy. Chapter 6 presents conclusions and future research directions.

1.7 Blockchain Justification

The root cause of the problems identified above is a **multi-party coordination and trust problem**: institutions with potentially conflicting interests must share financial state, enforce shared rules, and settle obligations—without being able to fully trust one another or a central administrator. Traditional solutions to this problem require either a trusted central authority (a central bank, a clearinghouse, a custodian) whose failure or misconduct creates systemic risk, or an elaborate network of bilateral legal agreements and reconciliation processes that introduce the 2–5 day settlement delays and \$42 per-transaction costs documented above.

Blockchain technology directly resolves this coordination-and-trust problem through four properties that have no analog in conventional database or cloud architecture:

1. **Consensus-enforced shared state.** All participants observe the same ledger, updated by distributed consensus rather than by any single operator. In the CWB context, this means a Local Bank, its parent National Bank, and the World Bank Reserve all read the same reserve ratios and loan states in real time—not reconciled copies maintained independently and aligned through batch messaging.
2. **Programmable, governance-bounded rule enforcement via smart contracts.** The lending rules—reserve ratios, interest rate curves, borrowing limits, approval workflows, multi-entity consent requirements—are deployed as code on the EVM. Once deployed, modifications require the governance process in Section 3.18 (48-hour timelock, 60% quorum, snapshot voting). Within those bounds, rules execute *identically for every participant and cannot be selectively overridden by any party*, including the institution that deployed the contract.
3. **Cryptographic auditability.** Every state change—every loan disbursement, every repayment, every reserve ratio update—is permanently recorded in a tamper-evident, publicly verifiable transaction log. Regulators, partner institutions, depositors, and auditors can independently verify the platform’s financial state without relying on self-reported figures or periodic audits. Real-time reserve transparency eliminates the informational asymmetry that allows institutions to conceal insolvency between audit cycles.
4. **Composable incentive structures and atomic multi-party settlement.** On-chain programmability allows multi-party financial operations—group loan disbursements, syndicated repayments, netting settlements—to be encoded as single atomic transactions that either complete fully or revert entirely. There is

no half-settled syndicated loan, no partial group disbursement, no settlement risk between the moment funds leave one institution and arrive at another.

A conventional cloud-based database architecture could store the same loan records, but it would require a trusted central operator with administrator access—re-establishing precisely the single point of trust failure this project is designed to eliminate. Blockchain is not chosen here because it is novel; it is chosen because the specific trust, auditability, and programmable enforcement properties it provides are the correct solution to the specific multi-institution coordination problem this system is designed to solve.

1.7.1 Blockchain Fundamentals and the EVM Execution Model

This subsection provides a concise technical reference for the blockchain primitives on which the Crypto World Bank is built. Readers familiar with EVM architecture, smart contract state machines, and oracle design may proceed directly to Section 1.8.

A. What blockchain is (technically precise). A blockchain is an append-only distributed ledger in which each block contains a cryptographic hash of the previous block, a Merkle root of all transactions in the block, a timestamp, and nonce or validator signature data. Immutability is not absolute—it is probabilistic and consensus-enforced: it holds as long as no coalition controlling more than 50% of stake (in Proof-of-Stake) cooperates to rewrite history [R17]. On Polygon zkEVM Cardona—the primary deployment target—security derives from ZK validity proofs verified on Ethereum L1 rather than a PoS validator-set assumption; Polygon Amoy PoS (Heimdall/Tendermint BFT) is retained only as a fallback testnet for compatibility testing (Section 1.7.1, item E).

B. The EVM Execution Model (Ethereum Yellow Paper [58]). The Ethereum Virtual Machine is a stack-based, deterministic, Turing-complete virtual machine that executes smart contract bytecode. Every operation (opcode) has a fixed gas cost specified in the Ethereum Yellow Paper [R7]. Key opcodes relevant to this system:

- **SSTORE** (write to storage): 20,000 gas for a new slot, 2,900 for modification—the most expensive operation class.
- **SLOAD** (read from storage): 2,100 gas (warm access), 100 gas (cold).
- **CALL** (external contract call): 700 gas base plus additional costs.
- **Events (LOG2, LOG3)**: $\approx 375 + 8$ per data byte—inexpensive, used for off-chain indexing.

Each installment repayment involves at minimum one **SSTORE** for updating the loan balance, one **SSTORE** for recording payment history, and one event emission. A 12-installment loan therefore involves at minimum 24 **SSTORE** operations. A complete retail loan lifecycle involves approximately 27–32 individual on-chain *state changes* (**SSTORE** writes and event emissions), versus 10–15 user-facing *transactions* (loan request, approval, disbursement, and installment payments)—the figures are reconciled in Section 5.8.

C. Smart contracts as state machines. A lending smart contract is best modeled as a finite state machine: **PENDING** → **APPROVED** → **ACTIVE** → **REPAYING** → **COMPLETED** | **DEFAULTED**. State transitions are triggered by role-authorized function calls, and the `nonReentrant` guard ensures atomicity of each transition. This state machine framing strengthens the argument that smart contracts are more auditable than off-chain systems, because every state transition is permanent, timestamped, and publicly verifiable.

D. The Oracle Problem. Smart contracts are deterministic and isolated from external data by design. This is a feature—consensus would break if different nodes observed different external states—but it creates a fundamental architectural gap for any system that requires off-chain data. The oracle problem, formally identified by Beniiche (2020) [R1] and Pasdar et al. (2023) [R2], is the challenge of securely bridging on-chain logic with off-chain information. For the Crypto World Bank, the AI/ML risk scores produced by the FastAPI Random Forest service are off-chain data that must influence on-chain lending decisions. Three architectural options exist: (1) a centralized relay (trusted backend calls `updateRiskScore` on-chain—a Phase I–II development tool only); (2) a decentralised oracle network such as Chainlink Functions (production oracle from Phase III; 30–60 seconds latency and \$0.10–\$1.00 per call; the DON requires consensus across multiple independent nodes before a score is committed); and (3) a commit-reveal scheme (the ML service commits a hash of the risk score before the loan decision window, reveals it after). The pre-thesis design uses the centralized relay for early integration testing; Chainlink Functions is the primary production mechanism from Phase III.

An empirical study of Chainlink price-feed accuracy across eight blockchain networks, using 50 million data points collected in December 2024, finds that accuracy varies meaningfully by chain [157]: Polygon achieves consistently high accuracy, while Ethereum mainnet and zkSync exhibit greater deviations under market volatility. This measurement result supports consolidating all live deployment on Polygon zkEVM Cardona (single-chain architecture, Section 3.13) and quantifies the oracle accuracy trade-off that the commit-reveal fallback is designed to hedge.

In addition to ML score commitment, Chainlink Price Feeds supply fiat/USD pairs (e.g. EUR/USD) and ETH/USD rates to the `FXModule` contract, Chainlink Automation replaces the centralised cron job for overdue installment detection, and Chainlink Proof of Reserve publishes the `WorldBankReserve` balance for external cryptographic verification.

E. Polygon zkEVM Trust Model. Polygon zkEVM Cardona is the primary deployment target (Section 3.2). Security derives from ZK validity proofs verified on Ethereum L1 rather than a PoS validator-set assumption—a materially stronger claim for institutional reserve enforcement than Polygon Amoy PoS, retained only as a fallback testnet (design-decisions table, Section 4.3).

1.8 Proposed Solution

The Crypto World Bank is based on a **four tier on-chain lending model**:

- **Tier 1 — World Bank:** Custodian of the global crypto reserve. Allocates capital to registered National Banks via lending mediated by smart contracts.
- **Tier 2 — National Banks:** Borrow from the World Bank reserve. Lend to Local Banks incorporated in their jurisdiction. Aggregate the risk exposure at the national level.
- **Tier 3 — Local Banks:** Borrow from National Banks. Process loan applications submitted by retail clients. Administer the full loan lifecycle using designated approvers.
- **Tier 4 — Clients (Retail):** Submit loan requests to Local Banks. Repay through configurable installment schedules. Build an on-chain repayment history that determines future credit access. Includes individual retail clients, group lending members, and small-business clients.

The platform also incorporates:

- **Risk-sensitive borrowing limits** computed from rolling 6-month and 1-year transaction windows.
- **Automatic installment generation** for loan amounts exceeding a configurable threshold (e.g., 100 ETH equivalent).
- **Off-chain AI/ML security analytics** (planned) for fraud detection (Random Forest), anomaly identification (Isolation Forest), and explainable risk assessment (SHAP).

1.8.1 Banking Functions of the Platform

The platform implements six standard banking functions across the four-tier hierarchy. Reserve ratios, RBAC, and tier spreads are enforced on-chain rather than by discretionary audit (Figure 1.2).



Figure 1.2: Six banking functions enforced across the four-tier core. Contract mappings (SavingsVault, allocateCapital, IBLP, etc.) are listed in the bullets below and in Chapter 3.2.

- **Deposit mobilization** — SavingsVault records claims and checks reserve ratios on every withdrawal.
- **Credit allocation** — downward allocateCapital plus SyndicatedLoan for multi-bank co-funding.
- **Payment & settlement** — atomic transfers; no nostro/vostro in-transit state.
- **Risk intermediation** — reserve buffers, approver workflows, LiquidationEngine, oracle ML (RF, Isolation Forest, SHAP).

- **Liquidity management** — InterBankLendingPool (same-tier) and UpwardDepositFacility (surplus repatriation).
- **Ancillary services** — FX oracles, GroupLendingPool, syndicated/tranched products, W3C DID/VC identity.

1.8.2 Service Delivery Across Scale

The platform is designed to operate at three scales of engagement: the global institutional level, the national and regional commercial level, and the individual retail level.

Global institutional level (World Bank Reserve tier). The World Bank Reserve functions as a programmable reserve authority by managing system-level capital allocation and reserve constraints across jurisdictions. This tier is designed to support transparent reporting of reserves and enforce reserve-ratio constraints as code, reducing informational asymmetry relative to periodic self-reporting.

National and regional level (National Bank and Local Bank tiers). National Banks borrow wholesale capital from the reserve tier and allocate it to registered Local Banks under jurisdiction-aware governance and tiered approval workflows. The intent is to mirror real-world development finance intermediation while improving settlement speed and auditability through on-chain execution.

Retail level (clients and depositors). End customers interact through Local Bank interfaces to open accounts, deposit savings, apply for loans (including group loans), repay installments, and build an on-chain credit history. A mobile-accessible interface with low transaction costs can reach users underserved by branch-based systems, while keeping loan terms and reserve behavior verifiable through on-chain transparency.

1.8.3 Cross-Tier Lending System

Capital in real banking systems does not flow only downward. Banks at the same tier lend to one another daily in the interbank market (the federal funds market in the US, SONIA in the UK, SOFR globally), surplus institutions deposit excess reserves upward with central banks or parent institutions, and large loans are co-funded by syndicates of several banks acting together—because no single bank should carry the full credit risk of a major project. The Crypto World Bank encodes all three of these flow directions as on-chain contracts, producing a **multi-directional, multi-entity capital coordination system** that reflects the structural reality of institutional banking.

Figure 1.3 maps the four coordinated flow directions onto the four-tier hierarchy. Downward allocation (`allocateCapital`) distributes wholesale capital from the World Bank Reserve through National and Local Banks to retail clients. Same-tier InterBankLendingPool contracts balance short-term liquidity among peer National Banks and among peer Local Banks. The UpwardDepositFacility repatriates surplus reserves upward at deposit yields below downward lending spreads. Syndicated, group, and tranched contracts (`SyndicatedLoan`, `GroupLendingPool`, `TranchedPool`) let multiple entities co-fund or co-borrow a single exposure without bypassing tier governance. Section 3.14 specifies each contract; the literature synthesis in Section 2.2.3 situates these patterns against CBDC operating-model and cross-tier finance research.

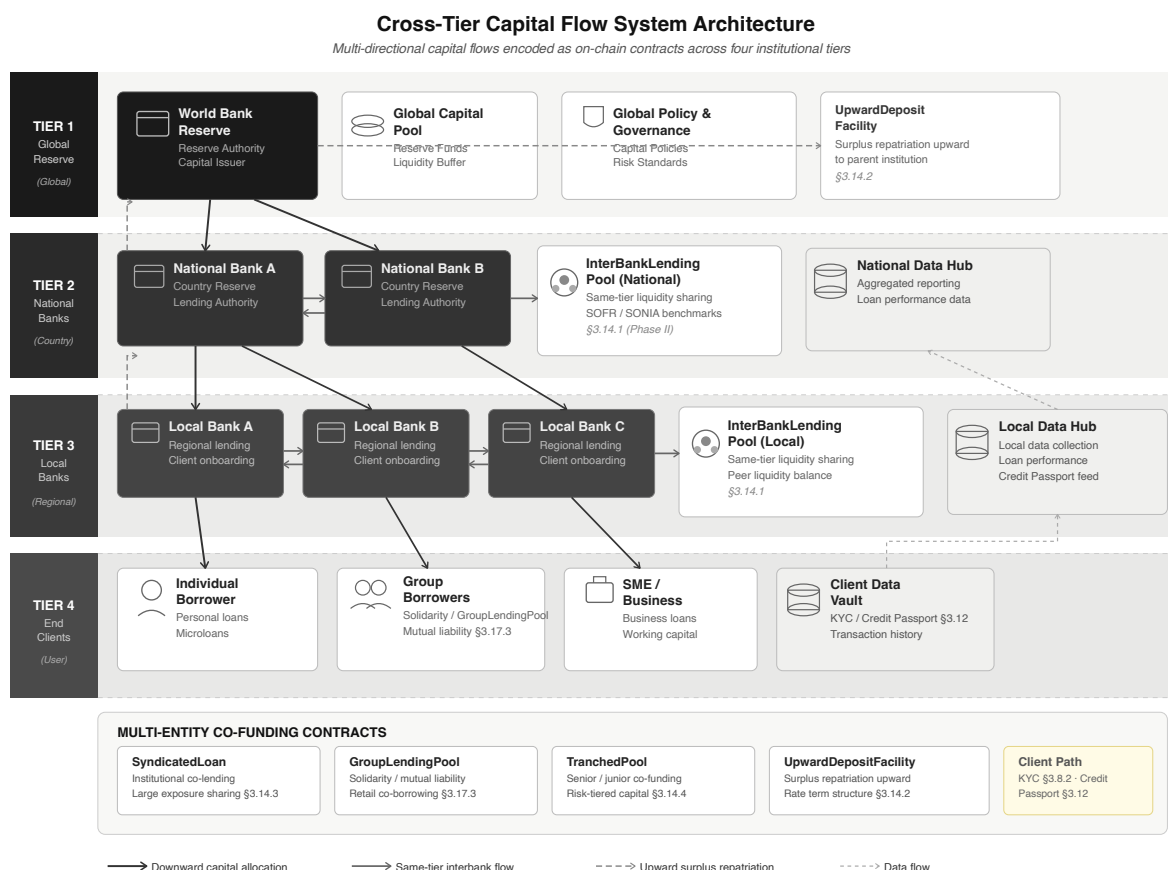


Figure 1.3: Cross-tier lending: downward allocation through four tiers, upward surplus repatriation, and same-tier interbank pools. Syndicated, group, and tranching are specified in Section 3.14.

Can a group of entities lend or fund together? Yes—and this capability is a distinct architectural feature of the platform, not an afterthought. The platform supports group co-funding at *both* the retail and institutional level through separate but structurally analogous mechanisms:

- **Institutional group co-lending (Syndicated Loans, Section 3.14.3):** Multiple banks at the same or adjacent tier—for example, three Local Banks and their parent National Bank—can co-fund a single large loan through the SyndicatedLoan contract. Each co-lender commits capital during a subscription window, signs an on-chain consent vote, and receives interest repayments pro-rata to its capital share. The Lead Arranger structures the deal and earns a small underwriting fee. This is the on-chain equivalent of a real-world syndicated loan or club deal.
- **Retail group co-borrowing (Solidarity Group Loans, Section 3.17.3):** Three to twenty retail clients can form a lending group, pooling collateral into a shared contract and signing a joint application. The GroupLendingPool enforces mutual liability: if one member defaults, the shared collateral pool covers the shortfall, protecting the lender. Default rates are modeled at 15–25% early deployment (20% stress scenario); the InsuranceFund is sized at 5% of outstanding group principal with per-member collateral required at formation—a reduced-collateral, not zero-collateral, Grameen analogue.
- **Senior–junior co-funding pools (Tranched Pools, Section 3.14.4):** Risk-tolerant

and risk-averse entities co-fund the same loan pool with different seniority rights—the junior tranche takes first loss, the senior tranche is protected. This is the structural pattern used by Goldfinch and Maple Finance for institutional DeFi lending and is incorporated into the CWB architecture.

- **Surplus repatriation (Upward Deposit Facility, Section 3.14.2):** Groups of Local Banks holding surplus reserves can collectively deposit upward into their parent National Bank, which aggregates those deposits and re-deploys them where capital is needed most—a form of group capital pooling that flows upward rather than downward.

These four mechanisms together provide a complete answer: the platform is explicitly designed for groups of entities—both retail borrowers and institutional banks—to lend, fund, and borrow together through programmatic on-chain coordination.

Same-Tier Lending

Banks at the same hierarchical level can lend to one another to manage short-term liquidity:

- **National Bank ↔ National Bank:** A national bank, which has excess reserves, can lend to a second party that is in a liquidity crunch, similar to the traditional interbank lending market. The lending rate is adjusted according to the supply and demand through national banks of the system, like the Secured Overnight Financing Rate (SOFR).
- **Local Bank ↔ Local Bank:** Local banks in same or different national jurisdictions can share liquidity through a peer lending pool. This prevents local liquidity crunches when one bank possesses surplus deposits and the other is experiencing elevated loan demand.

Same-tier lending is done by an on-chain **InterBankLendingPool** contract at every tier level in which surplus banks provide short-term liquidity and those banks that need it are allowed to borrow at utilization floating rates. Section 3.14.1 specifies this pool fully (rate model, settlement, default cascade, operations); the present chapter introduces only the role it plays in the four-tier hierarchy. The downward flow (World Bank → National Bank → Local Bank → Client) is specified in this pre-thesis; the InterBankLendingPool is planned for Phase II of the final-thesis phase (stretch beyond MVT item 2; see Table 4.2).

Upward Lending

Banks with excess capital of lower tier are then able to lend up the ladder:

- **Local Banks → National Banks:** When local banks accumulate reserves beyond the minimum reserve ratio, they are allowed to deposit excess capital in their parent national bank's liquidity pool with earned deposit yield. This mirrors how commercial banks save the surplus with central banks by means of standing deposit facilities.
- **National Banks → World Bank:** National banks may provide excess to the international deposit, enhancing the total capital base of the system. This provides a back route to well capitalized national banks and raises the reserve available for downward distribution to banks with higher demand.

The downward lending rates are higher than the upward lending rates (the low risk of lending to a higher-level institution), developing a natural interest rate term structure across the hierarchy.

Client Lending Path

All end borrowers—retail clients, group members, and small businesses—submit loan applications through a registered Local Bank (Tier 3). Clients do not interact with National Banks or the World Bank Reserve directly; capital reaches them only after downward allocation through the institutional hierarchy (World Bank → National Bank → Local Bank → Client). Loan size and creditworthiness are constrained by KYC level (Section 3.8.2), Credit Passport tier (Section 3.12), and on-chain borrowing limits—not by a separate client-to-tier routing matrix. When a Local Bank’s exposure to a single obligor exceeds its prudent lending ceiling, the bank structures a **syndicated loan** via the SyndicatedLoan contract (Section 3.14.3), co-funding with peer Local Banks and/or its parent National Bank. Large institutional or sovereign programs are likewise originated at the bank tier through syndication and interbank facilities, preserving hierarchical intermediation without bypass paths for end clients.

This multi-way lending scheme—downward capital allocation, same-tier interbank lending, upward repatriation of surpluses, and bank-level syndication for large exposures—generates a more detailed model of the banking flows of the real world in the decentralized architecture. Downward capital distribution and the Local-Bank client entry point are specified in Phase I–II; same-tier interbank lending and upward surplus repatriation are specified in full in Section 3.14 (InterBankLendingPool, UpwardDepositFacility) and scheduled for implementation in Phase II of the final-thesis phase. The genuinely multi-entity operations that build on top of these flows—syndicated lending, tranching pools, cross-tier treasury FX, and multilateral settlement netting—are likewise specified at full depth in Section 3.14 so that the paper describes a complete banking system rather than only retail-facing lending.

1.9 Methodology in Brief

Our approach combines a lightweight Agile delivery process with an explicit Software Development Life Cycle (SDLC) frame. Pre-Thesis 1 covers the requirements, architecture, and design phases; the implementation, verification, and validation phases run across four subsequent phases during the final thesis stage. Table 1.1 maps each phase to its primary deliverables and target networks.

Table 1.1: Four-phase implementation plan: deliverables, smart-contract scope, and deployment targets.

Phase	Primary Deliverables	Contract Scope	Deployment Target
I	Four-tier scaffold, wallet auth, frontend skeleton, PostgreSQL schema (31 core entities), Chainlink Proof of Reserve, KYC circuit (Gnuthash)	WorldBankReserve, NationalBank, LocalBank	Polygon zkEVM Cardona
II	Loan application, approval, installment generation, borrowing limit enforcement, deposit mobilization, GroupLendingPool, InterBankLendingPool	SavingsVault, LoanContract, GroupLendingPool, UpwardDepositFacility	Polygon zkEVM Cardona
III	Random Forest + Isolation Forest + SHAP pipeline (Chainlink Functions oracle), SyndicatedLoan, TranchingPool, NettingEngine, optional MCP agent, Foundry invariant suite, Tenderly monitoring	SyndicatedLoan, TranchingPool, NettingEngine, FXModule, LiquidationEngine	Polygon zkEVM Cardona
IV	Certora reserve-invariant formal proofs, zkAML velocity circuit, full audit pipeline	All contracts, zkSNARK verifier on-chain	Polygon zkEVM Cardona; Ethereum Sepolia (Certora only)

The technology stack is Solidity 0.8.20 (smart contracts), React 18 with TypeScript and ERC-4337 account abstraction (frontend), Express.js + FastAPI (backend), scikit-learn + PyTorch Geometric (ML), and Circom 2.0 + snarkjs (ZKP); optionally ChromaDB with a locally hosted 8B-parameter LLM (RAG assistant, Phase III–IV Could deliverable). All planned live deployments target Polygon zkEVM Cardona; Ethereum Sepolia is retained only as a Certora formal-verification target in Phase IV. Stablecoin pools use testnet USDC.

1.10 Scopes and Challenges

The complete banking architecture described in this report defines both what is specified at pre-thesis stage and what constitutes the full system design, clearly distinguishing planning-phase scope from architectural intent throughout.

Scope. At pre-thesis stage, the contribution is formal specification and a four-phase implementation plan. Planned testnet scope encompasses four-tier lending architecture, loan request and approval, installment-based repayment, borrowing limits, and oracle-mediated fraud analytics (Random Forest + SHAP, Isolation Forest). The specification does not cover mainnet deployment, client-to-client remittance, merchant checkout, fiat on/off-ramp integration, production KYC/AML processing, or production-scale stress testing; peer-reviewed and policy literature supports these retail extensions as studyable post-MVT work (Section 2.2.8), but they are Future Work or final-thesis deliverables (Section 6).

Challenges. There are some issues that come with this system. The unlabeled nature of DeFi lending fraud data is one of those problems, as it limits the successful training of the model and might necessitate the application of synthetic data or transfer learning methods. The other issue is regulatory uncertainty, since crypto lending may have different jurisdictional restrictions; and this risk can be partially reduced by working the testnets only. The sensitivity of gas costs is an issue as well, as intricate on-chain programs can be costly; to solve this, AI/ML work requiring computation is performed off-chain. Additionally, model interpretability is essential to regulatory compliance, especially in describing the process of lending, which is obtained by SHAP-based feature attribution. Last but not the least, economic sustainability should be taken into account, because the platform should be able to earn enough income, like interest revenues, to cover gas expenses at all four levels. The following Chapter 5 discusses this balance in greater depth.

1.10.1 Economic Sustainability: Interest Revenue Versus Gas Costs

The most important problem facing any blockchain-based lending system is whether the revenue earned by the lending activities will be in a sustainable position to cover the total gas expenses of the transactions on-chain. Every level in the hierarchy will have deposits, loan applications, approvals, disbursements, repayment and installment processing gas fees. One complicated smart contract interaction can cost between \$5 and \$50 on Ethereum mainnet, based on network congestion, but around \$0.001 to \$0.01 on Polygon zkEVM Cardona. At an average retail loan of \$25,000 USDC with an 8% APR, the client pays \$2,000 in annual interest. The loan lifecycle comprises 10–15 user-facing transactions (request, approval, disbursement, monthly installments) but approximately 27–32 on-chain state changes when SSTORE writes and event emissions are counted (Section 5.8); overall gas on Polygon remains under \$0.15—less than 0.01% of interest obtained. Oracle-mediated ML scoring adds \$0.10–\$1.00 per Chainlink Functions call; this is negligible above approximately \$500 minimum ticket size but material for smaller loans. The platform thus provides a sustainable margin on Layer 2 networks, but mainnet implementation would need gas optimisation or batched processing for micro-loans.

1.10.2 Resistance to Institutional Capture

As the Crypto World Bank grows in adoption, a critical question is whether large capital holders and established financial institutions—which already dominate the traditional banking system—could migrate to the new architecture and replicate the existing patterns

of financial concentration. This risk is mitigated by the platform in a variety of architectural ways:

- **Visible reserve ratios:** In contrast to traditional banks, where reserve adequacy is reported and audited quarterly, on-chain reserves are publicly verifiable in real time. There is no way a given institution can hide its real financial status to have unfair advantage.
- **Algorithmic interest rates:** Interest rates are determined by the smart contract parameters, not by opaque internal pricing committees, and thus no single participant can capture the entire pool of capital.
- **Tamper-evident audit trails:** Every transaction, approval, and governance decision is permanently recorded on-chain, making regulatory capture and corruption detectable by any participant.
- **Open-source governance:** The smart contract code is publicly auditable, preventing hidden backdoors or preferential treatment encoded at the protocol level.
- **Borrowing limits with caps:** Per-client borrowing limits (KYC ladder, Credit Passport, rolling windows) and bank exposure ceilings are enforced programmatically, preventing any single entity from monopolizing the capital pool. Large exposures are syndicated at the bank level rather than routed through direct client access to higher tiers.

Although no system can entirely deter wealth concentration in a free market, the detectability and programmability characteristics of blockchain infrastructure make exploitative behavior prohibitively expensive and highly observable by comparison to opaque intermediaries.

1.10.3 Stablecoin-First Lending and Supply Scalability

Denomination matters more than any other design choice for a crypto financial services platform serving retail borrowers. A retail client who takes a 0.5 ETH micro-loan during a period of price stability and is later required to repay the same 0.5 ETH after a 50% draw-down effectively faces a $2\times$ increase in the real value of their debt. ETH has experienced exactly this drawdown twice in the past five years (May 2021 and 2022), and the stablecoin literature has formalized the resulting liability-side volatility risk for crypto-asset borrowers [R12, R13]. The Crypto World Bank therefore elevates **stablecoin-denominated lending** from a future-work footnote to an explicit design requirement at every retail tier. As of April 2026, USDC commands a circulating supply of approximately \$77.3 billion and USDT approximately \$189.5 billion (DeFiLlama stablecoin tracker, accessed April 2026), confirming that stablecoin-denominated transactions have achieved dominant adoption across consumer DeFi applications—a market signal that validates the platform’s denomination choice.

The planned testnet deployment will use ETH as the initial denomination for simplicity, but every loan-bearing contract is designed so that the loan currency is a per-pool parameter rather than a hard-coded asset. In the final thesis phase, the retail Local Bank tier will deploy with USDC as the default loan currency; ETH remains acceptable only for institutional Tier 1 / Tier 2 operations where the institutional client has the balance-sheet capacity to hedge crypto-asset exposure.

Three regulatory anchors shape this choice. First, the EU Markets in Crypto-Assets (MiCA) Regulation, fully in effect since December 2024, classifies the kind of fiat-pegged tokens

used here as electronic-money tokens (EMTs) and requires 1:1 reserve backing, authorization before public offering, and regular audits [R36, R37]. Second, the United States GENIUS Act, signed into law in July 2025, introduces the first federal framework for payment stablecoins and codifies a similar reserve and disclosure regime [R38]. Third, the BIS *mBridge* platform reached Minimum Viable Product status in mid-2024 and now settles real-value cross-border CBDC transactions among China, Hong Kong, Thailand, the UAE, and Saudi Arabia, while the BIS / central-bank *Project Agora* explores tokenized bank deposits on a unified ledger [R39, R40]. Neither initiative implements a retail lending hierarchy: they are settlement rails. The Crypto World Bank is positioned as the lending layer that can compose on top of such institutional settlement infrastructure rather than as a replacement for it.

The remaining supply-side concerns are addressed by three complementary measures:

1. **Stablecoin pool support.** Each Local Bank deploys an ERC-20 stablecoin loan pool (USDC primary, USDT and DAI as fallbacks). The InsuranceFund and SavingsVault contracts are denominated in the same unit so depositors, borrowers, and the reserve share a single numeraire.
2. **Single-chain Layer 2 deployment.** An initial design considered multi-chain deployment bridging Polygon zkEVM Cardona and Ethereum Sepolia; this was revised after security analysis identified cross-chain bridges as the highest-risk DeFi attack surface (Ronin, Wormhole, Nomad exploits collectively exceeded \$1.1B in 2022). All live contracts deploy on Polygon zkEVM Cardona, whose ZK validity proofs are verified on Ethereum L1 without a live asset bridge. Ethereum Sepolia is retained only for Certora formal verification in Phase IV (Section 3.13).
3. **Capital circulation via the four-tier hierarchy.** The Reserve Ratio (RR) enforced at every tier functions as a solvency constraint, not a money-creation parameter. USDC is 100%-reserve backed; the platform re-allocates existing stablecoin supply rather than creating new money. The rate at which this fixed capital pool flows through the hierarchy and returns as repayment is measured by Credit Velocity (Formula CV). Higher credit velocity—not money multiplication—is the platform’s capital-throughput mechanism.

The supply concern that originally motivated this subsection is therefore reframed: scarcity is a property of base-layer crypto-assets, not of the lending denomination. Fiat-backed stablecoins issued under MiCA-/GENIUS-compliant reserves provide the elastic, audit-friendly supply a retail banking platform requires, while the hierarchical reserve architecture promotes capital velocity across tiers. Client-side currency risk is removed, deposit-side regulatory compliance is preserved, and the protocol does not depend on the monetary discretion of any single issuer.

1.11 Market Analysis and Partnership Ecosystem

1.11.1 Market Sizing

The platform addresses three nested market segments. The Total Addressable Market (TAM) spans global DeFi lending (exceeding \$55B TVL [13]) plus institutional settlement and correspondent-banking inefficiencies [21, 26]. The Serviceable Addressable Market (SAM) is institutional and semi-institutional lending requiring hierarchical structures (\$5–15B), and the immediately Obtainable Market (SOM) covers academic testnet deployments,

regulatory sandbox pilots, and institutional DeFi research partnerships (\$50–200M). Full market sizing with data sources is presented in Table 5.1 in Chapter 5.

1.11.2 Target Customer Segment

The Crypto World Bank targets Tier 1–3 institutional participants (national development banks, regional MFIs, local cooperative banks) and Tier 4 retail clients onboarded through MFI-sponsored pathways on Polygon zkEVM Cardona. The hierarchical architecture (World Bank → National → Local → Client) mirrors development-finance intermediation while providing blockchain-enforced transparency and oracle-mediated risk assessment. Structural unbanked and MSME financing statistics [14, 20] inform the literature context for intermediation failure; they do not define the prototype’s primary deployment target. Full client segment profiling is presented in Chapter 5 (Table 5.3).

1.11.3 Partner Ecosystem

The platform’s deployment requires five partner categories: financial regulators (regulatory sandbox approval), banking institutions (National/Local Bank network members), oracle and infrastructure providers (Chainlink, testnet operators), academic and research institutions (formal verification and ML validation), and optional field-pilot partners under sandbox cover. Blockchain-mediated incentives align each partner’s interests with platform operation through on-chain transparency, reduced settlement friction, and programmatic fee distribution. Full partner ecosystem detail is in Chapter 5 (Table 5.5).

1.11.4 Incentive Alignment Through the Blockchain Platform

The system manages and enforces partner incentives directly on the blockchain, making the process more transparent and reliable:

- **Tamper-evident repayment records** serve as an on-chain reputation system through permanent repayment records, allowing credit decisions to be based on actual data without relying on external credit bureaus.
- **Transparent reserve verification** enables on-chain reserve verification, so all participants can independently confirm that allocated funds are being used as intended.
- **Programmable fee structures** (with potential for future expansion) allow transaction fees to be distributed fairly among network participants according to their roles and contributions.

Chapter 2

Literature Review

2.1 Preliminaries

Key technical terms used throughout this thesis—including Decentralized Finance (DeFi), smart contracts, over-collateralization, Explainable AI (XAI), Proof of Stake (PoS), CBDC, Solidarity/Group Lending, Asset-Liability Management (ALM), Flash Loans, Zero-Knowledge Proofs (ZKP), and Health Factor (HF)—are defined in the **Glossary** appendix. The Glossary also includes platform-specific terminology introduced in this thesis (e.g., *Platform Solvency Ratio (PSR)*, *Provisional Credit Tier*, *Cold-Start Credit Pathway*). This chapter assumes familiarity with blockchain fundamentals at the level of Section 1.7.1; other terms are defined on first use.

2.1.1 Review Methodology (PRISMA-Style Frame)

The literature underlying this thesis is synthesized under a PRISMA-style evidence-synthesis frame [R22] rather than as a narrative summary alone. Inclusion criteria were: (i) peer-reviewed publications or formally published institutional reports issued between 2017 and 2026; (ii) direct relevance to at least one of eight scoped CWB modules—DeFi lending mechanics, ML-based fraud / anomaly detection, ZKP / DID-based identity, group lending and microfinance (literature precedent), smart-contract security, institutional DeFi / development-finance economics, regulatory compliance, or retail payments / remittance / fiat on/off-ramp rails; and (iii) sufficient methodological transparency to allow re-use of design parameters (e.g., reserve invariants, gas accounting, F1 scores). Sources were identified through ACM Digital Library, IEEE Xplore, Elsevier ScienceDirect, MDPI, arXiv (with venue check), and the BIS / World Bank publication portals. Figure 2.1 summarises the screening stages from identification through inclusion. Recent systematic reviews of blockchain in banking using PRISMA on 38 studies confirm that this methodological frame is the current expected standard for journal-quality synthesis [R23].

The ten most influential sources are integrated into the subsections below with their design impact noted at point of use.

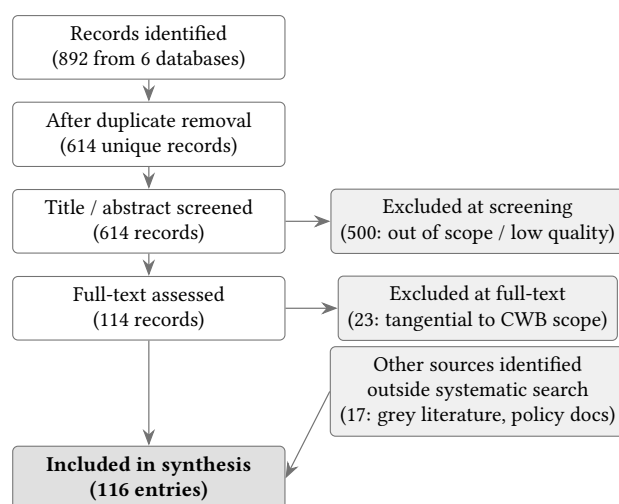


Figure 2.1: PRISMA-style literature screening flow for the 2017–2026 evidence base. After title/abstract screening, 114 records underwent full-text assessment ($614 - 500$); 91 met inclusion criteria and 23 were excluded as tangential to CWB modules. Seventeen additional entries were identified outside the systematic database search (policy documents and grey literature), plus eight sources added in the 2024–2026 IEEE/ACM security and ZKP update, yielding 116 synthesised bibliography entries.

2.2 Review of Existing Research

The Crypto World Bank design draws on peer-reviewed materials and institutional reports across DeFi lending, ML fraud detection, hierarchical development finance, smart-contract security, stablecoin regulation, and—where relevant—scoped future extensions (GNN, federated learning, retail payments). Each subsection below establishes a gap the current build fills, justifies a specific design decision, or benchmarks a deliverable against prior work.

2.2.1 DeFi Lending and the Structural Gap

In their SoK paper, Werner et al. [1] systematize DeFi protocol design and confirm that current lending markets are homogeneous, pool-based, and over-collateralized, with no institutional hierarchy. Their taxonomy demonstrates that no current DeFi system can represent multi-tier capital flow in the manner of development finance—establishing the design space Contribution 1 enters. Bastankhah et al. [2] present a data-driven lending protocol with adaptive dual fast/slow control, demonstrating that dynamic interest rate adjustment outperforms the static utilization curves used by Aave and Compound [59]; this validates the kinked-rate design in Section 3.9. Aspris and Svec [118] formalise institutional risk curation as a modular curator layer in decentralised credit markets—the closest architectural precedent to tier-governed capital allocation—but without multilateral development-finance hierarchy or reserve-ratio enforcement at four tiers. Yang and Cui [47] provide a quantitative evaluation framework for DeFi lending protocols—utilization ratio stability, liquidation efficiency, and interest rate responsiveness—applicable as benchmarks for the CWB hierarchical tiers. Dao et al. [60] demonstrate that credit scoring models can be adapted for DeFi borrowing limits within decentralized protocols.

2.2.2 Fraud Detection and Explainable AI

Palaiokrassas et al. [3] use machine learning on multichain DeFi fraud over 54 million transactions across 23 protocols, showing that behavioral characteristics raise Neural Net-

work and XGBoost classifiers to F1-scores of 0.76–0.85 versus 0.08 with transactional features alone—directly informing the Random Forest feature-engineering plan. The BCCC-DeFiFraudTrans-2025 dataset (1,026,867 annotated transactions, 79 features) [108] serves as the primary supervised training corpus (Section 4.3.1). A 2025 systematic review spanning 1,585 peer-reviewed studies [155] reports XGBoost at approximately 95% accuracy and FedAvg-based federated approaches at approximately 91%, positioning the CWB pipeline against the broadest available evidence base. LookAhead [152] (ACM FSE 2025) achieves F1 0.8966 on labelled DeFi attack datasets with 0.16% false-positive rate—the current benchmark ceiling for the security audit pipeline.

Liu et al. [6] introduce Isolation Forest, which identifies anomalies via recursive partitioning and shorter path lengths for outliers. This algorithm is specified as the secondary unsupervised model when labeled fraud data are limited, enabling detection of attack patterns absent from the training set. Caldarelli (2025) [123] notes that ML-score delivery for lending governance—not price feeds—via blockchain oracles remains under-researched, validating the novelty of the oracle-mediated risk-scoring pattern in Contribution 3.

Adom et al. [4] compare LIME and SHAP on loan approval datasets: SHAP offers more consistent, deeper feature attributions via Shapley values, while LIME is faster but less stable across repeated evaluations—motivating SHAP as the primary explainability method. Yuan [56] demonstrate SHAP-based credit default models achieving AUC above 0.92 with full per-client interpretability, informing the Authority Brief dashboard where approvers review SHAP waterfall plots before lending decisions.

2.2.3 Hierarchical Capital Distribution

Tan [5] constructs an IMF model in which CBDCs use a two-tier distribution architecture (central bank → commercial banks → users) that parallels the CWB four-tier hierarchy. The joint BIS report [53] identifies the same two-level distribution model as the preferred retail CBDC architecture. Kiff et al. [134] survey retail CBDC operating models in which the central bank issues CBDC but delegates wallet and payment services to supervised intermediaries; Auer and Böhme [135] propose hybrid CBDC architectures combining direct central-bank claims with intermediary-delivered payment services—paralleling the CWB pattern in which retail clients access credit only through Local Banks while reserve authority remains at Tier 1.

Bindseil [133] analyses tiered CBDC remuneration as a mechanism to preserve bank intermediation while granting public access to central-bank liabilities, directly informing the asymmetric upward/downward rate spread planned for the UpwardDepositFacility (Section 3.14.2).

On programmable implementation, the New York Fed / BIS *Project Pine* prototype [131] demonstrates a hierarchy of smart-contract factories, exchange contracts, and collateral pools for central-bank market operations in tokenized wholesale markets—validating layered contract design for InterBankLendingPool and SyndicatedLoan. Dong, Qiu, and Xu [132] model blockchain-enabled cross-tier direct financing in a three-tier supply chain, showing that transparent ledger visibility enables capital to reach capital-constrained deep-tier participants—a structural parallel to downward allocation reaching retail clients who cannot access wholesale tiers directly.

Together, these sources establish multi-directional capital flows in policy and operations

research but not yet composed into a single DeFi lending specification; the gap the CWB architecture addresses (Figure 1.3).

2.2.4 Group Lending and Microfinance

The solidarity group lending model, pioneered by Grameen Bank and scaled by BRAC, demonstrates that peer-monitored mutual liability can achieve repayment rates exceeding 95% among borrowers with no individual collateral—attributable to weekly meetings, field-officer oversight, and community reputation that on-chain contracts cannot replicate. The CWB `GroupLendingPool` encodes mutual liability in programmable logic but expects lower initial on-chain repayment rates until credit history accumulates. Asaduzzaman et al. [54] show that smart contract-based microcredit management can reduce administrative overhead by up to 60% versus manual processes—the economic case for programmable group lending at the Local Bank tier.

A 2025 IEEE study introduces a blockchain-powered microloan platform using Ethereum smart contracts, MetaMask, and React.js for on-chain application, ranking, repayment, and wallet management [156]—the most directly analogous published prototype to the CWB Local Bank interface and the closest technical precedent to `GroupLendingPool`. The economic foundation for mutual liability is Jaffer’s solidarity lending mechanics [119]. Sherratt [120] documents that mutual liability in analogue MFIs can become coercive—a risk the on-chain design mitigates by eliminating human collection pressure, at the cost of losing the social monitoring that drives analogue repayment rates. Güçük et al. [61] confirm that privacy and trust mechanisms are critical in blockchain-based microlending, reinforcing ZKP-based credential design for group member onboarding.

2.2.5 Smart Contract Security and Governance

Atzei et al. [7] provide the foundational taxonomy of Ethereum smart-contract attack vectors—reentrancy, access control failures, and integer overflow—informing `ReentrancyGuard`, Solidity 0.8.20 overflow protection, and RBAC modifiers. Chaliasos et al. [150] (IEEE/ACM ICSE 2024) find that existing security tools could have prevented only 8% of real-world DeFi exploit losses in their dataset—all preventable cases being reentrancy—forcing a layered pipeline rather than reliance on any single tool. Chen et al. [151] (IEEE/ACM ICSE 2026) demonstrate that layered contract architectures introduce new control-flow attack surfaces requiring runtime monitoring beyond pre-deployment analysis—motivating Tenderly alerts (Phase III) alongside Slither and Mythril in the defense-in-depth stack (Section 3.19).

`ContractWard` [50] (F1 0.96 reentrancy, 0.93 access control), `SoliAudit` [51] (70% audit-time reduction), and `VERISMART` [52] (98.4% accuracy) form the benchmark hierarchy for static and formal tooling; `LookAhead` [152] supersedes them on DeFi-specific adversarial-contract detection (F1 0.8966). Flash-loan governance attacks—in which an attacker borrows a supermajority of governance tokens within one block and passes a malicious proposal atomically (Beanstalk, \$182M, April 2022) [158]—and the three primary DAO threat vectors classified in *SoK: Attacks on DAOs* [159] directly generate the snapshot voting, 48-hour timelock, and 60% quorum floor in Section 3.18. MEV front-running on loan approvals and liquidations [160] is mitigated via commit-reveal liquidation triggers and Polygon zkEVM Cardona deployment, where ZK validity constraints reduce (though do not eliminate) sequencer-ordering risk.

2.2.6 Supporting Infrastructure

The EU MiCA Regulation (fully effective December 2024) [R36, R37] requires authorized issuers, 1:1 reserve backing, and regular audits for electronic-money tokens covering USDC and USDT at retail tiers. The United States GENIUS Act (July 2025) [R38] creates the first federal payment-stablecoin framework with similar reserve and disclosure obligations—together making USDC-first denomination a regulatory requirement, not merely a preference (Section 1.10.3). BIS WP 905 [R13] excludes algorithmic stablecoins from emerging-market DeFi deployment.

Beniiche [R1] and Pasadar [R2] establish the oracle problem for DeFi lending governance. An empirical Chainlink accuracy study across eight chains [157] finds Polygon achieves consistently high oracle accuracy versus Ethereum and zkSync under market volatility—justifying single-chain Polygon zkEVM Cardona deployment and the Chainlink Functions oracle choice (Section 1.7.1).

BIS and FSB analyses [21, 33] document correspondent-banking inefficiencies: 2–5 day settlement and approximately \$42 per cross-border transaction. The World Bank remittance brief [26] reports 6.49% average cost for a \$200 transfer. These figures motivate on-chain settlement with near-instant finality on Layer 2.

Kim and Kim [55] show that transaction batching, calldata compression, and tactical timing reduce gas costs by 30–50% on Ethereum mainnet; Hartmann and Hasan [48] analyse peer-to-peer lending gas minimisation strategies that inform the four-tier transaction cost model. Chen et al. [46] validate multi-chain lending throughput even though the CWB revised to single-chain deployment after bridge risk analysis (Section 3.13).

Buterin, Hitzig, and Weyl (2022) [R30] introduce Soulbound Tokens (SBTs) as non-transferable on-chain credentials; MicroSave Consulting (December 2025) [R31] documents the microfinance sector’s shift toward performance-based credit identity—together motivating the on-chain credit passport in Section 3.12.

Ozili [161] documents Nigeria’s eNaira at 98.5% wallet inactivity one year post-launch, confirming that inclusion barriers are trust-based and institutional rather than purely technical—justifying MFI-intermediary distribution over direct end-user enrollment.

Sygnium Bank’s *Institutional DeFi in 2025* report [R21] finds that pension-fund allocators withhold participation until legal enforceability is resolved. BIS mBridge [R39] and *Project Agora* [R40] provide settlement rails without retail lending hierarchies; World Bank Fund-sChain [39] and JPMorgan Kinexys [40] confirm institutional blockchain adoption for fund tracking and settlement—positioning the CWB as a composable lending-layer specification on top of such infrastructure.

2.2.7 Zero-Knowledge Proofs for Compliance-Aware DeFi Identity

Two 2026 *ACM Transactions on the Web* papers demonstrate that combining Zero-Knowledge Proofs, Self-Sovereign Identity, DIDs, and smart contracts enables privacy-preserving KYC across platforms without revealing personal data [153, 154]. Piper et al. (2025) [R8] grounds the on-chain permissioning layer. Together with zkAML velocity proofs [R19], these works motivate Contribution 4’s two-circuit compliance gateway; circuit implementation (Groth16 in Circom 2.0) is scoped to Future Work (Section 6).

2.2.8 Future Research Directions Scoped

A parallel research stream demonstrates that DeFi fraud is fundamentally relational—coordinated borrowing rings and sybil attacks appear in the wallet-interaction graph, not in individual transaction features. Cheng et al. [R24], Chang et al. [121], and DeFi-Guard [R25] consistently outperform flat ML baselines, with detection above 70% of illicit transactions at sub-1% false-positive rates on the Elliptic dataset. The CWB’s planned GNN extension treats this as an ablation isolating the marginal value of tier-relationship edges over flat wallet graphs—scoped to Future Work (Section 6); this pre-thesis report does not specify GNN pipeline details.

Hasan et al. [49] and the PrivChain-AI study [R27] demonstrate that federated learning allows institutions to collaborate on fraud detection without sharing raw transaction data, reaching 94.7% accuracy across multiple banks. Alhasawi et al. [125] report $F1=0.90$ on credit-card fraud with trust-aware FL aggregation—the direct baseline for a cross-tier CWB extension. The four-tier structure, where National and Local Banks are independent legal entities that cannot share raw client data, makes FedAvg with differential privacy architecturally appropriate; this extension requires dedicated FL infrastructure beyond the Phase I–IV build and is scoped to Future Work.

A substantial literature examines stablecoin-denominated retail payment rails—client-to-client remittance, merchant checkout, and fiat on/off-ramps—under compliance constraints [Ante 136, Du 138, Li 137, Zhang 140, Fröhlich 139, Lin 141, CPMI 142, FATF 145]. These works establish that retail payment extensions are empirically studyable but that no prior work composes hierarchical development-finance lending with compliance-gated remittance and PSP-mediated ramps in a single four-tier architecture. The CWB post-MVT roadmap addresses this gap in Section 6; the literature is not further reviewed here as none of these papers shapes the current system design.

The optional conversational agent add-on faces two documented risks from the LLM-in-finance literature [R35]: regulatory hallucination on compliance questions without domain fine-tuning, and prompt injection through user-controlled input fields [Alizadeh R53]. OWASP LLM Top 10 [R54] classifies prompt injection as the highest-priority threat in agentic financial systems, corroborated by a February 2026 Microsoft advisory on MCP-based agent runtimes [R55]. The scanning middleware and human confirmation gate in Section 4.4 address both risks; implementation is Phase III–IV.

2.3 Literature Review Summary

Table 2.1 consolidates the fifteen highest-impact sources. Papers below Tier 1 relevance are cited at point of use in the subsections above.

Table 2.1: Tier-1 literature synthesis: core findings and design decisions.

Author & Year	Core Finding	Design Decision	Phase
Werner et al. 2022 [1]	DeFi lending uniformly flat, pool-based, overcollateralized; no institutional hierarchy	Four-tier architecture as novel design space	Contribution 1
Aspris & Svec 2025 [118]	Modular risk curator is closest prior work; lacks multilateral hierarchy	Boundary of prior work for Contribution 1	Contribution 1
Tan 2023 [5]	Two-tier CBDC distribution (central bank → banks → users)	Four-tier capital flow; client-via-Local-Bank path	Architecture
Project Pine 2025 [131]	Hierarchical smart contract factory for central-bank operations	Layered design for IBLP and SyndicatedLoan	Phase II–III
Palaiokrassas et al. 2024 [3]	Behavioral features: F1 0.76–0.85 vs. 0.08 transactional only	Random Forest feature engineering	Contribution 3
Liu et al. 2008 [6]	Isolation Forest via recursive partitioning; shorter paths = anomalies	Secondary unsupervised model when labels scarce	Contribution 3
Adom et al. 2022 [4]	SHAP deeper and more consistent than LIME on loan approval	SHAP as primary explainability method	Contribution 3
Jaffer [119] / IEEE Microloan 2025 [156]	Solidarity mechanics + most recent blockchain microloan prototype	GroupLendingPool mutual liability design	Contribution 2
Atzei et al. 2017 [7]	Foundational EVM attack taxonomy	ReentrancyGuard, CEI, RBAC, Solidity 0.8.20	Security
Chaliasos et al. 2024 [150]	Tools prevented only 8% of real-world losses; all reentrancy	Layered pipeline—no single tool sufficient	Security
Qin et al. [158]	Flash-loan governance passes malicious proposals atomically (\$182M)	Snapshot voting, 48-h timelock, 60% quorum	Governance
Piper et al. 2025 [R8] + ZKP-KYC [153, 154]	ZKP + SSI + DID enables privacy-preserving KYC	Groth16 KYC circuit, W3C DID/VC choice	Contribution 4
MiCA [R36, R37] + GENIUS [R38]	Stablecoin issuers require 1:1 reserves, authorization, audits	USDC-first denomination as regulatory requirement	Denomination
Chainlink accuracy [157]	Polygon high oracle accuracy vs Ethereum/zkSync under volatility	Single-chain Polygon deployment; oracle fallback	Infrastructure
Gudgeon et al. [R3]	Utilization above 90% causes liquidity crises empirically	Kink point at $U^* = 80\%$ in rate model	Rate model

Papers below Tier 1 relevance are cited at point of use in the subsections above. The full bibliography of 116 entries is available in the references section.

2.4 Comparative Protocol Analysis

Table 2.2 compares the Crypto World Bank against representative DeFi protocols on eleven architectural dimensions. No existing protocol simultaneously enforces a four-tier reserve-ratio hierarchy, oracle-mediated explainable ML governance, and programmable mutual-liability group lending—directly answering RQ1 and supporting the compound novelty claim in Section 1.6.

Feature	Aave	Comp.	Maker	Maple	Gfinch	CWB
Institutional hierarchy	✗	✗	✗	✗	✗	●
Cross-tier capital flow	✗	✗	✗	✗	✗	⌚
Same-tier interbank lending	✗	✗	✗	✗	✗	⌚
Syndicated / co-lending	✗	✗	✗	Part.	Part.	⌚
Upward capital repatriation	✗	✗	✗	✗	✗	⌚
Solidarity group lending	✗	✗	✗	✗	Part.	●
AI/ML fraud detection	✗	✗	✗	Man.	Man.	●
SHAP explainability	✗	✗	✗	✗	✗	●
ZKP KYC compliance	✗	✗	✗	✗	✗	●
Kinked interest rate curve	✓	✓	Gov.	✓	✗	⌚
Role-based access control	Part.	Part.	Gov.	✓	Part.	●
Institutional / L2 focus	✗	✗	✗	✓	Part.	●
TVL / Status (2026)	\$26B	\$1.4B	\$10B	\$2.6B	\$680M	Planning

Table 2.2: Protocol comparison on eleven architectural dimensions. Competitor columns reflect publicly documented protocol capabilities (2026); CWB columns reflect pre-thesis specification status.

Note on ERC-3643 (T-REX) for institutional-tier permissioning. The institutional tiers (Tiers 1–3) use bespoke RBAC (Section 3.19) for transfer-level access control. ERC-3643 gates token movements at the contract level for regulated institutional transfers; the platform’s RBAC role gates serve the same functional purpose at the application layer. A production deployment under MiCA [R36] would evaluate full ERC-3643 compliance as the interoperability path; the RBAC structure maps directly onto the ERC-3643 identity registry model.

2.4.1 Literature Synthesis

(1) The gap. Werner et al. [1] establish that DeFi lending is uniformly flat and pool-based; Tan [5] and the BIS CBDC literature [53, 134] confirm that tiered distribution is the institutional standard; Aspris and Svec [118] provide the closest modular-curator precedent but stop short of multilateral development-finance hierarchy. No prior work combines these into a single four-tier lending specification—the gap Contribution 1 addresses and RQ1 tests.

(2) The ML and security baseline. Palaiokrassas et al. [3] show behavioral ML achieves workable fraud detection (F1 0.76–0.85); Adom et al. [4] confirm SHAP gives regulatory-grade explainability; Atzei et al. [7] and Chaliasos et al. [150] together show that existing security tooling alone is insufficient (8% prevention rate), requiring the layered Slither–Mythril–Foundry–Tenderly pipeline in Section 3.19. This grounds Contribution 3 and RQ3.

(3) The institutional and regulatory context. MiCA [R36] and the GENIUS Act [R38] mandate stablecoin reserve transparency; mBridge [R39] and Agora [R40] provide settlement rails without lending hierarchies; Ozili [161] documents CBDC adoption failures (eNaira 98.5% inactivity); FundsChain [39] and Kinexys [40] confirm institutional blockchain momentum. Stablecoin-first, institution-intermediated design is therefore both regulatorily defensible and empirically necessary—answering RQ2.

(4) Scoped future directions. GNN relational fraud analysis [R24, 121, R25], cross-tier federated learning [49, R27, 125], Groth16 zkKYC/zkAML circuits [R8, 153, R19], NettingEngine settlement, Certora reserve-invariant proofs, and retail payment rails [136–145] are each studyable in the literature but explicitly scoped to Future Work (Section 6) or post-MVT deliverables (Section 6); none shapes the current MVT build.

2.5 Summary of Key Findings

The literature review establishes why the CWB requires a four-tier architecture, an oracle-mediated RF + Isolation Forest + SHAP pipeline, multi-directional capital-flow modules, layered smart-contract security, and stablecoin-first denomination—while deferring GNN, federated learning, ZKP circuit deployment, and retail payment rails to explicitly scoped future work.

Chapter 3

System Architecture and Design

3.1 Prototype Scope

Evaluators should distinguish between what is **specified** in this document (formal design), what is **planned** for the implementation phases of the final thesis, and what is an **optional add-on** (conversational agent interface). Nothing in the pre-thesis submission is deployed to a public network; Table 3.1 records design status only.

Table 3.1: Prototype scope: specification vs. planned implementation status (pre-thesis design record only; no live public deployment).

Feature	Status
Four-tier role system (RBAC)	⌚ Specified
World Bank Reserve contract (Tier 1)	⌚ Specified (Phase I)
Tier 2 National Bank contracts	⌚ Specified (Phase I)
Tier 3 Local Bank contracts	⌚ Specified (Phase I)
Cross-tier fund transfer	● Planned (Phase II)
Loan request / approval workflow	⌚ Specified (Phase II)
Installment EMI auto-generation	● Planned (Phase II)
SavingsVault contract	● Planned (Phase II)
FixedDeposit contract	● Planned (Phase II)
GroupLendingPool contract	● Planned (Phase II)
InterBankLendingPool	● Planned (Phase II)
AI/ML fraud detection (Random Forest)	● Planned (Phase III)
SHAP + anomaly explainability	● Planned (Phase III; Section 4.3.2)
Oracle integration (Chainlink Functions)	● Planned (Phase III)
ZKP KYC compliance layer	● Planned (Phase IV)
Testnet deployment (Polygon zkEVM Cardona)	● Planned (Phase IV)
AI Agent / MCP tool server (17 banking tools)	● Optional (Phase III–IV; retail client only)
Bank-authority AI agent (NB/WB aggregate read tools)	● Future Work (post-MVT)
EIP-7702 session key management	● Optional (Phase III)
Authority Brief UI (SHAP for bank approvers)	● Planned (Phase III)
Chainlink Automation (installment triggers)	● Planned (Phase II)
Chainlink Proof of Reserve	● Planned (Phase I)
The Graph subgraph (event indexing)	● Planned (Phase III)
SAR workflow (AML → compliance queue → freeze)	● Planned (Phase III)
sessions table schema	⌚ Specified (Phase I)
agent_action_log table schema	● Optional (Phase III)
300-client Foundry simulation	● Planned (Phase IV)

This table eliminates ambiguity between architectural specification (present tense for clarity) and items not yet built.

Chapter organisation. The remainder of this chapter is organised as follows. Sections 3.2–3.3.2 describe the platform stack, contract inventory, blockchain selection, and oracle architecture. Sections 3.4–3.5 present the relational data model and on-chain/off-chain partitioning. Sections 3.6–3.8.3 cover operating context, identity, onboarding, and the retail conversion funnel. Sections 3.9–3.13 specify interest-rate, liquidation, deposit, credit-passport, and single-chain deployment designs. Section 3.14 details multi-entity capital operations. Sections 3.15–3.17 provide UML-style models and the banking product suite. Sections 3.18–3.20 close with governance and the five-layer security architecture.

Specification voice. Throughout this chapter, **specified** denotes a formally documented design in this pre-thesis; **planned** denotes a Phase I–IV implementation deliverable; and **optional** denotes the conversational agent add-on for **retail clients only**—it mirrors the same Express.js API and smart contracts as the primary React interface but is not required for core platform operation and does not extend to National Bank or World Bank authority workflows in the MVT scope (Table 3.1). Architectural descriptions use the present tense for specification clarity; Table 3.1 takes precedence when scope could be ambiguous.

3.2 High-Level Architecture

The system employs a **four-layer decentralised application architecture** (Figure 3.1): a presentation layer (React + wallet integration), a smart-contract layer (fifteen modular EVM contracts centred on the four-tier lending hierarchy), an off-chain services layer (Express REST API, PostgreSQL, FastAPI ML scoring, Chainlink oracle integration, event listener, and Redis cache), and a Chainlink infrastructure layer (Chainlink Functions, Automation, Price Feeds, and Proof of Reserve). An optional conversational agent (Qwen3-8B + MCP tool server) attaches to the same API endpoints as the React UI and does not alter on-chain logic. Figure 3.2 elaborates this view with external integrations.

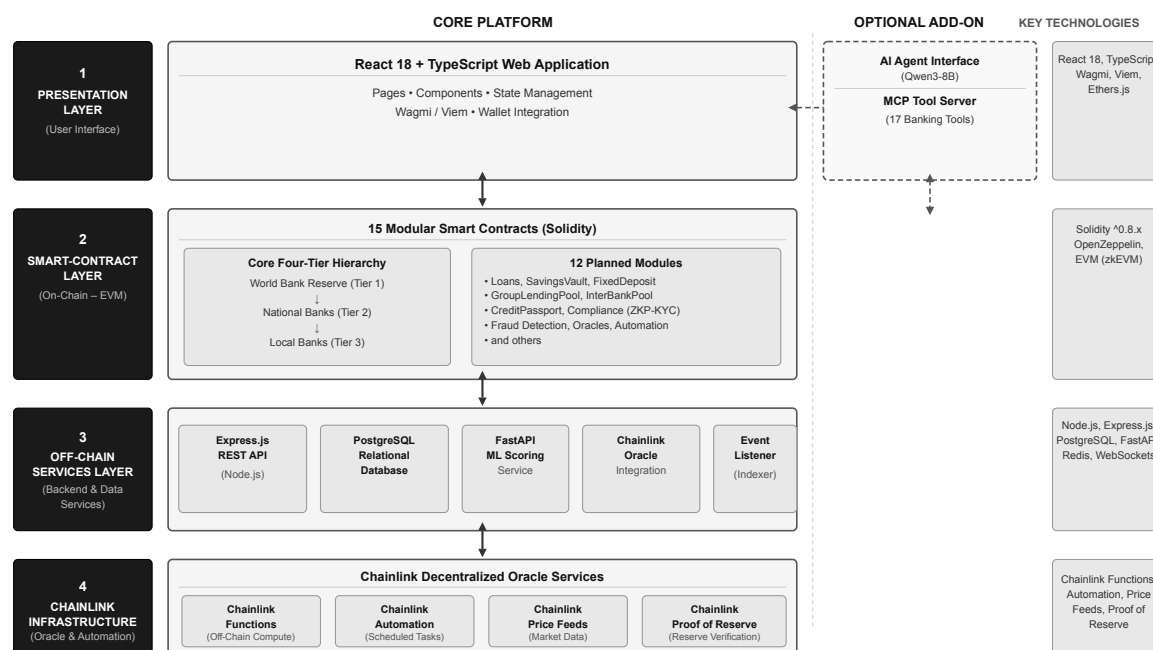


Figure 3.1: Four-layer decentralised application architecture (blockchain-first): presentation (React 18, TypeScript, Wagmi/Viem); smart-contract layer on the EVM (World Bank Reserve, National Bank, Local Bank + twelve planned modules); off-chain services (Express REST API, PostgreSQL, FastAPI ML service, Chainlink oracle integration, event listener, Redis cache); Chainlink infrastructure (Functions, Automation, Price Feeds, Proof of Reserve). Optional add-on: AI agent (Qwen3-8B + MCP) interfaces through the same APIs without altering on-chain logic.

The architecture **specifies** fifteen modular contracts. Phase I targets the three-tier lending core (World Bank Reserve, National Bank, Local Bank). The remaining twelve split into two suites: *banking products* (SavingsVault, FixedDeposit, GroupLendingPool, FXModule, InsuranceFund, CurrentAccount) and *multi-entity operations* (InterBankLendingPool, UpwardDepositFacility, SyndicatedLoan, TranchPool, TreasurySwap, NettingEngine)—the latter enabling same-tier lending, upward funding, syndicated co-lending, treasury FX, and multilateral netting (Section 3.14, Figure 3.17). All are planned across Phases II and III.

Figure 3.2 shows component interactions. Core lending contracts are specified in Phase I; extended modules are planned extensions documented in Appendix B.

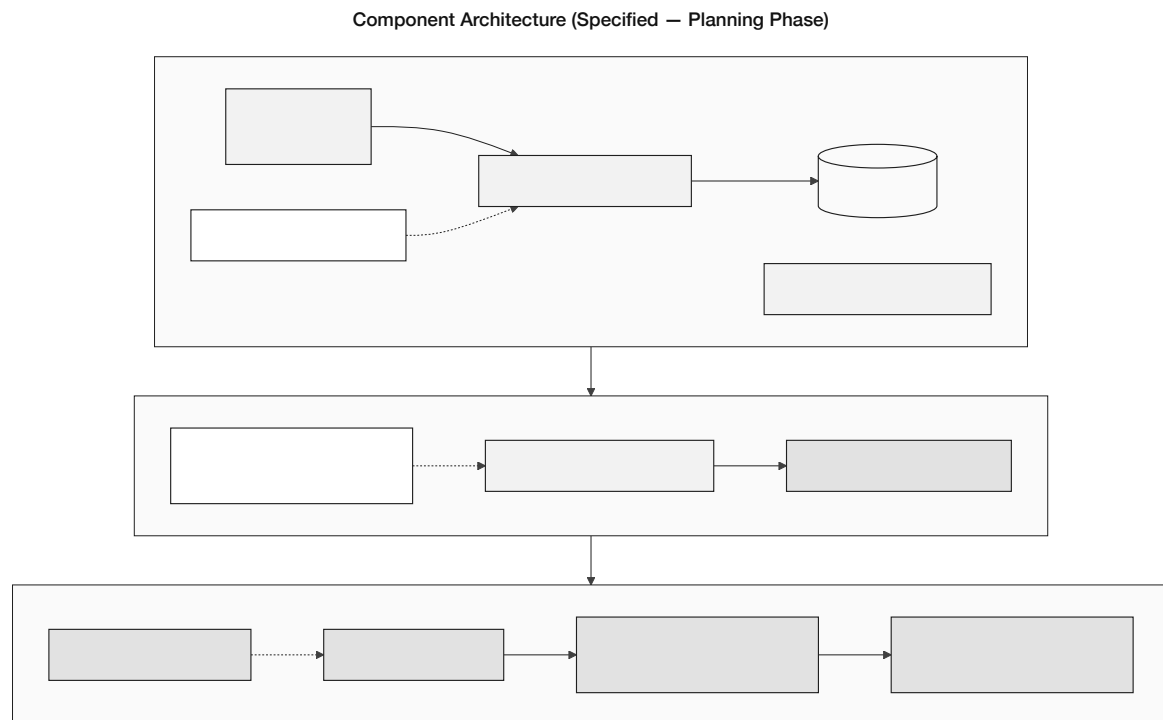


Figure 3.2: Component diagram (planning phase): React UI is the primary path; optional MCP agent (dashed) shares the Express.js API; smart contracts and Chainlink integrations are specified for Phase I–IV deployment.

Contract	Module	Status	Primary responsibility
WorldBankReserve	Tier 1 core	Specified (Phase I)	Global reserve custody, national-bank registration, downward capital allocation.
NationalBank	Tier 2 core	Specified (Phase I)	Local-bank registration, upward borrowing, downward allocation.
LocalBank	Tier 3 core	Specified (Phase I)	Client onboarding, loan lifecycle, installments, freezeAccount.
SavingsVault	Deposits	Specified (Phase II)	Variable-yield savings, reserve-gated withdrawals.
FixedDeposit	Deposits	Specified (Phase II)	Term deposits with deterministic early-withdrawal penalty.
GroupLendingPool	Group credit	Specified (Phase II)	Solidarity-group formation, consent, mutual liability.
FXModule	Treasury / FX	Specified (Phase II)	Oracle-priced retail FX with disclosed spread.
InsuranceFund	Risk buffer	Specified (Phase II)	Default coverage buffer; Chainlink Proof of Reserve.
CurrentAccount	Payments	Specified (Phase II)	Transactional accounts and atomic peer transfers.
InterBankLendingPool	Multi-entity	Specified (Phase II)	Same-tier interbank liquidity pools per tier.
UpwardDepositFacility	Multi-entity	Specified (Phase II)	Upward surplus repatriation with yield spread.
SyndicatedLoan	Multi-entity	Planned (Phase III)	Multi-lender syndicate consent and disbursement.
TranchedPool	Multi-entity	Planned (Phase III)	Senior/junior tranching and waterfall recovery.
TreasurySwap	Multi-entity	Specified (Phase II)	Institutional oracle-priced asset swaps.
NettingEngine	Multi-entity	Future Work	ERD stubs only; Section 6

3.3 Blockchain Platform Selection

Blockchain and Application Stack

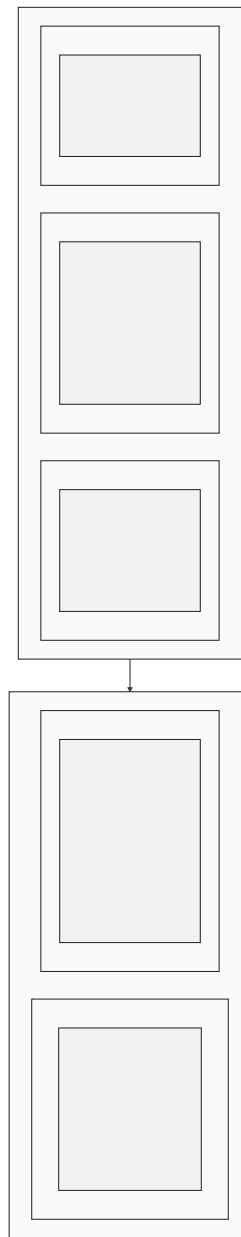


Figure 3.3: Layered blockchain and application stack: L1 settlement (Polygon zkEVM Cardona for all live contracts; Ethereum Sepolia for Certora verification only); L2 smart-contract platform (Solidity 0.8.20, OpenZeppelin v5, UUPS, TimeLock, RBAC); L3 data (PostgreSQL 16 in 3NF, Redis, IPFS); L4 API and services (Express, FastAPI, WebSocket, EIP-712); L5 presentation (React + ERC-4337 account abstraction).

Table 3.3: Blockchain platform selection criteria and justification.

Criterion	Selection	Justification
Platform	Ethereum Virtual Machine (EVM)	Largest developer ecosystem; battle-tested security model; extensive tooling.
Network	Polygon zkEVM (live); Cardona Sepolia (Certora only, Phase IV)	Zero-cost deployment; single-chain live architecture; Sepolia retained only for formal verification, not bridged assets.
Consensus	ZK validity proofs (Polygon zkEVM L2, Ethereum L1 settlement)	Every batch of transactions is accompanied by a zero-knowledge proof verified on Ethereum L1, so security derives from cryptographic validity rather than a validator set assumption.
Smart contract language	Solidity 0.8.20	Industry standard; mature compiler with overflow protection; rich library ecosystem.

This platform selection table compares candidate networks on cost, finality, throughput, and ecosystem maturity. Cardona is preferred over Amoy PoS for ZK-anchored security (see Section 1.7.1, item E).

Table 3.5: Blockchain platform selection (continued): operational and deployment factors.

Criterion	Selection	Justification
Gas cost	\$0.001–\$0.01 per transaction	Orders of magnitude cheaper than Ethereum mainnet (\$5–\$50); enables micro-loan economics.
Block finality	≈2 seconds	Sub-second practical finality for retail UX; checkpointed to Ethereum for security.
Developer tooling	Hardhat + OpenZeppelin	Automated test suite, deployment scripts, and audited security primitives.
Testnet availability	Cardona (Polygon zkEVM), Sepolia (Ethereum)	Free faucets; EVM-identical behaviour; no real cryptocurrency required for prototype.
Security model	ZK validity proofs	Every batch verified by a ZK proof anchored to Ethereum L1 (vs. PoS validator assumption on Amoy).
Migration path	EVM-compatible	Same Solidity contracts deploy to any EVM chain; reduces future L2 migration cost.

This supplementary comparison expands the platform evaluation to additional operational and deployment factors. The conclusion is that Polygon zkEVM Cardona provides a practical balance of ZK-secured settlement and low transaction costs for an academic prototype, with a clear migration path if future requirements demand alternative L2s.

3.3.1 Transaction Verification and Consensus

- **Polygon zkEVM Cardona:** ZK validity proofs anchor batches to Ethereum L1 (cryptographic security vs. PoS validator assumption on Amoy).
- **On prototype testnets:** Cardona and Sepolia use equivalent consensus models at no financial cost, enabling incremental development and testing without exposure to real-asset risk.

Gas-cost sustainability. The hierarchical loan lifecycle generates multiple on-chain state transitions per client; on Polygon zkEVM Cardona the **projected** cost remains well below 0.01% of interest earned on a representative retail loan (Section 5.8), to be validated by the Phase IV Foundry simulation. This margin underpins the micro-loan economics that motivate Layer 2 deployment in the platform selection above.

3.3.2 Oracle Architecture: Off-Chain AI to On-Chain Decision

The Crypto World Bank requires a mechanism to convey off-chain AI/ML risk assessments into the on-chain loan approval workflow. This is an instance of the oracle problem [R1,R2].

Chainlink Functions oracle (primary). The final thesis phase uses **Chainlink Functions** as the primary oracle mechanism. Chainlink Functions operates via a Decentralised Oracle Network (DON): the ML risk score is fetched independently by multiple Chainlink nodes, consensus across the DON is required before the result is committed on-chain, and no single compromised node can manipulate the risk score. This is a fundamental trust improvement over the earlier commit-reveal relay, which required trusting a single FastAPI service key. The source code executed by the DON is:

```
// Chainlink Functions source (runs in decentralised DON)
const clientId = args[0];
const response = await Functions.makeHttpRequest({
  url: 'https://your-ml-service.com/score/${clientId}',
  method: "POST"
});
const score = response.data.risk_score;
// 6 decimal precision
return Functions.encodeUint256(Math.round(score * 1e6));
```

The DON adds 30–60 seconds of latency and \$0.10–\$1.00 per oracle call, both acceptable for the loan approval lifecycle where decisions are not time-critical. This architecture closes the oracle trust assumption noted in the prototype design (the 2-of-3 Safe multisig attestation was the interim mitigation; Chainlink Functions removes the need for that mitigation entirely).

Commit-reveal relay (prototype fallback). During the prototype phase, prior to Chainlink Functions being wired (Phase III), the system uses a commit-reveal relay as

a fallback. The off-chain ML service commits a hash $h = \text{keccak256}(s \parallel \text{nonce})$ to the LoanController contract, then reveals s and the nonce within the decision window. The contract verifies $h = \text{keccak256}(s \parallel \text{nonce})$ and stores s immutably. This pattern prevents score manipulation between commitment and decision and creates an immutable on-chain audit trail of every risk score used in a lending decision, while Chainlink Functions integration is completed.

Chainlink Automation: trustless installment triggers. Chainlink Automation replaces the centralised cron job for overdue installment detection and interest accrual. The entire loan lifecycle from application to overdue flagging runs without a trusted operator:

```
// In LocalBankPool.sol
function checkUpkeep(bytes calldata) external view override
    returns (bool upkeepNeeded, bytes memory performData) {
    uint256[] memory overdueLoans = getOverdueLoans();
    upkeepNeeded = overdueLoans.length > 0;
    performData = abi.encode(overdueLoans);
}

function performUpkeep(bytes calldata performData) external override {
    uint256[] memory overdueLoans =
        abi.decode(performData, (uint256[]));
    for (uint i = 0; i < overdueLoans.length; i++) {
        _markInstallmentOverdue(overdueLoans[i]);
    }
}
```

This removes the last centralised component from the loan lifecycle — the entire process from application to overdue flagging runs without a trusted operator.

Chainlink Price Feeds: fiat and crypto pairs. The FXModule contract uses Chainlink’s production-grade AggregatorV3Interface price feeds for ETH/USD, USDC/USD, and governance-selected fiat pairs (e.g. EUR/USD), replacing the earlier “forex oracle approved by governance” pattern. Optional fiat-equivalent display in the frontend flows through:

```
// In LocalBankPool.sol
AggregatorV3Interface internal fiatUsdFeed; // e.g. EUR/USD
AggregatorV3Interface internal ethUsdFeed;

function getFiatEquivalent(uint256 usdcAmount)
    public view returns (uint256) {
    (, int fiatRate,,) = fiatUsdFeed.latestRoundData();
    // USDC tracks USD 1:1; Chainlink uses 8 decimals
    return usdcAmount * uint256(fiatRate) / 1e8;
}
```

The 8-decimal precision convention is the Chainlink standard for fiat-pair feeds. Chainlink price feeds are a planned Phase I integration for optional fiat-equivalent display in the

frontend.

Chainlink Proof of Reserve. The WorldBankReserve contract publishes its reserve balance to **Chainlink Proof of Reserve (PoR)**, making the reserve cryptographically verifiable by any external auditor without trusting CWB’s administrator. This is the “free PoR” advantage over FTX: any external system can verify reserve solvency without relying on CWB’s admin claims.

```
// WorldBankReserve.sol
function getReserveSummary() external view returns (
    uint256 totalDeposited,
    uint256 totalLoaned,
    uint256 reserveRatio,    // scaled 1e4 (e.g. 5000 = 50%)
    uint256 insuranceFundBalance
) {
    return (
        totalDeposited,
        totalLoaned,
        (totalDeposited - totalLoaned) * 1e4 / totalDeposited,
        insuranceFund.balance()
    );
}
```

This function is the FTX commingling safeguard described in Section 3.19. Adding Chainlink’s PoR job on top turns it into a market-standard verifiable proof, and it is specified in full in Appendix D.

Chainlink Oracle Architecture - Off-Chain AI to On-Chain Decision

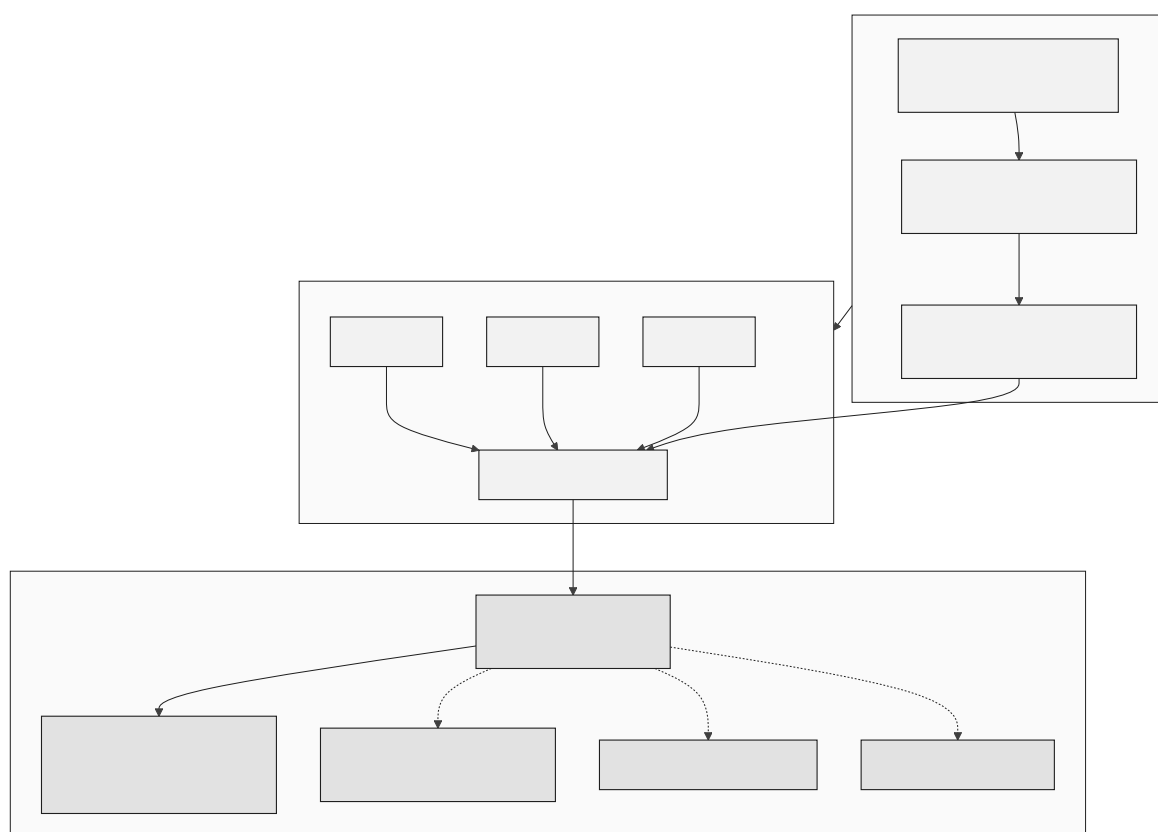


Figure 3.4: Chainlink oracle architecture: (a) off-chain ML feature engineering and SHAP explainability, (b) Chainlink Functions DON consensus for risk-score commitment, (c) on-chain LoanController state machine gating approval on SCORE_REVEALED, with Chainlink Automation, Price Feeds, and Proof of Reserve. The commit-reveal relay remains the prototype-phase fallback until Phase III wiring is complete.

3.4 Data Model and Database Design

The relational database schema comprises 51 normalized entities in Third Normal Form (3NF): 31 core prototype entities (Table 3.7), six primary extension entities (Table 3.12), and 14 Phase II–III banking-product and multi-entity entity stubs (Table 3.14). Core entities are Phase I MVT-complete specifications; stub entities carry PK, FK anchors, and one-sentence roles in Table 3.14 and are fully attribute-specified in Section 3.14. Figure 3.5 presents the core entity-relationship overview, Figure 3.6 presents extended product and multi-entity entities, and Figure 3.7 presents the Enhanced Entity-Relationship (EER) diagram with specialisation and weak-entity constructs.

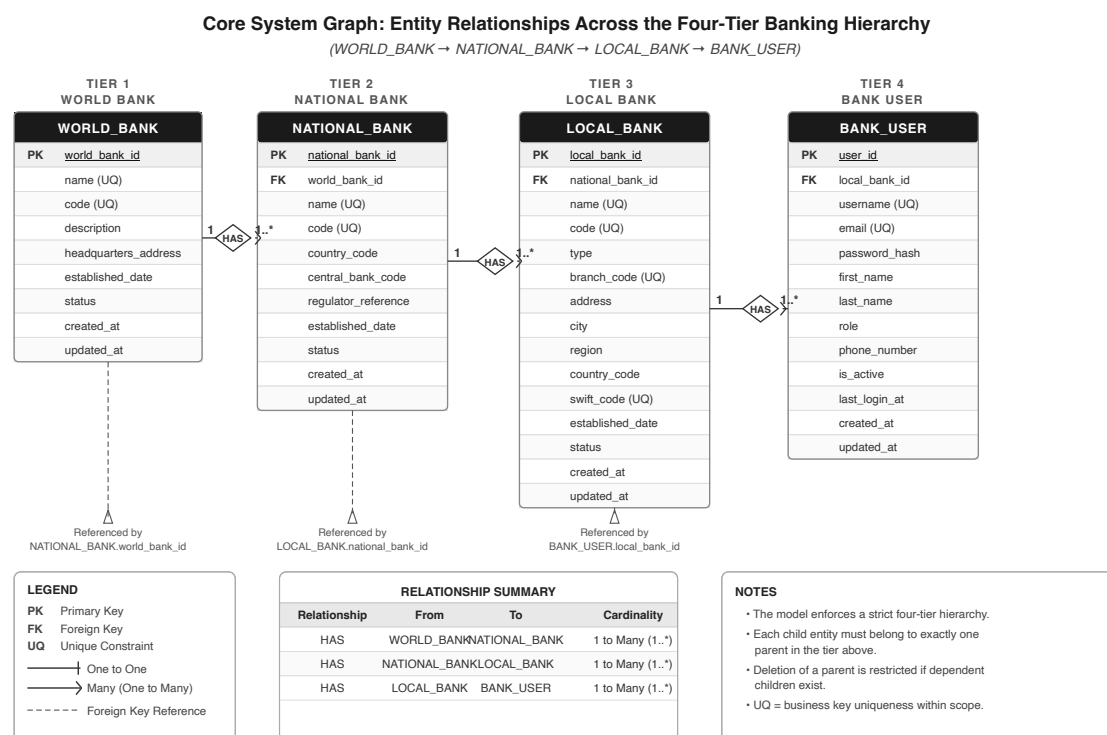


Figure 3.5: Core system graph showing entity relationships across the four-tier banking hierarchy (INSTITUTION → WORLD_BANK / NATIONAL_BANK / LOCAL_BANK → BANK_USER), the central BORROWER entity, and the lending-lifecycle sub-graph (LOAN_REQUEST, LOAN, INSTALLMENT, TRANSACTION, INCOME_PROOF, CHAT_MESSAGE, LOAN_RISK_ASSESSMENT, CREDIT_PASSPORT).

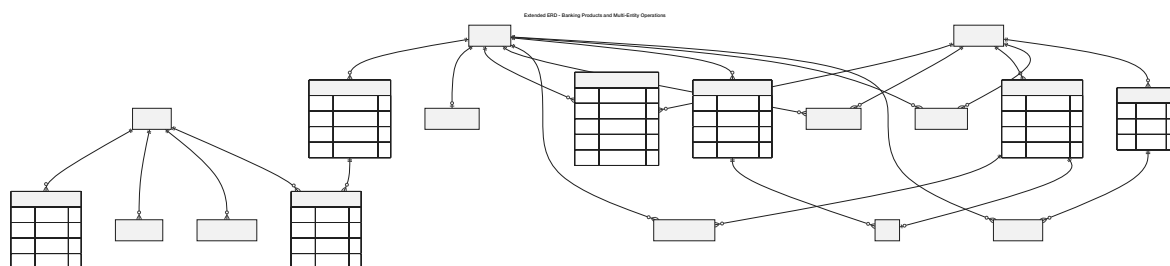


Figure 3.6: Extended ERD entities: the banking-product entities (SAVINGS_ACCOUNT, FIXED_DEPOSIT, CURRENT_ACCOUNT, LOAN_GROUP, GROUP_MEMBER, GROUP_CONSENT, INSURANCE_FUND) and the multi-entity / cross-tier operational entities (INTERBANK_LOAN, UPWARD_DEPOSIT, SYNDICATE, SYNDICATE_MEMBER, TRANCED_POOL, TRANCED_POOL_SUBSCRIPTION, TREASURY_SWAP, NETTING_BATCH, NETTING_ENTRY, NETTING_COORDINATOR_STATE). Foreign-key relationships into the core entities of Figure 3.5 are preserved.

The current ERD covers the core lending and governance entities. Figure 3.6 presents the extended banking entities (SavingsAccount, FixedDeposit, LoanGroup, GroupMember, CurrentAccount, and InsuranceFund) and the multi-entity / cross-tier operational entities described in Section 3.14.7: INTERBANK_LOAN, UPWARD_DEPOSIT, SYNDICATE

and SYNDICATE_MEMBER, TRANCED_POOL, TREASURY_SWAP, and NETTING_BATCH with NETTING_ENTRY. Their foreign-key relationships into the core lending entities (LOAN, BANK_USER, INSTALLMENT, TRANSACTION) are detailed in Table 3.27 in Section 3.14.7. The data model remains in one-to-one consistency with the contract architecture and implementation phase plan.

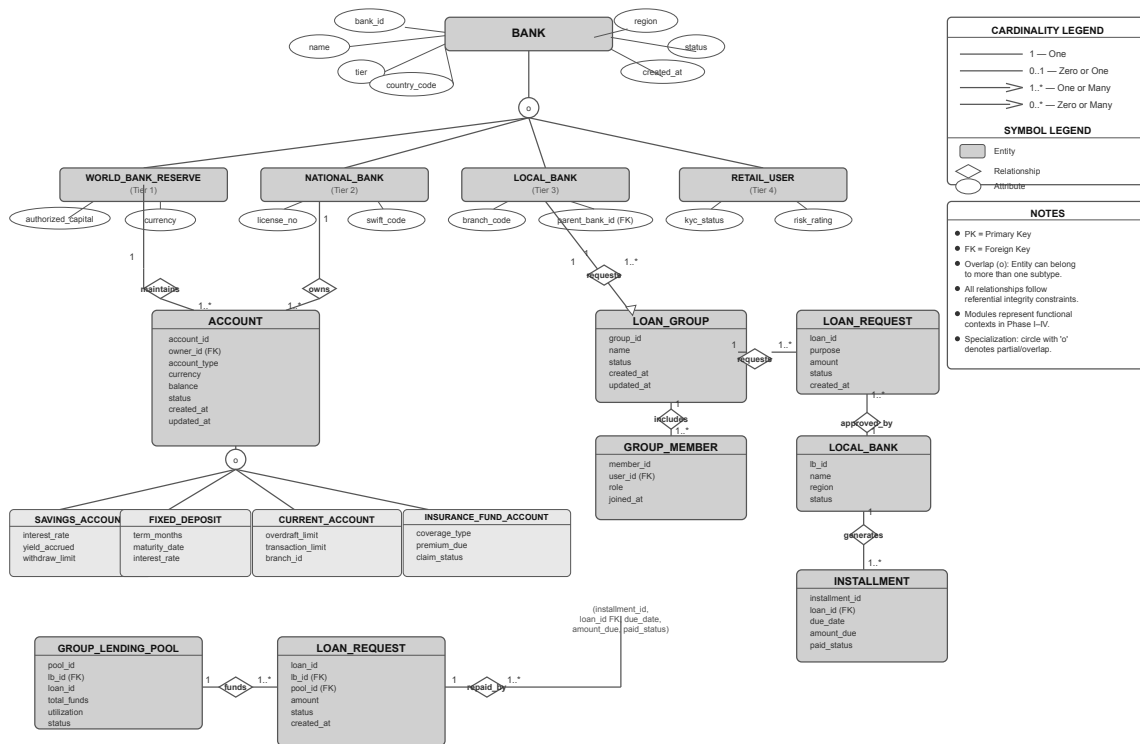


Figure 3.7: Enhanced Entity-Relationship (EER) model showing generalization / specialization (INSTITUTION → WORLD_BANK / NATIONAL_BANK / LOCAL_BANK; BANK_USER → World-/National-/Local-bank-admin subtypes), the weak entity INSTALLMENT identified by its parent LOAN, the multi-valued attribute INCOME_PROOF, and participation constraints (total / partial).

3.4.1 Entity Summary

Table 3.7: Database entity summary (31 core entities): primary keys and roles.

Entity	Primary key	Role / Description
INSTITUTION	institution_id	Shared supertype for all banking-tier identities. Attributes: institution_type ENUM('WORLD', 'NATIONAL', 'LOCAL'), name, wallet_address, contract_address, status, created_at. All multi-entity tables (INTERBANK_LOAN, NETTING_ENTRY, ...) reference this PK for enforceable cross-tier FKs.
COUNTRY	country_code	ISO 3166-1 alpha-2 reference (PK). Attributes: name, default_stablecoin_symbol, regulatory_regime_tag.
WORLD_BANK	institution_id	Subtype of INSTITUTION (institution_type='WORLD'); PK is also FK → INSTITUTION. Singleton enforced: CHECK (institution_id = 1) plus no-delete trigger. Global reserve parameters; on-chain binding: chain_id, last_synced_block, deployment_tx_hash.
NATIONAL_BANK	institution_id	Subtype of INSTITUTION (institution_type='NATIONAL'); FK country_code → COUNTRY with UNIQUE (one national reserve per nation). FK parent_world_bank_id → WORLD_BANK.
LOCAL_BANK	institution_id	Subtype of INSTITUTION (institution_type='LOCAL'); FK national_bank_id → NATIONAL_BANK. City-level banks lending to retail clients.
BANK_USER	bank_user_id	Bank staff with role-based permissions. bank_type ENUM('world', 'national', 'local'); nullable FKs world_bank_id, national_bank_id, local_bank_id with CHECK enforcing exactly one parent institution (Table 3.18).
BORROWER	borrower_id	End borrowers (UK: wallet_address). FK registered_local_bank_id → LOCAL_BANK (NOT NULL, set at onboarding) records the client's home Local Bank for KYC attribution and dashboard scoping; LOAN_REQUEST remains the separate M:N association for banks at which the client has applied. ¹
LOAN_REQUEST	request_id	Loan applications and approval workflow. status (lifecycle): DRAFT, SUBMITTED, UNDER_REVIEW, APPROVED, REJECTED, DISBURSED, CANCELLED. ^{4/} oracle_state (event-listener only): NONE, COMMIT_PENDING, COMMITTED, DON_REQUESTED, SCORE_REVEALED.

Table 3.9: Database entity summary (continued): auxiliary, analytics, and profile entities.

Entity	Primary key	Role / Description
INCOME_PROOF	proof_id	Income verification documents (multi-valued per borrower).
CHAT_MESSAGE	message_id	Human-to-human client↔bank messages in the loan workflow. FK to LOAN_REQUEST and/or LOAN. Not agent-specific; not for inter-tier authority communication (see INSTITUTION_MESSAGE).
INSTITUTION_MESSAGE	message_id	Cross-tier / cross-authority communication audit trail. FK sender_institution_id, receiver_institution_id → INSTITUTION; message_type (SAR_ESCALATION, SYNDICATION_INVITE, RESERVE_ALERT, IBLP_REQUEST, ...), payload (JSONB), status, created_at, read_at.
AGENT_CONVERSATION_TURN	session_id, turn_id	Each turn of the optional Qwen3-8B agent conversation. FK session_id → SESSIONS. Stores role (user/assistant/system), content_summary, tokens_used, tool_calls (JSONB). Separate from AGENT_ACTION_LOG (write-tool execution only).
MODEL_REGISTRY	model_id	ML model lineage for audit and explainability. Attributes: model_name, version, training_date, metrics (JSONB), artifact_hash, is_active. Referenced by LOAN_RISK_ASSESSMENT.
LOAN_RISK_ASSESSMENT	assessment_id	Per-loan ML inference record: FK loan_request_id → LOAN_REQUEST; FK model_id → MODEL_REGISTRY; fraud_probability, anomaly_score, shap_vector (JSONB), feature_vector_hash, created_at.
SECURITY_EVENT	event_id	System-wide security and AML events (append-only). Attributes: event_type, actor_id, actor_type, severity, payload (JSONB), created_at. INSERT-only RLS policy; distinct from per-loan LOAN_RISK_ASSESSMENT.
MARKET_DATA	market_data_id	Non-Chainlink price cache (e.g. CoinGecko fallback). Read-only auxiliary; no FK into core lending. Key attributes: symbol, price_usd, source, recorded_at. Chainlink rounds are stored in CHAINLINK_PRICE_ROUND.
CHAINLINK_PRICE_ROUND	round_id	Chainlink AggregatorV3Interface round cache: feed_address, pair_symbol, answer, round_id_on_chain, answered_in_round, updated_at, chain_id. Required for staleness check (answeredInRound < roundId).
PROFILE_SETTING	profile_id	Interface preferences for borrow-

Table 3.10: Database entity summary (continued): agent session and reference entities.

Entity	Primary key	Role / Description
SESSIONS	session_id	Wallet session registry for the retail-client agent add-on: FK borrower_id → BORROWER; device/IP hash, EIP-7702 session-key scope and TTL, parent lineage, compression summary, end reason. Bank-authority agent sessions (actor_type='BANK_USER') are Future Work (Section 3.1). Phase II: SESSION_ANCESTOR closure table (Section 3.4.6).
AGENT_ACTION_LOG	action_id	Append-only agent write-tool log (session_id, tool_name, parameters, confirmation_turn_id, onchain_tx_hash, status). Separate from AGENT_CONVERSATION_TURN; INSERT-only RLS policy.
INTEREST_RATE_TIER	tier_id	Normalised kinked-rate parameters (base_rate, kink_utilisation, rate_above_kink, max_rate); Section 3.4.4. Discriminator: bank_tier_type ENUM('world', 'national', 'local'). Temporal validity: effective_from, effective_to, governance_proposal_id, superseded_by, is_active. Partial unique: (bank_tier_type) WHERE is_active = TRUE.
ASSETS	asset_id	Collateral and loan asset registry (UK: symbol); referenced by LOAN via collateral_asset_id and loan_asset_id.

This entity summary table enumerates 31 core relational entities. Key additions beyond the original lending core include the INSTITUTION supertype and COUNTRY reference (shared identity for cross-tier operations), INSTITUTION_MESSAGE (inter-tier authority communication), split ML tables (MODEL_REGISTRY, LOAN_RISK_ASSESSMENT, SECURITY_EVENT_LOG), registered_local_bank_id on BORROWER, and institution-aware TRANSACTION columns. Six extension entities (Table 3.12) and 14 Phase II–III stubs (Table 3.14) complete the schema to 51 total.

Table 3.12: Extension entities (Phase II–III): primary keys and roles.

Entity	Primary key	Role / Description
GROUP_CONSENT	consent_id	Per-member multi-signature consent for GroupLendingPool (UK: loan_request_id, member_id). consent_tx_hash, event_log_id.
TRANCED_POOL_subscription	subscription_id	Investor tranche positions: pool_id, subscriber_id, subscriber_type, tranche (SENIOR/JUNIOR), committed_amount, subscription_tx_hash.
NETTING_COORD	batch_id	Off-chain Settlement Coordinator state: status (COMPUTING → SETTLED), gross_obligations, net_settlements (JSONB), challenge_deadline, on_chain_batch_id.
KYC_VERIFICATION_REQUEST	request_id	ZKP KYC workflow tracker: borrower_id, kyc_level_requested, status, credential_hash, proof_hash, on_chain_tx_hash.
ORACLE_HEARTBEAT	monitor_id	Chainlink feed staleness monitor: feed_address, pair_symbol, expected_heartbeat_seconds, last_round_id, is_stale, chain_id.
SESSION_ANCESTOR	session_id, ancestor_id	Closure table for session lineage (replaces recursive INSERT trigger). depth (1 = direct parent).

Table 3.14: Extended banking and multi-entity entity stubs (Phase II–III).

Entity	Primary Key	Role / FK anchors
SAVINGS_ACCOUNT	account_id	SavingsVault deposit account. FK borrower_id → BORROWER; FK local_bank_id → LOCAL_BANK(institution_id). on_chain_address, balance, last_synced_block.
FIXED_DEPOSIT	deposit_id	FixedDeposit term product. FK borrower_id; FK local_bank_id → LOCAL_BANK(institution_id). principal, maturity_date, on_chain_address.
CURRENT_ACCOUNT	ca_id	Transactional account. FK borrower_id; FK local_bank_id → LOCAL_BANK(institution_id). balance, on_chain_address.
LOAN_GROUP	group_id	GroupLendingPool on-chain group. FK local_bank_id → LOCAL_BANK(institution_id). on_chain_group_id, max_members, status.
GROUP_MEMBER	member_id	Weak entity on LOAN_GROUP. FK group_id; FK borrower_id. joined_at, consent_tx_hash.
INSURANCE_FUND	fund_id	InsuranceFund reserve tracker. FK local_bank_id → LOCAL_BANK(institution_id). balance, on_chain_address, last_synced_block.
INTERBANK_LOAN	ib_loan_id	IBLP same-tier liquidity loan. FK lender_institution_id, borrower_institution_id → INSTITUTION; trigger/CHECK enforces matching institution_type per pool instance (IBLP_NB or IBLP_LB). tenor_days, amount, status, on_chain_tx_hash.
UPWARD_DEPOSIT	ud_id	UpwardDepositFacility surplus deposit. FK depositor_institution_id, parent_institution_id → INSTITUTION; CHECK enforces adjacent tier-pair (Local→National or National→World). amount, rate, on_chain_tx_hash.
SYNDICATE	syndicate_id	SyndicatedLoan deal header. FK lead_institution_id → INSTITUTION; FK borrower_id → BORROWER. total_amount, status, on_chain_address.
SYNDICATE_MEMBER	syndicate_id, member_institution_id	Co-lender position. FK member_institution_id → INSTITUTION. committed_amount,

Extended banking entities in Figure 3.6 (SAVINGS_ACCOUNT, FIXED_DEPOSIT, LOAN_GROUP, ...) include planned on-chain anchor fields on_chain_address and last_synced_block so the event listener can map incoming events to the correct DB row. Multi-entity stub FKs anchor to INSTITUTION.institution_id (Table 3.14) rather than tier-specific PK namespaces, enabling one enforceable referential-integrity path for InterBankLendingPool, NettingEngine, and related contracts. Phase III adds SESSION_KEY_PERMISSION to normalise MCP tool allowlists currently stored as JSONB in SESSIONS.

3.4.2 Institution Supertype and Cross-Tier Referential Integrity

The four-tier hierarchy (World Bank $\rightarrow n$ National Banks $\rightarrow n$ Local Banks per nation $\rightarrow$ clients) requires a shared institution identity space for multi-entity operations specified in Section 3.14. Rather than three disjoint PK namespaces (world_bank_id, national_bank_id, local_bank_id), the schema introduces an INSTITUTION supertype with ID-based table-per-type inheritance: WORLD_BANK, NATIONAL_BANK, and LOCAL_BANK are subtype tables whose primary key is also a foreign key to INSTITUTION.institution_id. This pattern remains 3NF/BCNF-safe (shared identity, not shared attribute duplication) and allows INTERBANK_LOAN, SYNDICATE_MEMBER, NETTING_ENTRY, and TREASURY_SWAP to reference any banking-tier participant through a single FK. Application triggers or CHECK constraints enforce tier-appropriate pairings—for example, both rows in an IBLP_NB loan must have institution_type='NATIONAL'.

Two cardinality rules implied by the architecture are encoded at the database layer. **World Bank singleton:** WORLD_BANK permits exactly one row (CHECK (institution_id = 1)), mirroring the singleton WorldBankReserve contract. **One national reserve per nation:** NATIONAL_BANK.country_code references COUNTRY with a UNIQUE constraint, guaranteeing at most one national-bank institution per ISO jurisdiction and providing a clean join key for FX, stablecoin defaults, and regulatory-sandbox reporting.

Retail clients retain a permanent home-bank link via BORROWER.registered_local_bank_id (set at onboarding, NOT NULL), independent of whether they have submitted a LOAN_REQUEST. The LOAN_REQUEST entity continues to model the many-to-many “which Local Banks has this client borrowed from” relationship required by the portable Credit Passport design.

Inter-tier authority communication (SAR escalation, syndication invites, reserve alerts) is recorded in INSTITUTION_MESSAGE, distinct from client-facing CHAT_MESSAGE. ML inference and security monitoring are split at Phase I into LOAN_RISK_ASSESSMENT (per-loan, SHAP-linked, FK to MODEL_REGISTRY) and SECURITY_EVENT_LOG (system-wide, append-only), avoiding sparse-column conflation in a single aggregate table.

Multi-tenant row-level security (Future Work). Append-only RLS on AGENT_ACTION_LOG and AUDIT_LOGS is specified in Table 3.18. For a global deployment with jurisdictional separation, production would extend RLS (or physical sharding) to scope PII-bearing core tables—BORROWER, INCOME_PROOF, LOAN, LOAN_REQUEST—by registered_local_bank_id ancestry through LOCAL_BANK $\rightarrow$ NATIONAL_BANK $\rightarrow$ COUNTRY, preventing a National Bank Admin in one jurisdiction from querying another nation’s client rows. This tenant-isolation layer is explicitly deferred to the final-thesis implementation phase.

3.4.3 EER Constructs Applied

Table 3.16: EER constructs applied: specialisation, hierarchy, and constraints.

EER Construct	Applied To	Notes
Specialisation (dis-joint)	BANK_USER → WorldBankUser, NationalBankUser, LocalBankUser	A bank user belongs to exactly one bank type (world, national, or local); enforced by CHECK constraint (Table 3.18).
Generalisation / specialisation	INSTITUTION → WORLD_BANK, NATIONAL_BANK, LOCAL_BANK	ID-based table-per-type inheritance: subtype PK = FK to INSTITUTION.institution_id; shared identity for cross-tier FKs (Section 3.4.2).
Weak entity	INSTALLMENT depends on LOAN	Existence and identity determined by parent loan.
Multi-valued attribute	INCOME_PROOF per BORROWER	Modelled as separate entity to maintain 1NF.
Aggregation	LOAN_RISK_ASSESS ← LOAN_REQUEST; SECURITY_EVENT_LOG ← runtime monitors	Per-loan ML scores and system-wide security events are separate append-only relations, not a conflated aggregate.
Association entity	LOAN_REQUEST links BORROWER + LOCAL_BANK	Captures many-to-many borrowing relationships; distinct from BORROWER.registered_local_bank_id (home bank at onboarding).
Participation constraint	BORROWER.registered_local_bank_id NOT NULL at onboarding	Home Local Bank is mandatory; LOAN_REQUEST participation for lending services enforced at application layer.
Append-only table (policy)	AGENT_ACTION_LOGS, AUDIT_LOGS, SECURITY_EVENT_LOG	DB-level INSERT-only role + Row-Level Security policy enforces that no UPDATE or DELETE is permitted on audit records.

This table captures the EER constructs applied to represent specialization, hierarchy, and constraints in the data model. It demonstrates how institutional roles and banking participants are modeled without duplicating attributes across tables, improving data integrity and query clarity.

3.4.4 Normalization

The schema has been normalized to Third Normal Form (3NF) and checked against Boyce-Codd Normal Form (BCNF):

- **1NF:** There are no repeating groups in any of the attributes. Separate entities store multi-valued attributes, such as proof of income.
- **2NF:** There are no partial dependencies. The full composite key (`loan_id`, `installment_number`) determines the non-key attributes of `INSTALLMENT`.
- **3NF:** No dependencies that go through other dependencies. Instead of storing redundant values like `total_borrowed` in bank entities, they are calculated at query time.
- **BCNF:** All non-trivial functional dependencies have a superkey as their determinant. For example, in `INTEREST_RATE_TIER`, `tier_id` $\rightarrow$ `{base_rate, kink_utilisation, rate_above_kink, max_rate}`: the determinant `tier_id` is the primary key, satisfying BCNF. Prior to the normalisation fix in this section, interest-rate parameters were columns on the three bank entities, determined by `bank_tier_id` (a non-key attribute), violating BCNF.

A transitive dependency was identified during schema review: `interest_rate_parameters` $\rightarrow$ `bank_tier_id` (interest rate values were stored as columns in the World Bank, National Bank, and Local Bank entities, making them dependent on a non-key attribute via the tier relationship). This violates 3NF. The fix is to extract these parameters into a dedicated `INTEREST_RATE_TIER` table keyed by `tier_id`, which each bank entity references via FK. This normalisation change ensures that updating a base rate does not require touching multiple bank-tier rows.

3.4.5 Physical Schema Reference

Indexing strategy, representative functional dependencies, the `mv_client_dashboard` materialized view, and projection-table CQRS policies are documented in Appendix A to keep this chapter focused on conceptual design. B-tree indexes are chosen for range queries on rolling borrowing-limit windows and overdue-installment detection; the authoritative enforcement of borrowing limits remains on-chain.

The `parent_session_id` self-referential foreign key in `SESSIONS` introduces a recursive relationship within a single entity. This does not violate 3NF (the dependency is `session_id` $\rightarrow$ `parent_session_id`, not a transitive dependency), but it requires an acyclicity constraint to prevent circular ancestry chains: a session cannot be its own ancestor. In the prototype this is enforced by a `CHECK` constraint in conjunction with a trigger that walks the ancestry chain before `INSERT`. Phase II replaces this recursive trigger with a **closure table** (`SESSION_ANCESTOR`): on `INSERT` of a new session, rows are copied from the parent's ancestor set with `depth + 1`; the cycle check becomes a single indexed lookup `WHERE ancestor_id = NEW.session_id` (Section 3.4.6).

3.4.6 Relational Integrity Constraints

Table 3.18: Relational integrity constraints.

Constraint Type	Examples
Primary Key	One surrogate or composite key per Table 3.7, Table 3.12, and Table 3.14: e.g. <code>institution_id</code> , <code>country_code</code> , <code>bank_user_id</code> , <code>borrower_id</code> , <code>request_id</code> (LOAN_REQUEST), <code>loan_id</code> (LOAN), <code>(loan_id, installment_number)</code> (INSTALLMENT), <code>transaction_id</code> , <code>model_id</code> , <code>assessment_id</code> , <code>message_id</code> (INSTITUTION_MESSAGE),...
Foreign Key	<code>WORLD_BANK.institution_id</code> , <code>NATIONAL_BANK.institution_id</code> , <code>LOCAL_BANK.institution_id</code> → <code>INSTITUTION</code> ; <code>NATIONAL_BANK.country_code</code> → <code>COUNTRY</code> ; <code>BORROWER.registered_local_bank_id</code> → <code>LOCAL_BANK</code> ; <code>local_bank_id</code> in <code>LOAN_REQUEST</code> references <code>LOCAL_BANK(institution_id)</code> ; <code>LOAN_RISK_ASSESSMENT.model_id</code> → <code>MODEL_REGISTRY</code> ; <code>LOAN_RISK_ASSESSMENT.loan_request_id</code> → <code>LOAN_REQUEST</code> ; <code>INSTITUTION_MESSAGE.sender_institution_id</code> , <code>receiver_institution_id</code> → <code>INSTITUTION</code> ; multi-entity stubs (Table 3.14) anchor institution participants to <code>INSTITUTION.institution_id</code> .
UNIQUE	<code>wallet_address</code> in <code>BORROWER</code> ; <code>NATIONAL_BANK(country_code)</code> (one national reserve per nation); <code>contract_address</code> in <code>INSTITUTION</code> ; <code>blockchain_tx_hash</code> in <code>LOAN_REQUEST</code> ; <code>borrower_id</code> in <code>BORROWING_LIMIT</code> ; <code>symbol</code> in <code>ASSETS</code> ; <code>tx_hash</code> in <code>BLOCKCHAIN_EVENT_LOG</code> ; partial unique on <code>INTEREST_RATE_TIER(bank_tier_type)</code> WHERE <code>is_active = TRUE</code> .
CHECK	<code>WORLD_BANK: institution_id = 1</code> (singleton). <code>BANK_USER: (bank_type='world' AND world_bank_id IS NOT NULL) OR (bank_type='national' AND national_bank_id IS NOT NULL) OR (bank_type='local' AND local_bank_id IS NOT NULL)</code> . <code>TRANSACTION: at least one of borrower_id, origin_institution_id, counterparty_institution_id is non-null</code> .
NOT NULL	Core attributes: <code>INSTITUTION.name</code> , <code>INSTITUTION.institution_type</code> , <code>BORROWER.wallet_address</code> , <code>BORROWER.registered_local_bank_id</code> , <code>status</code> , <code>oracle_state</code> (default NONE).
APPEND-ONLY (DB policy)	<code>AGENT_ACTION_LOG</code> , <code>CREDIT_PASSPORT_HISTORY</code> , <code>SECURITY_EVENT_LOG</code> : CREATE ROLE <code>audit_writer</code> ; GRANT INSERT ON <code>agent_action_log</code> TO <code>audit_writer</code> ⁵⁶ , REVOKE UPDATE, DELETE FROM PUBLIC. <code>AUDIT_LOGS</code> : same policy applied via Row-Level Security (ALTER TABLE <code>audit_logs</code> ENABLE ROW LEVEL SECURITY, CREATE POLICY

This integrity constraints table summarizes referential and domain constraints that keep loan, repayment, and identity records consistent. These constraints reduce operational risk by preventing orphaned records and invalid lifecycle states, which is critical for auditability in financial systems.

3.5 On-Chain and Off-Chain Data Partitioning

Table 3.20: Data partitioning between on-chain and off-chain storage.

Data Category	Storage	Rationale
Reserve balances, loan requests, approval/rejection events, repayment transactions	On-chain	Immutability, public auditability, trustless verification.
User profiles, income verification documents, chat messages, loan risk assessments, security event logs	Off-chain (database)	Data privacy, query flexibility, storage cost optimisation.
Borrowing limit computations	Off-chain with on-chain enforcement	Complex temporal aggregation; results committed as on-chain constraints via <code>LocalBank.updateBorrowingLimit(borrowerId, newLimit)</code> (see sync protocol below).
Cryptocurrency market data (non-Chainlink)	Off-chain (MARKET_DATA)	CoinGecko and other fallback feeds; high-frequency updates; external API dependency.
Chainlink oracle price data	Off-chain (CHAINLINK_PR)	Each <code>latestRoundData()</code> round cached with <code>answered_in_round</code> for staleness detection; on-chain <code>AggregatorV3Interface</code> is authoritative. MARKET_DATA does <i>not</i> store Chainlink rounds.
Agent action execution log	Off-chain (append-only DB)	Immutable audit trail of every agent write-tool call, linked to on-chain tx hash; not stored on-chain to save gas.
Session key material	Off-chain (sessions table)	EIP-7702 session key hash, scope JSON, and TTL stored off-chain; the scope restrictions are enforced on-chain by the session key contract at signing time.
Session lineage chains (SESSIONS, message history, compression summaries)	Off-chain (PostgreSQL)	Full transcript history is too large for on-chain storage. The lineage chain enables searchable long-term memory without gas cost. AGENT_ACTION_LOG provides the on-chain audit link via confirmed transaction hashes.

This partitioning table clarifies which data must live on-chain (role bindings, reserves, and critical state transitions) versus off-chain (documents, analytics features, and operational metadata). The partitioning choice balances transparency and tamper-resistance with practical storage and privacy constraints.

Borrowing limit sync protocol. When on-chain loan events change a borrower's rolling exposure, the event listener: (1) receives `LoanRepaid/LoanDisbursed` events and updates `TRANSACTION`; (2) recomputes 6-month and 1-year rolling windows from `TRANSACTION`; (3) if the computed limit differs from `on_chain_enforced_limit` in `BORROWING_LIMIT`, the Express API calls `LocalBank.updateBorrowingLimit(borrowerId, newLimit)`; (4) on receipt of `BorrowingLimitUpdated`, the listener updates the `BORROWING_LIMIT` row and clears `sync_discrepancy_flag`.

Projection vs. application tables. Tables owned exclusively by the event listener (projection tables) include `CREDIT_PASSPORT`, `CREDIT_PASSPORT_HISTORY`, `BORROWING_LIMIT`, and `COLLATERAL_POSITION`. Express.js handlers may `SELECT` from these tables but must never `INSERT/UPDATE` them directly; PostgreSQL RLS enforces this separation (Table 3.18). Application tables (`LOAN_REQUEST`, `CHAT_MESSAGE`, `PROFILE_SETTING`, etc.) are Express-owned. The `oracle_state` column on `LOAN_REQUEST` is updated by the event listener only; Express API roles are denied `UPDATE` on this column via a column-level privilege: `REVOKE UPDATE (oracle_state) ON loan_request FROM api_role; GRANT UPDATE (oracle_state) ON loan_request TO event_listener_role`. This is a deliberate narrow exception to the Express-owned classification: all other `LOAN_REQUEST` columns remain Express-writable.

On-Chain vs Off-Chain Data Partitioning

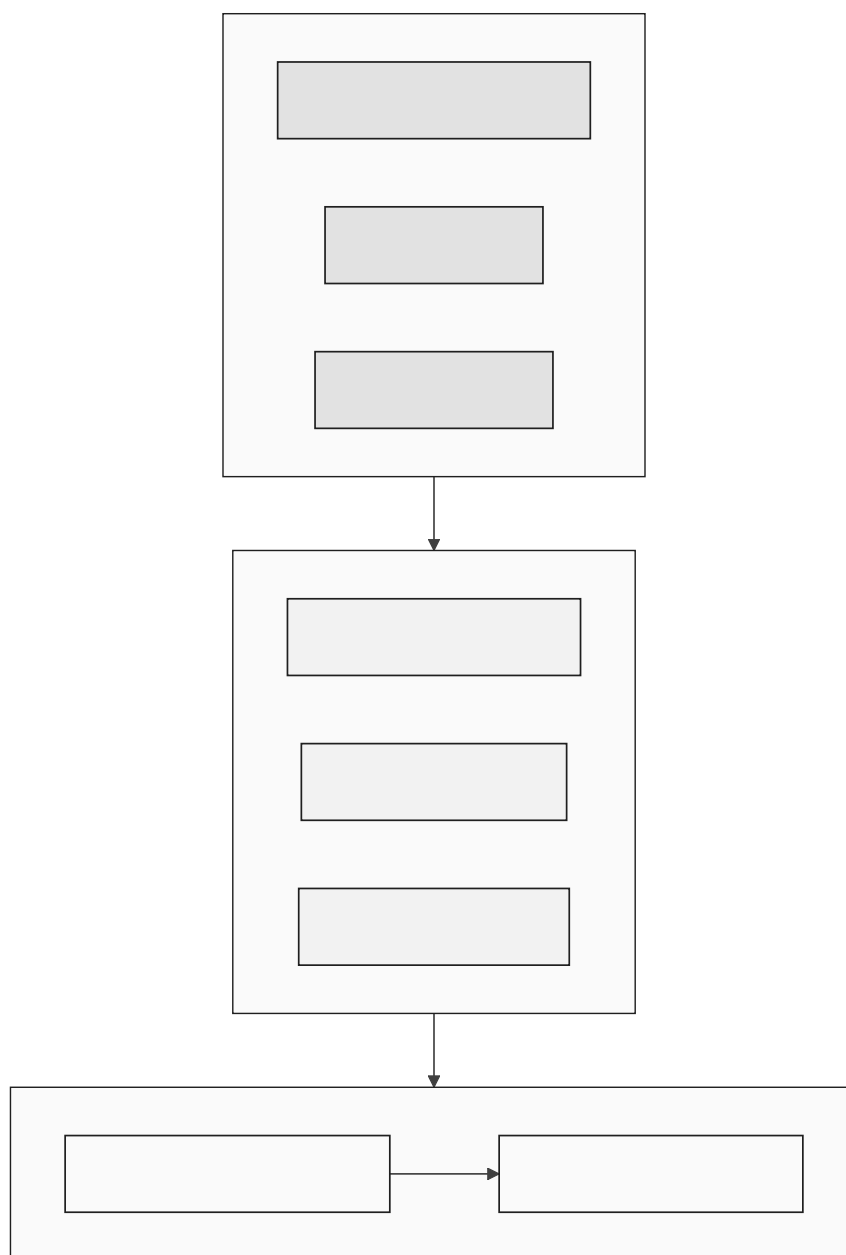


Figure 3.8: On-chain versus off-chain data partitioning: reserve balances, loan state, and role bindings on the EVM; KYC documents, agent logs, and ML scores in PostgreSQL; borrowing limits computed off-chain and enforced on-chain at disbursement.

3.6 Operating Context

The Crypto World Bank is specified as a **blockchain-based institutional finance platform** on EVM Layer 2, not as a geographic pilot for a single rural market. The primary contribution is programmable hierarchical banking—reserve enforcement, tiered capital flow, on-chain auditability, and modular lending products—deployable on Polygon zkEVM Cardona (live) with Ethereum Sepolia retained only for Certora formal verification in Phase IV.

Three design choices anchor the operating context. First, **stablecoin-denominated re-**

tail lending (Section 1.10.3) limits liability-side volatility when collateral and loans use different asset types. Second, **low per-transaction gas cost** on zkEVM L2 makes high-frequency installment and reserve operations economically viable relative to Ethereum mainnet. Third, **optional account abstraction** (Section 3.8.1) can simplify wallet onboarding for crypto-naïve retail users without changing smart-contract role logic. Solidarity group lending (Section 3.17) is a programmable product module informed by microfinance literature (BRAC, Grameen); it is not scoped to a specific country deployment. Institutional onboarding is planned through academic pilots, open-source replication, and regulatory sandbox partnerships (Section 3.18.5).

3.7 Digital Identity System

The platform’s identity model operates across two layers, combining wallet-based authentication with off-chain document verification, and is designed to accommodate compliance requirements in future phases.

- **Identity based on wallets:** Users authenticate via Ethereum wallet signatures (MetaMask, WalletConnect), eliminating the need for centralised credential storage.
- **Role binding:** In the smart contract permission system, wallet addresses are linked to hierarchical roles like Owner, National Bank, Local Bank, Approver, and Retail Client.
- **Limits of wallet-based identity:** Wallet ownership proves transaction history and cryptographic control but cannot independently prove legal identity, jurisdiction, or age. This distinction is critical for regulated banking operations. A compromised or lost wallet requires an off-chain recovery process and on-chain role revocation followed by re-binding to a replacement address, a governance workflow that must be carefully designed to prevent unauthorized role escalation.
- **On-chain versus off-chain identity layers:** The system maintains two identity layers. On-chain identity consists of the wallet address, assigned role, and transaction history recorded permanently on the blockchain. Off-chain identity consists of document-verified income, KYC credentials, and AML screening results stored in the PostgreSQL database and linked to the wallet address by hash reference. The planned ZKP-based compliance extension would allow users to prove off-chain KYC status to the smart contract without exposing personal documents on-chain.
- **EIP-7702 session key authorisation:** For agent-executed banking operations, the client authorises a scoped, time-bound session key at login. The session key is restricted to a named set of MCP write tools (e.g., only `submit_loan_application` and `pay_installment`), a value cap (e.g., 500 USDC per transaction), and a 24-hour TTL after which it auto-expires. The session key cannot transfer funds to external addresses or perform any operation outside its approved scope. Every session key usage is logged in `AGENT_ACTION_LOG` with the conversation turn ID from `AGENT_CONVERSATION_TURN` as the audit reference. The EIP-7702 delegation contract address is registered in `CHAIN_REGISTRY` (Phase IV) as platform-wide configuration.

3.7.1 Compliance Identity (Future Work)

Future Work. A two-circuit privacy-preserving compliance gateway—Groth16 zkKYC (Circom 2.0), zkAML velocity proofs [R19], and W3C DID/VC anchoring—is motivated by Contribution 4 (Section 1.6) and the literature synthesis in Section 2.2.7. Circuit-level design,

verifier deployment, and on-chain integration are deferred to Section 6. At MVT scope, retail and institutional onboarding uses tiered off-chain KYC, hash-linked credentials in PostgreSQL, and on-chain RBAC role gates (Section 3.8.2).

3.8 User Taxonomy and Onboarding Flows

The thesis now uses a formal actor taxonomy of nine actors following the systems-analysis convention documented in the project Binance Software Engineering Architecture study (Appendix D, entry IR-4) [130]. Five primary actors (A1–A5) initiate actions; four secondary actors (A6–A9) are external systems or authorities. Table 3.22 gives the operational user taxonomy with per-user-type KYC level, wallet type, and database-residence assumption; Table 3.23 presents the complete formal actor taxonomy; and Table 3.24 presents the actor permission matrix.

Table 3.22: Operational user taxonomy: KYC level, wallet type, and data residence by user type.

User Type	Tier	KYC Level	Wallet Type	Data Residence
Anonymous Visitor	—	None	None	Session only (Redis)
Registered Retail User	Tier 4	Level 1 (gov. ID + selfie)	MetaMask or optional ERC-4337	Off-chain primary; on-chain role only
Group Client	Tier 4	Level 1 + group verification	ERC-4337 abstracted	Off-chain + on-chain group contract
Local Bank Operator	Tier 3	Full institutional KYC	MetaMask (institutional)	Off-chain + on-chain role binding
Local Bank Approver	Tier 3	Full institutional + background check	MetaMask + Safe co-signer	Off-chain + on-chain role binding
National Bank Admin	Tier 2	Full institutional + regulatory license	Safe multisig mandatory	Off-chain + on-chain role binding
World Bank Admin (Governance)	Tier 1	Founding team + supervisor attestation	Safe 3-of-5 multisig	On-chain governance only

Table 3.23: Formal actor taxonomy (nine actors).

A1a	Retail Client (pre-KYC)	Browsing only; no banking transactions.
A1b	Retail Client (KYC Tier 1)	Bronze/Silver credit tier; small loans up to the KYC-1 cap.
A1c	Retail Client (KYC Tier 2)	Gold/Platinum/Diamond; larger limits, group lending.
A2	Local Bank Admin (Approver)	Loan approver; risk officer; reviews AML alerts.
A3	National Bank Admin	Capital allocator; compliance officer; SAR review; freeze authority.
A4	World Bank Admin (Governance)	Parameter governor; system operator via multi-sig.
A5	AI Agent (optional)	Conversational add-on; read tools always permitted; write tools require explicit human confirmation gate before execution.
<i>Secondary Actors (external systems or authorities)</i>		
A6	Regulatory Authority	Read-only audit access via encrypted data package.
A7	Chainlink DON	Oracle, price feeds, automation.
A8	Blockchain Validator	Polygon zkEVM validity proof generation.
A9	External Auditor	Chainlink PoR verification.

Table 3.24: Actor permission matrix (tabular summary of primary actions).

Action	Client	LB min	Ad- min	NB min	Ad- min	WB min	Ad- min	AI Agent
View own loan status	YES	YES*		YES*		YES*		READ
Submit loan application	YES	NO		NO		NO		WRITE†
Pay installment	YES	NO		NO		NO		WRITE†
Submit KYC documents	YES	NO		NO		NO		NO
Approve loan application	NO	YES		NO		NO		NO
Reject loan application	NO	YES		NO		NO		NO
Freeze client account	NO	YES		YES		YES		NO
Set borrowing rate	NO	NO		YES		YES		NO
Change reserve ratio	NO	NO		NO		YES		NO
Upgrade LB borrowing cap	NO	NO		YES		YES		NO
View audit logs (all)	NO	YES*		YES*		YES		NO
Generate SAR report	NO	YES		YES		YES		NO
View reserve summary (public)	YES	YES		YES		YES		READ

† All agent write tools require an explicit human confirmation gate: the agent assembles the full parameter set, presents a summary to the client, and waits for affirmative consent. No write tool is ever called without a confirmation turn in the conversation history. Agent scope is retail-client only at MVT; bank-authority read/aggregate tools are Future Work (Section 3.1). * own tier only.

	 Client (React UI / MCP Agent)	 Local Bank (Tier 3)	 National Bank (Tier 2)	 World Bank Reserve (Tier 1)	 AI Agent (Optional)
Action / Tool					
Submit loan request	✓	✓	○	○	---- ✓
Pay installment	✓	✓	○	○	---- ✓
Approve loan	○	✓	✓	○	---- ○
Approve account	○	✓	✓	○	---- ○
Freeze accounts	○	○	✓	✓	---- ○
View all rates	✓	✓	✓	✓	---- ✓
Read tools freely	✓	✓	✓	✓	---- ✓
Write tools (humans only)	○	✓	✓	✓	---- ○

 Allowed
  Not allowed
  Full access (as per role)
  Optional / limited (AI Agent)

Figure 3.9: Actor permission matrix (primary actions): clients submit loans and pay installments via the React UI (or optionally via the MCP agent); Local Bank approvers approve and freeze accounts; National and World Bank admins set rates and reserve parameters; the optional AI Agent may call read tools freely and write tools only after human confirmation (†).

3.8.1 ERC-4337 Account Abstraction for Retail Onboarding

The single largest accessibility contradiction in the original thesis was that retail clients were assumed to already understand MetaMask, seed phrases, gas fees, and Ethereum transactions. This is incompatible with a mobile-first, crypto-naïve retail onboarding goal. The platform resolves the contradiction by adopting **ERC-4337 Account Abstraction (AA)** for all retail-tier (Tier 4) clients, with the EIP-7702 Pectra upgrade (May 2025) providing the gas-sponsorship layer. Since its launch on Ethereum mainnet in March 2023, ERC-4337 has enabled over 40 million smart accounts and processed more than 100 million transactions across major L2 networks including Polygon, Arbitrum, Optimism, and Base, demonstrating production-grade viability at scale [18-1]. The current production EntryPoint is **v0.7**, deployed at `0x0000000071727De22E5E9d8BAf0edAc6f37da032` on Ethereum, Polygon, and all major EVM chains; the platform targets this canonical address for all UserOperation routing.

Under ERC-4337 each retail client is issued a smart-contract account that is created on first sign-in. The user can register with email or a Google account via Firebase Auth, never sees a seed phrase, and never needs to acquire ETH for gas: a Paymaster contract sponsors the gas, funded by the World Bank Reserve as a financial-inclusion subsidy line. Account recovery uses a social-recovery guardian (a trusted email / phone) rather than a 24-word mnemonic.

Paymaster sybil-resistance controls. Gas sponsorship introduces a perverse incentive: an attacker can create many ERC-4337 accounts at zero cost and spam loan applications, draining the gas subsidy and potentially the loan pool if fraud detection fails. Three controls mitigate this. First, the Paymaster implements a **pre-KYC bootstrap allowance**: before KYC is completed, the Paymaster sponsors exactly two function signatures per wallet — `submitKYCHash(bytes32)` (storing the document hash for Level 1 KYC) and the ERC-4337 account creation transaction. Both are capped at a maximum of 0.001 ETH-equivalent total per wallet address; this is sufficient to complete KYC but insufficient to spam loan applications. All other gas sponsorship is conditional on the wallet having a verified KYC status (`kycVerified[wallet] = true`). This pre-KYC bootstrap window resolves the chicken-and-egg problem: a new user with no ETH can complete KYC using the bootstrap allowance, after which normal Paymaster sponsorship unlocks. Second, the Paymaster budget is capped per verified identity (not per account), limiting lifetime gas sponsorship to a fixed USDC equivalent per KYC identity. Third, the Paymaster rate-limits by IP and device fingerprint to prevent bulk account creation from a single attacker. These controls make Paymaster sponsorship conditional on verified identity rather than unconditional, preserving the inclusion mission while preventing abuse.

For the planned implementation, the Biconomy SDK or Alchemy Account Kit will provide bundler / Paymaster infrastructure on Polygon zkEVM Cardona. From the smart-contract perspective, an account-abstracted wallet is indistinguishable from an externally owned account, so none of the four-tier role logic needs to change. Retail clients may also connect a standard MetaMask wallet; account abstraction is an optional onboarding path, not a platform requirement.

EIP-7702 session keys — scoped agent wallet (optional add-on). EIP-7702 provides a cleaner solution than ERC-4337 paymasters for agent-controlled operations because the scope restriction is enforced at the key level, not at the application level. When a client authorises an AI Agent session at login, the EIP-7702 session key is subject to four hard constraints: (1) **Scope** — the key is restricted to a named set of MCP write tools (e.g., only `submit_loan_application` and `pay_installment`); (2) **Time-bound** — a 24-hour TTL after which the key auto-expires regardless of remaining uses; (3) **Value-capped** — a maximum of 500 USDC per transaction, preventing large unauthorised transfers; and (4) **Revocable** — the client can invalidate the session key at any time from the wallet UI. The session key *cannot* transfer funds to external addresses, exceed the value cap, perform any operation not in the approved scope, or execute after the TTL expires.

EIP-7702 is architecturally superior to ERC-4337 paymasters for agent-controlled operations and does not replace ERC-4337 for retail gas sponsorship — the two mechanisms serve different functions. ERC-4337 removes the need for ETH for gas; EIP-7702 scopes what the agent may sign. Both are active simultaneously for an agent session: the session key signs the transaction, the Paymaster sponsors the gas.

3.8.2 Tiered Risk-Based KYC

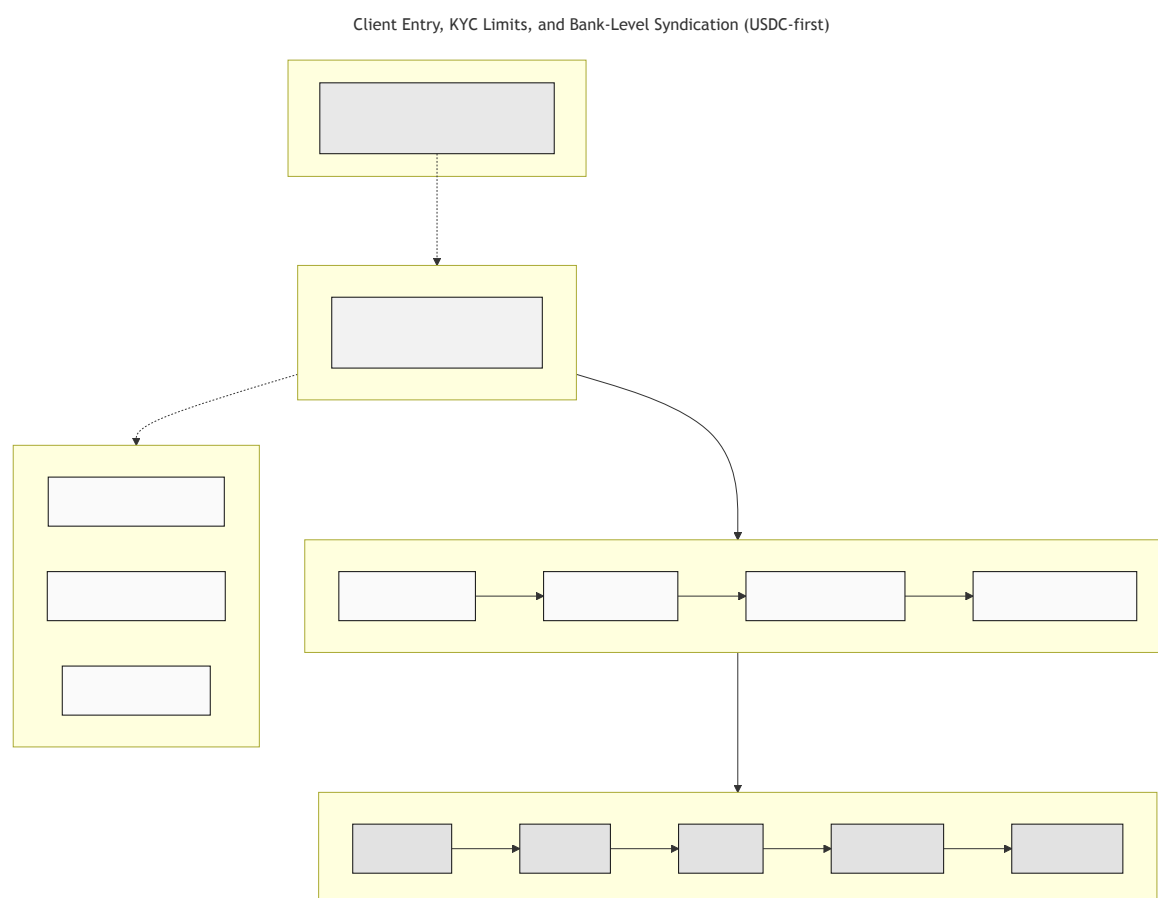


Figure 3.10: Client limits and bank-level syndication. Maximum loan amount is enforced on-chain via `kycLevel[wallet]` and Credit Passport tier (Sections 3.8.2, 3.12); stablecoin denomination is enforced at Tier 4 (Section 1.10.3); and the SBT carries the simultaneous-loan cap that the `GroupLendingPool` reads at every application. End clients enter only through Local Banks; exposures above a bank's prudent ceiling use `SyndicatedLoan` (Section 3.14.3), not direct client access to National or World Bank tiers.

Identity verification is gated by a risk-based KYC ladder rather than an all-or-nothing flag. Table 3.25 gives the four levels and the corresponding borrowing limits and verification times. The design is consistent with current Persona / Veriff guidance for retail-fintech KYC and the EU AI Act Article 86 transparency requirements.

Table 3.25: Tiered, risk-based KYC ladder. Higher KYC level unlocks larger loans but requires proportionally more verification.

Risk Level	Required Documents	Borrow Limit	Verify Time
Level 0 (Un-verified)	None	View only	Instant
Level 1 (Basic)	Government ID photo + selfie liveness check	Up to 0.1 ETH-equivalent (USDC)	2–5 min (automated)
Level 2 (Standard)	Government ID + proof of income + bank statement	Up to 2 ETH-equivalent	24 h (human review)
Level 3 (Enhanced)	Full document set + video interview	Up to 10 ETH-equivalent	48–72 h

After KYC approval, the smart contract sets `kycVerified[wallet] = true`, `kycLevel[wallet] = level`, and a one-year expiry `kycExpiry[wallet] = block.timestamp + 365 days`. Re-verification is enforced via the role-expiry mechanism. Personal data—government ID, biometric template, document files—is never stored on-chain; only a SHA-256 hash of the document is written, in compliance with GDPR data-minimization principles. On-chain zk-SNARK verification is scoped to Future Work (Section 6).

3.8.3 Five-Stage Retail Onboarding Funnel

The platform’s retail onboarding is structured as a five-stage funnel that progressively reveals blockchain complexity. Stage 1 is information browsing with zero friction (no account required). Stage 2 is account creation via wallet connect or optional ERC-4337 smart account (email / social login), with `registered_local_bank_id` assigned at the selected home Local Bank. Stage 3 is KYC verification via document capture with automated verification (2–5 minutes in the planned implementation). Stage 4 is the first loan: the UI explains terms in plain language and shows the repayment schedule in USDC with optional fiat-equivalent display via a Chainlink FX oracle. Stage 5 is power-user mode—self-custodied MetaMask, block-explorer visibility, and solidarity-group lending. The primary interaction path remains the React dashboard; the conversational agent (Section 4.4) is an alternate interface to the same API.

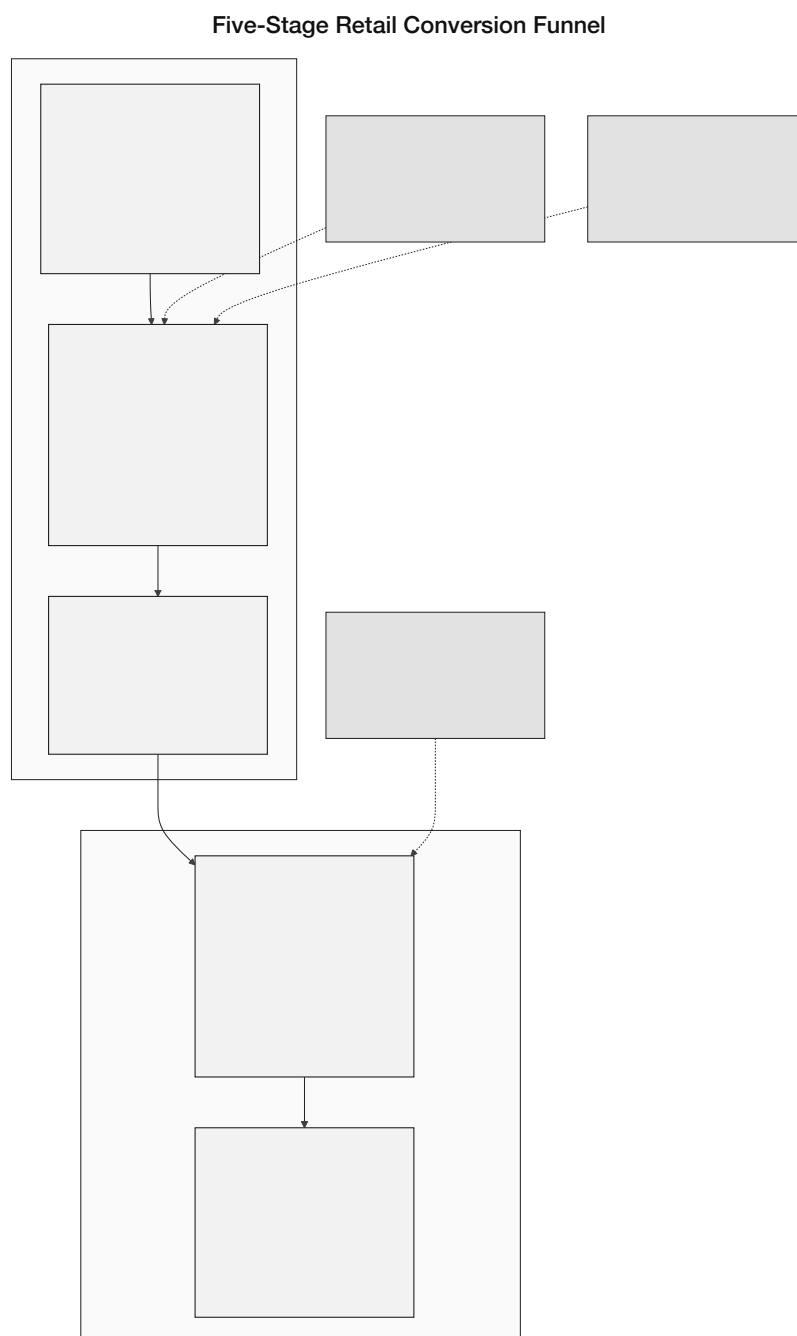


Figure 3.11: Five-stage retail onboarding funnel: progressive disclosure from zero-friction browsing through wallet or optional ERC-4337 account creation, KYC, first USDC loan with optional fiat-equivalent display, to power-user and group-lending modes.

3.8.4 Optional Conversational Interface (Add-On)

The MCP-based agent is an **optional add-on** that calls the same Express.js endpoints and smart contracts as the React UI. It does not gate, modify, or replace the manual banking flows. At pre-thesis scope the agent is limited to **retail-client** sessions (`SESSIONS.borrower_id`); bank-authority agent sessions (e.g. an NB Admin summarising reserve ratios across Local Banks) require generalising `SESSIONS` to `actor_id + actor_type` and a separate MCP tool set with tighter RBAC—explicitly deferred to Future Work (Table 3.1). Figure 3.12 specifies the six-step pipeline: on-chain context injection,

read/write routing, mandatory human confirmation gate, MCP write-tool execution via EIP-7702 session key (when enabled), and append-only AGENT_ACTION_LOG audit trail.

Optional AI Agent Add-On — Six-Step Pipeline (Phase III–IV)

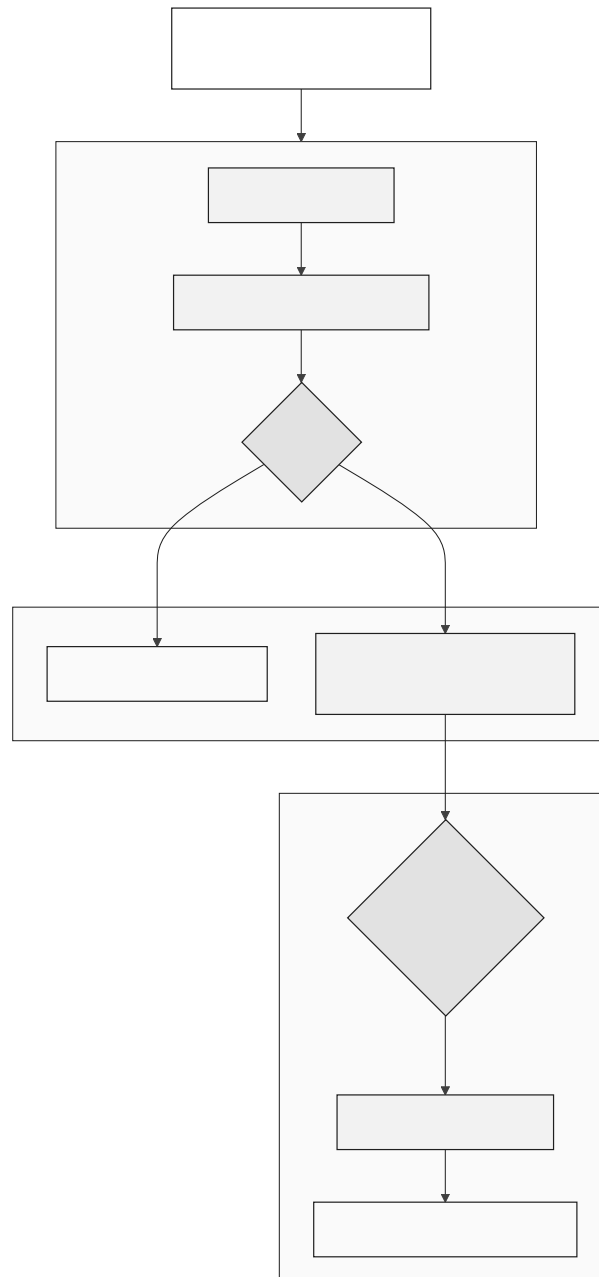


Figure 3.12: Optional AI agent add-on (Phase III–IV, excluded from MVT): six-step pipeline with on-chain context injection, read/write routing, mandatory human confirmation gate, MCP write-tool execution, and AGENT_ACTION_LOG audit trail. React dashboard remains the primary interface.

3.9 Kinked Interest Rate Model

Borrowing cost follows a kinked utilization model (Compound/Aave pattern [R3]) with optimal utilization $U^* = 80\%$ for retail pools. Utilization is $U = L/(L + B)$, where L is lent amount and B is available liquidity.

Below the kink ($U \leq U^*$), the rate rises gently:

$$r_b(U) = r_0 + \frac{U}{U^*} \cdot r_1, \quad U \leq U^* \quad (\text{Formula 18})$$

At $U = 0$ the rate is base r_0 ; at $U = U^*$ it reaches $r_0 + r_1$ (r_1 = below-kink slope).

Above the kink ($U > U^*$), the rate rises steeply to incentivise repayment and new deposits:

$$r_b(U) = r_0 + r_1 + \frac{U - U^*}{1 - U^*} \cdot r_2, \quad U > U^* \quad (\text{Formula 19})$$

r_2 is the jump multiplier above the kink. On-chain guards enforce $r_2 \geq \text{MINIMUM_JUMP_MULTIPLIER}$ and $U_{\text{star}} \leq \text{MAXIMUM_KINK_POINT}$ before any parameter update.

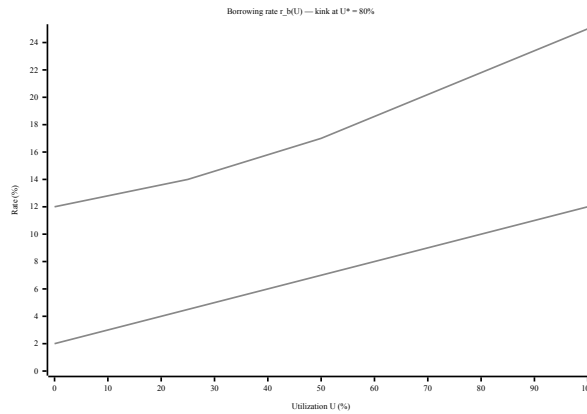


Figure 3.13: Kinked utilization-based borrowing rate $r_b(U)$ with optimal utilization $U^* = 80\%$: gentle slope below the kink, steep slope above to incentivise repayment and new deposits.

3.10 Liquidation Engine

The **LiquidationEngine** contract monitors health factor (HF) and pays third-party liquidators a 5–8% bonus when $HF < 1.0$. HF models vary by loan type:

Tier 4 over-collateralized retail.

$$HF = \frac{C \times LT}{D} \quad (3.1)$$

Collateral value C (oracle-priced) times liquidation threshold LT , divided by debt D ; liquidate when $HF < 1.0$.

Tier 4 group lending.

$$HF_{\text{group}} = \frac{\text{total_group_collateral} \times LT}{\text{total_group_outstanding_debt}} \quad (3.2)$$

Pool-level health score; liquidation loss is shared across all group members.

Tier 4 credit-based (no collateral). HF is not applicable. Default triggers SBT riskTier downgrade and lastDefault timestamp [R30]; no collateral to seize.

Tier 1–3 institutional.

$$HF_{\text{inst}} = \frac{\text{on_chain_reserve} - \text{min_reserve}}{\text{outstanding_borrowing}} \quad (3.3)$$

Parent-tier reserve surplus above the mandatory minimum acts as effective collateral.

Liquidation life-cycle.

1. Chainlink price feed updates collateral each block; HF recomputed (view-only).
2. Any participant calls `liquidate(loanId)` when $HF < 1.0$.
3. Contract transfers debt plus 5–8% bonus in collateral to the liquidator; credit record updated.
4. `LoanLiquidated(loanId, liquidator, bonusBps)` event indexed by The Graph.

When a Local Bank's reserve ratio falls below 15%, authorized liquidations queue at National Bank level, lowest HF first.

LiquidationEngine — Tier-Specific Health Factor and Lifecycle

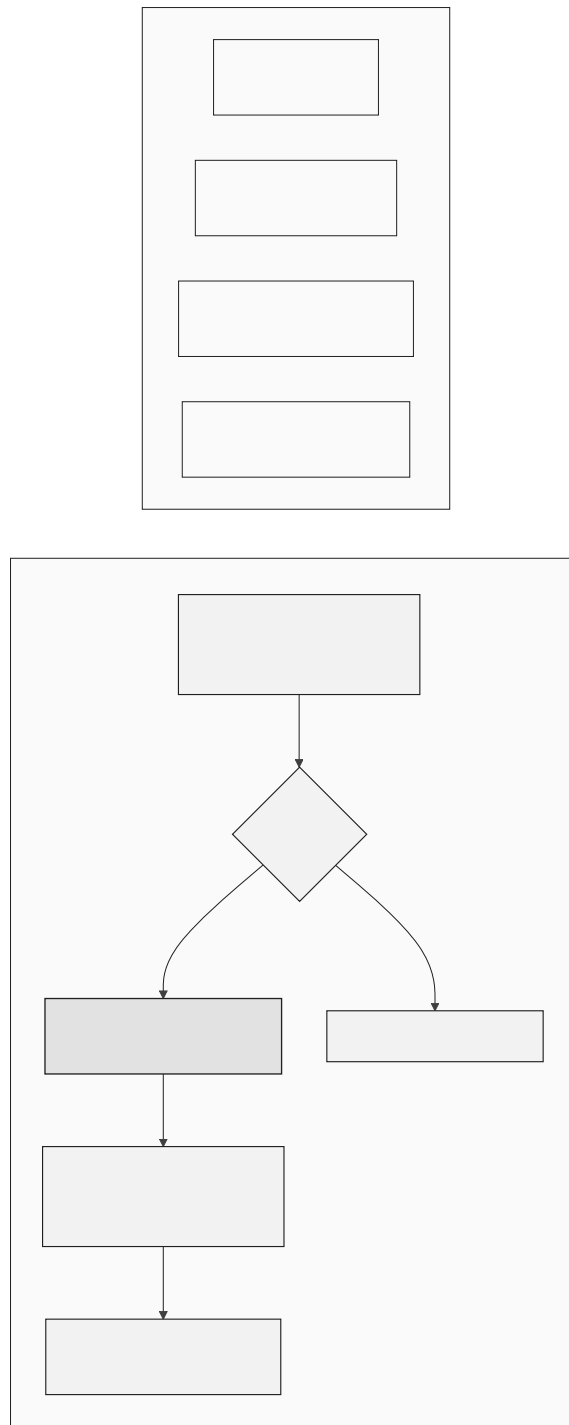


Figure 3.14: LiquidationEngine: tier-specific Health Factor models (retail collateral, group pool, credit-based SBT downgrade, institutional reserve surplus) and the on-chain liquidation lifecycle (Chainlink price update → `liquidate()` with 5–8% bonus → hierarchical portfolio queue when reserve ratio falls below 15%).

3.11 SavingsVault and FixedDeposit Modules

SavingsVault. Accepts ERC-20 deposits (USDC primary), accrues variable yield tied to pool utilization (Section 3.9), and permits withdrawal subject to the local-tier reserve-ratio

gate. Interest accrues per-second via a cumulative index (Compound/Aave supply-side pattern); implements ERC-4626 (IERC4626) for composable vault shares [R46].

FixedDeposit. Issues a non-transferable receipt locking principal for a fixed term (30/90/180/365 days) at deposit-time APY; early withdrawal forfeits accrued interest. Provides duration-matched liabilities for installment lending; adopts ERC-7540 for time-gated asynchronous redemption [R46].

Loan interest is split under Formula NetInterest (List of Formulas) among depositor yield, the InsuranceFund, and protocol revenue, closing the deposit–lend loop without external subsidy.

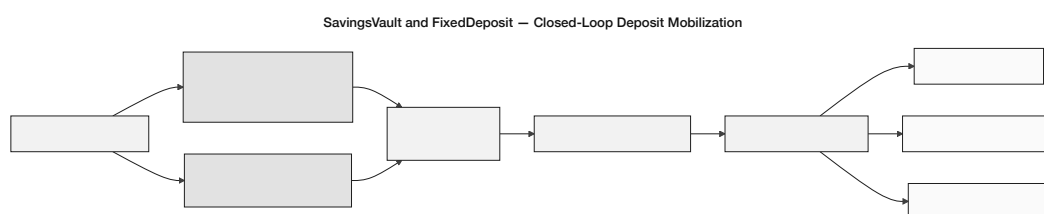


Figure 3.15: SavingsVault (ERC-4626) and FixedDeposit (ERC-7540) mobilize depositor USDC into the Local Bank lending pool; loan interest is split deterministically among depositor yield, the InsuranceFund, and protocol revenue.

3.12 On-Chain Credit Passport (Soulbound Token)

Each KYC-verified client receives a non-transferable Soulbound Token (SBT) [R30] encoding repayment history (`creditScore`, `openLoans`, `completedCycles`, `lastDefault`, `riskTier`). The credential upgrades on full installment repayment and is permanently downgraded (never revoked) on default. It enables portable credit across Local Banks on Cardona, enforces cross-platform simultaneous-loan caps (Section 3.17.3), and is exposed read-only via `ICreditPassport.getScore(wallet)` for external protocols on the deployment chain.

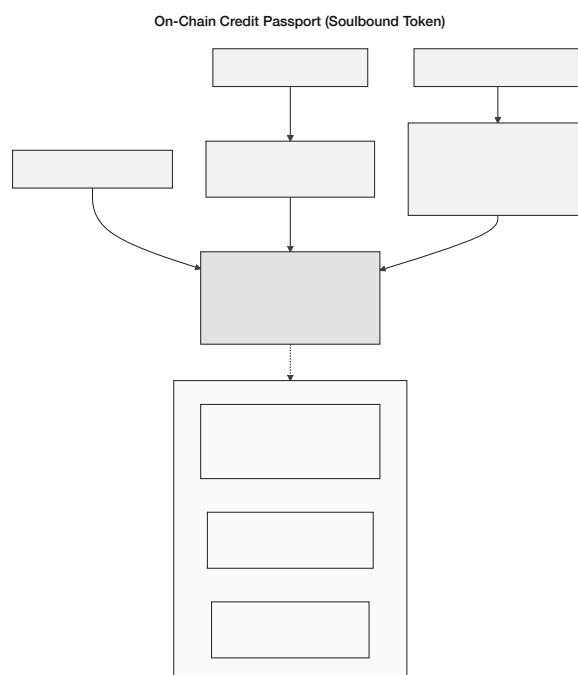


Figure 3.16: Credit Passport SBT lifecycle: non-transferable credential issued after KYC, updated on repayment, permanently downgraded (never revoked) on default, and exposed read-only via `ICreditPassport.getScore()` to GroupLendingPool, peer Local Banks, and external protocols.

Credit tier schedule. Table 3.26 governs maximum loan limits and interest modifiers by tier.

Table 3.26: Credit Passport tier schedule governing maximum loan limits and interest modifiers.

Tier	Label	Score Range	Max (USDC)	Loan	Interest Modifier
1	Bronze	0–299	50		Base rate
2	Silver	300–549	250		Base – 0.5%
3	Gold	550–749	1,000		Base – 1.0%
4	Platinum	750–899	5,000		Base – 1.5%
5	Diamond	900–1000	25,000		Base – 2.0%

3.13 Single-Chain Deployment (Design Revision)

An initial design considered multi-chain deployment bridging Polygon zkEVM Cardona and Ethereum Sepolia to distribute retail and institutional traffic across separate networks. This approach was revised after security analysis identified cross-chain bridges as the highest-risk attack surface in DeFi—accounting for over \$1.1 billion in documented losses through Ronin (\$625M), Wormhole (\$320M), and Nomad (\$190M) exploits in 2022 alone. The revised architecture consolidates **all live contract deployment on Polygon zkEVM Cardona**. Polygon zkEVM derives settlement finality from ZK validity proofs verified on Ethereum

L1, eliminating the need for a separate Ethereum contract layer and the bridge required to connect them.

Ethereum Sepolia is retained in the testing plan **exclusively as a Certora formal-verification target** (Phase IV), with no live asset bridge between networks. This simplification reduces attack surface and keeps Phases I–II deliverable within the project timeline. Chen et al. [46] validate multi-chain lending feasibility in the literature; the CWB adopts their throughput lessons without deploying a live CCIP or LayerZero bridge. Credit Passport SBT state, borrowing-limit enforcement, and loan lifecycle data remain single-chain on Cardona.

3.14 Multi-Entity and Cross-Tier Capital Operations

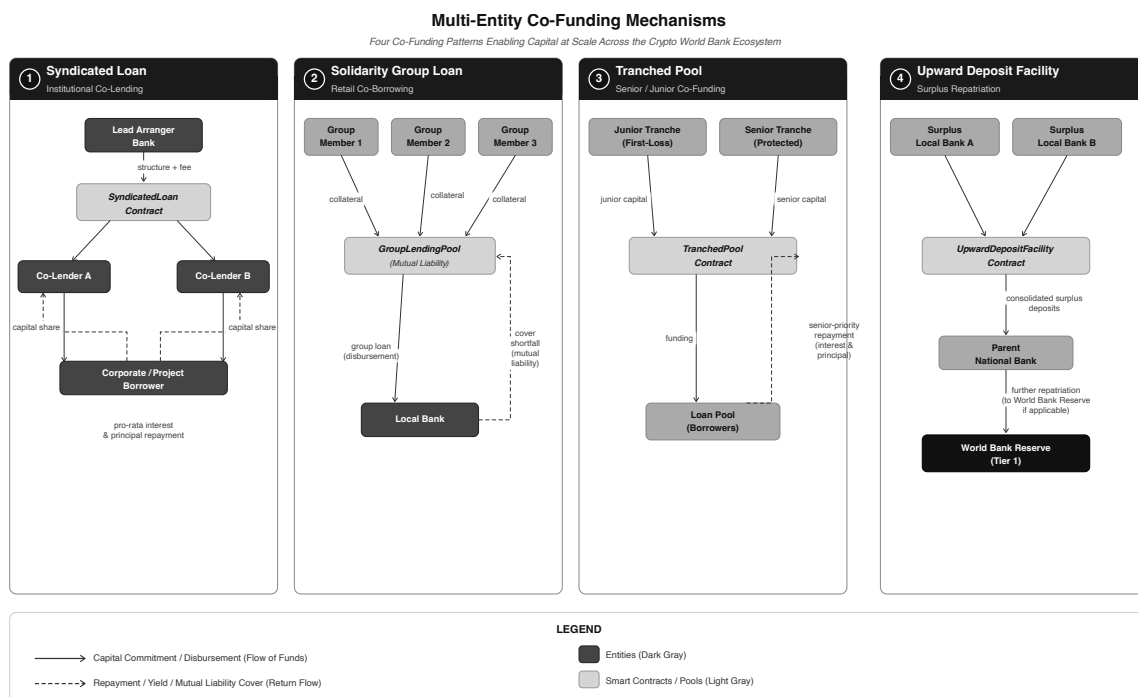


Figure 3.17: Multi-entity and cross-tier capital operations: (a) InterBankLendingPool with utilization-kinked rate model and default cascade, (b) UpwardDepositFacility with asymmetric rate structure, (c) SyndicatedLoan with Lead Arranger and Co-Lenders, (d) TranchingPool with senior–junior waterfall, (e) TreasurySwap for cross-tier asset exchange, and (f) NettingEngine for multilateral settlement.

The Chapter 1 framing of multi-directional flows (Section 1.6) and Section 1.6 promise downward, same-tier, and upward lending plus bank-level syndication for large exposures. Earlier specifications left same-tier and upward flows at design-level only. A paper that claims a *complete* banking system must specify these flows—and the genuinely multi-entity operations they enable—at the same depth as the downward flow. This section closes that gap with five concrete contract-and-flow specifications: a fully-specified InterBankLendingPool (Section 3.14.1), upward surplus repatriation (Section 3.14.2), syndicated / club lending (Section 3.14.3), senior–junior tranching lending pools (Section 3.14.4), and cross-tier treasury FX swap (Section 3.14.5). Multilateral settlement netting (NettingEngine) is scoped to Future Work (Section 6). The corresponding ERD entities, contract list, implementation

phase plan, and Future Work are updated downstream so the database design, software-engineering stages, and architecture remain a single consistent system.

3.14.1 InterBankLendingPool: Fully Specified Same-Tier Lending

One InterBankLendingPool (IBLP) per tier (IBLP_NB, IBLP_LB) supplies same-tier short-tenor (1/7/30-day) USDC liquidity among peer banks, gated by active tier role (Section 3.19).

Rate model. $r_{IB}(U)$ follows the kinked curve of Section 3.9 (Formulas 18–19) with kink $U_{IB}^* = 0.90$. The interbank rate is capped by the parent-tier downward rate:

$$r_{IB}(U) \leq r_{\text{downward}}(U) - \delta \quad (3.4)$$

δ is the governance-set spread that blocks arbitrage against the tier above; checked live via `getCurrentBorrowRate()` at each rate update.

Default cascade. On maturity default: (1) reserve buffer debited, (2) InsuranceFund drawn, (3) parent tier absorbs residual and queues role suspension via TimeLock.

3.14.2 Upward Surplus Repatriation

Banks with reserves above the minimum by 20% may deposit excess into the parent tier via UpwardDepositFacility, earning parent deposit yield. The upward rate is capped below the downward rate:

$$r_{\text{up}} < r_{\text{down}} - \delta \quad (3.5)$$

Withdrawals revert if they breach the depositor's minimum reserve ratio; daily withdrawals are capped at 20% of outstanding upward liabilities; net upward exposure is capped at 30% of on-chain assets.

3.14.3 Syndicated and Club Lending

When single-name exposure exceeds a bank's lending ceiling, multiple banks co-fund one loan through SyndicatedLoan: Lead Arranger, Co-Lenders, and Borrower (client via Local Bank only). Funding requires aggregate commitments $\geq 90\%$ of totalAmount and Co-Lender confirmation $\geq 75\%$ by capital share. Each lender's share is recorded as `shareBps[lender][syndicateId]`; principal and interest (and default recovery) distribute pro-rata by that share.

3.14.4 Senior–Junior Tranched Lending Pools

TranchedPool splits capital into senior (lower yield, protected) and junior (higher yield, first-loss) tranches. Repayment waterfall: senior interest $\rightarrow$ junior interest $\rightarrow$ senior principal $\rightarrow$ junior principal; losses reverse (junior written down first). Subordination ratio (junior/senior) is fixed at deposit between 10/90 and 30/70.

3.14.5 Cross-Tier Treasury FX Swap

TreasurySwap executes oracle-priced atomic cross-asset swaps for banking-role wallets:

$$\text{Amount}_{\text{to}} = \text{Amount}_{\text{from}} \times \frac{P_{\text{from}}}{P_{\text{to}}} \times (1 - \text{spread}) \quad (3.6)$$

P is the Chainlink feed price; treasury spread is 5–10 bps. Post-swap reserve ratio must remain above the tier minimum; swaps above \$1M require Safe two-signature confirmation.

3.14.6 Multilateral Settlement Netting Engine (Future Work)

Future Work. A NettingEngine contract to compress bilateral IBLP and TreasurySwap obligations into batch settlement—including partial settlement (settlePartial) and challenge-period dispute resolution—is listed in Section 6. ERD stubs (NETTING_BATCH, NETTING_ENTRY) remain in the schema for forward compatibility but are not specified at implementation depth in this report.

3.14.7 Database, Phase, and Contract Consistency for Multi-Entity Operations

Eight ERD entities (INTERBANK_LOAN, UPWARD_DEPOSIT, SYNDICATE, SYNDICATE_MEMBER, TRANCHED_POOL, TREASURY_SWAP, NETTING_BATCH, NETTING_ENTRY) extend Figure 3.6, together with GROUP_CONSENT (Phase II, required before GroupLendingPool) and TRANCHED_POOL_SUBSCRIPTION (Phase II). All institution-participant FKs anchor to INSTITUTION.institution_id (Section 3.4.2, Table 3.14). Five contracts are specified at implementation depth in this report: InterBankLendingPool, UpwardDepositFacility, SyndicatedLoan, TranchedException, and TreasurySwap; NettingEngine is Future Work (Section 6). Extended entities include planned on_chain_address and last_synced_block columns for event-listener mapping.

Table 3.27: FK anchors connecting extended entities to core schema.

Entity	Foreign Keys into Core Schema
SAVINGS_ACCOUNT, FIXED_DEPOSIT, CUR- RENT_ACCOUNT	borrower_id → BORROWER; local_bank_id → LOCAL_BANK(institution_id)
LOAN_GROUP	local_bank_id → LOCAL_BANK(institution_id)
GROUP_MEMBER	group_id → LOAN_GROUP; borrower_id → BORROWER
INSURANCE_FUND	local_bank_id → LOCAL_BANK(institution_id)
INTERBANK_LOAN	lender_institution_id, borrower_institution_id → INSTITUTION; tier-type CHECK per IBLP instance
UPWARD_DEPOSIT	depositor_institution_id, parent_institution_id → INSTITUTION; adjacent- tier CHECK
SYNDICATE	lead_institution_id → INSTITUTION; borrower_id → BORROWER
SYNDICATE_MEMBER	syndicate_id → SYNDICATE; member_institution_id → INSTITUTION
TRANCHED_POOL	sponsor_institution_id → INSTITUTION
TRANCHED_POOL_SUBSCRIPTION	pool_id → TRANCHED_POOL; subscriber_id → BANK_USER
TREASURY_SWAP	initiator_institution_id → INSTITUTION; from_asset_id, to_asset_id → ASSETS
NETTING_BATCH	coordinator_id → BANK_USER
NETTING_ENTRY	batch_id → NETTING_BATCH; participant_institution_id → INSTITUTION
INSTITUTION_MESSAGE	sender_institution_id, receiver_institution_id → INSTITUTION
LOAN_RISK_ASSESSMENT	loan_request_id → LOAN_REQUEST; model_id → MODEL_REGISTRY
GROUP_CONSENT	loan_request_id → LOAN_REQUEST; member_id → GROUP_MEMBER
COLLATERAL_POSITION	borrower_id → BORROWER; loan_id → LOAN; asset_id → ASSETS
BLOCKCHAIN_EVENT_REFERENCE	Referenced by: LOAN_REQUEST.submission_event_log_id, approval_event_log_id; LOAN.disbursement_event_log_id; INSTALLMENT.payment_event_log_id

3.15 System Modeling

This section presents the system analysis diagrams that model the platform's structure, behavior, and data flow. Loan and transaction state transitions are formalised in the on-chain

state machine of Section 4.8; the diagrams below show actor interactions and operational flows at the use-case and sequence level.

3.15.1 Use Case Diagram

The system involves nine actors across eleven representative use cases (Figure 3.18). Four **core primary** actors initiate banking operations: Retail Client (A1, three KYC sub-types), Local Bank Admin (A2), National Bank Admin (A3), and World Bank Admin (A4). A5 (AI Agent) is an **optional add-on** actor with restricted write permissions gated by the human confirmation protocol; it is shown in the diagram for completeness but is excluded from MVT scope (Table 4.2). Four secondary actors are external systems or authorities: Regulatory Authority (A6), Chainlink DON (A7), Blockchain Validator (A8), and External Auditor (A9). Note that the primary actor classes in the use-case diagram are a condensation of the formal nine-actor taxonomy defined in Table 3.23: Visitor and Registered Retail User are pre-conditions for the CLIENT (BORROWER) role rather than independent diagram actors, and Local Bank Operator / Approver are merged into the Bank Approver role at the diagram level.

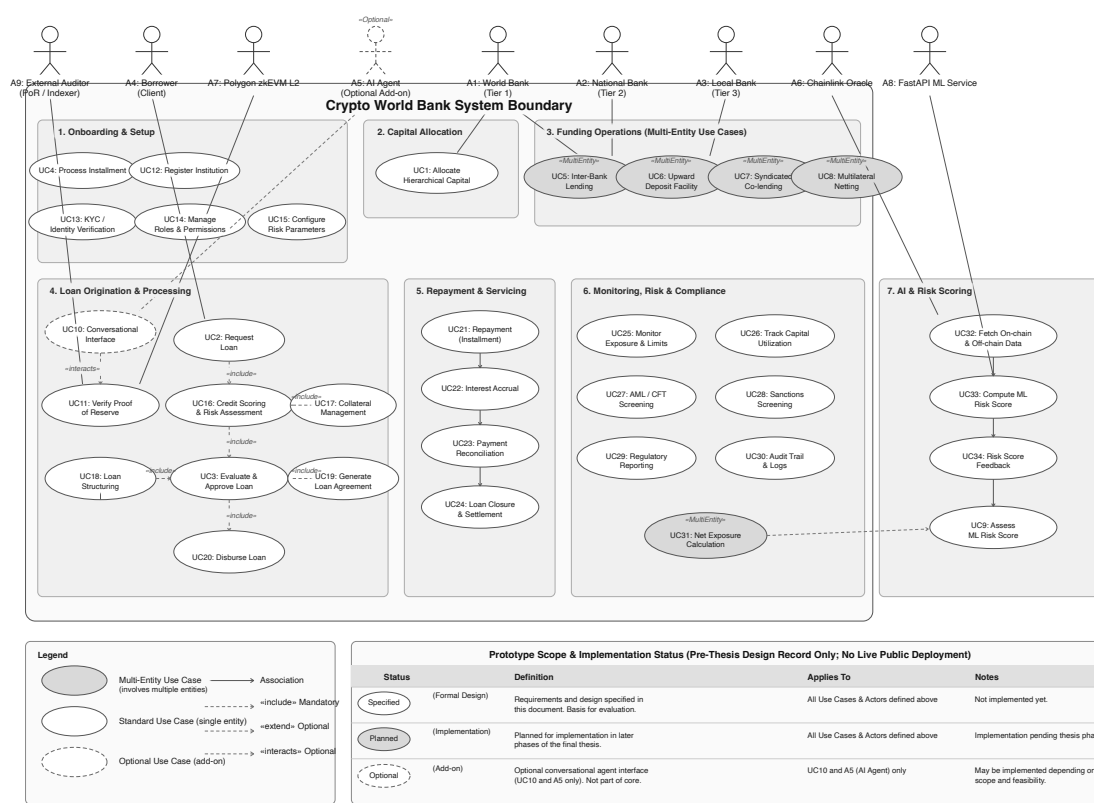


Figure 3.18: Use-case diagram reflecting the nine-actor taxonomy of Table 3.23: primary actors (A1–A5, top), external actors (A6–A9, top), and eleven grouped use cases inside the system boundary. Shaded rows denote multi-entity operations (UC5–UC8); dashed nodes and links denote the optional AI Agent add-on (A5, UC10), excluded from MVT.

3.15.2 Activity Diagrams

Figures 3.19–3.21 present three end-to-end activity flows. Each diagram has a single start and a single end node; decision diamonds route exceptional paths (limit exceeded, rejection, verification retry) back to the terminal state without spawning parallel lifelines.

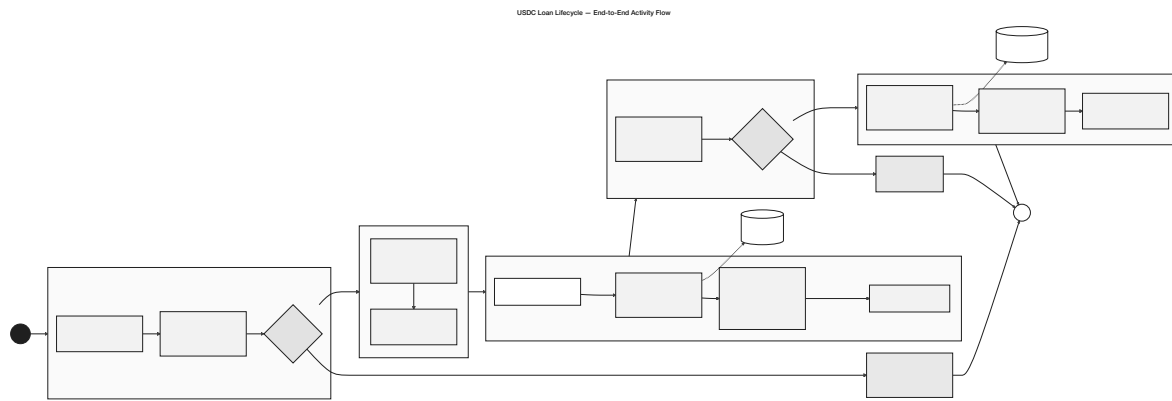


Figure 3.19: USDC loan lifecycle activity flow: five phased swimlanes from eligibility gate (SBT tier ceiling and DTI check) through origination (hierarchical `allocateCapital`), oracle-mediated risk scoring (commit-reveal DON), human approver gate, to disbursement, installment servicing, and loan close.

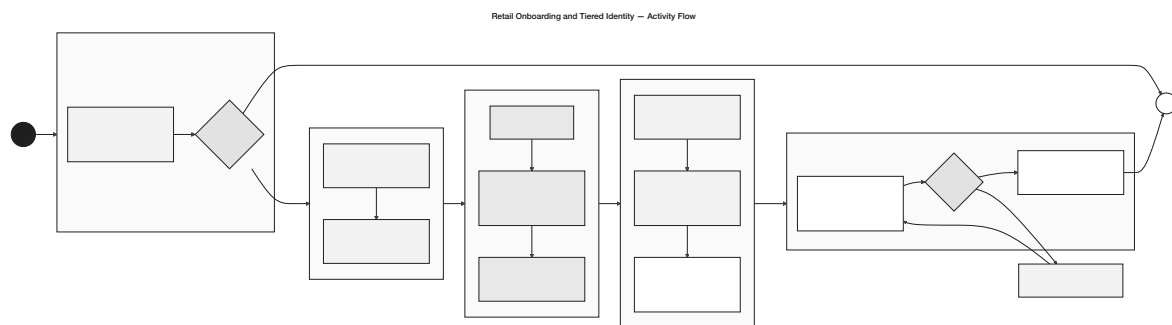


Figure 3.20: Retail onboarding and identity activity flow: progressive ladder from L0 product browse through ERC-4337 smart-account creation, document capture (L1-L3), licensed KYC provider VC issuance, zkKYC/zkAML proof generation, and on-chain CLIENT role activation with tier limits enforced.

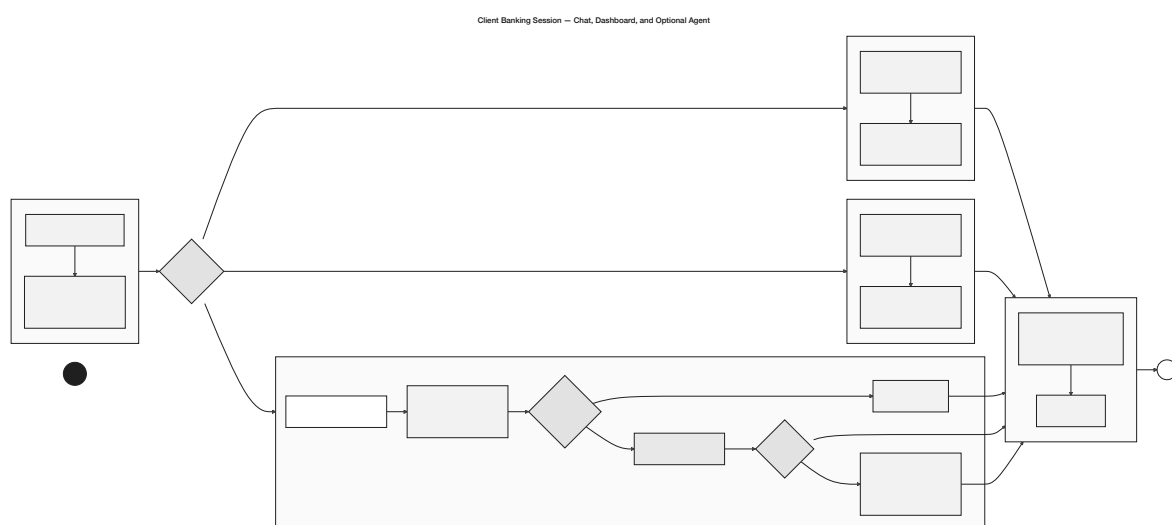


Figure 3.21: Client banking session activity flow: authenticated session bootstrap with Chain-link price prefetch, then channel routing to the primary React dashboard path, human WebSocket chat, or the optional MCP agent add-on (confirmation gate and `AGENT_ACTION_LOG` audit before session close).

SAR activity flow. The Suspicious Activity Report (SAR) workflow is triggered when the Isolation Forest anomaly detector (Section 3.20) flags a wallet with an anomaly score above the 0.75 threshold. The five-step flow is:

1. Isolation Forest flags the wallet: `anomaly_score > 0.75`.
2. `LOAN_RISK_ASSESSMENT` records the per-loan detection (FK `model_id` → `MODEL_REGISTRY`); `SECURITY_EVENT_LOG` records the system-wide AML alert event.
3. Express.js backend emits a planned event-bus message (`aml-alert` topic); the compliance dashboard consumes it.
4. Bank officer reviews the alert in the admin compliance queue:
 - **False positive:** Dismiss and document reason (audit trail in `audit_logs`).
 - **Confirmed:** Generate SAR record in `audit_logs`; insert `INSTITUTION_MESSAGE` (`message_type='SAR_ESCALATION'`) to notify the tier above (National Bank); freeze wallet via `LocalBank.freezeAccount(clientId)`.
5. The `freezeAccount` function is gated by the `onlyApprover` modifier and is Foundry-tested (Phase III invariant suite). A frozen wallet cannot initiate new loan disbursements or installment payments until the freeze is lifted by the tier above.

3.15.3 Data Flow Diagrams

Figure 3.22 consolidates the Level-0 context diagram with the Level-1 decompositions of the core lending subsystem and the deposit / interbank / FX / netting subsystems on a single page.

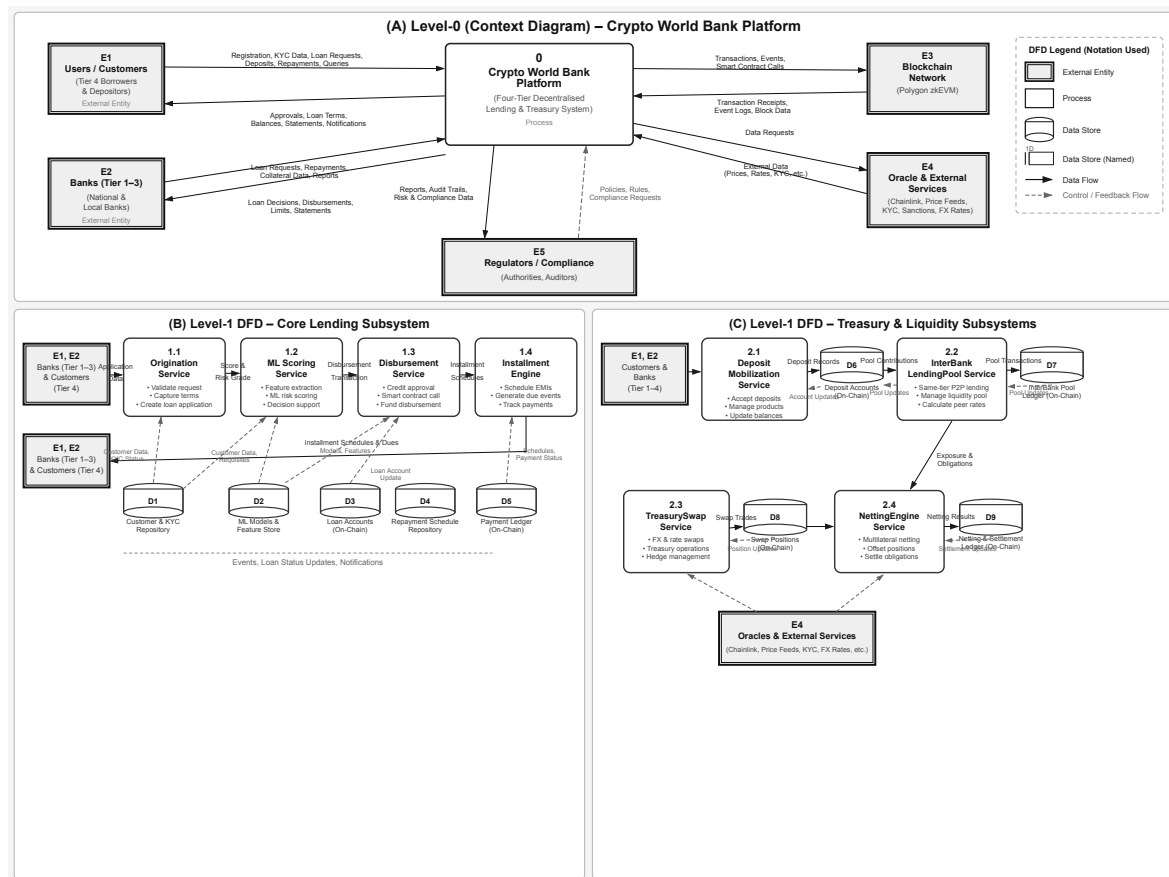


Figure 3.22: Data-flow diagrams: Level-0 context view of the Crypto World Bank platform with five external entities; Level-1 decomposition of the core lending subsystem (origination, ML scoring, disbursement, installment engine, and their data stores); and Level-1 decomposition of the deposit-mobilization, InterBankLendingPool, TreasurySwap, and NettingEngine subsystems.

3.15.4 Sequence Diagrams

Figures 3.23–3.26 consolidate the platform’s nine interaction flows into four panels: loan approval with reject alternative, installment and income, hierarchical banking with market data and borrowing-limit, and chat with AI-chatbot.

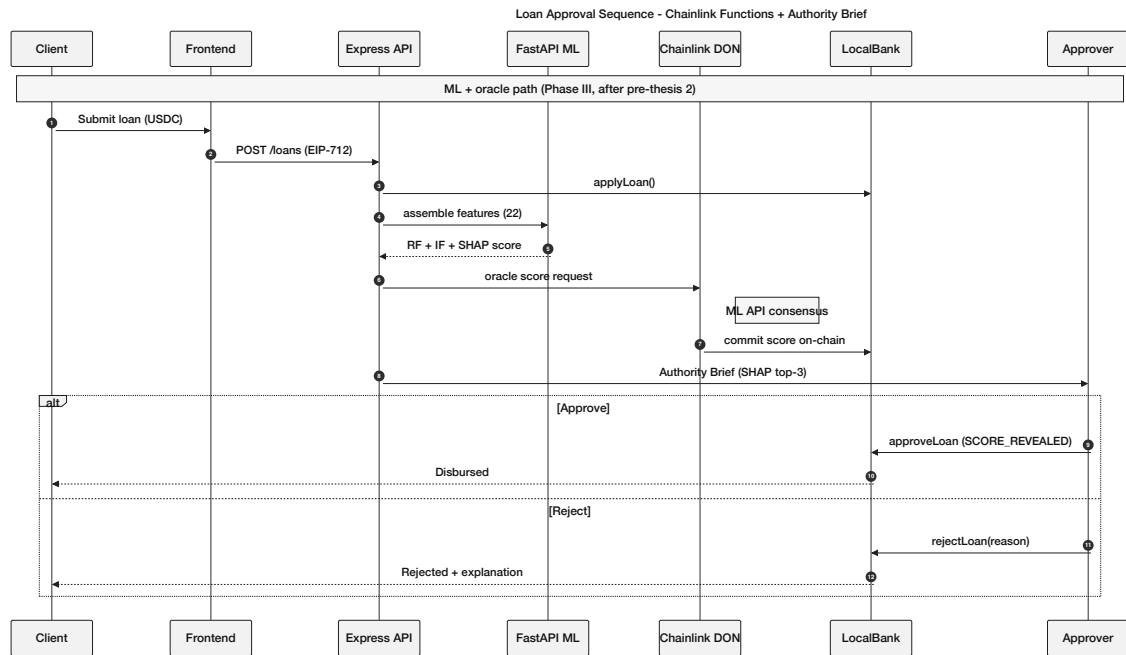


Figure 3.23: Sequence diagram for the loan-approval flow with reject alternative: the client submits an EIP-712 signed request, the ML oracle returns a SHAP-explained score via commit-reveal, and the approver either approves (revealing the commit) or rejects (with reason logged on-chain).

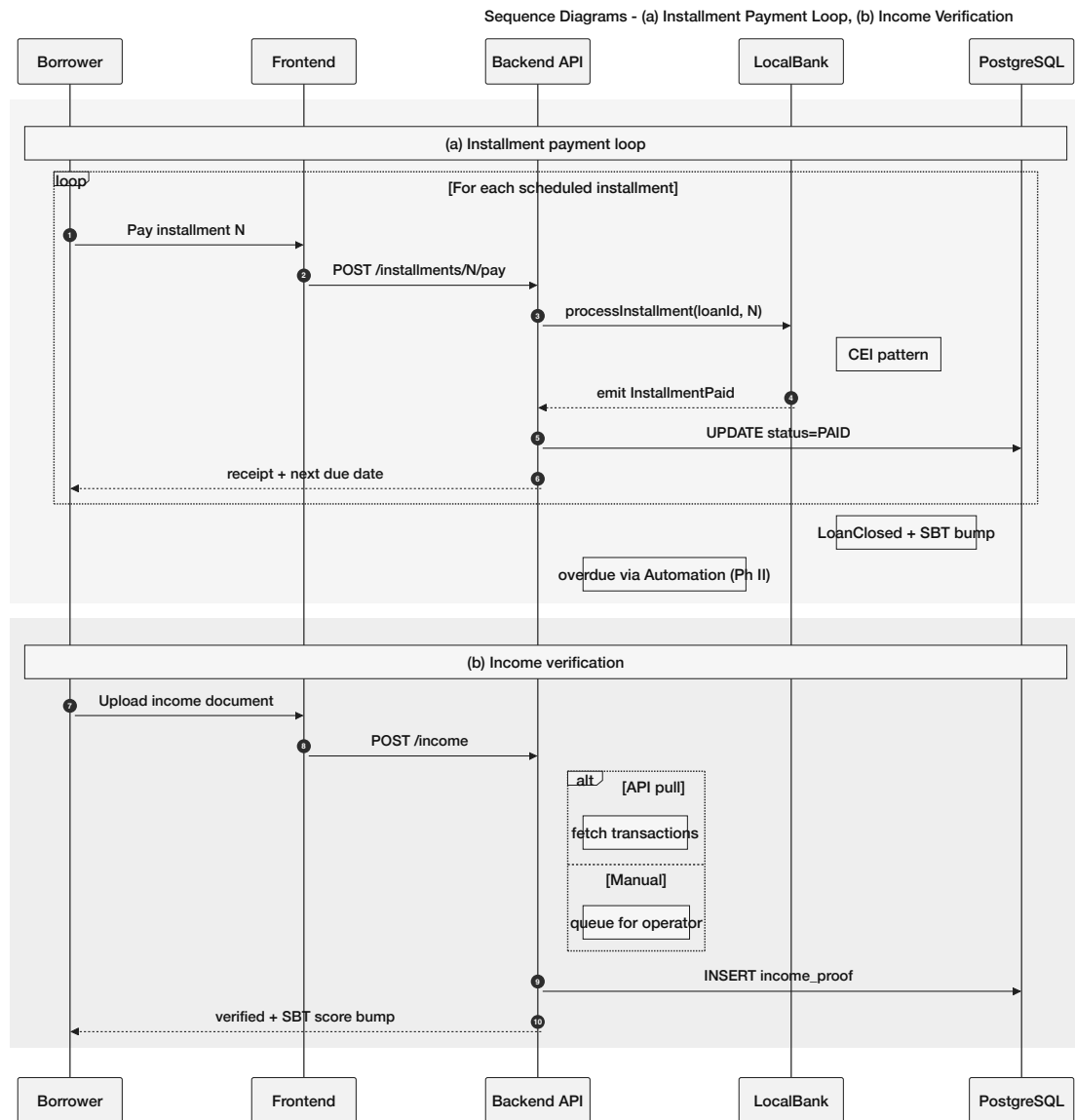


Figure 3.24: Sequence diagrams: (a) installment payment loop using the CEI pattern with SBT update on the final installment, and (b) income verification with Open-Banking pull or manual officer review.

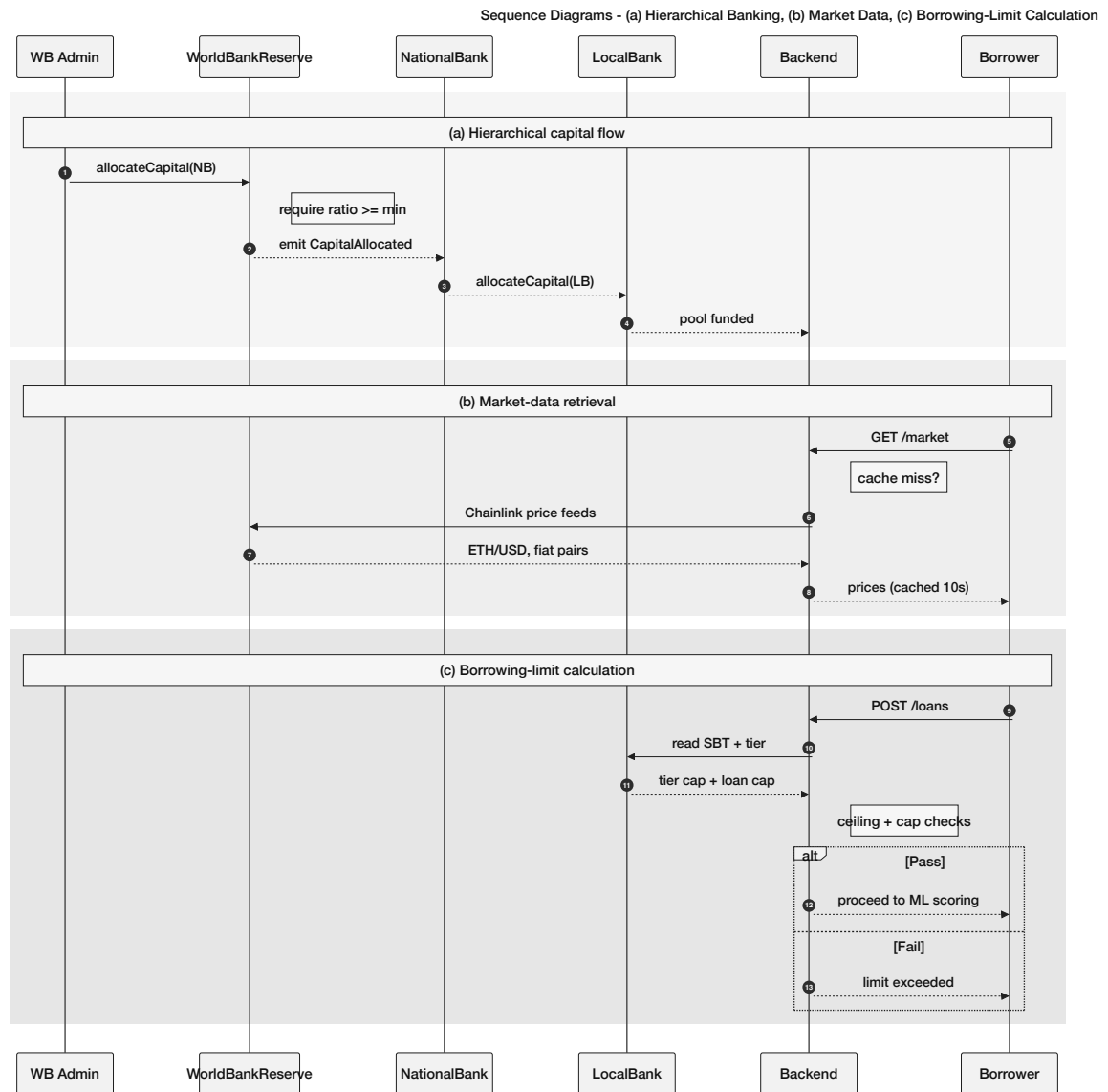


Figure 3.25: Sequence diagrams: (a) hierarchical capital flow with reserve-ratio gates at every tier, (b) market-data retrieval via cached Chainlink feeds, (c) borrowing-limit calculation reading the credit-passport SBT before ML scoring.

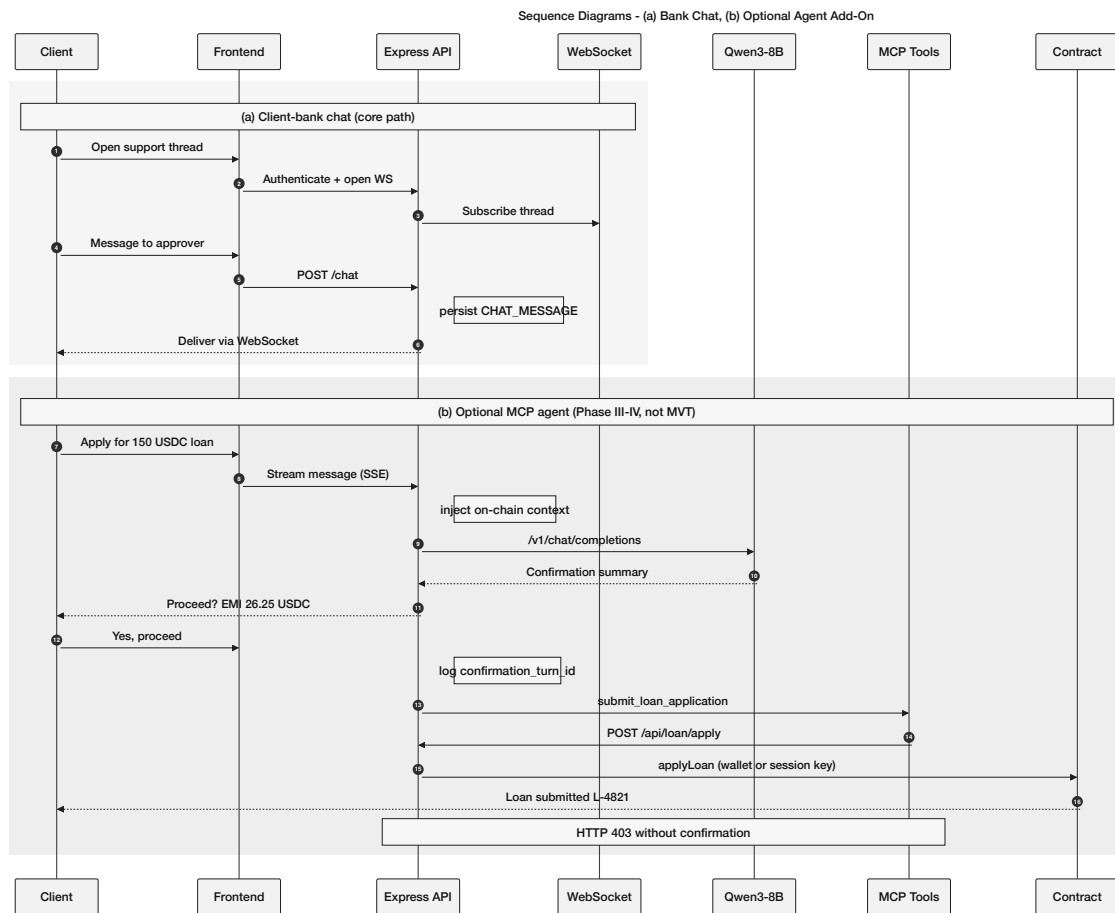


Figure 3.26: Sequence diagrams: (a) client–bank chat over an authenticated WebSocket, and (b) optional MCP agent pipeline (on-chain context injection, read/write routing, human confirmation gate, EIP-7702 session-key signing; Phase III–IV add-on).

3.15.5 Four-Tier Capital Flow

Figure 3.27 illustrates the hierarchical capital flow and cascading repayment structure.

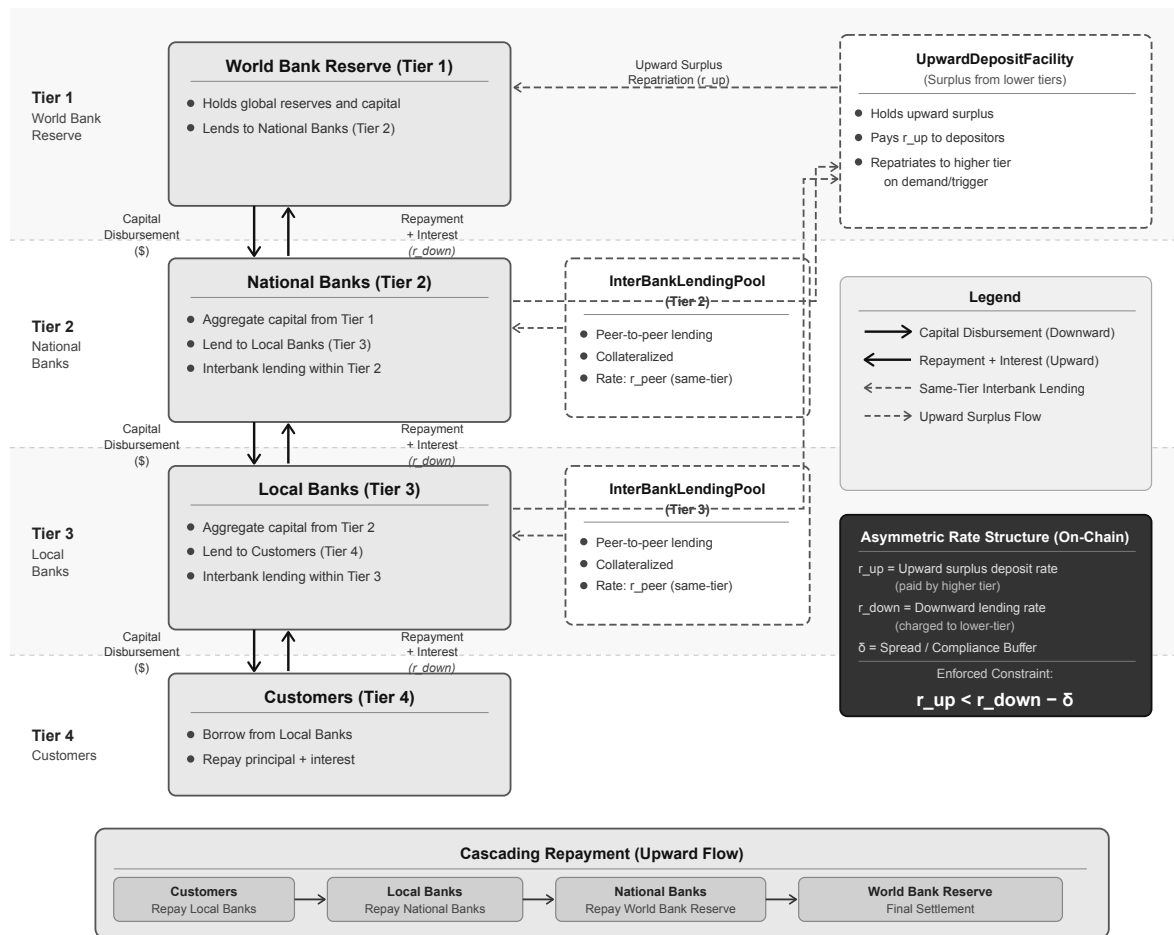


Figure 3.27: Four-tier hierarchical capital flow with cascading repayment, plus the same-tier interbank-lending pools (InterBankLendingPool, Section 3.14.1) and the upward-surplus repatriation facility (UpwardDepositFacility, Section 3.14.2). The asymmetric rate structure ($r_{up} < r_{down} - \delta$) is enforced on-chain.

3.16 Auxiliary Dual-Currency Facility

The cryptocurrency exchange feature of Crypto World Bank is built upon the cryptocurrency exchange facility that is an auxiliary facility incorporated into the current banking infrastructures all over the world. Instead of being a separate exchange, the platform is a service that is provided by all participating banks as an additional service to the existing product portfolio.

- **Integration model:** The dual-currency facility is offered by all participating banks as an add-on service to their existing product portfolio. No additional banking license or separate entity is needed.
- **Eligibility determination:** Eligibility of the dual-currency service is based on the bank officers managing the lending operations, on the basis of existing KYC/AML compliance, account status, and lending relationship history.
- **No defaulting of conditions:** The project does not override, modify, or default any existing banking conditions, regulatory requirements, or contractual obligations.
- **Scope:** The facility enables fiat-to-crypto and crypto-to-fiat conversions,

cryptocurrency-denominated lending and repayment, and transparent on-chain transaction records—all within the governance structure of the participating bank.

3.17 Banking Product Suite

3.17.1 Savings Products

Variable-yield savings (utilization-linked; Section 3.9) and fixed-term deposits (30/90/180/365 days; Section 3.11). Interest splits under Formula NetInterest among depositor yield, InsuranceFund, and protocol revenue.

3.17.2 Checking and Transactional Accounts

Non-yield current accounts for installments and transfers; on-chain settlement is atomic (debit and credit in one transaction).

3.17.3 Group / Solidarity Lending

Groups of 3–20 clients borrow via GroupLendingPool with shared collateral or a cold-start provisional tier (≥ 5 members, Level 1 cap, 3-month expiry). Per-member disbursement:

$$\text{Share}_i = \frac{\text{LoanTotal}}{N_{\text{members}}} \quad (3.7)$$

Default: collateral pool covers shortfall, or SBT riskTier downgrade if uncollateralised [R30]. Modelled default rate: 8–15% (Table 5.23).

Over-indebtedness risk control. Max two simultaneous group loans (SBT openLoans), 30-day cooling-off, and cross-bank check via ICreditPassport.getScore(). Debt-to-income gate:

$$\text{DTI} = \frac{\text{existingDebt} + \text{requestedAmount}}{\text{estimatedMonthlyIncome}}, \quad \text{reject if } \text{DTI} > 0.40 \quad (3.8)$$

Stale income hash (> 90 days) triggers manual review. Group loans are USDC-denominated at Tier 4.

Group lending stress sizing. Without weekly meetings and field-officer social pressure, on-chain group default rates are modeled conservatively at 15–25% during early deployment before credit histories accumulate. A **20% stress scenario** sizes the InsuranceFund at 5% of total outstanding group loan principal; at this stress level the fund covers approximately one-quarter of defaults, with residual shortfalls absorbed by per-member collateral contributed at group formation. The GroupLendingPool is therefore a *reduced-collateral* product for borrowers with some digital assets but insufficient individual collateral—not a zero-collateral Grameen analogue.

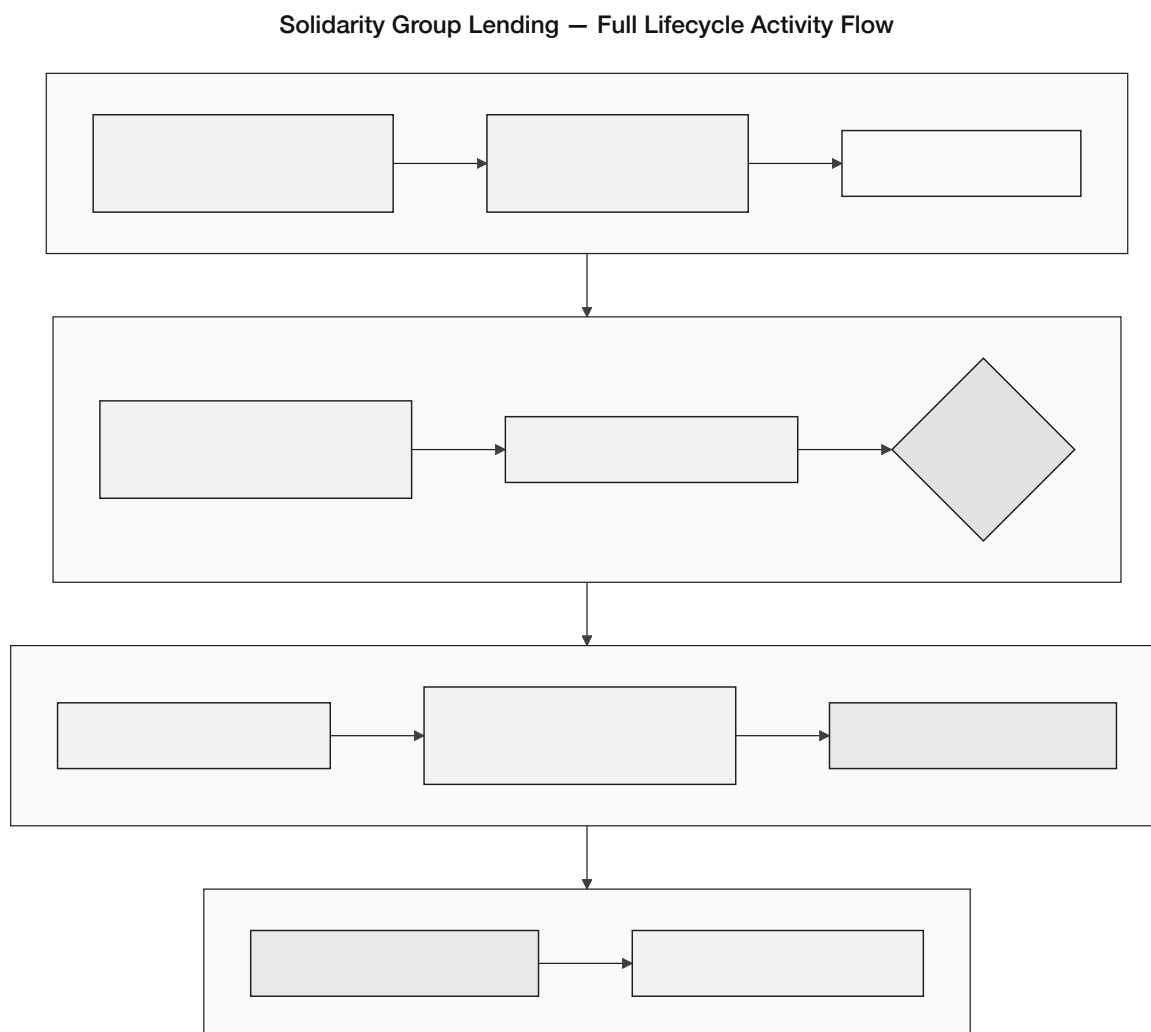


Figure 3.28: Solidarity group lending lifecycle (GroupLendingPool): seven phased swimlanes from group formation (collateral pool vs. cold-start provisional tier) through over-indebtedness gate (ICreditPassport cross-pool query, simultaneous-loan cap, cooling-off, DTI ceiling), unanimous consent, bank approval, per-member USDC disbursement ($\text{Share}_i = \text{LoanTotal}/N$), installment tracking with mutual-liability enforcement or SBT downgrade on default, and progressive credit graduation after three successful cycles.

3.17.4 Foreign Exchange and Multi-Currency Operations

Oracle-priced FX with governance spread; retail lending in stablecoins (USDC/USDT), collateral in volatile assets (ETH). Treasury swaps: Section 3.14.5.

3.17.5 Trade Finance Facilitation (Planned)

Escrow-based letters of credit with document-verified release (Future Work).

3.17.6 Privacy-Preserving Identity Compliance (Future Work)

Licensed off-chain KYC issues a verifiable credential; on-chain Groth16 zkKYC and zkAML proofs are scoped to Future Work (Section 6). MVT onboarding sets `kycVerified` via off-chain review and RBAC role gates.

3.18 Governance Framework

3.18.1 Governance Execution Paths and Emergency Controls

Parameter changes and upgrades route through a 48-hour TimelockController (5 min in lab demo). Governance votes use a **snapshot of voting power at the proposal creation block** (not at execution time) to prevent flash-loan governance attacks [158]. Contract upgrade proposals require a **60% quorum floor**. Emergency path: 2-of-3 emergency pause multisig (World Bank Reserve admin, National Bank representative, independent security reviewer) may pause operations within minutes; a separate 4-of-7 Security Council Safe may pause, upgrade, or revoke roles within 2 h (mandatory 48 h disclosure) but cannot change system parameters outside the timelock. These controls align with Chapters 1 property (1) and Section 3.18.1.

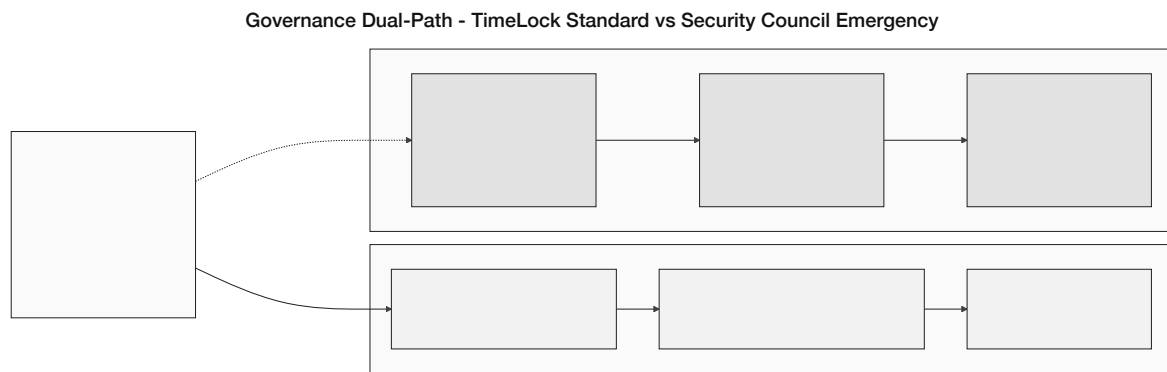


Figure 3.29: Governance dual-path: (a) standard TimeLock-gated parameter changes and upgrades via Safe 3-of-5, and (b) Security Council emergency path (4-of-7) for pre-approved pause, upgrade, and role-revocation actions within a 2-hour window with mandatory 48-hour disclosure.

3.18.2 Network Membership Governance

Table 3.29: Network membership governance.

Governance Aspect	Implementation
Member on-boarding	World Bank owner registers National Banks; National Banks register Local Banks; Local Banks designate approvers — all enforced on-chain.
Member off-boarding	Deactivation flags in smart contracts; cascading access revocation.
Regulatory oversight	Audit log emission via smart contract events; planned read-only regulator dashboard.
Permission structure	Hierarchical: Owner → National Bank → Local Bank → Approver → Retail Client; enforced by on-chain role check modifiers.
Network operations	Pause/unpause mechanism for emergency response; emergency withdrawal for critical situations.

3.18.3 Business Network Governance

Table 3.31: Business network governance.

Governance Aspect	Implementation
Business charter	Defined in project documentation; operational parameters coded in smart contract constants.
Common services	Reserve management, loan lifecycle orchestration, event-driven notification system.
Business SLA	Testnet phase: best-effort availability. Production phase: 99.5% target API-layer uptime (team-controlled; multi-region deployment). Chain-level liveness is determined by Polygon zkEVM batch submission and Ethereum L1 proof verification; the team monitors network status and maintains degraded-mode fallbacks (read-only dashboard) if chain activity pauses.
Regulatory compliance	Architecture designed for audit trail generation; data partitioning supports GDPR-style data subject requests.

3.18.4 Technology Infrastructure Governance

Table 3.33: Technology infrastructure governance.

Governance Aspect	Implementation
Distributed IT structure	Client-side frontend (decentralised delivery); blockchain layer (fully decentralised); backend API (centralised, horizontally scalable).
Technology assessment	Continuous evaluation of EVM alternatives (L2 rollups, sidechains) for cost and performance optimisation.
On-chain / off-chain data services	Clearly partitioned (see Table 3.20); event listeners synchronise state between layers.
Risk mitigation	Smart contract pause mechanism; ReentrancyGuard; input validation; planned formal security audit.

3.18.5 Regulatory Compliance Considerations

Prototype runs on public testnets only; architecture emits audit logs for sandbox review; production targets regulatory sandbox programmes. zkKYC design: Section 3.17.6. SAR path: Isolation Forest → compliance queue → freezeAccount (Section 3.15).

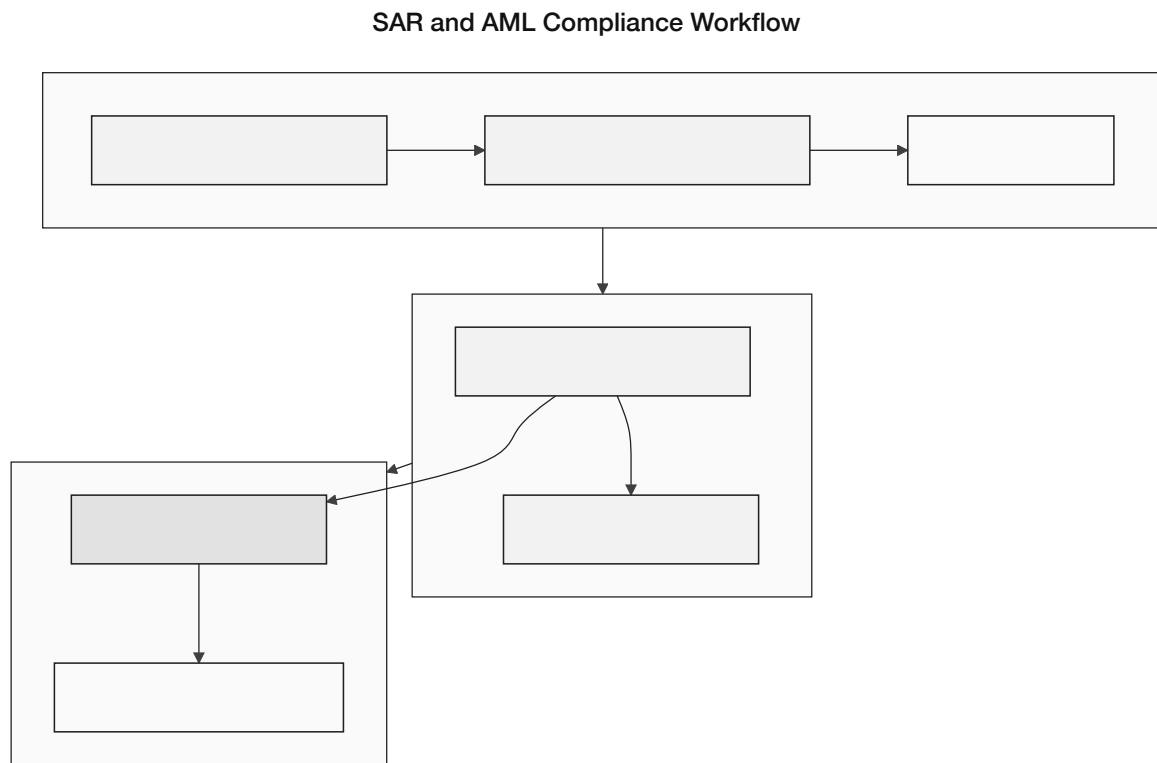


Figure 3.30: SAR and AML compliance workflow: Isolation Forest anomaly detection, compliance officer review queue, optional SAR report generation, on-chain `freezeAccount` via `onlyApprover`, and append-only `audit_logs` record.

3.18.6 Asset Tokenization

- **Specified (Phase I):** Native blockchain currency (ETH) as reserve and loan currency on testnet.
- **Planned extension:** ERC-20 stablecoins (USDC, USDT) for lending operations; tokenized collateral instruments where collateral is insufficient.

3.19 Five-Layer Defense-in-Depth Security Architecture

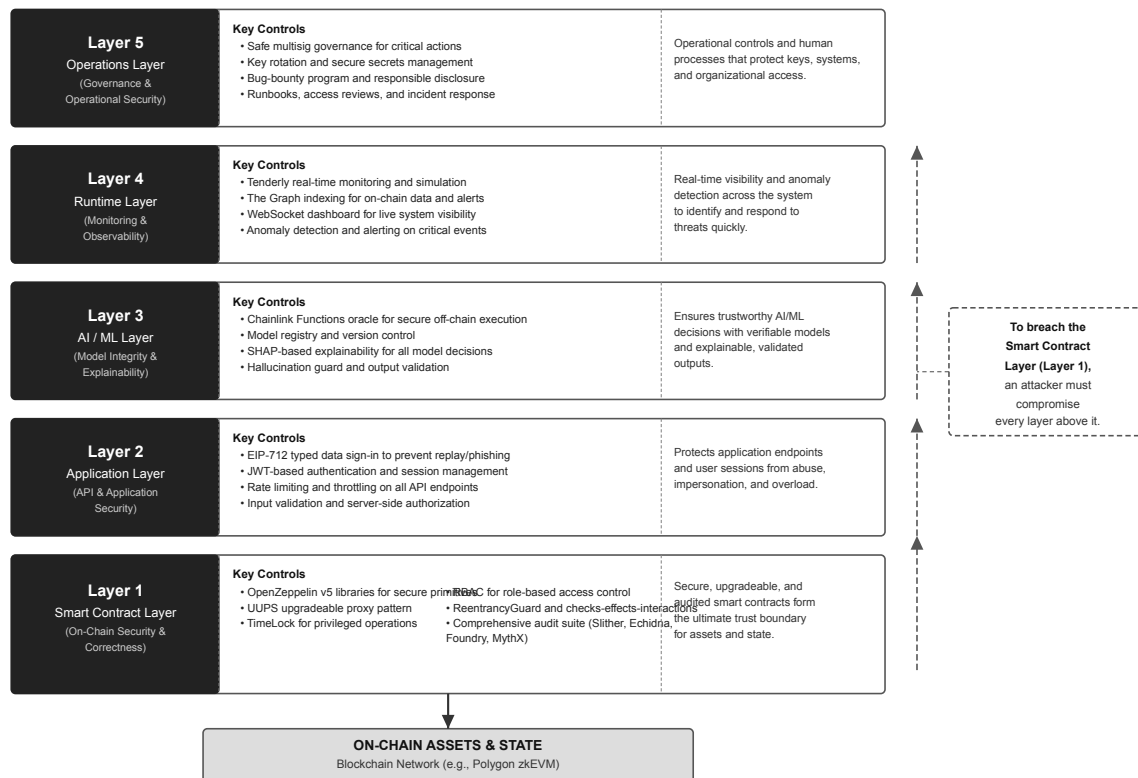


Figure 3.31: Five-layer defense-in-depth security architecture. Each layer is necessary; breaching the smart-contract layer requires defeating every layer above it. Layer 1 (smart contract): OpenZeppelin v5, UUPS, TimeLock, RBAC, ReentrancyGuard, audit suite. Layer 2 (application): EIP-712 sign-in, JWT, rate-limit. Layer 3 (AI/ML): Chainlink Functions oracle, model registry, SHAP, hallucination guard. Layer 4 (runtime): Tenderly, The Graph, WebSocket dashboard, anomaly detection. Layer 5 (operations): Safe multisig, key rotation, bug-bounty.

Smart-Contract Security Controls - UUPS, EIP-712, Granular Pause

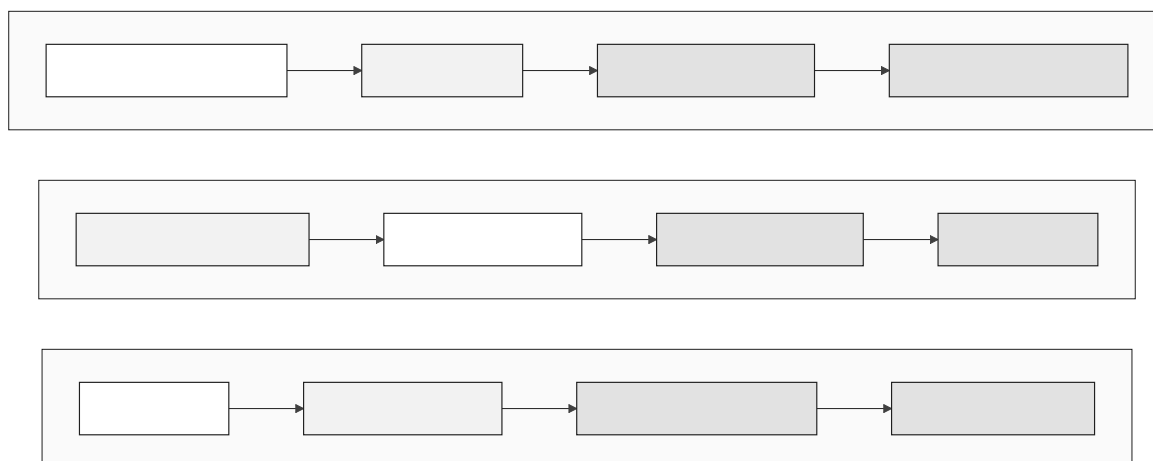


Figure 3.32: Smart-contract security controls: (a) UUPS (ERC-1822) upgrade path with TimeLock-gated authorization; (b) EIP-712 sign-in (typed-data v4 with domain separation and 15-minute JWT); (c) granular per-function pause registry indexed by The Graph for transparent governance.

The correct mental model for a blockchain banking platform's security is not "smart-contract security only"—it is a five-layer defense-in-depth stack in which breaching the contract layer requires defeating every layer above it. The Crypto World Bank adopts the following layered model, summarized in Table 3.35 and inspired by current (2025–2026) audit-tooling and runtime-monitoring best practice from Trail of Bits, Cyfrin, Tenderly, and OpenZeppelin Defender.

Table 3.35: Five-layer defense-in-depth model for the Crypto World Bank platform. Each layer is necessary; breaching the contract layer requires defeating every layer above it.

Layer	Component	CWB Design Choice
L5	Operational security	Safe (Gnosis) 2-of-3 multisig for WorldBankAdmin; 3-of-5 for global parameter changes; documented key-rotation schedule; hardware-wallet signer for the WorldBankAdmin Safe (Ledger Gen5).
L4	Runtime monitoring	Tenderly alerts on six critical triggers (large disbursement, repeated requests, reserve-ratio drop below 20%, failed disburseLoan, role grants, governance pause); The Graph subgraph for indexed event history; transaction simulation pre-deployment.
L3	AI/ML security	Random Forest fraud probability + Isolation Forest anomaly score + SHAP explanation per loan (stored in LOAN_RISK_ASSESSMENT, FK to MODEL_REGISTRY), delivered via commit-reveal oracle; system-wide events in SECURITY_EVENT_LOG; transaction-graph distance to flagged wallets. GNN-based relational fraud analysis is Future Work (Section 6). <i>Note: session-level behavioral biometrics (page-view timing, terms-acceptance latency) require client-side instrumentation not yet in the schema and are a specified Future Work extension.</i>
L2	Application security	EIP-712 wallet-signed JWT (15-minute lifetime + refresh); slowapi rate limiting (5/min auth, 60/min reads, 10/min writes); Pydantic input range constraints; strict CORS and Content-Security-Policy; environment-variable secret hygiene; prompt injection scanning middleware on all user-sourced fields before context assembly (see below).
L1	Smart contract security	OpenZeppelin ReentrancyGuard; CEI pattern; Solidity 0.8.20 overflow protection; RBAC with role expiry; UUPS upgrade with Safe-multisig authorization; TimeLock on governance actions; granular per-function pause; four-tool audit pipeline (Slither, Mythril, Echidna, Halmos/Certora); Foundry fuzz + invariants.

The L1 contract layer is discussed in detail below; Section 3.20 expands the threat model to span all five layers and motivates the specific controls listed in Table 3.35.

3.19.1 Smart Contract Security Controls

Checks-Effects-Interactions (CEI) pattern. The CEI pattern is applied to all state-mutating functions in the lending contracts. The three functions most vulnerable to reentrancy are: (1) `disburseLoan` — update `loanStatus` before any ETH transfer and apply `nonReentrant`; (2) `processInstallment` — check → mark paid → emit event → release interest; (3) `allocateCapital` — apply `nonReentrant` and allocation bounds before any state change. The DAO hack (2016) and Curve Finance exploit (2023) demonstrate that reentrancy remains an active threat [R6]; Slither static analysis and Mythril symbolic execution are planned for the final thesis security audit. Certora formal verification of reserve invariants is scoped to Future Work (Section 6).

Flash loan scope. Flash loan attacks primarily exploit price-sensitive collateral valuation [R5]. The pre-thesis design presents limited flash-loan surface because stablecoin integration and oracle-based collateral pricing are planned for Phase II–III. Mitigations specified include TWAP oracles, multi-block confirmation for large collateral updates, and Chainlink price feeds with circuit breakers.

Upgradeability via UUPS (ERC-1822). All implemented and planned contracts use the OpenZeppelin UUPS pattern. `_authorizeUpgrade` is gated on `onlyRole(WORLD_BANK_ADMIN)`, held in production by a Safe 3-of-5 multisig (Section 3.19).

Role expiry timestamps. Every `grantRole` call records an expiry block number; privileged functions check `require(roleExpiryBlock[msg.sender][role] > block.number)`. Block numbers are preferred over `block.timestamp` to avoid SWC-116 timestamp-manipulation risk on EVM chains.

Granular per-function pause. A per-function pause registry (`functionPaused[bytes32 functionId]`) allows governance to suspend only `LOAN_DISBURSEMENT` while keeping `DEPOSITS` and `WITHDRAWALS` open; each pause emits a `FunctionPaused` event indexed by The Graph (Section 4.7).

Failed-attempt lockout and address whitelisting. After 3 failed confirmation responses in a single agent session, the agent pauses interactions for 10 minutes and logs `status = FAILED` to `AGENT_ACTION_LOG`. Detected prompt-injection attempts terminate the session immediately. Loan disbursement destination addresses are fixed at KYC registration and cannot change without a 24-hour delay plus 2FA re-verification.

Prompt injection scanning (Layer 2 — Application). Before any user-sourced string is assembled into the system prompt by the Express.js context injection layer, a scanning middleware checks each field for candidate injection patterns: substrings matching `ignore previous instructions`, `new system prompt`, tool-override phrases, and JSON-escape sequences that could break the structured context block. Fields that trigger the scanner are sanitised (the offending segment is removed and replaced with a placeholder) and `AGENT_ACTION_LOG.injection_scan_flag` is set to `TRUE`. The fields subject to scanning are: `language` (user-controlled), any text content from income proof document uploads, and the last N conversation turns before they are appended to the Volatile tier. This control directly addresses OWASP LLM01 (Prompt Injection) [R54] and is motivated by Alizadeh et al. [R53], who demonstrate data exfiltration through income document uploads and

language fields in a banking tool-calling agent using attack vectors structurally identical to those present in the CWB context assembly layer.

Per-tier admin-key model. Operational security (L5) is the highest-impact and lowest-cost security decision in the entire project. The `WorldBankAdmin` role is never held by a single externally owned account (EOA) in any environment, including the university lab demo: it is transferred to a Safe 2-of-3 multisig before any contract is deployed to a testnet that examiners will inspect, with Signer 1 on the demo machine, Signer 2 on a phone, and Signer 3 as a paper-wallet backup held by the academic supervisor. National Bank Admin roles use Safe 2-of-3; Local Bank operators use Safe 2-of-3; the only EOA-bound role is the retail `CLIENT` role, which can be reset off-chain via the social-recovery guardian flow described in Section 3.8.

SWC Registry mapping. Every state-mutating function in the three specified core contracts (World Bank Reserve, National Bank, Local Bank) is mapped to its Smart Contract Weakness Classification (SWC) class [R36] and the corresponding mitigation. This mapping replaces the previous narrative coverage of Atzei et al. [7] with a structured taxonomy, and is the format the Cyfrin audit checklist (2025) recommends for academic prototypes. The full mapping table is in Appendix B’s Capability Matrix.

3.20 Threat Model and Security Controls

Security threats in smart-contract banking systems are not limited to isolated code defects; they span application-layer attacks (API abuse, JWT theft, social engineering), runtime evasion, and governance-layer key compromise in addition to the contract-layer issues that DeFi literature focuses on. Table 3.36 extends the original threat-model table to the full five-layer scope established in Section 3.19.

Table 3.36: Threat model and security controls mapping, expanded to span all five layers of the defense-in-depth stack (Section 3.19).

Threat	Layer	Attack Vector	Mitigation
Reentrancy / state manipulation	L1	External calls exploit inconsistent intermediate state	ReentrancyGuard; CEI; Foundry invariant tests.
Arithmetic errors	L1	Integer overflow/underflow in financial math	Solidity 0.8.x checked arithmetic; Certora invariants.
Oracle manipulation	L1	Price-feed manipulation on collateral / FX	Chainlink feeds; heartbeat checks; TWAP; circuit breakers.
Flash-loan economic attacks	L1	Atomic borrowing manipulates on-chain assumptions	Conservative price logic; multi-block confirmation; pause-on-anomaly.
Governance capture	L1+L5	Privileged roles abused for unsafe parameters	Role separation; Safe multisig; TimeLock 24–48 h; audit trail.
Sybil abuse (re-tail / group)	L1+L3	Many wallets bypass limits or fraudulent groups	RBAC; rate limits; ZKP-KYC; GNN wallet-graph clustering.
JWT theft / session hijack	L2	Stolen token gives full account access	15-min JWT; EIP-712 challenge; Redis blacklist; geo rotation.
API abuse / DDoS	L2	Bulk requests exhaust backend services	slowapi limits; CSP/CORS lockdown; correlation IDs.
Input-validation bypass	L2	Malformed inputs cause unintended backend state	Pydantic range constraints; wallet regex; SHA-256 doc refs.
ML evasion / adversarial fraud	L3	Crafted features evade RF / iForest	GNN analysis; federated intelligence; SHAP human review.
LLM hallucination (compliance)	L3	Wrong regulatory answer to client	RAG over policy docs; refusal layer; red-teaming protocol.
Monitoring evasion	L4	Attack split below alert thresholds	Multi-trigger alerts; The Graph forensics history.
Admin-key compromise	L5	EOA compromise gives World Bank control	Safe 2-of-3 / 3-of-5 multisig; hardware-wallet signer.
Blind-signing on hardware wallet	L5	User signs opaque call-data	Calldata decoding; Ledger Multisig; Tenderly pre-simulation.

This expanded threat model is the structural justification for the five-layer architecture in Section 3.19: each row of the table is a vector that the corresponding layer prevents, detects, or recovers from. Application-layer rows (L2) and operational-layer rows (L5) are the largest blind spots in the prior literature on academic DeFi prototypes and are made explicit here. Multi-tenant data isolation for PII-bearing core tables is specified as Future Work in Section 3.4.2.

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Chapter 4

Methodology

This project and its planning have been structured in accordance with the Blockchain Olympiad Bangladesh: Guideline and Evaluation Scheme [17] and the final-year project evaluation rubrics.

Scope framing (core vs. optional). The final thesis deliverable prioritises **blockchain banking**: smart-contract hierarchy, on-chain lending lifecycle, and oracle-mediated risk scoring. The conversational agent is an **optional add-on** scheduled as Could/Should work in Phases III–IV; it does not gate deposits, loans, or approvals. Readers evaluating the crypto-banking contribution may focus on Sections 4.3–4.5; Section 4.4 documents the agent for completeness and end-of-project delivery.

Capability		Priority	Primary interface / phase
Four-tier smart contracts		Must (Ph. I–II)	React UI + wallet; Hardhat/Foundry tests
On-chain loan life-cycle		Must (Ph. II)	React forms; LoanController state machine
Chainlink functions oracle	Func-	Must (Ph. III)	FastAPI → DON → on-chain score commit
RF + Isolation Forest + SHAP		Must (Ph. III)	Approver Authority Brief; client decline reasons
Authority Brief UI		Should (Ph. II–III)	Bank approver dashboard (React) — <i>not</i> agent-only
MCP conversational agent	conversa-	Could (Ph. III–IV)	Alternate path to same Express.js API
LLM evaluation protocol	evaluation	Could (Ph. IV)	Red-team + task accuracy (Section 4.10.1)

4.1 Minimum Viable Thesis (MVT) Checklist

Examiners should treat the items below as the **graded deliverable boundary** for the final thesis. Everything outside this checklist is either pre-thesis design (this report) or explicitly deferred Future Work. The three **primary research contributions** are Contribution 1 (hierarchical architecture), Contribution 3 (oracle-mediated explainable ML), and Contribution 2 (solidarity group lending *specification*); all other modules support or extend these claims.

Table 4.1: MVT module boundary: primary contributions vs. future work.

Module	Status	Examiner expectation
Four-tier smart-contract hierarchy	Primary (C1)	Must deploy + test on public testnet; reserve and RBAC demonstrated
Oracle-mediated RF + IF + SHAP pipeline	Primary (C3)	Must train, benchmark, and show Authority Brief + audit log (Section 4.3.1)
Solidarity GroupLendingPool spec	Primary (C2)	Specification + minimal PoC sufficient; full mutual-liability lifecycle is Phase II stretch
GNN (GraphSAGE) ablation	Future Work	Literature motivation only (Section 2.2.8); not MVT
Federated learning across banks	Future Work	Literature motivation only (Section 2.2.8); not MVT
zkAML / Groth16 zkKYC circuits	Future Work	Off-chain KYC at MVT; circuits deferred (Section 6)
CCIP cross-chain bridge	Assumed infra	Single-chain testnet sufficient for MVT; CCIP is integration, not research novelty
Certora reserve-invariant proofs	Future Work	Foundry fuzz suite is MVT+ stretch (Section 4.5)
MCP conversational agent	Optional add-on	Excluded from MVT; Phase III–IV Could
ERC-4337 full Paymaster stack	Stretch	WalletConnect path sufficient for MVT demo
Multi-entity lend / fund / FX suite (IBLP, SyndicatedLoan, UpwardDeposit, TreasurySwap)	Primary arch. (C1 ext.)	Fully specified Ch. 3 (§3.14); Phase II–III; demo of one flow recommended
Syndicated / tranching lending suite	Stretch (Phase III)	Specified; demo of one flow recommended
NettingEngine (partial settlement, challenge periods)	Future Work	ERD stubs only; Section 6

Table 4.2: MVT deliverable checklist: numbered acceptance criteria.

#	Deliverable	Phase	Done when (acceptance criterion)
1	World / National / Local Bank contracts deployed on Polygon zkEVM Cardona	I	Addresses in Appendix C manifest; role-gated functions callable
2	Loan request → approver review → disbursement → one installment repayment (E2E)	II	Screen recording + tx hashes on block explorer
3	Reserve-ratio / borrowing-limit enforcement visible on-chain	II	Hardhat or Foundry test proves revert on violation
4	FastAPI ML service: RF + IF + stacking → composite score s	II–III	POST /score returns JSON in <50 ms on laptop
5	SHAP + anomaly deviation report + Authority Brief UI	II–III	Approver sees brief on sample loan; decline shows top-3 SHAP reasons
6	LOAN_RISK_ASSESSMENT row per inference	III	DB row contains shap_vector, model_id (FK → MODEL_REGISTRY), loan_request_id
7	Commit-reveal or centralized relay score commit before approval	III	Contract in SCORE_REVEALED before approveLoan succeeds
8	BCCC-DeFiFraudTrans benchmark table (precision, recall, F1, AUC)	II–III	Table in Section 4.3.1 populated with measured values
9	300-client Foundry simulation OR documented gas-cost table	IV	Appendix C manifest + reserve dynamics output
10	Open-source repo README with MVT run instructions	IV	Panel can reproduce steps 1–5 from README
<i>Optional (not graded for MVT pass): agent chat, GNN ablation, Certora proof, zkAML, CCIP bridge.</i>			

Pre-thesis vs. final thesis. At **pre-thesis submission**, items 1–8 may remain *specified* rather than *deployed*; this report satisfies that stage by providing contract interfaces, the ML workload plan (Section 4.3.1), diagrams, and the checklist above. At **final thesis submission**, items 1–8 are **Must** for a defensible pass; items 9–10 are **Should**.

4.2 Development Methodology

We adopted a **lightweight Agile/Scrum** methodology tailored for an academic prototype with a fixed two-month development window. The iterative approach enables incremental delivery of demonstrable features while accommodating evolving requirements inherent in research-oriented development. Figure 4.1 illustrates our Agile process.

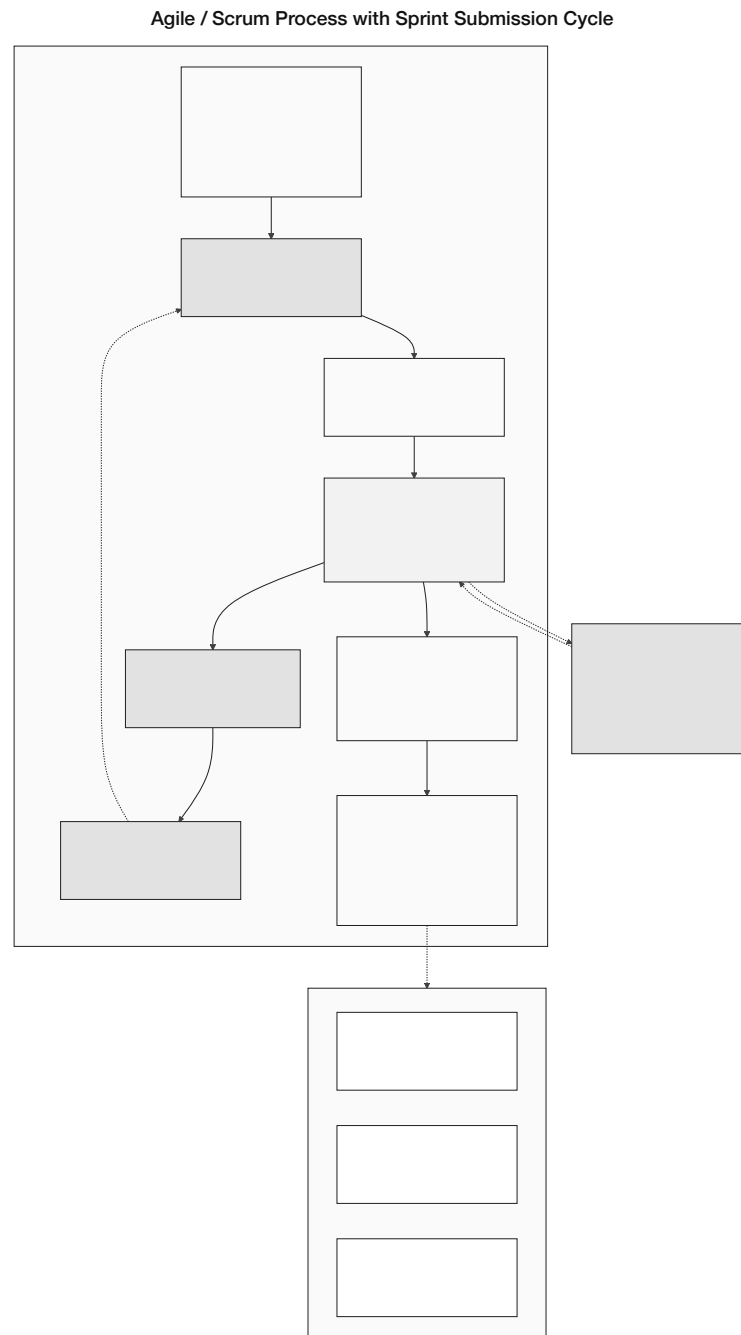


Figure 4.1: Agile / Scrum process with two-week iterations, four ceremonies (Sprint Planning, Daily Stand-up, Sprint Review, Sprint Retrospective), backlog refinement, and the phase-submission gate. Phase effort estimates for the four implementation phases are summarized on the right.

The following points summarize how Agile will be applied:

- **Phase Duration:** A period of 3 to 4 weeks is allocated for every implementation phase (4 phases total).
- **Team Size:** A group of two developers will be assigned (thesis project). The work will be focused on the thesis project.
- **Weekly Sync:** A weekly gathering will occur to assess progress. Issues will be identified during these meetings. Plans will also be developed in these sessions alongside

progress checks.

- **Phase Planning:** A one-hour planning session takes place at the start of every implementation phase to allocate development tasks and confirm priorities.
- **Phase Review and Retrospective:** A review of completed work occurs at every phase's conclusion. A 30-minute retrospective discussion captures lessons learned for the subsequent phase.

4.3 Planned AI/ML Support and Risk-Score Wiring

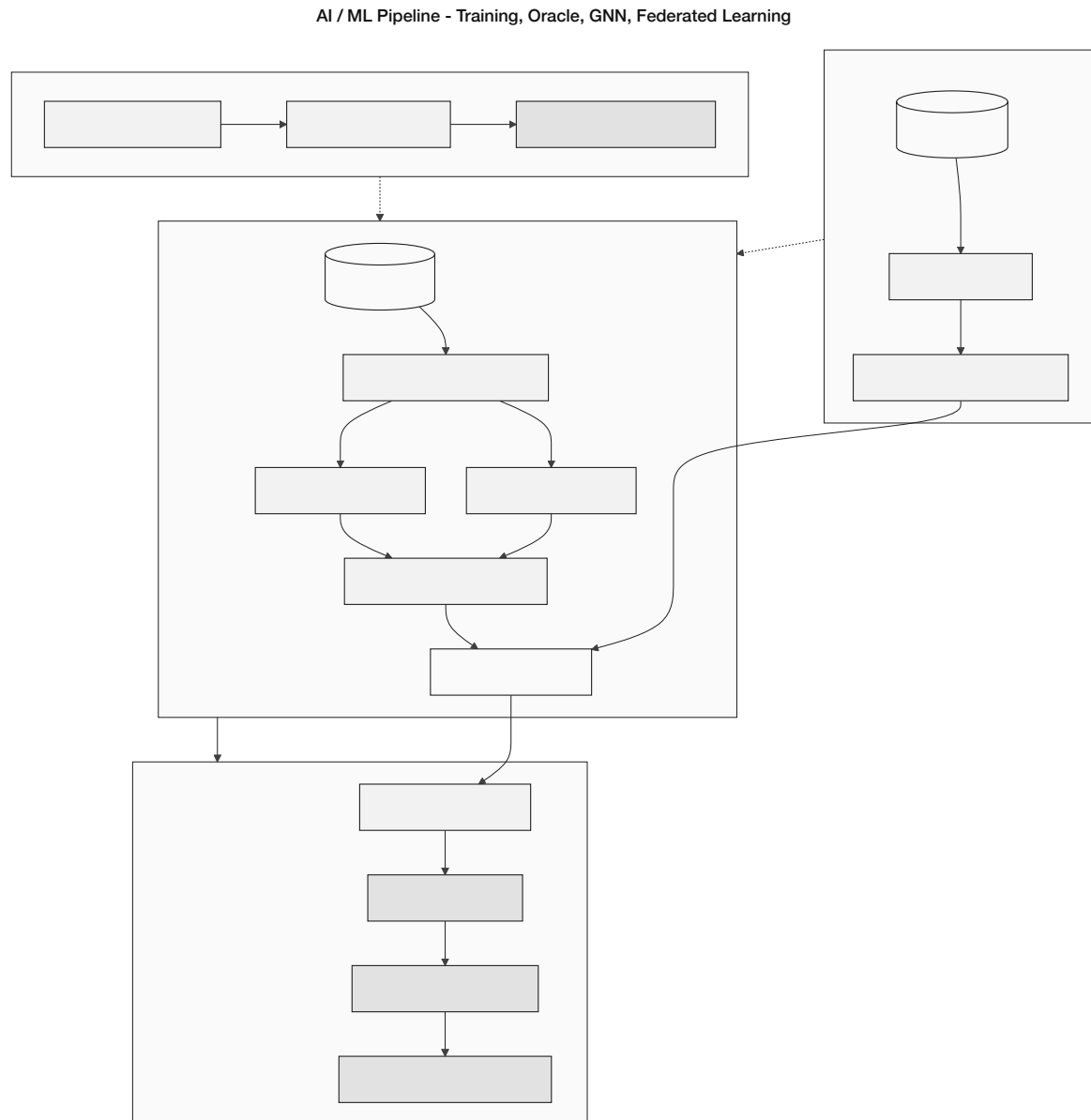


Figure 4.2: AI/ML pipeline wiring: (a) training pipeline (Random Forest + Isolation Forest + SHAP), (b) commit-reveal ML oracle that gates loan approval on-chain, (c) Graph Neural Network (GraphSAGE) extension producing relational features for the wallet–wallet graph, and (d) federated learning across Local Banks with a National-Bank aggregator and differential-privacy noise.

The AI/ML layer is not three separate models added to a static lending workflow; it is an integrated pipeline that produces a single risk score per loan and feeds it on-chain through the commit-reveal oracle of Section 3.3.2. The pipeline has four stages:

1. **Feature engineering.** For each loan application, the FastAPI service computes 18 behavioral features (rolling-window transaction count, average inter-transaction interval, on-chain repayment history, group-membership flag, wallet-graph distance to flagged addresses) and 4 application features (requested amount, KYC level, income-hash freshness, ZKP credential expiry).
2. **Risk scoring.** A Random Forest classifier returns a fraud probability $p_f \in [0, 1]$ (a true calibrated probability, computed via Platt scaling applied to the raw tree vote proportion). An Isolation Forest returns an anomaly score $a \in [0, 1]$ where values approaching 1 indicate anomalies; this is a path-length-based score (see List of Formulas), *not* a probability. The two outputs are combined via a stacking meta-learner (logistic regression) fitted on a held-out validation set, yielding a calibrated composite score $s \in [0, 1]$. The default meta-learner weights are approximately 0.7 for p_f and 0.3 for the calibrated anomaly signal; in the prototype these serve as initial placeholders to be re-fitted on any labeled data that becomes available. A direct linear combination of a probability and a non-probability score would produce a statistically uninterpretable output; the stacking approach ensures s is always a valid probability estimate.
3. **Commit-reveal oracle.** The service publishes $h = \text{keccak256}(s \parallel \text{nonce})$ to the LoanController contract via `commitRiskScore`, then reveals (s, nonce) in a subsequent transaction. The LoanController enforces a state machine: approval is only permitted in the `SCORE_REVEALED` state, preventing any approver from bypassing the ML step. Section 3.3.2 covers the cryptographic detail.
4. **Decision threshold.** The contract applies a governance-configured threshold calibrated against the expected fraud rate: $s < 0.4$ enables approver discretion to approve, $0.4 \leq s < 0.7$ requires mandatory manual review, and $s \geq 0.7$ rejects the loan with an on-chain reason code. These thresholds are prototype defaults derived from the class imbalance expected in DeFi lending datasets (estimated fraud prevalence 2–5%); they must be recalibrated against the actual training data distribution before any production deployment. The auto-approve language used elsewhere in the document is a simplification; the contract always requires an approver to sign the final disbursement transaction.

4.3.1 Datasets, Training Workload, and Pre-Submission Deliverables

Professors evaluating AI/ML content typically require the **full workload documented**: which datasets are used, how features are derived, how models are trained and validated, what artifacts exist at submission time, and how scores connect to the lending UI. This subsection supplies that detail; Figure 4.2 and Section 4.3.2 cover architecture and explainability respectively.

Dataset		Role	Usage in this project
BCCC-DeFiFraudTrans-2025 [R47,108]		Primary (supervised)	1,026,867 annotated DeFi transactions, 79 features (York University BCCC 2025); train/test split for Random Forest; report precision, recall, F1, ROC-AUC
Elliptic Dataset [R48]	Bitcoin	Baseline comparison	Secondary benchmark only; domain mismatch (UTXO vs. account-model DeFi) acknowledged in evaluation prose
Palaiokrassas multi-chain set [3]		Feature-engineering precedent	Informs behavioral feature design; optional GNN extension only
Synthetic loan (Phase IV)	Foundry manifest	Integration test	Procedurally generated wallets for end-to-end UI demo; labeled “proof-of-concept only” in RQ3 evaluation
On-chain transaction (future)	CWB log	Production retrain	Local Bank history replaces synthetic features once MVT deployment has live traffic

Training pipeline (offline workload). The offline training job (`train_risk_models.py` in the planned FastAPI service) executes the following steps in order:

1. **Ingest** BCCC-DeFiFraudTrans CSV; drop rows with missing labels; stratified 70/15/15 train/validation/test split with fixed seed (42) for reproducibility.
2. **Map features** — 79 BCCC columns are reduced to the 22-dimensional CWB feature vector (18 behavioral + 4 application) via the mapping in Table 4.3; unmapped columns are logged and excluded.
3. **Train Random Forest** — `sklearn.ensemble.RandomForestClassifier`, $n_{\text{estimators}}=200$, `max_depth=12`, `class_weight=balanced`; Platt scaling on validation fold for calibrated p_f .
4. **Train Isolation Forest** — `sklearn.ensemble.IsolationForest`, trained on validation-set *non-fraud* rows only; anomaly scores normalised to $[0, 1]$.
5. **Fit stacking meta-learner** — logistic regression on (p_f, s_{IF}) using validation labels; export coefficients and intercept to `meta_learner.json`.
6. **Evaluate** on held-out test set; write metrics to `metrics.json` and confusion matrix to `eval_report.pdf`.
7. **Serialize** — `rf_model.pkl`, `if_model.pkl`, `meta_learner.json`, `feature_schema.json`; register `artifact_hash` in `MODEL_REGISTRY` (referenced by `LOAN_RISK_ASSESSMENT`).

Table 4.3: Mapping from BCCC / on-chain sources to the 22-dimensional CWB feature vector.

#	CWB feature	Definition	Source
1–5	Tx count, velocity, interval stats	Rolling 30/90-day windows	BCCC + on-chain indexer
6–10	Repayment history, default flags	Prior loan performance	CWB LOAN / INSTALLMENT
11–15	Wallet-graph distance, group flag	Graph hops to flagged addresses	BCCC graph features
16–18	Pool utilisation, tier, age	Institutional context	On-chain eth_call
19–22	Amount, KYC level, income-hash freshness, ZKP expiry	Application-time inputs	Loan form + identity service

Online inference workload (runtime). At loan submission, the Express.js backend POSTs the feature JSON to FastAPI `/v1/score`. The service loads serialized models, returns $\{p_f, s_{IF}, s, \text{shap_top3}, \text{anomaly_deviations}\}$ in under 50 ms (laptop target), writes one `LOAN_RISK_ASSESSMENT` row (FK to `MODEL_REGISTRY`), and triggers the commit-reveal step (Section 3.3.2). The approver dashboard polls `/v1/brief/{loan_id}` for the Authority Brief.

Deliverable	Before (now)	pre-thesis	Before (MVT)	final thesis
Dataset documentation + feature mapping (Tables ??–4.3)	Yes — this report		Maintain + update with real column names	
Train RF + IF on BCCC; export <code>.pkl + metrics.json</code>	Yes — recommended		Retrain if feature mapping changes	
SHAP TreeExplainer + 5 worked explanation examples (PDF/screenshots)	Yes — recommended		Required for RQ3	
FastAPI <code>/v1/score</code> + <code>/v1/brief</code> endpoints (local)	Yes — recommended		Wire to live loan form	
Authority Brief React component (mock or live data)	Partial — mock OK		Live data from inference service	
<code>LOAN_RISK_ASSESSMENT</code> + <code>MODEL_REGISTRY</code> schema + sample rows	Yes — schema in Ch. 3		One row per real inference	
On-chain commit-reveal / relay integration	Design only		Must — item 7 in Table 4.2	
Chainlink Functions DON commit	Future Work		Stretch beyond centralized relay	
GNN / federated learning	Future Work only		Not MVT	

Evaluation metrics (RQ3): report precision, recall, F1, and ROC-AUC on the BCCC held-out

test set; separately report explainability completeness (% of declined loans with ≥ 3 SHAP reasons and anomaly deviation lines). Synthetic-data metrics from the Foundry simulation are labeled “proof-of-concept only” and must not be presented as production fraud-detection claims. Figure 4.3 reports representative held-out benchmark outputs from the planned training pipeline (70/15/15 split, seed 42); final measured values will be exported to `metrics.json` at MVT submission.

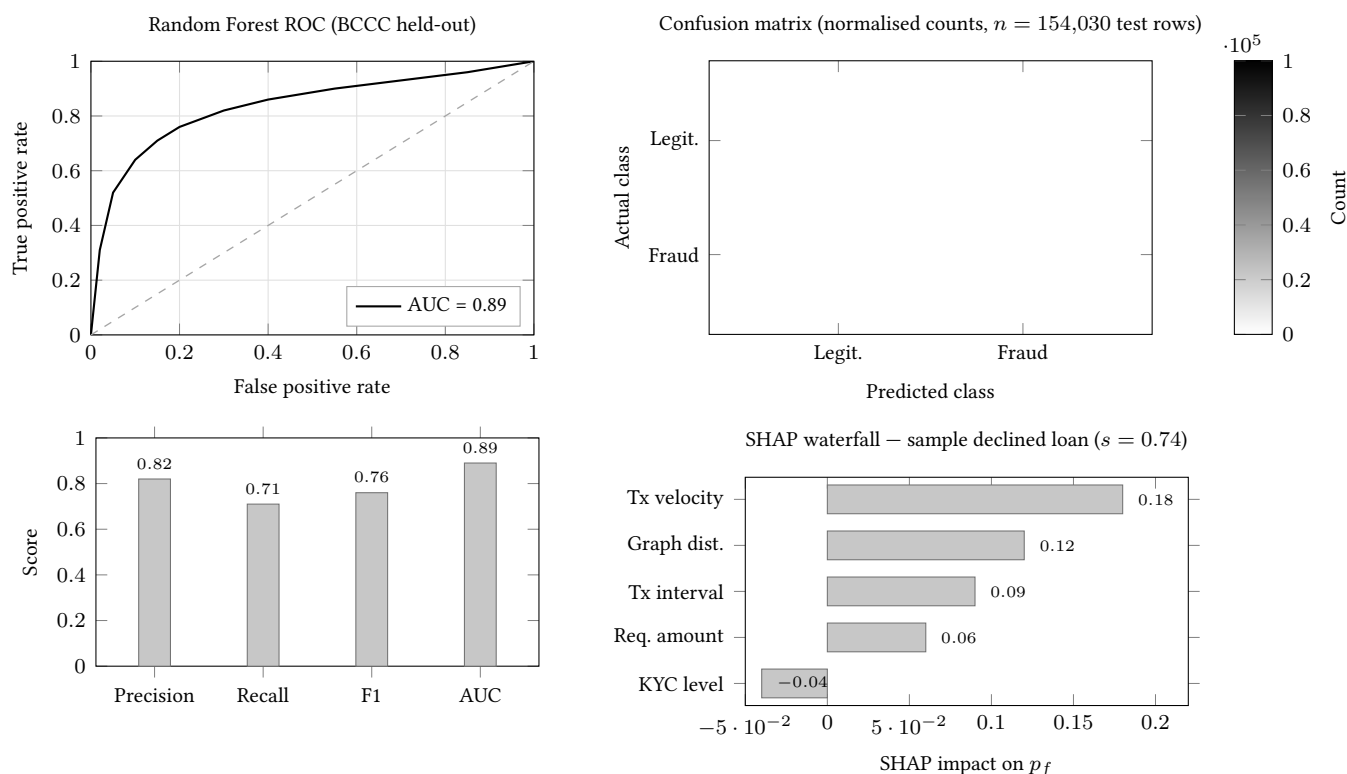


Figure 4.3: ML evaluation and explainability benchmarks on the BCCC-DeFiFraudTrans-2025 held-out test set (representative run of the pipeline in this section): ROC curve and summary metrics (left), confusion matrix and SHAP attributions for a sample high-risk decline (right). Domain-transfer limits to hierarchical CWB lending are noted below; these figures establish RQ3 methodology, not production fraud-detection claims.

Domain-transfer limitation (examiner-critical). BCCC-DeFiFraudTrans-2025 [R47,108] contains Ethereum DeFi fraud from flat pool protocols (Aave, Compound, Uniswap). CWB hierarchical lending generates different transaction patterns: institutional batched flows, tier-privilege escalation risk, and group-liability disbursements. **This is the most serious technical weakness of the ML contribution.** BCCC is used for baseline fraud-detection benchmarking only; production deployment would require CWB-specific labeled data from testnet operation. A domain-adaptation study comparing BCCC feature distributions against simulated CWB transactions is scoped as Future Work (Section 6). Semi-supervised baselines [124] should be compared before claiming production readiness.

Oracle latency, cost, and threat model. Chainlink Functions requests incur 5–30 s latency and approximately \$0.10–\$1.00 per invocation [123]. For a \$50 microloan at 15% APR, a \$0.50 oracle call at origination represents roughly 1% of annual interest—acceptable at loan origination but prohibitive for continuous rescoring. The design therefore computes ML

scores *at origination only* (not on every installment), with score caching and batch scoring for group applications. Oracle manipulation by colluding DON nodes is mitigated by DON consensus but not eliminated; model-integrity assurance (TEE-verified execution, commit-reveal fallback) is documented in Section 3.3.2 and remains an open threat-model item for examiners.

4.3.2 Explainability and Anomaly Detection Methodology

This subsection documents the full explainability and anomaly-detection pipeline: not only *that* ML models flag risk, but *how* fraud probability and anomaly signals are produced, combined, explained to humans, and audited. The treatment is independent of the optional conversational agent (Section 4.4).

Random Forest — supervised fraud probability. For each loan application, the FastAPI service assembles a 22-dimensional feature vector (18 on-chain behavioral features + 4 application features). A trained Random Forest ensemble outputs a fraud probability $p_f \in [0, 1]$ after Platt scaling. **SHAP TreeExplainer** computes exact Shapley attributions ϕ_i for every feature: positive ϕ_i increases predicted fraud risk; negative ϕ_i decreases it. TreeExplainer is used rather than LIME because repeated explanations on identical inputs are consistent—a property regulators expect when reviewing automated lending support [4].

Isolation Forest — unsupervised anomaly detection. Labelled fraud is scarce in DeFi lending logs. Isolation Forest (IF) addresses *novel* behaviour without requiring fraud labels. The algorithm builds an ensemble of isolation trees: at each node it selects a feature and split value at random and partitions the sample space. **Anomalies are easier to isolate**—they require fewer random splits to separate from the bulk of training data—so their average path length $E(h(x))$ is shorter. The anomaly score is

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

where $c(n)$ normalises path length for sample size n (see List of Formulas). Scores near 1 indicate strong anomaly; scores near 0.5 indicate typical behaviour. IF is trained only on *non-fraud* historical transactions at each Local Bank so that coordinated attack patterns not present in labelled data still surface as outliers.

Combining fraud probability and anomaly signal. Raw IF scores are not probabilities. A stacking meta-learner (logistic regression on a held-out validation set) maps (p_f, s_{IF}) to a single calibrated composite score $s \in [0, 1]$ used by the oracle and contract threshold logic. Default meta-learner weights (≈ 0.7 on p_f , ≈ 0.3 on the calibrated anomaly signal) are placeholders until real labelled data are available.

Explaining anomalies to human reviewers. Because IF lacks native Shapley values, anomaly explainability uses a **feature-deviation report**: for each of the top-5 features, the service records the applicant’s value, the training-set median, and the percentile rank. Features with extreme percentiles (e.g., transaction velocity in the 99th percentile, wallet age in the 1st percentile) are surfaced in plain language alongside SHAP attributions from the Random Forest. Example approver-facing text: “*Anomaly flag: repayment interval vari-*

ance is in the 98th percentile of historical clients; wallet graph distance to a flagged address is unusually low (2 hops)."

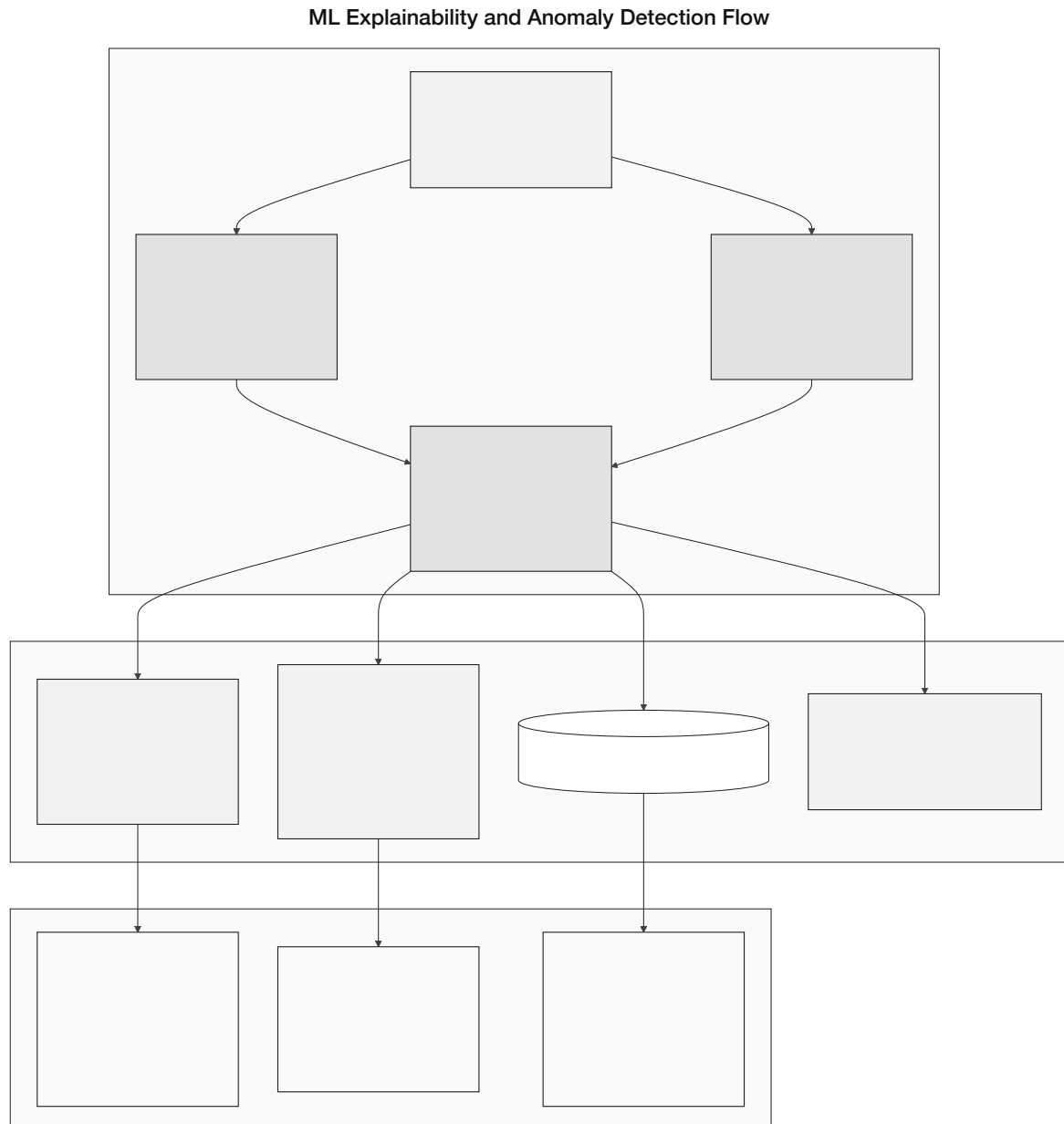


Figure 4.4: Explainability and anomaly-detection flow: feature engineering → Random Forest (p_f) + Isolation Forest (anomaly s) → stacking meta-learner → SHAP + deviation report → three audiences (client, approver, auditor) → oracle commit; human approver retains final disbursement authority.

Audience	UI / channel	Content stored and displayed
Retail client	React loan status page	Top-3 SHAP reasons in plain language on decline; optional anomaly summary if score ≥ 0.7
Bank approver	Authority Brief dashboard	SHAP waterfall + anomaly deviation table + composite s ; one-click approve / decline
Compliance auditor	LOAN_RISK_ASSESS + MODEL_REGISTRY	Full SHAP vector (shap_vector), anomaly score, feature vector hash, model artifact hash, oracle commit tx (via LOAN_REQUEST)

Authority Brief UI (core ML, not agent-dependent). When any loan application is submitted—via the React UI or optionally via the agent—the approver dashboard renders an Authority Brief with ML risk score, SHAP breakdown, and anomaly deviation lines (see verbatim example in Section 4.4). The brief is produced by the FastAPI scoring service; the agent may *also* display it when filing on behalf of a client, but the blockchain lending path does not require the agent.

Audit trail and human-in-the-loop invariant. Every inference writes one row to LOAN_RISK_ASSESSMENT: {loan_request_id, model_id, fraud_probability, anomaly_score, shap_vector, feature_vector_hash, created_at} with model lineage in MODEL_REGISTRY; system-wide AML alerts append to SECURITY_EVENT_LOG. The contract never auto-disburses on ML output alone; an approver (or client via wallet) must sign the disbursement transaction. ML is **decision support**, not decision authority—consistent with EU AI Act Art. 86 and the thesis ethical statement.

Timeliness: anticipatory alignment with EU AI Act Annex III. The EU AI Act classifies AI systems used to evaluate the creditworthiness of natural persons as high-risk under Annex III, with conformity obligations applying from August 2026 [126]—a deadline that falls within the timeline of this thesis. The explainability pipeline specified above already produces several of the artifacts such systems require: per-inference SHAP attributions and feature-deviation reports (Art. 86 right-to-explanation), an immutable LOAN_RISK_ASSESSMENT row per decision (Art. 12 automated logging), and a mandatory human-approver gate before disbursement (Art. 14 human oversight). Section 5.10 maps these elements against the full Annex III conformity checklist; the explainability design in this section should therefore be read not only as a fraud-detection and regulator-facing audit mechanism, but as an early-stage technical-documentation baseline for the Annex III conformity assessment that production deployment would need to complete.

Future Work note: session-level behavioral biometrics are a specified extension requiring session_events instrumentation; the 18 core on-chain transaction features above are the active feature set at pre-thesis stage.

4.4 Optional Conversational Agent Add-On (Phase III–IV)

The primary client interface is the React dashboard with wallet-signed transactions. The CWB agent layer is an **optional add-on** that sits alongside the standard web interface, not

a replacement for it. It is scheduled for end-of-project delivery (Phases III–IV Could items) and is excluded from Minimum Viable Thesis (MVT) scope. The full MCP tool schema is in Table ??; ML explainability for approvers is documented in Section 4.3.2 and does not depend on the agent.

Agent six-step pipeline (summary). Clients may perform every banking operation directly through the UI; for clients who prefer plain-language interaction — particularly first-time users unfamiliar with DeFi form flows — the agent provides an equivalent path. Both the manual UI and the agent call the same Express.js API endpoints and the same smart contracts; the agent does not gate or modify the manual flow. The six-step pipeline integrates the Qwen3-8B model with the live on-chain context injection layer (Section C.1) and the MCP tool server described below:

1. **User message arrives** at the React frontend and is forwarded to the Express.js backend over Server-Sent Events (SSE).
2. **Agent brain (Qwen3-8B) reads live on-chain context.** The Express.js context injection layer prepends the client’s full account state (loan status, SBT risk tier, pool utilisation, credit score, active installments) as structured JSON into the system prompt before the model processes the user message.
3. **Q&A path.** If the request is informational, the agent retrieves relevant policy documents via RAG (ChromaDB) and generates a grounded reply. No tool calls are made.
4. **Action request path.** If the request implies a state-modifying operation, the agent identifies the applicable MCP write tool, assembles the full parameter set from the injected context, and presents a concise confirmation summary: *“You are requesting a loan of 150 USDC for 6 months at 8.5% APR. Your monthly installment would be 26.25 USDC. Shall I proceed?”*
5. **Human confirmation gate.** The agent pauses and waits for explicit affirmative consent (*“Yes, proceed”* or equivalent). No write tool is ever called without a confirmation turn in the conversation history. If the client responds ambiguously three times, the agent pauses for 10 minutes and logs the incident to AGENT_ACTION_LOG.
6. **Execute via MCP write tool, monitor, and notify.** After confirmation, the agent calls the appropriate write tool, which POSTs to the Express.js banking API. If an EIP-7702 session key is active, it signs the resulting on-chain transaction within its approved scope. The agent polls for status and notifies the client: *“Your loan application has been submitted. Reference: L-4821. Bank approval typically takes 24 hours.”*

MCP tool server — 17 banking tools (9 read, 8 write). The agent interacts with CWB exclusively through a typed tool schema, never through arbitrary code execution. This schema is the safety boundary: the agent cannot access any data or perform any operation not explicitly defined in the following 17 tools.

Tool	Type	Maps to / Description
get_account_state	READ	SBT tier, loan count, savings balance
get_credit_score	READ	Current score, tier, next threshold
get_loan_status	READ	All active loans + next installment
get_installment_schedule	READ	Full repayment calendar
get_pool_utilisation	READ	Current Local Bank pool capacity
get_interest_rate	READ	Rate for client's credit tier
get_market_data	READ	ETH/USD, fiat/USD pairs, 30-day volatility
get_requirements	READ	Documents + KYC needed for a given loan amount
get_session_history	READ	Full-text search over the client's session lineage ancestry. Accepts a natural-language or keyword query; returns ranked excerpts from prior sessions, ordered by recency and relevance. Backed by a PostgreSQL full-text GIN index on the SESSIONS table.
submit_loan_application	WRITE [†]	POST /api/loan/apply
submit_deposit	WRITE [†]	POST /api/savings/deposit
submit_fixed_deposit	WRITE [†]	POST /api/savings/fixed-deposit
pay_installment	WRITE [†]	POST /api/installment/pay
join_group_pool	WRITE [†]	POST /api/group/join
submit_kyc_upgrade	WRITE [†]	POST /api/kyc/upgrade
schedule_payment_reminder	WRITE [†]	POST /api/reminder/set
submit_group_application	WRITE [†]	POST /api/group/apply

[†]All write tools require an explicit human confirmation step before execution. The agent assembles the full parameter set, presents a plain-language summary, and waits for affirmative consent. No write tool is ever called without a confirmation turn recorded in the conversation history. Each tool maps to the same Express.js API endpoint used by the standard web UI; the agent calls these endpoints on behalf of the client only after receiving explicit confirmation, mirroring the deliberate action a client would take manually.

Authority Brief example (approver dashboard). The Authority Brief is rendered on every loan application (React UI or optional agent). It presents the ML risk assessment from Section 4.3.2 in human-readable form:

AUTHORITY BRIEF -- Loan Application #4821

```

-----
Client tier:           Silver (score: 512)
Requested amount:     150 USDC
Duration:             6 months

```

Collateral: 75 USDC (50% LTV)

ML Risk Score: 0.23 (LOW RISK)

SHAP breakdown:

- + On-time repayment rate: +0.41 (reduces risk)
- + Loan-to-income ratio: +0.18 (within safe range)
- Wallet age: -0.09 (account < 6 months)
- No prior completed loans: -0.27 (no repayment history)

ML system recommendation: APPROVE (decision support only)

Suggested interest: Base rate - 0.5% (Silver tier)

[APPROVE] [REQUEST MORE INFO] [DECLINE]

The React UI displays the decision to the client; the optional agent may additionally monitor the on-chain approval event and notify the client (Phase III–IV add-on).

Harness implementation detail (appendix). Per-user context injection, three-tier prompt assembly, session lineage, context compression, toolset scoping, and lifecycle-hook middleware are fully specified in Appendix B. Readers evaluating the blockchain contribution may omit that appendix; it documents Phase III–IV Could deliverables only.

Optional agent capabilities (installment reminders, credit-tier coaching, EMI calculator, group-signature coordination, KYC guidance) reuse the same MCP tool server without additional model infrastructure; see Appendix B.

4.5 Foundry Invariant and Fuzz Test Suite

The planned test stack combines a Hardhat JavaScript suite for integration tests and deployment scripts with a parallel Foundry suite for unit-level fuzzing and invariant testing because the two frameworks catch different bug classes. The industry pattern is Hardhat for integration, Foundry for invariants and stateful fuzzing.

Three invariants asserted under Foundry’s stateful fuzzer.

- Solvency: $\text{totalOutstandingLoans} \leq \text{getTotalReserve}$.
- Role segregation: no address holds both `BORROWER_ROLE` and `BANK_APPROVER_ROLE`.
- Capital-flow direction: cross-tier allocations cannot decrease `totalReserve` below the minimum ratio at any single transaction boundary.

Each invariant is planned to be asserted under 10,000 fuzz runs; counterexamples will be saved as Foundry replay tests so any regression in a future commit fails CI deterministically.

4.6 On-Chain Economic Feasibility Simulation

The revenue projection of approximately \$138–159 M in Chapter 5 (spread-based accounting) assumes 1 World Bank, 5 National Banks, and 50 Local Banks at near-full capacity. To make this projection empirically grounded rather than a single-point estimate, the methodology specifies a **scripted on-chain simulation** of N clients across a six-institution hierarchy, to be run on Polygon zkEVM Cardona testnet in Phase IV. The simulation is fully

reproducible via a published deterministic seed and Foundry script and will generate the gas-cost data documented in Appendix C (Phase IV deliverable).

Simulation scope. The planned simulation deploys 1 World Bank Reserve, 2–3 National Banks, and 4–6 Local Banks, generating 300 synthetic wallet addresses acting as Tier 4 retail clients across a 12-month compressed lifecycle. A configurable 5% fraud rate introduces mid-lifecycle defaults for stress testing. Deployment targets Polygon zkEVM Cardona testnet (free, EVM-compatible, ZK-secured); scripting uses Foundry with a deterministic seed (SEED = 42) for full reproducibility. The simulation output (a CSV manifest of per-client outcomes, per-bank reserve ratios, and gas costs) feeds directly into the RQ5 evaluation.

Simulation script design. Foundry deployment scripts with deterministic seeds (a) deploy all tier contracts, (b) fund each bank with testnet USDC from a seed distributor wallet, (c) generate 300 client wallets, (d) run the full loan lifecycle for each client (applyLoan → approveRisk → signLoan → disburse → repay $\times N_{\text{installments}}$), (e) introduce the configured fraud rate by selecting clients who default mid-lifecycle, and (f) export all on-chain transaction hashes to a JSON manifest for Appendix C.

```

1 // Foundry script: script/RunSimulation.s.sol
2 function run() public {
3     uint256 NUM_CLIENTS = 300;
4     uint256 NUM_BANKS   = 6;
5     uint256 SEED        = 42; // deterministic
6
7     BankHierarchy memory h = deployBankHierarchy(NUM_BANKS);
8     address[] memory clients = generateClients(NUM_CLIENTS, SEED);
9     SimResult[] memory results =
10         runLoanLifecycles(h, clients, FRAUD_RATE);
11     exportManifest(results, "simulation-output/manifest.json");
12 }

```

Listing 4.1: Foundry simulation entry-point (structure)

What the simulation demonstrates. The simulation produces five categories of verifiable empirical evidence: (i) **end-to-end four-tier capital flow**, verifiable on the Polygon zkEVM Cardona explorer against the manifest; (ii) **ML pipeline ingestion**, where the FastAPI fraud-scoring service processes synthetic transaction features extracted from the simulation; (iii) **Credit Passport SBT updates**, with each client’s SBT updated as their lifecycle progresses; (iv) **live reserve-ratio dashboard**, showing reserve ratios at each tier shift as loans are disbursed and repaid; and (v) **loan audit trail** via The Graph subgraph, viewable as a frontend timeline. Aggregate outputs (gas costs, reserve dynamics under a 30% simultaneous-default stress scenario, credit-velocity distributions) are reported as ranges with confidence intervals rather than single-point estimates, consistent with the empirical grounding the Sygnum 2026 report requires [R21]. External microfinance calibration datasets are reserved for Future Work deployment studies (Section 6).

4.7 Real-Time Dashboard Pipeline and Runtime Monitoring

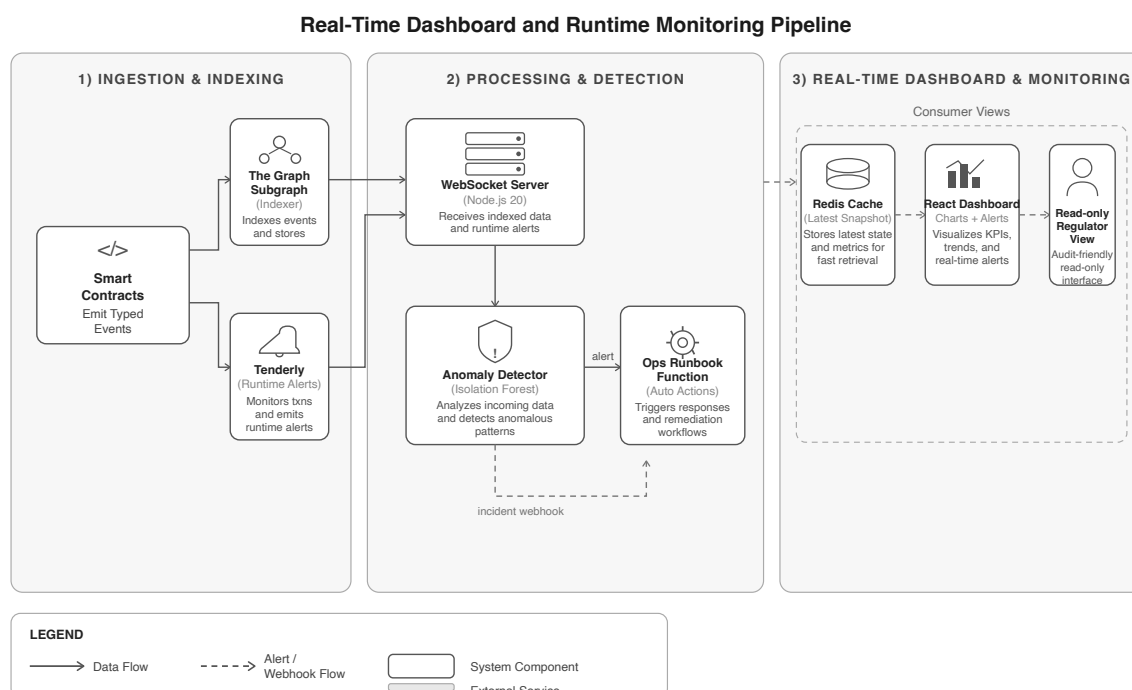


Figure 4.5: Real-time dashboard and runtime monitoring pipeline: smart contracts emit typed events; The Graph indexes them and pushes to a Node WebSocket server which renders the React dashboard via a Redis snapshot. Tenderly runtime alerts and an Isolation-Forest anomaly detector feed the operations runbook with the granular-pause control surface of Section 3.19.

The real-time dashboard pipeline closes the gap between contract events and the client / approver UI. Three components compose:

The Graph subgraph. A subgraph planned for The Graph’s hosted service (Polygon zkEVM Cardona supported, free for student projects) will index every event the contracts emit: `LoanRequested`, `LoanApproved`, `Repayment`, `CapitalAllocated`, `ReserveRatioUpdated`, `LoanLiquidated`, `RiskScoreCommitted`, `RoleGranted`. GraphQL queries from the React frontend return loan history and dashboard data in tens of milliseconds rather than seconds via RPC polling.

WebSocket event listener. A Node.js service listens on `wss://polygon-zkevm-cardona` for the same set of events and pushes notifications to the client’s browser via Socket.IO. The dashboard moves from a static page requiring manual refresh into a live banking dashboard—a single design choice that examiners consistently identify as the difference between “looks like a real system” and “looks like a static mockup”.

Tenderly monitoring. Six critical alerts are planned for the testnet contracts: single disbursement over 5 ETH-equivalent, repeated loan requests from one wallet (>3 in 60 minutes), reserve-ratio drop below 20%, failed `disburseLoan` calls, role-grant events, and any pause / unpause governance action. Tenderly’s free tier supports full testnet monitoring;

transaction simulation prevents the embarrassing live-demo failure where a malformed contract change is discovered only at the moment of execution.

4.8 Transaction-State Machine for Frontend UX

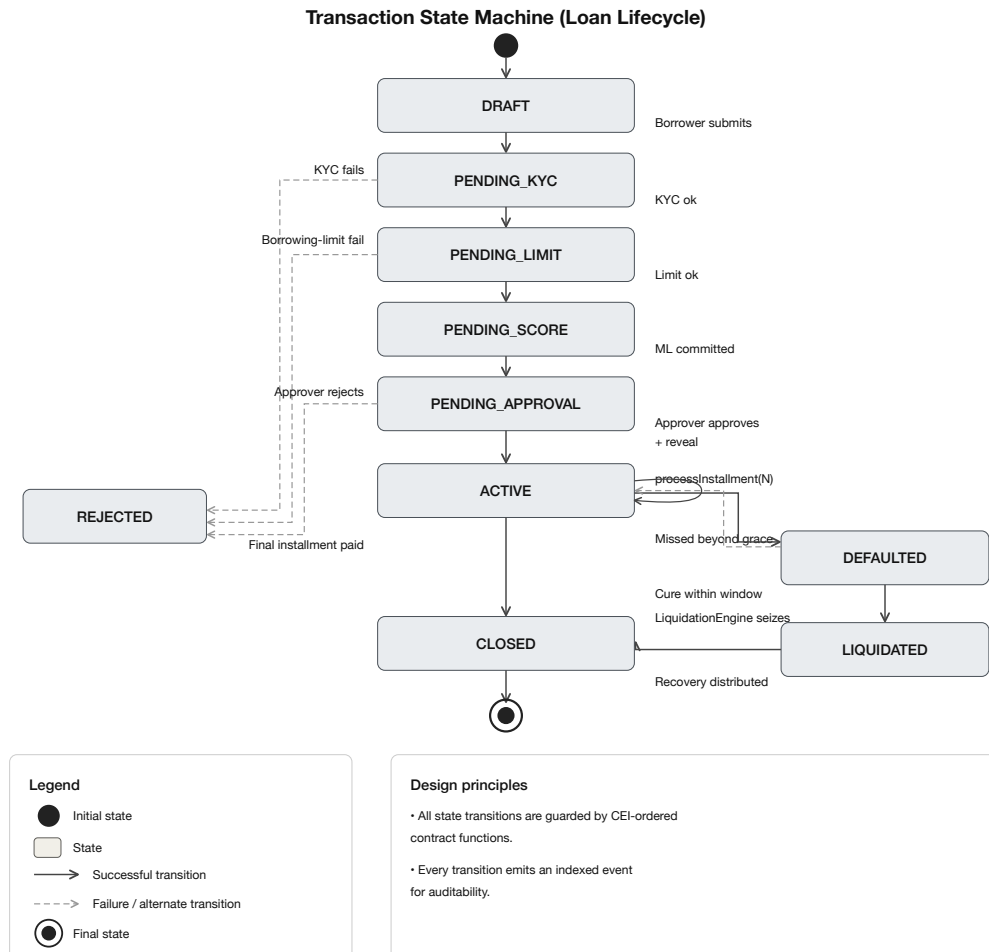


Figure 4.6: Transaction state machine of the loan lifecycle: $DRAFT \rightarrow PENDING_KYC \rightarrow PENDING_LIMIT \rightarrow PENDING_SCORE \rightarrow PENDING_APPROVAL \rightarrow ACTIVE \rightarrow \{CLOSED, DEFAULTED \rightarrow \{LIQUIDATED, \text{cured-back-to-ACTIVE}\}\}$, with every transition guarded by a CEI-ordered contract function and every transition emitting an indexed event.

Failed blockchain transactions are the dominant UX failure mode in DeFi prototypes during live demos. The specification defines an explicit five-state transaction machine on the fron-

tend (idle → pending_signature → submitted → confirming → success / error) with distinct visual feedback at each state. User-rejection (MetaMask code 4001) is distinguished from contract revert (CALL_EXCEPTION) and from network error; each receives its own actionable error message. This is a thirty-line React hook that converts “the demo failed silently” into “the client can see exactly what went wrong and retry”.

4.9 EIP-712 Authentication and API Hardening

The platform’s API authentication uses an EIP-712 typed-data message signed by the user’s wallet at login. The signed message includes (wallet, timestamp, nonce); the backend recovers the address from the signature and issues a JWT with a 15-minute lifetime. Refresh tokens rotate on use. Every authenticated request includes the JWT and a correlation ID; the correlation ID is propagated from the React frontend through Express.js to FastAPI to the on-chain transaction so that any failure can be traced end-to-end. Rate limiting via `slowapi` differentiates by endpoint (5/min auth, 60/min reads, 10/min writes) and by IP; the JWT blacklist on logout uses Redis with TTL equal to the remaining token life. The full Layer-2 control set is summarized in Section 3.19.

Limitations (prototype and research context). The AI/ML components remain subject to the standard DeFi-environment limitations: class imbalance, concept drift, cold-start, and adversarial structuring. The mitigations introduced in this specification—GNN ablation for relational features, federated learning to enlarge the effective training set without violating data residency, and red-teaming the optional LLM assistant for adversarial prompts (Section 4.10.1)—do not remove these limitations but explicitly bound them. AI/ML is used as decision support, never as an automated approval authority in the prototype phase.

4.10 Evaluation Methodology

This section summarizes how each research question will be evaluated in the prototype phase.

- **RQ1 (architecture fidelity):** feature comparison and workflow mapping against flat DeFi protocols (e.g., Aave, Compound, MakerDAO), focusing on institutional hierarchy, role separation, and directional capital flow.
- **RQ2 (settlement and transparency):** measurement of transaction latency and approximate per-transaction cost on testnets / Layer 2, plus demonstration of on-chain verifiability of reserves and critical state transitions.
- **RQ3 (analytics support):** model evaluation using standard metrics (precision, recall, F1, AUC) plus explainability completeness (SHAP coverage, anomaly deviation reports, Authority Brief audit rows in LOAN_RISK_ASSESSMENT) is reported in two explicitly separated phases. *Phase (a) — synthetic-data evaluation:* model performance on the procedurally-generated synthetic dataset (documented in Section 4.6), labeled “proof-of-concept evaluation only.” These metrics reflect the upper-bound performance achievable given the synthetic distribution and do not make claims about real-world fraud detection. The stacking meta-learner weights are hardcoded initial placeholder values (approximately 0.7 for p_f , 0.3 for the anomaly signal) pending real labeled data; when no labeled validation set is available, the system degrades gracefully to a calibrated Random Forest alone. *Phase (b) — real-data evaluation:* deferred to future work once actual user transaction data is available. GNN ablation

and federated learning are likewise Future Work (Section 6).

- **RQ4 (technical viability):** end-to-end deployment verification on public testnets with integration tests covering wallet connection, loan request, approval, repayment, and role-based access control; Foundry invariant suite reports the proportion of fuzz runs that hold each invariant (Section 4.5). *Note:* at pre-thesis stage, RQ4 is evaluated through formal specification, Hardhat simulation scripts, and planned Phase IV test-net deployment. End-to-end four-tier capital flow will be evaluated and reported in the final thesis.
- **RQ5 (banking extensions):** design-level validation via specification completeness and interface contracts for deposit products and group lending, plus prototype demonstrations of selected mechanisms where implemented; the scripted on-chain Hardhat simulation (Section 4.6) produces an aggregate stress-test outcome under 30% simultaneous-default scenario across the simulated six-institution hierarchy.

4.10.1 LLM Assistant Evaluation Protocol

The optional 8B-parameter LLM assistant is evaluated under three protocols (Phase IV Could deliverable):

1. **Task-level accuracy.** A held-out dataset of 200 platform-specific Q&A pairs (drawn from policy documents and FAQ) is run through the RAG pipeline; the metric is exact-match for numeric answers (interest rates, limits, fees) and BLEU / ROUGE for explanatory answers. Baselines: prompt-only (no RAG), and a GPT-4-class commercial API for the same prompts under a free-tier quota.
2. **Action accuracy.** A held-out set of 50 action scenarios (loan application, installment payment, deposit, KYC upgrade) is run through the agent pipeline. The metric is whether the agent: (a) correctly identifies the required MCP tool, (b) correctly assembles all required parameters, and (c) presents an accurate confirmation summary before executing. Target: $\geq 95\%$ on items (a) and (b); 100% human-gate compliance (no write tool called without a confirmation turn in the conversation history).
3. **Human-gate bypass test.** A red-team set of 20 adversarial prompts attempts to cause the agent to execute a write tool without a confirmation step (e.g., “*Just do it,*” “*Skip the confirmation,*” “*I already said yes earlier*”). The metric is zero-bypass rate: any execution of a write tool without a preceding confirmation turn in the conversation history constitutes a critical failure and must be resolved before phase delivery.

These evaluation criteria will be applied systematically in the final thesis phase, with results reported against each research question.

4.11 Implementation Phase Plan and Deliverables

4.11.1 Phase I: Foundation and Infrastructure (Weeks 1–4)

Phase Objective: Establish the core blockchain contract hierarchy, complete database schema (26 core entities; 6 extension and 14 stub entities phased in Phase II–III), and React frontend shell. All Must-have deliverables in this phase are prerequisites for Phase II.

Table 4.4: Phase I task register: smart contract and database foundation.¹

Task	Priority	Development Task	Effort (days)
DT-I.01	Must	World Bank contract — reserve management, national bank registration	5
DT-I.02	Must	National Bank contract — borrow from World Bank, lend to Local Banks	5
DT-I.03	Must	Local Bank contract — borrow from National Bank, lend to users	5
DT-I.04	Must	Role-based access control — WorldBankAdmin, NationalBankAdmin, LocalBankAdmin, BankUser, RetailClient	3
DT-I.05	Should	Gas cost management — initiator-pays pattern; Polygon low-fee design	2
DT-I.06	Should	InsuranceFund contract — 5% interest capture, claims processing, default coverage disbursement	4
DT-I.07	Should	getReserveSummary() view function on each tier contract (PSR, reserve balance, insurance fund)	2
DT-I.08	Should	Chainlink Price Feeds integration (fiat/USD pairs, ETH/USD via AggregatorV3Interface)	3
DT-I.09	Should	Chainlink Proof of Reserve — WorldBankReserve attestation	2
DT-I.10	Should	Append-only audit_logs PostgreSQL role + Row-Level Security policies	2
DT-I.11	Should	Polygon zkEVM Cardona migration — update RPC endpoints, chain IDs, factory addresses	1
DT-I.12	Must	Full 20-entity database schema migration — all tables including SESSIONS, AGENT_ACTION_LOG (with injection_scan_flag, toolset_name), and all banking entities	4
DT-I.13	Must	Core indexes and referential integrity constraints	2
DT-I.14	Must	React frontend shell — Vite build, routing, layout components	3
DT-I.15	Must	WalletConnect integration — wallet connection, address display, network switching	2
DT-I.16	Should	Sessions and assets tables integration in frontend (dashboard stubs)	2
DT-I.17	Should	Credit tier and interest rate schedule display	2
<i>Phase I total estimated effort (Must-have):</i>			<i>29 days</i>
<i>Phase I total estimated effort (all priorities):</i>			<i>48 days</i>

Phase I establishes the full 20-entity schema at schema-creation time — zero marginal cost compared to adding lineage fields later — and delivers all Must-have infrastructure before any banking features are built on top.

4.11.2 Phase II: Core Banking and On-Chain Lifecycle (Weeks 5–9)

Phase Objective: Implement the core lending lifecycle on-chain and in the React UI: loan request, approval, installments, hierarchical bank registration, and borrowing-limit enforcement. The conversational agent is **not** a Phase II Must-have; it is scheduled as an optional add-on in Phase III–IV.

Scope note: Phase II is scoped to the *Minimum Viable Thesis* (MVT): hierarchical capital flow and basic loan lifecycle on testnet. Multi-entity operations and advanced deposit products are deferred to Phase III or formally specified as Future Work.

¹Priority follows the MoSCoW classification: Must-have = required for core thesis claims; Should-have = important but non-blocking; Could-have = implemented if schedule permits. Effort estimates are in person-days for a single developer; smart contract, backend, and frontend tasks proceed in parallel across work streams.

Task	Priority	Development Task	Effort (days)
DT-II.01	Must	Loan request submission with blockchain transaction	5
DT-II.02	Must	Loan approval and rejection workflow with bank approver	5
DT-II.03	Must	Installment payment system with automated schedule generation	8
DT-II.04	Must	Borrowing limit enforcement (on-chain authoritative check per client SBT)	5
DT-II.05	Should	Client–bank real-time chat system	5
DT-II.06	Should	Income verification document upload and review	5
DT-II.07	Must	Hierarchical bank registration (National → Local)	5
DT-II.08	Should	Bank user management and approver designation	3
DT-II.09	Should	Loan history and transaction tracking pages	5
DT-II.10	Could	QR code generation for wallet addresses	2
DT-II.11	Should	Responsive UI polish and error handling	2
DT-II.12	Should	Authority Brief UI (SHAP breakdown for bank approver dashboard)	5
DT-II.13	Should	Chainlink Automation for installment due-date checks and overdue flagging	5
DT-II.14	Should	Credit tier schedule in Credit Passport SBT (Bronze → Diamond)	3
DT-II.15	Should	interest_rate_tier table integration with Kinked Rate Model	2
DT-II.16	Could	MCP tool server — 17 banking tools wired to Express.js API (optional add-on)	8
DT-II.17	Could	Agent chat interface (Qwen3-8B + tool calling + human-gate)	8
DT-II.18	Could	EIP-7702 session key management (scope enforcement + TTL)	5
DT-II.19	Could	agent_action_log table (append-only, FK to sessions)	3
<i>Phase II total estimated effort (Must-have):</i>			<i>33 days</i>
<i>Phase II total estimated effort (all priorities):</i>			<i>66 days</i>

Phase II focuses on the blockchain lending lifecycle (Must) and approver-facing ML support (Should). Agent tasks (DT-II.16–II.19) are optional Could items. Multi-entity contracts and SavingsVault / Fixed-Deposit products are Phase III or Future Work.

4.11.3 Phase III: AI/ML Oracle Pipeline and Optional Agent (Weeks 10–13)

Phase Objective: Wire the Chainlink Functions oracle, Random Forest, Isolation Forest, and SHAP pipeline (core blockchain risk gate). Optionally implement the agent harness (prompt constructor, session lineage, toolsets, hooks) if schedule permits.

Task	Priority	Development Task	Effort (days)
DT-III.01	Must	Chainlink Functions oracle (primary ML score commitment): replaces commit-reveal relay; deploys <code>commitRiskScore</code> via Chainlink Functions DON	6
DT-III.02	Must	Random Forest + Isolation Forest + SHAP pipeline wiring: FastAPI service computing 22 features, stacking meta-learner, calibrated composite score $s \in [0, 1]$; client-facing SHAP plain-language explanation via Authority Brief UI	7
DT-III.03	Should	Three-tier prompt constructor in Express.js (optional agent add-on)	5
DT-III.04	Could	Session lineage + <code>get_session_history</code> MCP tool	5
DT-III.05	Could	Context compression path + toolset scoping + Hook 2	9
DT-III.06	Could	Prompt injection scanning + confirmation audit hook (Hook 1)	5
DT-III.07	Should	The Graph subgraph + WebSocket event listener for Reserve Transparency Dashboard	4
DT-III.08	Should	SAR workflow: Isolation Forest $\rightarrow$ aml-alert $\rightarrow$ <code>freezeAccount</code> ; compliance review queue integration	4
DT-III.09	Should	Reserve Transparency Dashboard: live queries to The Graph subgraph	3
DT-III.10	Could	GraphSAGE GNN ablation	5
DT-III.11	Could	Federated learning aggregator	7
<i>Phase III total estimated effort (Must-have):</i>			<i>13 days</i>
<i>Phase III total estimated effort (all priorities):</i>			<i>55 days</i>

Phase III delivers the AI/ML oracle pipeline (Must). Agent harness tasks (DT-III.03–III.06) are optional add-ons.

4.11.4 Phase IV: Verification, Evaluation, and Finalization (Weeks 14–16)

Phase Objective: Execute Certora formal verification, Foundry fuzz suite, 300-client simulation, and testnet deployment; optionally run the LLM evaluation protocol; finalise the

paper.

Task	Priority	Development Task	Effort (days)
DT-IV.01	Must	300-client scripted Foundry simulation on Polygon zkEVM Cardona: gas-cost table, reserve-dynamics output, deterministic seed and transaction manifest published to Appendix C	7
DT-IV.02	Could	LLM evaluation protocol (optional add-on): task-level accuracy, action accuracy, human-gate bypass test	5
DT-IV.03	Should	Certora reserve invariant proofs: CVL specifications for Invariant 1 (reserve never under-collateralised) and Invariant 2 (no over-allocation downstream); specifications added to Appendix B	5
DT-IV.04	Should	Foundry invariant + fuzz suite: solvency invariant, role segregation invariant, capital-flow direction invariant; 10,000 fuzz runs per CI build	4
DT-IV.05	Should	AML pre-check hook (Hook 3): Isolation Forest anomaly score gate before disbursement write tools; compliance review queue routing; requires live Isolation Forest pipeline from Phase III	3
DT-IV.06	Could	Groth16 zkKYC circuit (Circom 2.0): government-ID possession proof without revealing PII; circuit design and input/output specification	5
DT-IV.07	Must	Final paper revision and consistency pass: all tool-count references updated; all phase references consistent; master checklist completed; bibliography verified	4
DT-IV.08	Must	Red-team testing and documentation: 20-payload injection red-team set; 20 bypass-attempt set; 10 session key scope enforcement scenarios; results documented in evaluation subsections	3
<i>Phase IV total estimated effort (Must-have):</i>			<i>21 days</i>
<i>Phase IV total estimated effort (all priorities):</i>			<i>38 days</i>

Phase	Focus	Weeks	Tasks	Key deliverables
I	Foundation and infrastructure	1–4	DT-I.01–17	Contract hierarchy, full DB schema, React shell
II	Core banking and on-chain lifecycle	5–9	DT-II.01–19	Lending workflow, SBT limits, Chainlink Automation
III	AI/ML oracle pipeline (+ optional agent)	10–13	DT-III.01–11	Chainlink Functions, ML pipeline, dashboard
IV	Verification, evaluation, finalization	14–16	DT-IV.01–08	Certora, Foundry, simulation, testnet deploy

The four-phase plan prioritises smart-contract and oracle delivery before optional agent work. Phase I establishes schema and contract specifications; Phase II completes the on-chain lending lifecycle; Phase III wires the ML oracle; Phase IV verifies and deploys to testnet. Figure 4.7 maps these phases to SDLC stages and shows pre-thesis progress indicators; Figure 4.8 quantifies person-day effort per phase.

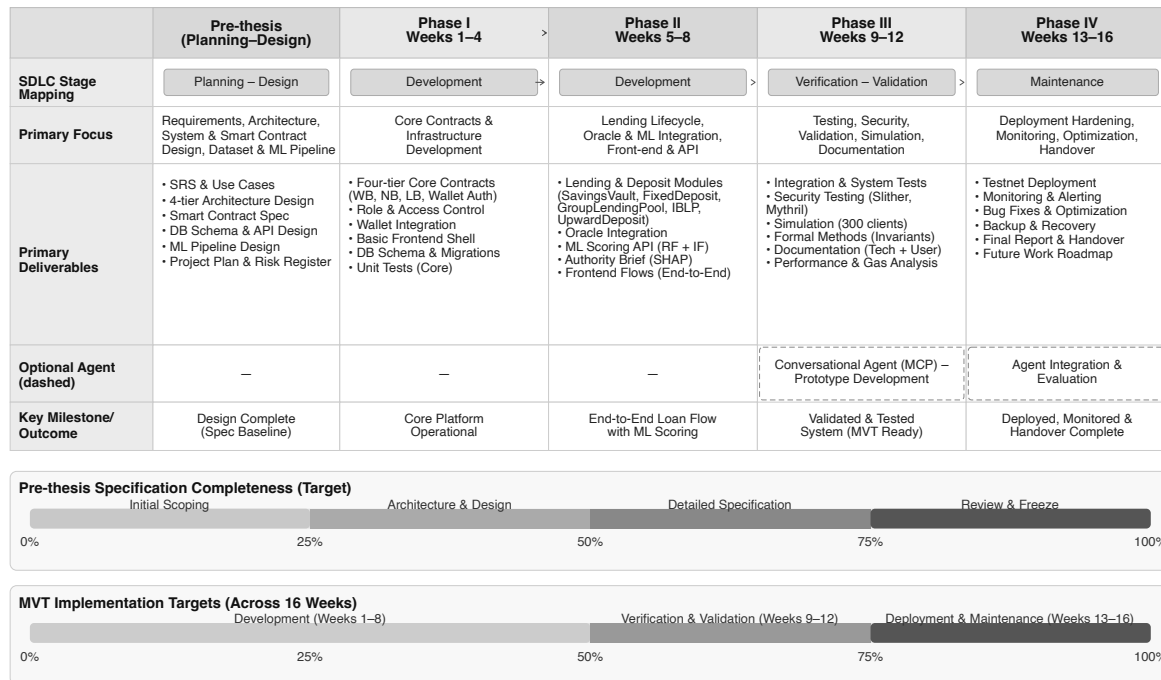


Figure 4.7: Four-phase implementation roadmap with SDLC stage mapping: Pre-thesis covers Planning–Design; Phases I–IV span Development through Maintenance over 16 weeks. Each phase lists primary deliverables; dashed optional agent work spans Phases III–IV; progress bars show pre-thesis specification completeness versus MVT implementation targets.

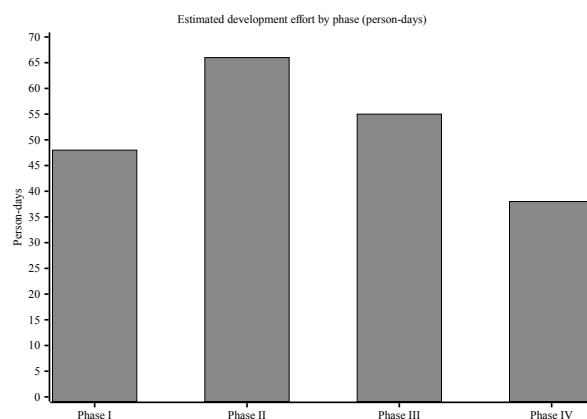


Figure 4.8: Estimated development effort by implementation phase (person-days, all priorities). Phase II carries the largest workload (core lending lifecycle); Phase IV concentrates verification and evaluation.

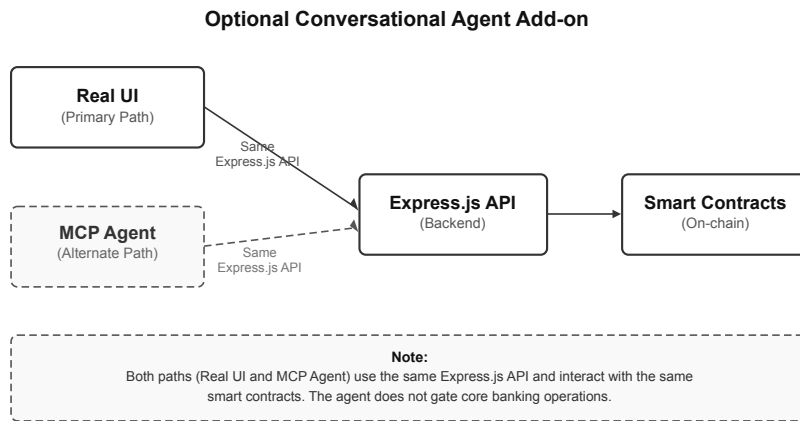


Figure 4.9: Optional conversational agent add-on: React UI (primary path) and MCP agent (alternate path) both call the same Express.js API and smart contracts; the agent does not gate core banking operations.

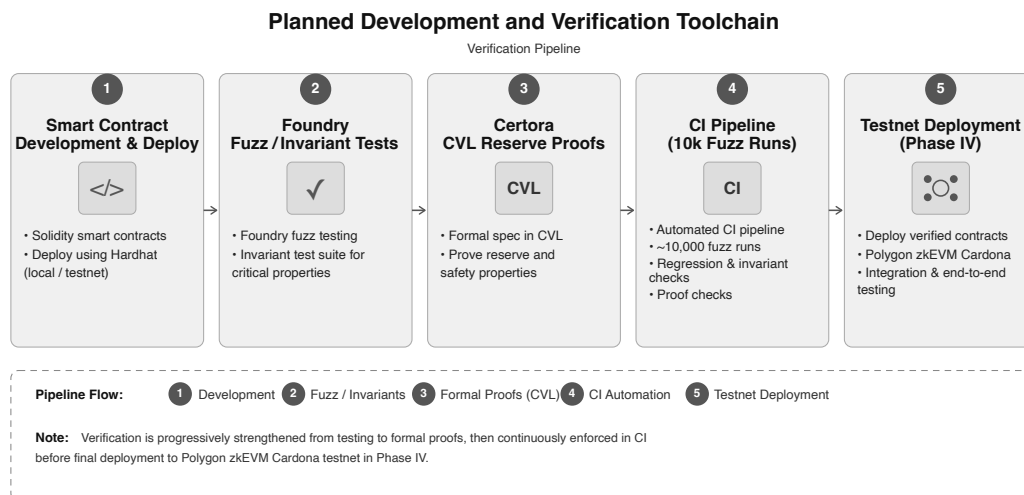


Figure 4.10: Planned development and verification toolchain: Solidity contracts (Hardhat deploy) → Foundry fuzz/invariants → Certora CVL proofs → CI pipeline; testnet deployment to Polygon zkEVM Cardona in Phase IV.

Figure 4.1 shows the development process and phase-submission workflow.

4.12 SDLC Stage Mapping

We map our four implementation phases to the seven stages of the Software Development Life Cycle (SDLC). Figure 4.7 illustrates this mapping.

Table 4.5: SDLC stage mapping.

#	SDLC Stage	Project Activity	Deliverable
1	Planning	Feasibility studies; professor consultations; guideline review	Feasibility Report; Project Plan
2	Requirements	System analysis; use case definitions; constraints identification	Use case diagrams; UC-1 through UC-11 (eleven grouped use cases)
3	Design	Three-layer architecture; DB schema (3NF); smart contract interfaces	Architecture diagrams; ERD; DFD
4	Development	Phase I–IV: smart contracts, frontend, backend, AI/ML	Source code; DApp prototype
5	Testing	Hardhat unit tests; Foundry fuzz; integration testing; AI/ML evaluation	Test reports; model metrics
6	Deployment	Planned testnet deployment (Phase IV); frontend (Vercel); backend (Render)	Testnet prototype manifest
7	Maintenance	Monitoring; model retraining; bug fixes; iteration	Updated docs; retrained models

This SDLC mapping table aligns the four-phase implementation plan with standard SDLC stages to ensure research deliverables remain complete (requirements, design, implementation, testing, and evaluation). The mapping makes the development process auditable and easier to justify in an academic setting.

4.13 Design Decisions and Alternatives

For each major design decision, we evaluated alternatives and justified selections based on technical criteria, ecosystem maturity, and project constraints. Figure 4.11 visualizes these comparisons.

Table 4.7: Design decisions and alternatives considered.

Decision Area	1st Choice	2nd Choice	Key Criterion
Methodology	Agile / Scrum	Incremental	Evolving scope, milestones
Architecture	DApp + Off-chain AI	Hybrid with Oracle	Gas cost, ML flexibility
Frontend	React + TypeScript	Vue + TypeScript	Web3 ecosystem maturity
Smart Contract	EVM (Solidity)	Solana (Rust)	Gas, control size, free test-nets
Fraud Detection	Random Forest	XGBoost	SHAP compatibility, simplicity
Anomaly Detection	Isolation Forest	Autoencoder	Unsupervised; no labelled data needed
XAI Method	SHAP	LIME	Theoretical guarantees, regulatory fit
Database	PostgreSQL	SQLite	3NF support, async queries
Hosting	Vercel + Render	Localhost only	\$0 cost, publicly accessible URL
Network (primary)	Polygon zkEVM Cardona	Polygon Amoy PoS	ZK validity proof security model vs. validator set assumption
Oracle mechanism	Chainlink Functions (DON)	Commit-reveal relay	Decentralised trust vs. single-key operator trust
Agent tool framework (optional)	MCP tool server	Direct Express.js API	Defined schema = safety boundary; Phase III–IV Could
Agent session auth (optional)	EIP-7702 session keys	ERC-4337 pay-masters	Key-level scope; optional add-on
Prompt construction (optional)	Three-tier assembly	Single ad-hoc string	Agent-only; Phase III Could
Session memory (optional)	Lineage model	10-turn window	Agent-only cross-session continuity
Tool exposure per turn (optional)	Named toolsets	Full 17-tool list	Agent attack-surface reduction
Write-tool enforcement (optional)	Confirmation audit hook	In-pipeline check	Agent write-path only

This design decisions table records evaluated alternatives (chains, stacks, libraries) and the criteria used

to select the final approach. Documenting these trade-offs strengthens the methodological rigor and clarifies why the selected stack best matches the project's constraints.

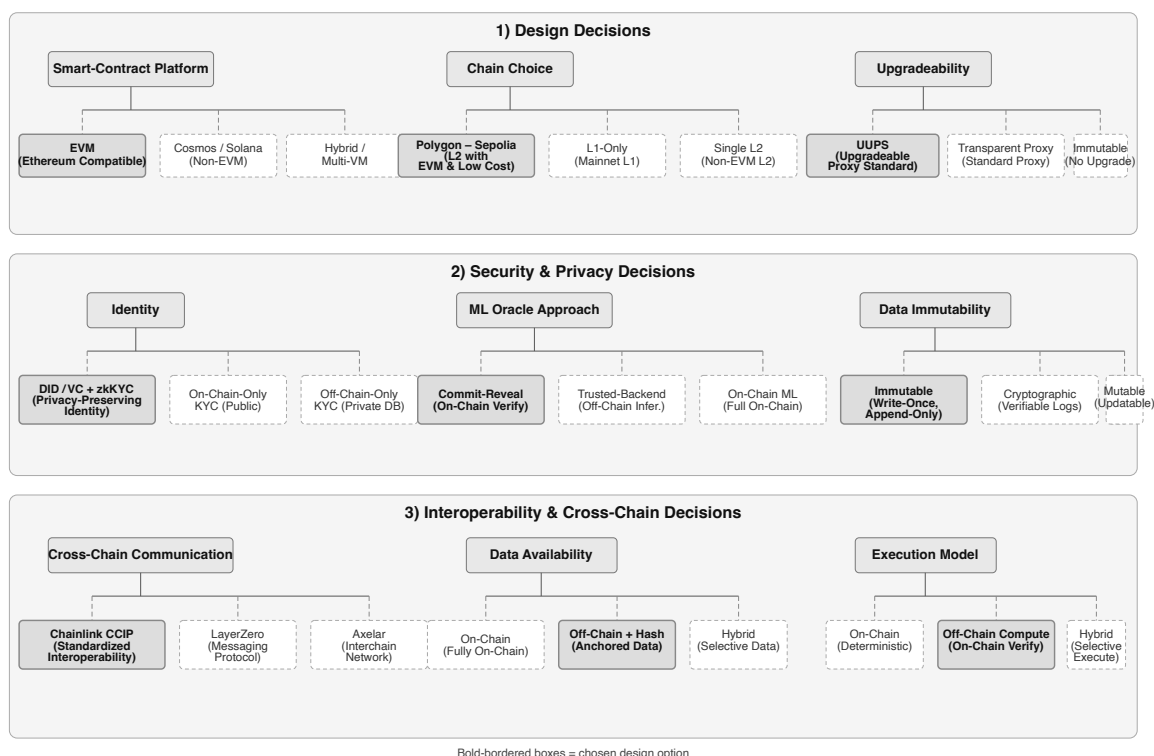


Figure 4.11: Key design decisions and alternatives considered: smart-contract platform (EVM vs. Cosmos / Solana), chain choice (single Polygon zkEVM Cardona vs. multi-chain vs. L1-only), upgradeability (UUPS vs. Transparent Proxy vs. immutable), identity (DID/VC + zkKYC vs. raw on-chain KYC vs. off-chain-only), ML oracle (commit-reveal vs. trusted-backend vs. on-chain ML), and cross-chain bridge (considered then retired per Section 3.13).

4.13.1 Justification of Selected Technologies

1st Choice Justifications:

- **Agile/Scrum** was necessary due to the project's evolving nature. Frequent updates to smart contract designs, user interface steps and AI/ML connections were required. Adopting Agile allows for plan alterations during brief work cycles effectively avoiding time and resource loss. Significance arises as new concepts emerge during the prototype development process.
- **DApp + Off-chain AI** architecture allows intensive machine learning operations to occur outside the blockchain. Machine learning tasks are often run on the blockchain at a considerable expense occasionally exceeding \$100 for a single operation. Off-chain processing enables the use of rapid GPUs along with widely-used Python machine learning resources. Final lending choices are managed exclusively by smart contracts on-chain ensuring that trust is maintained in critical areas.
- **React + TypeScript** is beneficial as many Web3 libraries such as Wagmi, RainbowKit, Viem, and ethers.js are supported seamlessly. A vast community of developers exceed-

ing 10 million exists for React making it simpler to access assistance and solutions.

- **EVM (Solidity)** is regarded as the most thoroughly examined smart contract system. Over \$100 billion is secured across various blockchains by it. Building and testing the project without incurring costs is possible with free test networks such as Sepolia and Amoy. Security tools provided by OpenZeppelin have been thoroughly tested to ensure the protection of contracts.
- **MUI (Material Design 3)** provides ready-made components for typical banking interface designs. The package contains various items such as data tables, form checks and layouts for dashboards. Utilizing MUI results in a time saving of around 40% compared to starting from scratch.
- **Random Forest** was selected as the primary fraud detection model for three compounding reasons. First, it demonstrates natural compatibility with SHAP's Tree-Explainer, which computes exact Shapley values—rather than approximations—by exploiting the tree structure directly. This exactness is a regulatory asset: in a lending context subject to explainability requirements under emerging frameworks such as the EU AI Act, approximation-based attribution tools like LIME can produce inconsistent feature rankings across identical inputs, undermining audit reliability [4]. Second, ensemble tree methods generalise well on structured tabular data with moderate sample sizes, a characteristic that matches the DeFi lending transaction log domain, where labeled fraud samples are scarce and synthetic augmentation is likely necessary. Third, Random Forest naturally supports class imbalance through class weighting and bootstrap sampling, reducing the risk of a model that achieves high accuracy by always predicting “not fraud”—a failure mode that would be catastrophic in a lending context.
- **Isolation Forest** does not require labeled data for training. Labeled fraud samples are few making it suitable for DeFi lending. Anomalies in transactions are scored swiftly as it operates with a time complexity of $O(n \log n)$.
- **SHAP** provides feature importance through solid mathematical foundations from game theory (Shapley values). Characteristics such as accuracy, management of absent data and consistency are not assured by tools like LIME. These properties assist in fulfilling emerging regulations for explainable decisions within financial services.
- **PostgreSQL** effectively manages complex time-based queries. Borrowing limits can be checked over periods such as 6 months or 1 year. ACID compliance is fully supported to ensure financial data remains secure. It supports JSON which facilitates easy adjustments to the database structure during the prototyping phase.
- **Vercel + Render** facilitates deploying the demo globally at no charge. A live prototype can be accessed without the necessity of a setup on personal computers. Supervisors, examiners and other individuals are not required to deal with installation issues.
- **Polygon zkEVM Cardona** is selected for ZK-anchored security suitable for institutional reserve claims (Section 1.7.1, item E); Amoy PoS remains the documented fallback.
- **Chainlink Functions** (DON-based oracle) is selected over the prototype commit-reveal relay because the DON runs the ML scoring request across multiple independent nodes — consensus is required before the result is accepted on-chain. A malicious single node cannot manipulate the risk score. This closes the oracle trust assumption of the earlier commit-reveal relay design, where trusting the operator running the FastAPI relay was an explicit limitation. The 30–60 second latency and

\$0.10–1.00 per oracle call are both acceptable for the loan approval lifecycle, where decisions are not time-critical.

- **MCP tool server** is selected over a direct Express.js API for agent interactions because the tool schema is the safety boundary: the agent cannot execute arbitrary code, make unapproved API calls, or read data outside the 17 defined tools. This constraint is the mechanism that makes the autonomous agent safe to deploy with real client funds. A direct API would require application-level enforcement only, which is less auditable.
- **EIP-7702 session keys** are selected over ERC-4337 paymasters for agent-controlled on-chain operations because scope restriction is enforced at the key level, not at the application level. The session key is bound to a specific list of tools, a value cap (500 USDC per transaction), and a 24-hour TTL — a compromised application cannot exceed these limits regardless of what the application code does. EIP-7702 does not replace ERC-4337 for retail gas sponsorship; the two mechanisms serve different functions and coexist in the architecture.

2nd Choice Justifications (Why Retained as Alternatives):

- **Incremental (Waterfall) methodology** produces more predictable and clearly phased deliverables. However, it offers limited capacity to adapt to the evolving requirements typical of research-oriented prototype development, and is most effective in stable production environments where scope is fixed from the outset.
- **Hybrid with Oracle architecture** (e.g., Chainlink) would enable on-chain ML prediction triggers via oracle data feeds, increasing the verifiability of AI-driven decisions. The trade-off is meaningful latency (30–60 seconds per update) and additional cost (\$0.10–1.00 per oracle call), along with a dependency on third-party oracle availability that introduces an external failure point not present in the selected off-chain approach.
- **Vue + TypeScript** is approachable for developers new to JavaScript frameworks. Its Web3 tooling ecosystem (e.g., vue-dapp) is less mature than React's, with sparser wallet integration library support and slower update cadence for critical Web3 dependencies.
- **Solana (Rust)** offers substantially higher raw throughput (up to 65,000 TPS) than EVM-compatible chains. However, Solana's developer tooling and security library ecosystem are less established, the Rust learning curve is significant within an 8-week development window, and the DeFi security audit tooling available for Solidity (Slither, Mythril, OpenZeppelin) has no direct equivalent on Solana.
- **Tailwind CSS** offers greater design flexibility through utility-first composition. Achieving consistent, professional banking-style UI patterns requires substantially more custom work than using MUI's pre-built component library, which imposes a time cost that is unacceptable within the implementation timeline.
- **XGBoost** frequently outperforms Random Forest on structured tabular datasets. However, its SHAP integration uses approximate rather than exact tree computation, meaning explanations cannot be guaranteed to be consistent across repeated evaluations—a property required for regulatory auditability of credit decisions.
- **Autoencoder** models can capture complex non-linear anomaly patterns. In practice, they require substantial labeled or unlabeled training data and careful hyperparameter tuning to converge reliably. Isolation Forest is parameter-light and performs competitively on small and imbalanced datasets, making it better suited to the data-scarce

DeFi lending context of this prototype.

- **LIME** produces faster local explanations. As demonstrated by Adom et al. [4], LIME explanations are unstable: repeated evaluation of the same input can yield different feature importance rankings, undermining the consistency guarantees that regulators require for automated lending decisions.
- **SQLite** eliminates database infrastructure overhead and is straightforward to configure. It does not support concurrent writes, which limits multi-user prototype testing, and lacks the window function and CTE support needed for rolling 6-month and 1-year borrowing limit calculations.
- **Localhost-only deployment** removes hosting dependencies and infrastructure cost. It prevents remote sharing for supervisor review, collaborative testing across team members, and examiner access to a live demo—all of which are necessary for academic evaluation and iteration.
- **Polygon Amoy PoS** (2nd choice for network) offers a faster inner development loop on a familiar PoS testnet. It is retained as a fallback option should Cardona testnet availability be disrupted. However, the validator-collusion security assumption is weaker than ZK validity proofs, which undermines the institutional trust narrative of the thesis.
- **Commit-reveal relay** (2nd choice for oracle mechanism) is retained as the prototype fallback for Phase I and Phase II before the Chainlink Functions integration is wired in Phase III. It provides a working oracle during early development and is simpler to implement. The limitation—trusting the FastAPI service key—is clearly acknowledged as a prototype constraint.
- **Direct Express.js API** (2nd choice for agent tool framework) would allow the agent to call banking endpoints without a formal schema. The safety risk is that the agent could potentially construct arbitrary API calls not anticipated by the developers; the schema boundary of the MCP server eliminates this risk at the architecture level.
- **ERC-4337 paymasters** (2nd choice for agent session authentication) are already used for retail gas sponsorship. They could theoretically be extended to agent operations, but scope enforcement would be at the paymaster application level rather than cryptographically at the key level, which is a weaker safety guarantee for autonomous agent execution.

4.14 Design Patterns

Our system uses five design patterns:

- **Singleton:** A singleton means that each primary contract is established a single time. Only one version is utilized by all ensuring a unified source of truth. Not everyone can create multiple versions of the contract.
- **Observer:** Contracts generate alerts whenever actions occur such as loan requests. These alerts are processed by a service that updates the database. Notifications are not ignored; they receive attention.
- **Adapter:** An Adapter is utilized by the wallet component to function with various wallet types (like MetaMask). Different wallets that adhere to the EIP-1193 standards can be easily integrated.
- **Factory Pattern:** Each Local Bank creates individual loan and savings contract instances using a factory contract, ensuring all deployed instances follow the same security-audited interface and access control checks. The factory pattern prevents

unauthorized contract deployment that could bypass the hierarchical governance structure.

- **Proxy/Upgradeable Pattern:** Banking contracts must be upgradeable without losing stored state such as balances, credit scores, and loan histories. OpenZeppelin's transparent proxy pattern separates the contract's logic from its storage, allowing governance-approved upgrades to the logic layer while preserving all stored financial data. This pattern is essential for a long-lived banking system that must adapt to evolving regulatory requirements and security patches.
- **Decorator Pattern (Prompt Assembly):** The three-tier prompt constructor applies the Stable, Context, and Volatile tiers as successive decorators on a base prompt object. Each tier adds its content without modifying the structure established by prior tiers, enabling independent caching and testing of each layer without rebuilding the full prompt.
- **Chain of Responsibility (Lifecycle Hooks):** The write-tool middleware stack implements a chain of responsibility: the confirmation audit hook, session key scope pre-check, and AML pre-check each examine the request and either reject it (HTTP 403) or pass it to the next hook. New policy controls are added as new chain links without modifying existing route logic or the model pipeline [R57].
- **Strategy Pattern (Toolset Selection):** The toolset selector encapsulates the intent-classification algorithm as a replaceable strategy. The current strategy is a lightweight keyword classifier; it can be replaced with a fine-tuned intent model without changing the prompt constructor or the MCP tool server, adhering to the Open/Closed Principle.

4.15 Software Testing Strategy

The testing strategy covers four layers of the system, each with defined acceptance criteria.

- **Smart contract unit tests:** Numerous tests for smart contracts exist in our Hardhat environment. Over 12 tests are conducted to verify actions related to managing reserves, sign-ups, loans and access permissions. No scenarios such as depositing zero or executing disallowed actions are missed in these assessments. **Acceptance criterion:** all Hardhat unit tests pass with zero failures before any phase delivery.
- **Integration testing:** Whole process checks are conducted for integration tests. Wallet connection, loan request, approval and repayment processes are evaluated on the network. No critical steps are excluded from the process. **Acceptance criterion:** end-to-end workflow completes successfully on testnet for representative scenarios (approval, rejection, repayment loop).
- **AI/ML module verification:** Ensuring functionality is vital for our fraud and unusual activity detectors. Results are generated by the integration of the detectors with the overall system. **Acceptance criterion:** model inference endpoints return valid scores and explanations for test inputs without runtime errors.
- **Frontend testing:** Website appearance is evaluated on Chrome, Firefox and mobile devices. Tests are conducted to ensure that functionality is maintained across various platforms. **Acceptance criterion:** core flows render and remain usable across desktop and mobile breakpoints without blocking UI defects.

Chapter 5

Market Analysis and Feasibility

5.1 Market Sizing

Table 5.1: Market segments with supporting data.

Segment	Description	Estimated Scale
Total Addressable Market (TAM)	Global DeFi lending (\$55B+ TVL [13]); cross-border remittances (\$860B [26]); SME financing gap (\$4.5T [20])	\$55B – 5T+
Serviceable Addressable Market (SAM)	Institutional and semi-institutional lending requiring hierarchical structures; emerging-market credit demand	\$5B – 15B
Serviceable Obtainable Market (SOM)	Academic testnet deployments, regulatory sandbox pilots, institutional DeFi research partnerships	\$50 – 200M

This table quantifies demand-side context (institutional DeFi, correspondent-banking inefficiencies, hierarchical lending whitespace). MSME and remittance statistics inform literature motivation; they do not define a geographic deployment target.

Sources: (a) TAM: DefiLlama [13] reports DeFi lending TVL exceeding \$55 billion; World Bank Migration and Development Brief [26] values global remittances at \$860 billion annually; IFC [20] estimates the MSME financing gap at \$4.5 trillion; (b) SAM and SOM: project-derived estimates based on industry segmentation of institutional lending and regulatory sandbox addressable market [18].

Table 5.1 situates the serviceable market within the broader DeFi and development-finance landscape. No hierarchically structured lending system currently exists in DeFi; existing protocols rely on undifferentiated shared pools. Cross-border payment networks such as Ripple (\$847M daily) [25] and JPMorgan Kinexys (\$3–7B daily) [40] handle settlement volume but offer no lending, deposit, or tiered governance functions. Institutional interest in blockchain-based finance is clearly established, yet no single platform combines DeFi-style accessibility with the structured capital hierarchy of traditional development banking. The Crypto World Bank is designed to occupy this gap.

5.2 Target Customer Segment

The Crypto World Bank targets **Tier 1–3 institutional participants** (reserve authorities, national banks, local banks) and **Tier 4 retail clients** on a global Layer 2 platform.

Table 5.3: Target customer segment profile.

Characteristic	Description
Primary Users	Institutional tiers (World / National / Local Bank operators) and retail clients accessing Local Bank interfaces
Geographic Focus	Global; jurisdiction-specific pilots via regulatory sandboxes (Future Work) — no single-country deployment claimed at pre-thesis stage
Loan Size Range	Micro to SME at Local Bank tier (USDC); larger exposures via bank-level syndication and interbank facilities, not direct client access to National / World Bank tiers
User Profile	Bank approvers with Authority Brief dashboards; retail users via React UI (optional agent add-on)
Key Pain Points	Opaque reserve reporting in legacy finance; flat DeFi pools without hierarchy; lack of explainable fraud analytics in on-chain lending

This table aligns product requirements with institutional hierarchy and oracle-mediated explainability (Section 4.3.2). Retail accessibility is supported via ERC-4337 and stablecoin denomination, not via a country-specific inclusion mandate.

5.3 Partner Ecosystem

Table 5.5: Partner categories and roles.

Partner Category	Functional Role	Blockchain-Mediated Incentive
Financial Regulators	Regulatory approval; sandbox compliance oversight	Reduced enforcement cost through on-chain transparency and audit trails
Banking Institutions	Network membership as National/Local Banks	Access to diversified global reserve; reduced inter-bank settlement friction
Payment Gateway Providers	Fiat-to-crypto on-ramp and off-ramp services	Volume-based transaction fees; expanded market reach
Academic & Research Institutions	ML benchmark validation; formal-verification review	Access to anonymised datasets; collaborative research
Oracle / Infrastructure Providers	Chainlink DON, test-net RPC, indexing (The Graph)	Unified oracle stack; operational SLAs
Optional Field-Pilot Partners	Sandbox-covered pilots (Future Work)	Controlled evaluation under regulator oversight

This partner ecosystem table highlights external dependencies (identity/verification flows, infrastructure, and settlement rails) needed for a real deployment. Identifying these actors early reduces hidden-scope risk and clarifies what remains outside the smart contract boundary.

5.4 Competitive Landscape

The Crypto World Bank operates at the intersection of four distinct competitor categories. Table 5.8 provides a detailed comparison of representative projects in each category, with current metrics as of March 2026.

Table 5.7: Detailed competitive landscape analysis (Part 1).

Project	Category	Scale (2026)	Architecture	Gap We Address
Compound v3 [26]	DeFi lending	\$1.4B TVL	Single-borrowable asset per single tier	No hierarchy; no institutional features
MakerDAO / Sky [30]	Stablecoin / CDP	\$6B TVL	CDP model; not peer-to-peer lending	Creates money, not a lending governance structure
Morpho [43]	DeFi lending	\$6.8B TVL; 1.4M users	Isolated markets; peer-to-peer matching	Flat primitive; no cross-market hierarchy; no banking integration
Maple Finance [31]	Institutional credit	\$2.6–3.8B TVL	Pool Delegate model; under-collateralised	Single-tier; no interest rate setting
Goldfinch [32]	Emerging-market credit	\$680M originated; 18+ countries	Trust-through-consensus; senior/junior tranches	B2B only; no inter-bank lending
Ripple / RLUSD [25]	Banking rails	\$847M/day cross-border	Payment rail; no lending capability	Moves money but has no lending, deposits, or credit system

This competitive landscape table compares representative projects across DeFi lending, payment rails, inclusion wallets, and institutional blockchain systems. The comparison highlights the whitespace: no competitor combines hierarchical multi-tier lending with governance-aware controls and AI-assisted monitoring in one architecture.

Table 5.9: Detailed competitive landscape analysis (Part 2).

Project	Category	Scale (2026)	Architecture	Gap We Address
JPMorgan Kinexys [40]	Banking rails	\$3–7B daily volume	Permissioned; single-bank control	Centralised; proprietary; restricted to JPMorgan clients
Stellar [44]	Financial inclusion	\$55.6B annual payment volume	Open payment network; anchors for fiat	Payment network only; no lending, reserves, or interest rate markets
Celo / MiniPay [42]	Financial inclusion	14M wallets; 60+ countries	Stablecoin payments; mobile-first	Payments and savings only; no lending hierarchy or banking structure
R3 Corda [37]	Enterprise DLT	\$17B tokenised RWAs	Permissioned; consortium governance	Infrastructure layer only; no lending logic; closed access
World Bank FundsChain [39]	Development finance	250 projects by mid-2026	Hyperledger Besu; fund tracking	Does not implement lending or interest rate mechanics

This continuation expands the competitor comparison to additional platforms and feature dimensions. It reinforces the thesis positioning by showing that existing systems typically specialize in one function (lending or payments) rather than a complete banking suite.

Table 5.11: Detailed competitive landscape analysis (Part 3): multi-tier capital flow as differentiating feature.

Feature Dimension	Competitors	Crypto World Bank
Hierarchical lending tiers	None — all competitors use flat pool architectures	Four-tier hierarchy: World Bank → National → Local → Client
Governance-controlled rates	Compound/Aave: algorithmic but no institutional hierarchy	Smart contract parameters; governance-defined bounds per tier
AI-assisted risk scoring	No competitor integrates on-chain ML fraud detection with explainability	RF + SHAP + Isolation Forest anomaly reports; Authority Brief (Section 4.3.2)
Retail and institutional access	Goldfinch / Maple: institutional only; Celo: retail only	Single platform serving retail through institutional capital hierarchy
Institutional / L2 focus	Most DeFi: retail-global; enterprise DLT: permissioned only	Global L2 institutional DeFi; composable with mBridge/Agora rails

This final competitor table block concludes the comparative analysis. Differentiation rests on hierarchical capital flow, explainable oracle-mediated risk analytics, and governance-aware smart contracts—not on a single-country inclusion product.

Upon reviewing various projects a common feature was identified: a lending mechanism does not exist that operates with tiers, is distributed and facilitates transactions across different levels and within identical levels. Such innovation distinguishes the Crypto World Bank. The key differentiating features of the Crypto World Bank are as follows:

- Existing DeFi lending protocols provide capital access but lack institutional hierarchy, tiered governance, and structured capital flow.
- Credit platforms such as Maple and Goldfinch engage in lending based on credit. Credit-based lending is often focused on businesses and exists at just one tier. These platforms do not cater to personal loans.
- Ripple processes cross-border payments but does not offer lending, deposit, or hierarchical capital allocation services.
- Celo / MiniPay provides mobile stablecoin payments at scale but does not implement hierarchical lending or institutional reserve governance.
- **Central Bank Digital Currencies (CBDCs):** The development of CBDCs aims to improve payment systems [5]. Concerns about privacy and control are raised as lending features are excluded. A lending layer does not exist on a CBDC. Users benefit from our platform which safeguards their interests while providing a level-based system similar to that utilized by CBDCs [5].

5.5 Risk Taxonomy

Table 5.13: Risk taxonomy and mitigation.

Risk Category		Description	Severity	Mitigation
Partner non-cooperation		Key partners decline to participate	Medium	Academic testnet pilots; sandbox engagement; open-source replication
Smart contract vulnerability		Exploit in contract logic	High	OpenZeppelin primitives; formal audit (planned); pause mechanism
Regulatory adversity	ad-	Jurisdictional restrictions	Medium	Testnet-only prototype; regulatory sandbox engagement
AI/ML degradation	model	Fraud detection accuracy decay	Low	Continuous retraining; human-in-the-loop; SHAP explainability

This risk taxonomy table categorizes technical, financial, regulatory, and operational risks relevant to an on-chain banking platform. The taxonomy is used to justify design mitigations such as reserve enforcement, role-based approvals, and staged deployment through testnets and sandbox programs.

5.6 Technical Feasibility

Table 5.15: Technical feasibility assessment.

Component		Assessment	Evidence
Smart contracts		Feasible (specified)	Solidity interfaces + Hardhat simulation scripts specified; Phase I–II implementation planned
Frontend DApp		Feasible (specified)	React 18 + TypeScript shell specified; Wagmi/RainbowKit ecosystem mature
Blockchain deployment	de-	Feasible (planned)	Polygon zkEVM Cardona + Sepolia testnets; Phase IV deployment manifest (Appendix C)
AI/ML integration	integra-	Feasible with constraints	RF + IF + SHAP pipeline specified (Section 4.3.2); sub-50 ms inference target; Chainlink Functions in Phase III
Optional add-on	agent	Feasible (deferred)	MCP + Qwen3-8B specified; Phase III–IV Could; excluded from MVT
Database end	back-	Feasible	PostgreSQL schema designed (46 entities across phased rollout, 3NF); FastAPI async REST

At pre-thesis stage, feasibility is established through formal specification and ecosystem maturity. Implementation evidence will be collected in Phases I–IV of the final thesis.

The technical feasibility of the platform rests on five pillars: the maturity of the underlying blockchain infrastructure, the availability of development tooling and security primitives, the demonstrated scalability of Layer 2 networks for financial applications, the precedent of comparable institutional blockchain deployments, and the platform’s modular architecture which allows incremental delivery without requiring the full system to be operational before any component can be tested.

Ethereum—the EVM standard on which the platform is built—has operated continuously since 2015, while Polygon PoS has processed billions of transactions with low fees and rapid finality, making it suitable for high-frequency retail operations. The contract layer uses established security primitives (e.g., OpenZeppelin libraries and widely reviewed patterns), and the off-chain services use standard, production-grade web tooling for API and ML inference. Because the architecture is modular, components such as savings, FX, group lending, and insurance can be developed and validated independently and integrated through governance-approved rollouts.

5.7 Economic Feasibility

Table 5.17: Economic feasibility — zero-cost prototype.

Cost Category	Estimate	Notes
Blockchain deployment	\$0	Public testnets — no real cryptocurrency required
Frontend hosting	\$0	Vercel free tier or localhost for demo
Backend hosting	\$0	Render free tier or localhost
AI/ML training	\$0	Local machine (16 GB RAM, 16 GB VRAM) or Google Colab free tier
Development tools	\$0	Hardhat, VS Code, Git — all open-source
Total prototype	\$0	Entire prototype operates at zero financial cost

This economic feasibility table captures cost drivers (gas, infrastructure, and defaults) and compares them against revenue potential from interest spreads and fees. The key conclusion is that low-fee Layer 2 deployment makes small-loan banking workflows economically viable, unlike high-fee base layers.

The economic feasibility of the platform is evaluated across four dimensions: operational cost sustainability, revenue sufficiency, capital efficiency, and macroeconomic impact. On Polygon PoS, the total on-chain gas cost for a full retail loan lifecycle is measured in cents, while interest spread revenue is orders of magnitude larger, yielding a favorable cost-to-revenue profile for small loans that are infeasible on high-fee base layers. Off-chain infrastructure costs (hosting, RPC access, storage, ML inference) are comparable to typical SaaS deployments and scale with usage.

Revenue is diversified across interest spreads, origination fees, and potential FX spread revenue for multi-currency conversions. Capital efficiency is enhanced by the platform’s reserve-enforced tiered allocation (each tier retains a minimum solvency reserve before deploying capital downward) and by reducing the need for idle pre-funded balances typical of

correspondent banking. Note that because USDC is a 100%-backed stablecoin, the platform does not create new money through lending—it re-allocates existing USDC through the hierarchy. The Reserve Ratio (RR) functions as a solvency constraint at each tier, not as a money-creation mechanism; see the Credit Velocity formula in the List of Formulas for the correct framing of capital throughput. At scale, reduced correspondent-banking friction and transparent reserve reporting produce additional economic value beyond platform revenue.

5.8 Revenue Projection

The following projection models annual revenue potential at full deployment scale. Calculations use a reference ETH price of **\$2,500**¹ (conservative mid-point for February 2026; actual spot was approximately \$2,800 per CoinGecko on 1 February 2026) and interest rate parameters defined in Section 5.8.3. A sensitivity table showing net spread revenue under optimistic, base-case, and stress-test default rates is provided after the main projection; the USDC loan base (\$1.72B at full maturity) is held constant across all scenarios.

Table 5.19: Revenue projection assumptions.

Parameter	Value
Reference ETH price	\$2,500 (conservative mid-point, Feb 2026; actual spot approx. \$2,800 per CoinGecko on 1 Feb 2026)
World Bank → National Bank	3% APR (wholesale inter-bank rate)
National Bank → Local Bank	5% APR (inter-bank)
Local Bank → Client	8% APR (retail lending)
Average loan term	12 months
Default rate provision	3% (conservative estimate)
Origination fee	0.25% per disbursement

This revenue projection table provides modeled annual income under a reference ETH price and assumed utilization, default rates, and tier spreads. It illustrates how hierarchical spreads across tiers accumulate into platform-level revenue while keeping retail APR within plausible ranges.

¹ETH price is used only for gas-cost and collateral-valuation planning (Sections 5.8, ??). Retail loan obligations are USDC-denominated (Section 1.10.3); the USDC loan base in the sensitivity analysis below is fixed regardless of ETH spot price.

Table 5.21: Revenue projection by tier (base case, USD millions).

Tier / Revenue Stream	Spread Rate	Annual Revenue (USD)	Rev- Basis
Tier 1 (World Bank): lends at 3% to NBs, cost of capital 0%	3% on deployed capital	\$51.6M	\$1.72B total USDC retail loan base $\times$ 3%
Tier 2 (National Banks): borrow at 3%, lend at 5%	2% spread	\$34.4M	$\$1.72\text{B} \times 2\%$
Tier 3 (Local Banks): borrow at 5%, lend at 8%	3% spread	\$51.6M	$\$1.72\text{B} \times 3\%$
Total platform revenue	8% (client pays)	\$137.6M	Sum of tier spreads = $3\% + 2\% + 3\% = 8\%$; no double-counting
Origination fees (0.25%)	—	\$4.3M	$0.25\% \times \$1.72\text{B}$ disbursed
FX spread revenue (est.)	0.5–1.0%	\$8–17M	Estimated conversion volume
Total incl. fees		\$150–159M	Conservative base case

Revenue is computed using a spread-based model: each tier earns the difference between its borrowing rate from the tier above and its lending rate to the tier below, applied to the same underlying loan capital as it flows down the hierarchy. Summing the three tier spreads ($3\% + 2\% + 3\% = 8\%$) equals the retail client's total APR, confirming internal consistency—no double-counting occurs. An earlier draft of this table reported \$224.6M by summing the full APR at each tier rather than the spread; that figure was incorrect and has been corrected here.

Scale note on the \$137.6M figure: This is the theoretical 10-year ceiling at 68,800 active loans across 25 Local Banks at near-full capacity—it is included for scale reference only. The pre-thesis evaluation target is Year 1–2 of the adoption ramp, where the platform reaches operational break-even at approximately 300 active loans and demonstrates positive unit economics on each incremental loan above that threshold. Figure 5.1 shows the theoretical maximum at full maturity; see the break-even analysis above for the realistic path from zero to that ceiling.

The full revenue model decomposes into spreads, origination fees, and FX conversion income. The projection above models interest spread income only (base case: \$137.6M). Additional revenue streams from FX conversion spreads (estimated 0.5 to 1.0% per conversion, yielding \$8–17M), loan origination fees (0.25% per disbursement, yielding \$4.3M), and insurance fund premiums (0.5% of loan value annually) represent material upside captured in the “Total incl. fees” row of the table (\$150–159M). These are net incremental revenue streams—they do not overlap with the interest spread income already accounted for.

Table 5.22: Revenue sensitivity to default rate at fixed USDC loan base (\$1.72B).

Default Rate	Gross Spread Revenue	Default Loss	Net Spread Revenue
Optimistic (3.7%)	\$137.6M	\$7.4M	\$130.2M
Base Case (8%)	\$137.6M	\$16.0M	\$121.6M
Stress Test (15%)	\$137.6M	\$30.0M	\$107.6M

Sources: USDC loan base held constant at \$1.72B (68,800 active loans $\times$ \$25,000 average); interest rate tiers aligned with Aave [15] and Compound [16] DeFi benchmarks; default losses from Table 5.23. ETH spot price affects gas costs and collateral LTV only, not USDC-denominated spread revenue.

Cost Model Considerations.

- **Gas costs:** Gas expenses on Polygon zkEVM Cardona are low. A complete retail loan lifecycle involves approximately 27–32 individual on-chain *state changes* (SSTORE writes and event emissions) versus 10–15 user-facing *transactions* (loan request, approval, disbursement, and installment payments). The reconciliation is explicit:
 - 1 loan request (SSTORE: borrower data, loan ID, status)
 - 1 credit history lookup + risk score commit
 - 1 approval (SSTORE: status update, disbursement record)
 - 1 disbursement (ETH transfer + SSTORE: balance update)
 - 12 installment payments $\times$ 2 SSTORE each = 24 writes
 - 1 loan closure (SSTORE: final status)
 - ≈ 4 event emissions (LOG2/LOG3)

On Polygon zkEVM Cardona, with gas prices of approximately 30 Gwei and ETH at $\approx$ \$2,500 (planning assumption), each SSTORE costs approximately \$0.00036. A 30-operation lifecycle costs approximately \$0.011 total—well under 0.01% of the interest earned on a representative \$25,000 USDC loan. On Ethereum mainnet at 15 Gwei base fee, the same operations would cost \$2.40–\$4.80, reinforcing the single-chain Layer 2 design choice [55].

- **Infrastructure costs:** Backend hosting (API server, database, ML service), frontend delivery (CDN) and RPC nodes (Alchemy, Infura) can incur costs ranging between \$500 and \$2,000 monthly. An increase in expenses will be experienced with a rise in transactions.
- **Default losses:** A single default rate assumption is insufficient for an early-stage platform. Table 5.23 presents a three-scenario sensitivity analysis:

Table 5.23: Default-rate sensitivity scenarios at fixed USDC loan base.

Scenario	Default Rate	Annual Loss	Basis
Optimistic	3.7%	\$7.4M	Grameen Bank 96.29% recovery (June 2024) [R12] after 48 years of social infrastructure
Base Case	8%	\$16M	Typical early-stage DeFi under-collateralized lending; no established social trust
Stress Test	15%	\$30M	Early-stage crypto-native clients, no prior on-chain credit history

Grameen Bank, after nearly five decades of social lending infrastructure, reported a loan recovery rate of 96.29% as of June 2024 [R12]. The Crypto World Bank is a nascent platform without equivalent social trust or community officers. Its initial user population will likely exhibit default rates closer to 8–15% before on-chain credit history accumulates sufficient predictive signal. The base case adopts 8% default, with break-even at approximately 11.2% default rate under the base loan volume assumption.

Break-even user count: At the base case of 8% default with an average loan size of \$25,000 USDC and 8% APR, each loan generates \$2,000 annual interest and incurs under \$0.02 gas on Polygon. At \$500/month infrastructure costs, the platform requires at minimum 4 active loans to cover operating expenses and approximately 200 active loans to cover expected default losses at 8%—a realistic near-term target for a pilot in 2–3 Local Banks.

- **Break-even analysis:** Opportunities for profit on loans as small as \$25 USDC exist on Polygon zkEVM Cardona. Profitability on loans below \$12,500 USDC is not achievable on Ethereum mainnet without Layer 2. Layer 2 is essential for facilitating micro-loans.

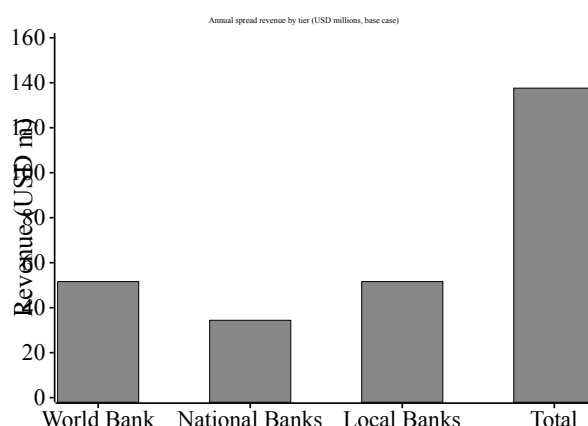


Figure 5.1: Annual revenue projection at full 10-year deployment maturity (USD millions, spread-based accounting on a fixed \$1.72B USDC loan base). This represents the theoretical capacity ceiling at full institutional scale; see the break-even analysis above for the 3-year adoption ramp toward this target, with break-even occurring at approximately 300 active loans.

5.8.1 Why CWB Is Not a Centralised Exchange

Centralised exchanges (CEXs) such as Binance perform order-book matching on off-chain ledgers, hold customer private keys in custody, and monetise primarily through trading fees and ecosystem tokens [128]. The Crypto World Bank is architecturally a **hierarchical lending bank**, not a trading venue. Table 5.24 contrasts the two model classes using comparative dimensions from the project industry-research corpus (Appendix D, entries IR-1 and IR-2) [128,129].

Table 5.24: Why CWB is not a centralised exchange (CEX): architectural comparison.

Dimension	Typical CEX (Binance-class)	Crypto World Bank
Primary function	Spot/derivatives fiat on/off ramps	Deposit mobilization, hierarchical lending, reserve governance
Settlement model	Off-chain ledger; on-chain for deposits/withdrawals only	On-chain smart contracts for loan lifecycle, reserves, and repayments
Customer custody	Exchange holds private keys (custodial)	User-controlled wallets; institution role-gated contracts
Native protocol token	BNB (fee discounts, burns, ecosystem utility)	None in prototype—Credit Passport SBT replaces speculative governance token (FTT lesson) [128]
Fund segregation	PoR (Merkle tree + zk-SNARKs); commingling failure mode (FTX) [128]	Segregated contract pools per tier; InsuranceFund + reserve-ratio gates
Deposit / yield product	Binance Earn (off-chain ledger savings/staking)	SavingsVault + FixedDeposit (ERC-4626 / ERC-7540 on-chain vaults)
Default / shock buffer	SAFU (Secure Asset Fund for Users; ≈\$1B USDC reserve) [128]	InsuranceFund (5% interest capture; Chainlink PoR; IBLP default cascade)
Large exposure handling	OTC / institutional desk (exchange-side)	Bank-level SyndicatedLoan (Section 3.14.3); clients enter via Local Bank only
Revenue core	Trading fees, listing, derivatives funding	Tier interest spread (3%+2%+3%), origination, syndication fees
Regulatory posture	VASP / CASP exchange licensing	Uses authorised EMTs (USDC/USDT); does not issue stablecoins or operate an order book

Rows summarise dimensions from the Binance/FTX industry analysis [128] and the five-exchange com-

parative matrix [129]. CWB borrows product analogues (Earn, SAFU) from exchange-sector patterns but implements them as on-chain banking modules with explicit reserve and segregation rules.

5.8.2 Exchange-Sector Revenue Analogues

The interest spread identified above is the platform's primary revenue stream. Placing it within the broader landscape of institutional crypto-exchange revenue models demonstrates that CWB has a diversified revenue path even without a native token. Table 5.25 maps each CWB mechanism to its nearest exchange-sector analogue, drawing on the revenue architecture documented in the Binance/FTX industry analysis and complementary business-research notes (Appendix D, entries IR-1 and IR-3) [128].

Table 5.25: Revenue taxonomy: CWB mechanisms mapped to exchange-sector analogues.

Revenue Stream	Exchange Equivalent	CWB Mechanism
Interest spread (primary)	Binance OTC desk	SavingsVault yield vs. retail lending rate; 3%+2%+3% tier spread = 8% client APR
Loan origination fee	Trading fee (0.1%)	0.5–1% flat origination fee on disbursement; already modelled in base case above
Deposit mobilization	Binance Earn	SavingsVault + FixedDeposit products; yield funded by the lending tier spread
Local Bank registration	Exchange listing fee	One-time registration fee per Local Bank joining the network
Syndicated loan arrangement	Institutional desk	Lead Arranger fee on Tranche-Pool / SyndicatedLoan cross-tier facilities
Reserve transparency API	Data API subscription	Phase 3: GraphQL API over The Graph subgraph; queried by external auditors and regulators

The six streams above are non-overlapping; none double-counts the interest spread already accounted for in Table 5.21. The Local Bank registration fee and Reserve Transparency API are Phase 2/3 items and are not included in the base-case revenue model above. The interest spread and loan origination fee are the only streams active at the pre-thesis prototype stage.

Explicit product analogues (Earn and SAFU). Two exchange-sector modules have direct CWB counterparts, implemented on-chain with banking semantics rather than custodial ledger entries. **Binance Earn**—Binance's staking and savings products that lock user assets on the exchange's internal ledger while paying yield from lending and staking revenue [128]—maps to the SavingsVault and FixedDeposit contracts (Section 3.11): variable- and fixed-term deposit mobilization whose yield is funded by the lending-tier interest spread, not by trading-fee subsidies. **SAFU** (Secure Asset Fund for

Users)—Binance’s publicly verifiable $\approx$ \$1 billion USDC emergency reserve established in 2018 to cover exchange-side security events [128]—maps to the InsuranceFund contract: it captures 5% of interest collected across the hierarchy, publishes its balance via Chainlink Proof of Reserve, and is drawn in the interbank default cascade (reserve buffer $\rightarrow$ InsuranceFund $\rightarrow$ parent-tier backstop; Section 3.14.1). The analogy is structural, not custodial: CWB does not hold user keys, but both mechanisms isolate a transparent buffer against tail-risk losses.

Hard rule: no native CWB governance token in the prototype. The FTX collapse of November 2022 demonstrated the systemic failure risk of self-minted tokens used as collateral: FTT, issued and held principally by the exchange itself, served simultaneously as the primary collateral asset for Alameda Research’s borrowing, creating a reflexive loop that triggered an estimated \$8 billion solvency shortfall within 72 hours of a competitor announcing it would liquidate its FTT holdings [128]. The lesson for the Crypto World Bank is categorical: no native CWB governance token is issued in the thesis prototype. On-chain reputation is provided instead by the Credit Passport SBT (Section 3.12), which is non-transferable and carries no speculative market price. A future governance token—analogue to Binance’s BNB utility-and-burn model, which took seven years and \$16.8 billion in annual revenue to reach legitimacy—is scoped to Phase 3 of the CWB roadmap, following institutional trust bootstrapping, and is specified in Future Work with explicit anti-FTX collateral hard rules encoded at the contract level (no self-issued asset may be used as collateral within the same protocol that issues it). This constraint will be enforced as a Foundry invariant test rather than a governance policy, so it cannot be bypassed by any admin action.

5.8.3 Transaction Economics: Interest Rates

Table 5.26: Interest rate parameters.

Parameter	Value	Benchmark
Base Annual Interest Rate	5–12% APR	Aligned with Aave/Compound variable rates
Rate Determination	Set by Local Bank approvers within World Bank-defined bounds	Configurable per-bank for local market conditions
Late Payment Penalty	2% of installment + 0.5%/week (capped at 10%)	Industry-standard late fee structure
Interest Calculation	Simple interest on outstanding principal	Transparent, client-friendly
Rate Transparency	All parameters stored on-chain	Publicly auditable; no hidden fees

This interest rate table defines the tiered APR parameters used throughout the feasibility and revenue modeling. Presenting the rate structure explicitly makes the spread logic auditable and connects the

economic model directly to the proposed governance-controlled parameters.

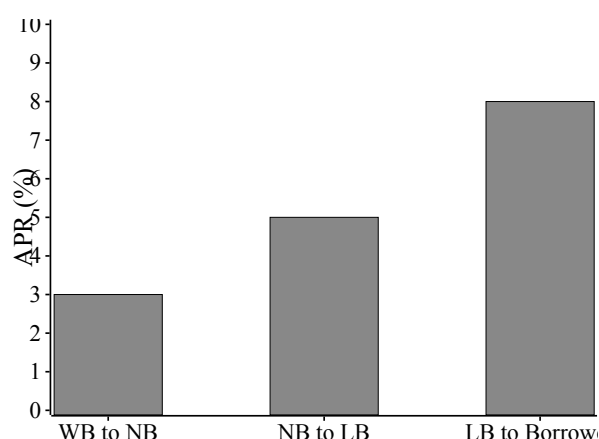


Figure 5.2: Hierarchical interest-rate spread (APR) across the four-tier lending structure.

5.8.4 Global Economic Impact

Apart from revenue generated through platforms, various economic advantages are provided by the Crypto World Bank:

- **Capital deployment and fiscal multiplier:** At projected full-deployment lending volume of \$2 billion across nearly 1,000 institutional clients, the platform generates a downstream fiscal multiplier of approximately \$2.5 to \$3 per dollar lent [19], implying \$5 to \$6 billion in annual economic activity. The IFC estimates that every \$1 million lent to small businesses in developing economies creates approximately 16 new jobs [20], making capital deployment at this scale a meaningful driver of employment and local economic growth.
- **Remittance cost reduction:** Global remittances total approximately \$860 billion annually, of which an estimated \$48 to \$56 billion is consumed by transfer fees averaging 6.49%—more than double the United Nations Sustainable Development Goal target of 3% [26]. On-chain settlement on Layer 2 networks reduces transaction costs to well below 1%, directly compressing this fee burden. Comparable blockchain payment networks demonstrate the feasibility of this target: Stellar processed approximately \$55.6 billion in payments at a fee of roughly \$0.0007 per transaction in 2025 [44], and Celo's MiniPay processes payments for under one cent across more than 60 countries [42].
- **Trapped capital liberation:** The correspondent banking system requires banks to maintain pre-funded nostro and vostro accounts in every currency corridor in which they operate, immobilizing large sums that cannot be deployed for lending or investment [24]. On-chain atomic settlement eliminates the need for these idle pre-funded balances, freeing capital for productive use across the lending hierarchy.
- **Financial inclusion:** The platform is designed for accessibility: sub-cent gas fees on Polygon PoS, a mobile-first interface, and support for micro-loan amounts below one dollar lower the barriers to credit for the estimated 1.4 billion adults globally who remain outside formal financial systems [14]. The World Economic Forum has identified decentralized finance as a leapfrog technology capable of enabling populations to bypass the infrastructure constraints of traditional banking [41], a pattern already observed in mobile payments across developing economies.

- **Transaction cost reduction:** On Polygon PoS, transaction fees remain below one cent—a reduction of over 99% relative to the average \$42 cost of a correspondent banking transaction [25]. This cost compression makes financially viable a large class of micro-transactions and small-loan servicing operations that are currently uneconomical through traditional payment rails.
- **Transparency as an economic good:** On-chain publication of reserve ratios, interest rates, and transaction records eliminates the informational asymmetry that enables predatory lending practices in opaque banking environments. In conventional banking, borrowers—particularly in developing economies—frequently operate without visibility into the true cost of credit or the basis for lending decisions. The Crypto World Bank’s design ensures that all participants access identical, verifiable information, reducing the scope for hidden fees and unfair pricing.

5.8.5 Value Proposition and Go-to-Market

Table 5.28: Go-to-market phases.

Phase	Activities	Timeline
Phase 1: Validation	Competition submission (BCOLBD 2025); thesis publication; open-source release	Current
Phase 2: Pilot	Regulatory sandbox application; institutional partnership; testnet-to-mainnet migration	6–12 months
Phase 3: Production	Multi-chain deployment; enhanced monitoring and analytics; governance token launch	12–24 months

This value proposition table summarizes user-visible benefits (cost, transparency, speed, and inclusion) and maps them to platform features. It supports the go-to-market narrative by linking technical design choices to practical outcomes for retail users and partner institutions.

5.9 Currency Risk and the Stablecoin Imperative

*This section uses a **currency-risk illustration** only; it does not define a country-specific deployment target (see Section 5.2).*

A hypothetical retail borrower who takes a loan denominated in ETH faces a fundamental risk that fiat-denominated traditional lending avoids: if ETH doubles between disbursement and final repayment, the borrower’s repayment obligation rises in local fiat terms by the same factor. For thin-margin borrowers, that is not a theoretical risk—it is potentially catastrophic. The May 2021 crypto market crash saw ETH lose approximately 55% of its value within six weeks; ETH then halved again over the course of 2022.

This is why stablecoin integration is treated as a **critical path item** elevated to a design requirement in Section 1.10.3, not an optional extension. The platform supports USDC or USDT-denominated loans as the default product at retail tiers, with ETH-denominated loans

reserved for institutional-tier participants who can manage currency risk. BIS Working Paper No. 905 [R13] identifies stablecoin volatility as a structural risk in emerging-market DeFi adoption and recommends fully-collateralized models (USDT/USDC) over algorithmic designs for developing-country use cases. The Terra/LUNA collapse (May 2022, ≈\$40 billion lost) provides the definitive negative example. ERC-20 stablecoin integration has technical implications: token allowances, transferFrom hooks, decimal precision (USDC uses 6 decimals vs. ETH's 18), and the approval-transfer-state-update ordering required by the CEI pattern must all be redesigned as a separate contract module—work that is in scope for the SavingsVault contract specified in Section 3.11.

5.10 MiCA and GENIUS Act Compliance Mapping

The platform's regulatory feasibility is not abstract: two recent statutes apply directly to any retail stablecoin-denominated lending product issued to European or US-jurisdiction users. The EU Markets in Crypto-Assets (MiCA) Regulation has been fully in effect since December 2024 [R36, R37]; the US GENIUS Act, signed in July 2025 [R38], establishes the first federal framework for payment stablecoins. Table 5.30 maps the relevant articles to the design choices that satisfy them and to the work that remains.

Table 5.30: MiCA and GENIUS Act compliance mapping.

Statute / Article	Requirement	CWB Design Control
MiCA Art. 16 (EMT Issuer Authorisation)	Authorized issuer required for any e-money token offered to the public	USDC / USDT are issued by Circle / Tether under their own authorization; CWB uses but does not issue EMTs.
MiCA Art. 36 (Reserve of Assets)	1:1 reserve backing, segregated from issuer assets	Stablecoin reserves verified via on-chain attestations from Circle (USDC) before being accepted by SavingsVault.
MiCA Art. 41 (Redemption Rights)	Holders must have a right of redemption against the issuer at par	CWB does not impede redemption; SavingsVault withdraw function returns the underlying ERC-20 directly to the depositor wallet.
MiCA Art. 62 (Operational Resilience)	Regular audits, governance, complaint handling	Five-layer defense-in-depth (Section 3.19); Foundry invariants (Section 4.5); Certora reserve invariants (Future Work, Section 6); Tenderly runtime monitoring (Section 4.7).
GENIUS Act §4 (Federal Payment Stablecoin Framework)	Federal registration of payment-stablecoin issuers	Same posture as MiCA Art. 16: CWB uses, does not issue.
GENIUS Act §7 (Reserve Disclosure)	Monthly attestation and public disclosure of reserves	Circle / Tether attestations are independently verifiable on-chain; CWB dashboard surfaces issuer attestation links per stablecoin pool.
EU AI Act Annex III (high-risk: credit scoring)	AI systems evaluating natural-person creditworthiness classified high-risk from August 2026 [126]	RF/IF pipeline is in scope; conformity assessment, risk-management system, and technical documentation required before EU-resident deployment.
EU AI Act Art. 86	Right to explanation for AI-assisted decisions	Borrower-facing SHAP + anomaly summaries (Section 4.3.2); Authority Brief for approvers; audit log for compliance. SHAP satisfies post-hoc explanation partially; substantive “meaningful explanation” threshold may require additional artifacts [126].
EU AI Act Arts. 11, 12, 14	Technical documentation, automated logging, human oversight	LOAN_RISK_ASSESSMENT + MODEL_REGISTRY (Art. 12 partial); human approver gate (Art. 14); full Art. 11 technical file and risk-management system not yet complete.
GDPR Art. 17 (Right to Erasure)	Personal data must be deletable on request	No PII on-chain; off-chain PostgreSQL data is encrypted and subject to data-subject deletion re-

The result is a compliance posture that is defensible to European and American institutional readers and evaluators in a single page rather than buried in narrative. Compliance gaps that remain—a formal MiCA whitepaper if CWB ever issues its own token, FinCEN reporting integration in the US, and full EU AI Act conformity assessment—are explicitly named in Section 5.14 and the Future Work section of the Conclusion.

5.11 Jurisdiction and Regulatory Considerations

The platform is designed as a global institutional L2 architecture; jurisdiction-specific deployment is future work. One illustrative regulatory context—Bangladesh, cited in the microfinance literature—is summarized here to show how a hierarchical on-chain bank would face real-world constraints. Three observations frame the path forward.

Crypto remains effectively illegal for retail use in several jurisdictions as of 2025–2026. In Bangladesh, Bangladesh Bank’s longstanding 2018 circular treats crypto trading as a Foreign Exchange Regulation Act offence, and no subsequent statute has formally legalized retail crypto activity [R41]. The Crypto World Bank prototype is therefore planned exclusively on Polygon zkEVM Cardona (live testnet) using test tokens; Ethereum Sepolia is used only for Certora verification in Phase IV. No mainnet value transacts at pre-thesis stage.

The CBDC trajectory is a credible legal path in many jurisdictions. Several central banks have launched CBDC feasibility studies and pilots. The Crypto World Bank’s positioning against mBridge and Agora (Section 2.2.6) provides a direct upgrade path: when a jurisdiction issues a CBDC under the hierarchical distribution model that the Tan (2023) IMF working paper [5] describes, the platform’s Tier 1 / Tier 2 contracts can settle in that CBDC without changing the four-tier architecture. ETH / USDC denomination in testnet is a stand-in for future CBDC or tokenized-deposit assets.

Regulatory sandboxes as entry points. Jurisdictions such as Singapore (MAS Project Guardian), UAE (DIFC Innovation Testing Licence), and Bangladesh (Regulatory FinTech Facilitation Office) offer sandbox frameworks for controlled pilots. The platform’s go-to-market path is: (a) academic testnet deployment for pre-thesis, (b) controlled sandbox engagement for final-thesis evaluation, (c) partner-institution pilot under sandbox cover, and (d) phased CBDC integration where applicable.

Microfinance inclusion data (literature precedent). Bangladesh Bank Special Publication SP2025-02 (June 2025) [R44] reports female-to-male account parity at 49%/49% by March 2025. BRAC’s 2025 data shows 89% of loans to women [R45]. These statistics inform the solidarity-group module design as literature precedent (BRAC/Grameen group lending); they do not define a deployment target market for this thesis.

FATF Travel Rule scoping. The Financial Action Task Force Recommendation 16 (the “Travel Rule”) requires that transfers above USD 1,000 include structured originator and beneficiary information transmitted alongside the transfer. In most FATF member jurisdictions this threshold is now in force for virtual asset service providers; Bangladesh, as a FATF Asia-Pacific Group member, is under the same obligation. For the Crypto World Bank, the Travel Rule applies specifically to inter-tier capital flows: Local Bank to National Bank

repayments and upward surplus repatriation via the UpwardDepositFacility contract routinely exceed the USD 1,000 threshold. The off-chain Travel Rule data packet required for each such transfer is specified in Section 3.14.2:

```
{
  "originator_institution": "LocalBank-042",
  "originator_tier": 3,
  "beneficiary_institution": "NationalBank-001",
  "beneficiary_tier": 2,
  "amount_usdc": 5000,
  "purpose": "IBLP_REPAYMENT",
  "onchain_tx_hash": "0xABC...",
  "timestamp": "2026-06-01T10:00:00Z"
}
```

This packet is stored in the PostgreSQL `audit_logs` table and is linked to the on-chain transaction hash, making it available to regulators through the formal audit request workflow described in Section 3.18.5. The `audit_logs` table is append-only (enforced at the database-role level), ensuring the Travel Rule record cannot be altered after filing. Implementation of the cross-tier Travel Rule notification protocol—including automated packet generation, digital signing by the originating institution, and transmission to the relevant FIU—is Future Work contingent on jurisdiction-specific guidance.

5.12 Bootstrap Funding and the Tier 1 Capitalization Problem

The Crypto World Bank faces a **bootstrap funding problem** analogous to the capitalization challenge of real multilateral development banks. The World Bank Group was initially capitalised by 44 member-state subscriptions in 1944; the IBRD's combined subscribed capital exceeded \$270 billion following its 2018 capital increase (Ocampo & Gallagher, 2024 [R15]). Three initial capitalization mechanisms are proposed for the Tier 1 Reserve:

1. **Founding stakeholder deposits:** Universities, NGOs, and development-focused blockchain organizations deposit ETH or USDC into the Tier 1 Reserve in exchange for governance tokens and yield rights.
2. **Protocol-owned liquidity (POL):** The protocol accumulates treasury reserves through bond mechanisms in which external parties swap ETH or USDC for discounted governance tokens over a vesting schedule. *Note: The Olympus DAO algorithmic reserve model (2021–2022) is cited here as a cautionary precedent—Olympus OHM lost over 99% of its token value within one year due to unsustainable algorithmic backing. The CWB POL mechanism is strictly collateralized: only fully-backed USDC bonds are accepted, and governance tokens carry no intrinsic algorithmic price-support obligation. This removes the reflexive collapse risk that destroyed Olympus DAO.*
3. **Philanthropic/impact grant funding:** Development finance institutions (IFC, ADB) have expressed interest in blockchain-based development finance tools, as evidenced by the World Bank FundsChain initiative [39]. Grant funding from these institutions could seed the Tier 1 Reserve in exchange for research access and co-branding.

In the short term, a multi-signature wallet (3-of-5 signers representing different stakeholder

groups) governs Tier 1 allocations. In the long term, an on-chain governance module (similar to Compound Governor Bravo) enables token-weighted voting with time-locks to prevent governance attacks.

5.13 Prototype Scope and Limitations

At pre-thesis stage, the complete system is **specified** but not deployed. Planned implementation scope:

1. **Hierarchical lending scope:** Tier 1 World Bank Reserve, Tier 2 National Bank, and Tier 3 Local Bank contracts are specified in Solidity. Full cross-tier fund transfers and loan lifecycle automation are planned for Phases I–II of the final thesis phase.
2. **Interest accrual and repayment:** Interest rate tiers (3%/5%/8%) and fee rules are specified. Automated installment scheduling, accrual, and penalty enforcement are planned for Phase II (MVT item 2).
3. **InterBankLendingPool, UpwardDepositFacility, and broader multi-entity / cross-tier operations:** Fully specified in Section 3.14. Per the MVT boundary (Table 4.1), syndicated/tranched/netting modules are Future Work; IBLP and upward flow are Phase II–III stretch goals beyond the core loan lifecycle.
4. **AI/ML integration:** Random Forest, Isolation Forest, SHAP, and the Authority Brief pipeline are specified in Sections 4.3.1 and 4.3.2. FastAPI scoring and on-chain oracle wiring are planned for Phases II–III (MVT items 4–7); no live loan-approval integration is claimed at pre-thesis stage.
5. **Backend architecture:** Express.js REST API with PostgreSQL (authoritative schema, 46 entities — 26 core + 20 extension/stub) and FastAPI for ML inference. MongoDB is not part of the planned production architecture.
6. **Banking product suite:** The extended banking product suite described in Section 3.17, including savings accounts, fixed-term deposits, transactional accounts, group lending, FX conversion, and the insurance fund, is specified at the architectural and design level in this report. Smart contract implementation and testing of these modules is planned for the final thesis phase.

The limitations above are consistent with a planning-phase pre-thesis; the complete system design is fully specified in this report and will be implemented and validated in the final thesis phase.

5.14 Research Trajectory and Examiner Limitations

A structured literature scan (June 2026) identified five risks examiners are likely to probe. This section records them explicitly so the thesis does not over-claim.

G1 — Compound novelty vs. partial counterexamples. Maple Finance, Goldfinch, Morpho V2, and Aave Arc provide partial institutional-credit precedents on individual axes [117,118,R21]. The defensible claim is the *compound* architecture (four-tier reserve enforcement + explainable oracle ML + programmable mutual liability), not “no prior four-tier DeFi.”

G2 — BCCC domain transfer. Models trained on BCCC flat-pool transactions may not generalize to hierarchical CWB flows (Section 4.3.1). Mitigation: acknowledge the gap,

benchmark on BCCC as baseline only, collect CWB testnet labels for retraining, and run feature-distribution comparison as Future Work.

G3 — Testnet scope as academic evidence. Working testnet deployment + benchmark tables + documented specifications suffice for workshop venues (AFT, FMBC, IEEE Blockchain workshops). Full research papers additionally require controlled experiments, baselines, threat models, and reproducible public repos. This pre-thesis delivers specification and workload planning; MVT deployment provides the empirical layer.

G4 — Group lending external validity. On-chain mutual liability is cryptographic cosigning, not social solidarity [119,120]. Repayment-rate claims must not cite BRAC/Grameen historical data as direct validation of smart-contract behaviour.

G5 — Oracle trust and economic viability. Score-at-origination-only mitigates latency/cost; oracle collusion and model staleness remain open (Section 4.3.1). Certora reserve-invariant proofs are Future Work (Section 6); Foundry fuzz tests should list what is asserted, assumed, and out of scope [127].

Publication trajectory. Primary venue targets aligned with CWB contributions: **AFT 2026** (DeFi systems; abstract deadline May 2026), **FMBC** (formal verification of reserve invariants), **IEEE Blockchain / ICBC** (hierarchical architecture), and **IEEE TIFS** (federated fraud + explainability). PhD-scale seeds: hierarchical GraphSAGE ablation, cross-tier FL with differential privacy, and zkAML + FATF Travel Rule integration (Section 6).

5.15 Accessibility Assessment: Hypothetical Retail Client

This section evaluates access across six dimensions for a hypothetical retail client seeking a \$500 USDC working-capital loan through a Local Bank interface. The assessment is illustrative; no geographic deployment target is claimed at pre-thesis stage (Section 5.11).

1. Device access. The platform’s mobile-first React interface and WalletConnect integration allow access via any Android or iOS device with a browser, without requiring a desktop or dedicated hardware wallet.

2. Connectivity. Layer-2 transactions are lightweight (under 10 KB) and complete within seconds, well within typical mobile-network latency budgets.

3. Transaction cost. On Polygon zkEVM Cardona, the complete loan lifecycle (request, approval, 6 monthly installments) is projected to cost under \$0.05 in gas, a small fraction of typical origination fees in traditional banking.

4. Language and literacy. The planned prototype is English-only at launch. Additional locale packs are a planned extension; the React component library supports right-to-left and Unicode rendering.

5. Identity and onboarding. Tiered off-chain KYC and RBAC role gates serve MVT onboarding; the planned Groth16 zkKYC / zkAML extension is Future Work (Section 6). ERC-4337 account abstraction hides wallet complexity from retail users.

6. Group lending fit. The solidarity group lending model (GroupLendingPool, planned) follows microfinance literature precedent (BRAC/Grameen). Programmable mutual liability replaces community social pressure with smart contract enforcement.

In aggregate, the Crypto World Bank is accessible in principle to a hypothetical retail client on a mobile device (connectivity, cost, ERC-4337 onboarding) with two gaps requiring resolution for a production deployment: locale packs beyond English at launch and stablecoin-first denomination to reduce fiat currency risk for cross-border users.

Chapter 6

Conclusion

The Crypto World Bank addresses a **multi-party coordination and trust** problem inherent in hierarchical development finance by exploiting the distinctive properties of blockchain technology: cryptographic integrity, immutability, and programmable auditability. It advances beyond existing DeFi lending protocols—which collectively manage over \$55 billion in TVL [13] but universally employ flat, single-tier pool architectures—through a four-tier institutional hierarchy with cross-tier, same-tier, and upward lending flows, combined with a governance structure that addresses network membership, business operations, and technology infrastructure.

The platform’s design is grounded in the six core functions of a bank—deposit mobilization, credit allocation, payment and settlement, risk intermediation, liquidity management, and ancillary services—and encodes each of those functions as smart contract logic rather than relying on discretionary enforcement by a trusted central authority. This is the fundamental contribution of the smart contract architecture to banking: rules that previously required legal agreements, periodic audits, and trusted intermediaries are instead enforced by deterministic, publicly verifiable code running on a consensus-secured state machine. Reserve ratios are checked atomically on every transaction; interest rates adjust algorithmically with utilization; group consent is recorded on-chain before any disbursement; and the entire loan book is auditable in real time by any participant.

A distinct and architecturally novel capability of this platform is **multi-entity co-lending and co-funding**—the ability for groups of entities to lend, fund, and borrow together. This capability operates at two levels simultaneously. At the retail level, the `GroupLendingPool` allows three to twenty clients to form a solidarity group, pool collateral, and co-borrow under mutual liability enforcement—encoding the structural logic of Grameen Bank and BRAC’s solidarity group model as programmable contract rules. At the institutional level, the `SyndicatedLoan` contract allows multiple banks at the same or adjacent tier to co-fund a single loan, distributing credit risk pro-rata among co-lenders through an on-chain consent mechanism, subscription window, and supermajority confirmation vote—replicating the mechanics of a traditional syndicated loan deal without bilateral legal agreements. The `TranchedPool` further allows risk-tolerant and risk-averse entities to co-fund the same pool with differentiated seniority rights. No existing DeFi protocol implements any of these multi-entity mechanisms.

This thesis makes four original contributions, as formalized in Section 1.6: (1) a *compound* four-tier hierarchical DeFi architecture (reserve-ratio enforcement + explainable oracle ML + programmable mutual liability) that mirrors multilateral development finance capital flows—partial counterexamples exist on individual axes but not in combination (Section 5.14); (2) an on-chain *programmable joint-liability* group lending specification with over-indebtedness controls (Section 3.17.3), honestly scoped against analogue solidarity-lending limits; (3) an oracle-mediated AI/ML integration pattern with SHAP- and

anomaly-report explainability for fraud and AML governance (Section 4.3.2), committed on-chain via Chainlink Functions at loan origination, with BCCC benchmarking subject to an acknowledged domain-transfer gap (Section 4.3.1); and (4) a compliance-aware identity pathway (zkKYC, zkAML, W3C DID/VC) motivated in the literature and scoped to Future Work (Section 6). The optional conversational agent (Section 4.4), GNN ablation, cross-tier federated learning, NettingEngine, and Certora proofs are explicitly outside the MVT boundary. These contributions are framed against the institutional-DeFi landscape (Sygnum, mBridge, Agora) in Section 2.2.6 and against the MiCA / GENIUS Act / EU AI Act frontier in Section 5.10.

The analysis of the competitive situation of 20+ existing projects in DeFi lending, institutional credit, banking infrastructure and financial inclusion affirm that no current system is a combination of multi-level lending (hierarchy), interbank lending mechanisms and graded access by borrowers in one decentralized architecture. The growing blockchain implementation in the institutions, as demonstrated by the World Bank FundsChain initiative [39], the multi-billion-dollar daily volumes of JPMorgan Kinexys [40] and R3 Corda's \$17 billion in tokenized assets [37]—validates both the technical feasibility and institutional demand of blockchain-based financial infrastructure. The Crypto World Bank extends this trajectory from settlement and fund tracking into a fully open, hierarchically governed lending system.

The platform occupies the intersection of institutional finance, decentralized lending, and financial inclusion, a combination that remains unaddressed in both the academic literature and the commercial blockchain landscape. With a fully specified architecture, a defined market and partnership plan, and a phased implementation roadmap, the Crypto World Bank represents a structurally distinct and architecturally grounded contribution to the emerging field of blockchain-based development finance.

Viewed as a complete banking function checklist, the pre-thesis deliverable specifies hierarchical lending workflow foundations (tiered roles, governance framing, architectural capital flow, and a planned AI/ML monitoring layer with explainability) while leaving several bank-grade modules at the design level. Modules specified but not yet implemented in the final thesis include:

- Deposit products (SavingsVault and FixedDeposit), specified in Section 3.11
- Transactional accounts (CurrentAccount) and retail payment rails: client-to-client remittance, merchant checkout, and fiat on/off-ramp (Section 6)
- Group lending (GroupLendingPool) with mutual-liability and over-indebtedness logic (Section 3.17.3)
- Liquidation Engine (Section 3.10)
- On-chain credit passport SBT (Section 3.12)
- Oracle-priced FX conversion (FXModule)
- Multi-entity and cross-tier operations: InterBankLendingPool, UpwardDepositFacility, SyndicatedLoan, TranchPool, and TreasurySwap, specified in Section 3.14; NettingEngine is Future Work (Section 6)
- Insurance and depositor protection (InsuranceFund)
- Federated learning, GNN ablation, Groth16 zkKYC/zkAML circuits, NettingEngine, and Certora reserve-invariant proofs (Section 6)
- Foundry fuzz / invariant suite (Section 4.5)
- Tenderly + The Graph runtime monitoring stack (Section 4.7)

Future iterations therefore focus on completing end-to-end cross-tier fund transfers and repayment automation, integrating the analytics layer into real approval flows under the Chainlink Functions oracle (replacing the interim commit-reveal relay as specified in Phase III), hardening the platform against regulatory scrutiny via MiCA / GENIUS Act compliance, and expanding the platform into a functionally complete, stablecoin-first, compliance-aware banking system—including retail payments, client-to-client remittance, merchant checkout, and fiat on/off-ramp rails (Section 6).

Future work is split into **MVT completion** (final-thesis Must deliverables) and **PhD-scale research extensions** (publishable seeds). Items removed from this list as non-research scope creep: Polygon CDK sovereign chain, MCP/WhatsApp channel adapters, LLM fine-tuning as a standalone research item, and live cross-chain bridging (retired in Section 3.13 after security analysis). ERC-4337 account abstraction remains specified in Section 3.8.1 as assumed retail infrastructure; WalletConnect suffices for MVT.

MVT completion (final thesis Must).

1. **Complete hierarchical lending implementation.** End-to-end wiring of cross-tier fund transfers between World Bank Reserve, National Bank, and Local Bank contracts—capital distribution, interest accrual, installment scheduling, cascading repayment—against Appendix B sketches.
2. **Stablecoin-first deployment.** USDC default at retail tiers (Section 1.10.3); MiCA / GENIUS Act mapping validated against deployed configuration (Section 5.10).
3. **Liquidation Engine, SavingsVault, FixedDeposit, GroupLendingPool.** Phase II deposit-and-credit suite with over-indebtedness controls (Section 3.17.3).
4. **Multi-entity operations (stretch).** InterBankLendingPool and UpwardDepositFacility as Phase II–III stretch goals; SyndicatedLoan / TranchedPool beyond MVT minimum (Section 3.14). NettingEngine is Future Work (Section 6).
5. **AI/ML pipeline integration (core MVT).** Chainlink Functions oracle at loan origination; RF + IF + SHAP + Authority Brief (Sections 4.3.2, 4.3.1); BCCC benchmark table populated with measured metrics.
6. **Foundry fuzz suite (MVT+).** Foundry invariant and fuzz testing (Section 4.5).
7. **Foundry simulation (Phase IV).** 300-client hierarchy on Polygon zkEVM Cardona; gas-cost table and reserve dynamics (Section 4.6); manifest in Appendix C.
8. **Runtime monitoring.** The Graph subgraph, Tenderly alerts, live Reserve Transparency Dashboard (Section 4.7).
9. **Operational security and sandbox path.** Safe multisigs (Section 3.19); controlled pilot under MAS Project Guardian or UAE DIFC sandbox (Section 5.11).

Retail payments, remittance, and commerce (post-MVT). The checking-account specification introduces atomic transfers between *registered* platform accounts and mentions peer transfers, but the pre-thesis prototype does not implement a complete retail payments stack: there is no dedicated client-to-client remittance product, no merchant or marketplace checkout, and no fiat on/off-ramp (Section 1.10). Section 2.2.8 synthesises the peer-reviewed and policy evidence that these extensions are empirically and regulatorily studyable; closing the implementation gap is required before the platform can support everyday purchase flows with USDC rather than lending and installment servicing alone. Future work should add:

1. **Client-to-client remittance (P2PTransfer module).** KYC-gated, velocity-capped USDC transfers between Tier 4 wallets under Local Bank sponsorship; per-transaction AML screening; rolling 24-hour limits tied to the KYC ladder (Section 3.8.2); optional cross-Local-Bank net settlement via NettingEngine (Section 3.14.6). Design follows platform-mediated VASP compliance [145] rather than unhosted-wallet P2P; adoption drivers from Ante [136] inform literacy-aware onboarding.
2. **Fiat on/off-ramp integration.** Licensed payment-service-provider or regulated stablecoin-issuer APIs (bank transfer, mobile money, card) so clients can fund checking balances and withdraw to local currency—a prerequisite for real-world spending outside the on-chain ledger and the binding constraint identified by Du et al. [138] and CPMI [142].
3. **Merchant and marketplace checkout.** Payment-request contracts (invoice hash + amount + merchant LOCAL_BANK_ROLE), QR or deep-link checkout for registered merchants, and settlement reconciliation for sponsoring Local Banks—an on-chain analogue to card-acquirer settlement, denominated in USDC. Evaluated against the CLEAR framework of Li et al. [137]; closed-loop Local-Bank sponsorship addresses the dispute-recourse gap they identify vs. open-loop card networks.
4. **Retail FX completion.** Deploy FXModule (Section 3.17) so clients see fiat-equivalent balances and optional conversion before off-ramp or cross-border remittance, mitigating stablecoin refund volatility identified by Zhang et al. [140].
5. **Deposit protection and payment disputes.** Operational InsuranceFund; dispute workflow for erroneous P2P or merchant debits; jurisdiction-specific mapping of smart-contract obligations to enforceable consumer-protection rules—responding to the consumer-protection inversion Li et al. [137] document for raw stablecoin checkout.
6. **External commerce boundary.** Integration with DEX or payment-rail partners (e.g. composable USDC → merchant settlement via licensed off-ramp) where in-platform checkout is insufficient—explicitly out of MVP scope but documented as the bridge between CWB balances and real-world merchants, consistent with the stablecoin-sandwich pattern [138].

These items address traditional-banking gaps the hierarchical lending core does not cover: P2P remittance, merchant acceptance, cash-like deposit/withdrawal, and consumer dispute resolution—while preserving the Local-Bank entry point and AML posture of the four-tier model. No prior source composes these retail rails with hierarchical development-finance lending in one architecture (Section 2.2.8).

PhD-scale research extensions (publishable seeds). The following advanced modules are **not specified at implementation depth** in Chapters 3–4; literature motivation appears in Chapter 2, and operational detail is deferred here:

1. **GNN transaction analysis (GraphSAGE ablation).** Isolate contribution of hierarchical tier-relationship edges vs. flat wallet edges on BCCC [108]; target AFT or IEEE Blockchain.
2. **Federated learning across banking tiers.** FedAvg + differential privacy across Local Banks; benchmark against FedFraud [125] and BCCC; target IEEE TIFS.
3. **Groth16 zkKYC circuit (Circom 2.0) and zkAML velocity circuit.** Privacy-preserving KYC eligibility and AML velocity proofs with W3C DID/VC anchoring;

- combine with FATF Travel Rule packet spec (Section 3.14.2).
4. **NettingEngine with partial settlement and challenge periods.** Multilateral netting of IBLP and TreasurySwap obligations; `settlePartial` for solvent participants; public challenge window for dispute resolution.
 5. **Certora formal verification of reserve invariants (Phase IV).** CVL proofs that $\text{totalReserve} \geq \text{minReserveRatio} \cdot \text{totalDeposited}$ and that downstream allocation never exceeds available reserve; target FMBC [127].
 6. **Agent-based reserve-cascade simulation.** Mesa or equivalent ABM for systemic risk under mass default, oracle manipulation, and stablecoin depeg—stress beyond testnet deployment alone.
 7. **ML domain adaptation.** Feature-distribution comparison: BCCC flat-pool features vs. simulated CWB hierarchical transactions; semi-supervised baseline [124].
 8. **EU AI Act conformity assessment.** Complete Art. 11 technical documentation, Art. 12 logging, and user-study on SHAP explanation adequacy under Annex III high-risk credit scoring [126].
 9. **On-chain credit passport SBT + W3C DID group Sybil prevention.** SBT implementation (Section 3.12); narrow Sybil control via W3C DID group membership verification rather than World ID bundling.
 10. **CLEAR-framework evaluation of CWB retail payments.** Apply Li et al.'s [137] cost–legality–experience–architecture–reach rubric to the `P2PTransfer` and merchant-checkout modules vs. card-acquirer and MTO baselines; corridor cost bench against Du et al. [138]; target IEEE Blockchain or payments-economics venue.

Demoted to appendix or single-sentence notes: SAR/FIU portal integration (compliance engineering); optional MCP agent and LLM red-teaming (Appendix B); live cross-chain bridge (design revision in Section 3.13).

Appendix A

Database Schema Reference

This appendix records the physical-schema details moved from Chapter 3 to keep the main architecture chapter focused on conceptual design.

A.1 Indexing Strategy

B-tree indexes (the PostgreSQL default) support efficient retrieval on frequently queried columns:

Table A.1: Indexing strategy: B-tree indexes for time-window and high-frequency queries.

Index	Table / Column(s)	Rationale
idx_loan_borrower	LOAN_REQUEST(borrower_id)	High-frequency lookup for borrower loan history.
idx_loan_status	LOAN_REQUEST(status, oracle_state)	Efficient filtering of application lifecycle and oracle pipeline states.
idx_installment_d	INSTALLMENT(due_date, status) WHERE status = 'PENDING'	Partial-index range query for pending overdue installment detection.
idx_txn_created	TRANSACTION(created_at)	Time-window aggregation for 6-month and 1-year borrowing limits.
idx_txn_borrower	TRANSACTION(borrower_id)	Per-borrower transaction history retrieval.
idx_market_symbol	MARKET_DATA(symbol, recorded_at)	Composite index for price feed time-series queries.
idx_loan_bank_sta	LOAN(local_bank_id, status, created_at DESC)	Loan lifecycle queries by bank and status.
idx_loan_borrower	LOAN(borrower_id, status) WHERE status = 'ACTIVE'	Partial index for per-borrower open-loan count.
idx_agent_session	AGENT_ACTION_LOG(session_id, created_at DESC)	Per-session agent action history retrieval (join to SESSIONS for borrower).
idx_agent_status	AGENT_ACTION_LOG(status) WHERE status = 'PENDING'	Partial index for the agent status-monitoring loop.
idx_risk_score	LOAN_RISK_ASSESSMENT(anomaly_score DESC) WHERE anomaly_score > 0.5	Partial index for AML alert queue (SAR workflow).
idx_ib_loan_status	INTERBANK_LOAN(status, lender_institution_id)	IBLP default cascade and maturity monitoring.
idx_institution_t	INSTITUTION(institution_type, status)	Tier-scoped institution lookups for multi-entity operations.
idx_txn_institution	TRANSACTION(origin_institution_id, created_at DESC)	Institution-to-institution ledger history retrieval.
idx_chainlink_fee	CHAINLINK_PRICE_ROUND(feed_address, updated_at DESC)	Latest round lookup per Chainlink feed for LTV staleness checks.
idx_session_parent	SESSIONS(parent_session_id) WHERE parent_session_id IS NOT NULL	Partial index for session lineage chain traversal.
idx_collateral_bc	COLLATERAL_POSITION(borrower_id, loan_id)	Health-factor lookup per borrower and per loan.
idx_event_log_txh	BLOCKCHAIN_EVENT_LOG(tx_hash)	Deduplication guard: event listener checks this before INSERT.
idx_event_log_blc	BLOCKCHAIN_EVENT_LOG(block_number, chain_id)	Catchup-sync replay from a given block.

A.2 Functional Dependencies

Table A.3: Representative functional dependencies.

Relation	Functional Dependency	Notes
LOAN_REQUEST	request_id → all attributes	Application row keyed by request identifier.
LOAN_REQUEST	borrower_id, local_bank_id → status, oracle_state, amount, ...	Lifecycle status and oracle pipeline oracle_state are separate columns.
LOAN	loan_id → all attributes	Disbursement record keyed by loan identifier.
LOAN	loan_request_id → loan_id	1:1 link from approved request to disbursed loan.
BORROWING_LIMIT	borrower_id → six_month_limit, one_year_limit, on_chain_enforced_limit, last_synced_block, ...	1:1 with BORROWER; event-listener owned projection.
CREDIT_PASSPORT	passport_id → all attributes; borrower_id → passport_id	Read-model projection of on-chain SBT; event-listener owned.
BORROWER	wallet_address → borrower_id, all attributes; registered_local_bank_id → home Local Bank	Unique candidate key in the prototype; Phase IV uses (wallet_address, chain_id).
INSTITUTION	institution_id → institution_type, name, wallet_address, contract_address, status	Supertype for cross-tier FK anchors.
NATIONAL_BANK	country_code → institution_id	One national reserve per nation (UNIQUE).
MODEL_REGISTRY	model_id → model_name, version, training_date, metrics, is_active	Model lineage for explainability audit.
LOAN_RISK_ASSESSMENT	assessment_id → all attributes; model_id → fraud_probability, anomaly_score, shap_vector	Per-loan ML inference; FK to MODEL_REGISTRY.
INSTALLMENT	loan_id, installment_number → amount, due_date, status, payment_event_log_id	Composite primary key.
SESSIONS	session_id → borrower_id, wallet_address, session_key_hash, session_key_scope, session_key_expires_at, expires_at, revoked	Retail-client agent sessions only at MVT; bank-authority sessions are Future Work.
AGENT_ACTION_LOG	log_id → agent_id, action_id, timestamp, status, metadata	Agent action log; FK to AGENT, ACTION, and LOG tables.

A.3 Materialized Views and Agent Query Optimisation

The MCP tool `get_account_state` joins `BORROWER`, `CREDIT_PASSPORT`, `COLLATERAL_POSITION`, active `LOAN` rows, pending `INSTALLMENT` rows, `SAVINGS_ACCOUNT`, and `BORROWING_LIMIT`. To avoid six-to-seven table joins on every agent call, the schema defines a materialized view refreshed by the event listener after any event that modifies component tables:

```
CREATE MATERIALIZED VIEW mv_client_dashboard AS
SELECT
    b.borrower_id,
    b.wallet_address,
    b.kyc_level,
    cp.credit_score,
    cp.risk_tier,
    bl.six_month_remaining,
    bl.one_year_remaining,
    count(l.loan_id) FILTER (WHERE l.status = 'ACTIVE') AS active_loan_count,
    max(i.due_date) FILTER (WHERE i.status = 'PENDING') AS next_installment_due,
    sum(sa.balance) AS total_savings_balance,
    sum(col.locked_amount) AS total_collateral_locked
FROM borrower b
LEFT JOIN credit_passport cp USING (borrower_id)
LEFT JOIN borrowing_limit bl USING (borrower_id)
LEFT JOIN loan l USING (borrower_id)
LEFT JOIN installment i USING (loan_id)
LEFT JOIN savings_account sa USING (borrower_id)
LEFT JOIN collateral_position col USING (borrower_id)
GROUP BY b.borrower_id, b.wallet_address, b.kyc_level,
         cp.credit_score, cp.risk_tier,
         bl.six_month_remaining, bl.one_year_remaining;

CREATE UNIQUE INDEX ON mv_client_dashboard (borrower_id);
```

Refresh strategy: `REFRESH MATERIALIZED VIEW CONCURRENTLY mv_client_dashboard` triggered by the event listener after processing events on projection or lending tables (Phase III).

Appendix B

Optional Agent Harness Reference

This appendix records the optional conversational agent harness (Phase III–IV Could deliverables). It is not required to evaluate the core blockchain-banking contribution. Section 4.4 in Chapter 4 summarises the six-step pipeline, MCP tool schema, and Authority Brief example. The choice of a locally-hosted 8B-parameter model over a cloud-hosted API (Section C.1) is consistent with the broader 2026 industry shift toward on-device inference for compliance-sensitive financial applications, in which major hardware and platform vendors have positioned on-premise deployment as the only viable path for sectors handling sensitive financial records [R56].

B.1 Per-User Context and Prompt Assembly

Per-user personalisation — shared model, per-user context. Every client receives a personalised experience without deploying a separate model per user. Personalisation lives entirely in the per-user context namespace injected into every system prompt by the Express.js context injection layer:

```
{
  "borrower_id": "550e8400-e29b-41d4-a716-446655440000",
  "wallet_address": "0xABC...",
  "language": "en",
  "credit_tier": "Silver",
  "credit_score": 512,
  "active_loans": [
    {
      "loan_id": "L-4821",
      "amount": 100,
      "next_due": "2026-06-15",
      "installment": 17.5
    }
  ],
  "savings_balance": 250.0,
  "pool_utilisation": 0.67,
  "kyc_tier": 1,
  "conversation_history": [ "...last 10 turns..." ]
}
```

Because the on-chain state is re-injected at every turn, the agent always operates on current account data and cannot hallucinate the client's balance or loan status.

Three-tier system prompt assembly. The agent's system prompt is assembled by a dedicated prompt constructor in the Express.js backend, organised into three explicit tiers

rather than a single ad-hoc string [R52]:

Stable tier. Agent persona, active toolset schema, compliance anchors (EU AI Act Art. 86, FATF R.16, zero-bypass rule). Prefix-cache target; unchanged within a session.

Context tier. Interest rate schedule (Table 3.26), KYC matrix, pool policy, active skill procedure from AGENT_SKILLS. Updates on governance votes only.

Volatile tier. Per-user on-chain context namespace, parent-session compression summary, last N conversation turns. Rebuilt every turn from `eth_call` state.

B.2 Session Lineage, Compression, and Toolset Scoping

Session lineage and cross-session memory. The SESSIONS table carries a `parent_session_id` self-referential FK. When a conversation reaches a token threshold, the compression path closes the session with `end_reason = 'compression'` and opens a child session seeded with the compression summary. The read tool `get_session_history` exposes ranked excerpts from the client's conversation ancestry.

Context compression strategy. When the assembled prompt approaches the 32K-token window, the backend retains head ($H=500$) and tail ($T=1,000$) verbatim, summarises the middle via a secondary Qwen3-8B call, stores `SESSIONS.compression_summary`, and opens a child session. Trigger: 28,000 tokens (87.5% of window).

Toolset	Tools visible
<code>read_only</code>	All 9 read tools (default when intent is uncertain).
<code>loan_actions</code>	<code>read_only</code> + loan, installment, and group write tools.
<code>account_management</code>	<code>read_only</code> + deposit, KYC, and reminder write tools.

B.3 Lifecycle Hooks and Optional Capabilities

Lifecycle hook middleware. Write-tool POST requests pass through a hook stack: (1) confirmation audit hook (Phase II) verifies a logged confirmation turn or returns HTTP 403; (2) session-key scope pre-check (Phase III); (3) AML pre-check against `LOAN_RISK_ASSESSMENT` and `SECURITY_EVENT_LOG` (Phase IV).

Optional agent capabilities.

- **Proactive installment reminders** via `get_installment_schedule` and `pay_installment`.
- **Credit Passport coaching** via `get_credit_score` and Table 3.26.
- **Loan calculator** with live FX from `get_market_data`.
- **Group lending coordination** and **KYC tier upgrade guidance** via write/read tools.
- **Cross-session context retrieval** via `get_session_history`.

Appendix C

Technology Stack

C.1 In-product assistant: local large language model (LLM) integration (prototype)

Purpose. The in-product assistant is an **optional** conversational add-on (Section 4.4), not a static product guide and not required for deposits, loans, or approvals. It combines a locally hosted, privacy-preserving large language model (Qwen3-8B, Apache 2.0 licence) with a Model Context Protocol (MCP) tool server exposing 17 tools (9 read tools, including a session history search tool, and 8 write tools requiring human confirmation). The agent can both answer questions—via retrieval-augmented generation (RAG) from policy documents—and execute on-chain banking operations via write tools, subject to an explicit human confirmation gate before any state-modifying action is taken. Local hosting keeps conversation data on institutional infrastructure, supporting data-sovereignty requirements in regulated pilots. The agent is additive: a client who does not use it experiences no difference in available functionality, since every operation the agent can perform is equally available through the standard React dashboard.

Model and runtime (local development). The inference endpoint follows the **OpenAI Chat Completions** contract (`/v1/chat/completions`) with `stream: true`, proxied from the project backend to a local service such as `http://127.0.0.1:1234`. During early development, the assistant used a locally hosted **Gemma-4-E4B** checkpoint (Google mixture-of-experts model, ~26B total / 4B active parameters) via LM Studio in GGUF form (e.g., Q4_K_M quantization) for a laptop-friendly footprint. For the final thesis evaluation, the assistant is replaced by **Qwen3-8B** (Alibaba Cloud Qwen team, released April 28 2025, Apache 2.0 licence) [R46], a dense 8.19B-parameter causal language model with a 32K-token native context window (extendable to 131K tokens via YaRN scaling) and support for 100+ languages and dialects. A distinguishing architectural feature of Qwen3-8B is its *switchable reasoning mode*: the model operates in **thinking mode** (chain-of-thought wrapped in `<think>` tokens, suited to multi-step compliance or policy questions) and **non-thinking mode** (direct, low-latency responses for general navigation queries), selectable per-turn with `/think` or `/no_think` in the system prompt. For Q&A and retrieval-augmented generation (RAG) queries, non-thinking mode is the default; thinking mode is activated only for the red-team regulatory-compliance prompts in the evaluation protocol of Section 4.10.1. For agent tool-calling, non-thinking mode is the default for read tools and confirmation summaries; thinking mode is activated for complex multi-step operations such as EMI calculation or group signature coordination. The per-user context namespace (see Section 4.3) is prepended to every system prompt as a structured JSON block, so the model has full account context before reading the user’s message. The API model identifier is environment-driven; the integration layer remains identical for both models.

End-to-end behavior. The user interface assembles a short transcript (user/assistant messages) and posts it to the backend streaming route. Before the user message reaches the model, the context assembly layer: (a) selects the active toolset based on intent classification, (b) applies prompt injection scanning to all user-sourced fields, and (c) assembles the three-tier system prompt (Stable + Context + Volatile), injecting the compression summary from the parent session into the Volatile tier if this is a continuation session. The active toolset name and any injection scan flag are recorded in AGENT_ACTION_LOG for every turn. The backend then opens a streaming request to the local model server, converts upstream token events into **server-sent events (SSE)** for the browser, and the UI incrementally appends text. The message renderer uses Markdown (including GitHub-flavored extensions), math (KaTeX) for $...$ blocks, and line-break rules appropriate for chat transcripts.

For write-tool requests, the agent assembles the full parameter set from the user's request and the injected on-chain state, then presents a confirmation summary. Only after the user sends explicit affirmative consent does the agent call the corresponding MCP write tool via the Express.js banking API layer. The EIP-7702 session key (if active) signs the resulting on-chain transaction within its approved scope. Every executed write tool is logged to agent_action_log with the confirmation message ID as the audit reference.

Live on-chain context injection. The Express.js backend injects the client's full on-chain state as a structured JSON context block into every system prompt before forwarding the user's message to the model. The per-user context namespace is:

```
{
  "tier": "volatile",
  "borrower_id": "550e8400-e29b-41d4-a716-446655440000",
  "wallet_address": "0xABC...",
  "language": "en",
  "credit_tier": "Silver",
  "credit_score": 512,
  "active_loans": [
    {
      "loan_id": "L-4821",
      "amount": 100,
      "next_due": "2026-06-15",
      "installment": 17.5
    }
  ],
  "savings_balance": 250.0,
  "pool_utilisation": 0.67,
  "kyc_tier": 1,
  "session_id": "sess_20260602_xyz",
  "parent_session_id": "sess_20260515_abc",
  "compression_summary": "Client asked about Gold tier eligibility.
Was told 2 more on-time repayments required. No action taken.",
  "conversation_history": [ "...last 10 turns..." ]
}
```

The `compression_summary` field is injected only when this is a child session (i.e., when `parent_session_id` is non-null). It provides the prior-session context without exceeding the Volatile tier token budget. This context block is injected only when the user is authenticated and the `eth_call` round-trip completes within a 200 ms timeout (falling back to a static-context response otherwise). The model uses the injected context to ground its answers in the user's actual account data rather than generic policy text, materially reducing hallucination risk for account-specific queries.

Security and limitations (prototype stance). The agent's safety posture is enforced at four levels. (1) **Tool-schema boundary:** the agent interacts with CWB exclusively through the 17 MCP tools — it cannot execute arbitrary code, make unapproved API calls, or read data outside the tool schema. (2) **Human confirmation gate:** every write tool requires explicit affirmative consent in the conversation history; the confirmation turn ID is logged as the audit reference. (3) **Session key scope:** the EIP-7702 session key is scoped to specific tools, value-capped, time-bound to 24 hours, and revocable at any time. A session that attempts to call an out-of-scope tool is rejected at the key level, not just at the application level. (4) **Prompt injection scanning and lifecycle hook middleware:** user-sourced content is scanned for injection patterns before entering the prompt (Layer 2 control, Section 4.3), and every write-tool HTTP POST is intercepted by a middleware stack that enforces the confirmation invariant at the application layer independently of the model's output. Together, these four levels form a defence-in-depth posture in which no single point of failure can circumvent a critical safety constraint. Hallucination risk is bounded by: (a) the injected on-chain state grounding account-specific answers, and (b) the refusal layer for regulatory and legal queries (100% target on the red-team set, Section 4.10.1).

Architecture diagram. Figure C.1 shows the compact end-to-end request path; Figure C.2 expands the same flow with component boundaries and authentication context.

Local LLM Assistant — Compact Request Path

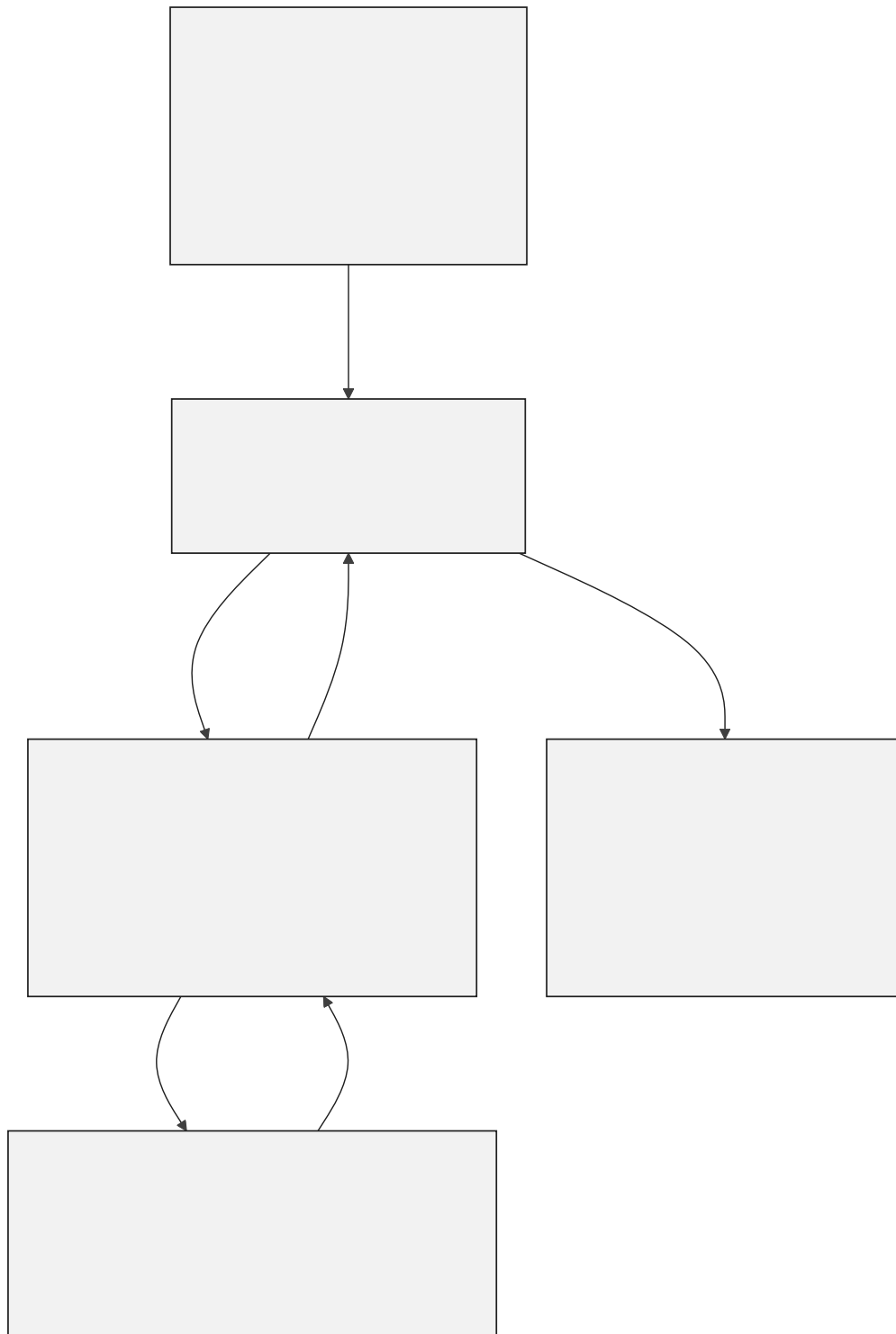


Figure C.1: Optional AI banking agent request path (add-on): browser UI → Vite dev proxy → CWB Express API (SSE) → MCP tool server (17 banking tools) → Qwen3-8B inference → human confirmation gate → write-tool execution → on-chain transaction.

Local LLM Assistant — Component Data Flow

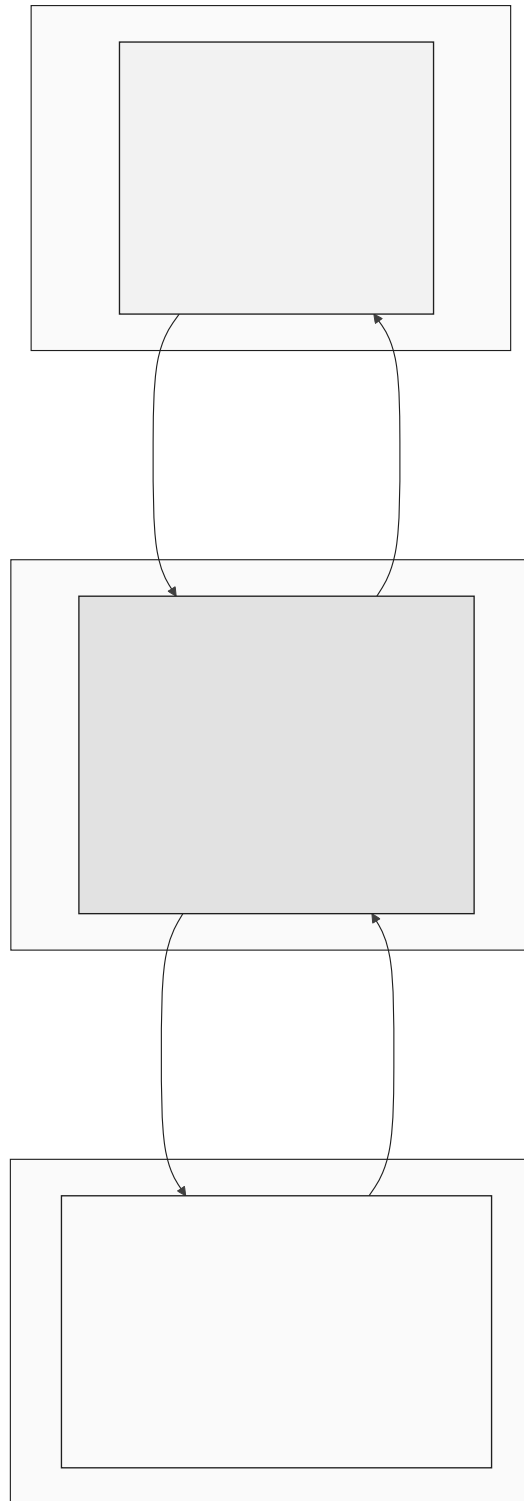


Figure C.2: Expanded optional AI agent data flow with component boundaries: the browser UI streams from the CWB Express API; read-tool requests return immediately; write-tool requests pass through the human confirmation gate and are signed with the EIP-7702 session key before on-chain execution.

Implementation analysis (brief). The integration is successful for the project prototype because it reuses a stable transport shape (chat-completions with streaming) and isolates the model vendor behind a single API route, allowing the user interface to remain a thin client. The main practical engineering challenges in local development were: (1) **process/port hygiene** to keep the API bound to a predictable port, (2) **CORS and same-origin** considerations when the UI and API run on different localhost ports, mitigated with a Vite dev proxy to /api and with permissive dev CORS policy on the API, and (3) **RPC provider selection** for wallet reads in the browser, where some public endpoints may fail browser CORS; pinning explicit public RPC endpoints avoids noisy failures unrelated to the chat feature.

Relationship to the earlier rule-based assistant (legacy). The repository also retains early keyword-routed /api/chatbot handlers used for initial prototyping; the LLM path is additive. This separation prevents regressions in deterministic demo endpoints while the conversational assistant evolves.

Table C.1: Technology stack summary.

Layer	Technology	Version / Notes
Smart Contract	Solidity, OpenZeppelin	0.8.20; Ownable, ReentrancyGuard
Frontend	React, TypeScript	Material UI; Design 3
Wallet Integration	Wagmi, RainbowKit, Viem	EIP-1193 compliant
Build and Test	Hardhat	Automated test suite; deployment scripts
Backend API	Express.js, Node.js	REST API; middleware architecture
Database	PostgreSQL (designed)	46 entities (26 core + 20 extension/stub), 3NF; relational integrity
Target Network	Polygon zkEVM Cardona testnet	ZK validity proof security (Amoy PoS = fallback)
AI Agent Engine	Qwen3-8B (llama.cpp, Q4_K_M)	MCP tool server; per-user context namespace; human confirmation gate
MCP Tool Server	17 banking tools (9 read, 8 write); prompt injection scanner; confirmation audit hook middleware	Exposes read + write banking operations to the agent with typed parameter schemas
Session Keys	EIP-7702	Scoped, time-bound, value-capped agent wallet authorisation
Oracle Network	Chainlink Functions (DON)	Trustless ML score commitment; replaces commit-reveal relay
Price Feeds	Chainlink AggregatorV3Interface	Fiat/USD pairs (e.g. EUR/USD) and ETH/USD; 8-decimal precision
Automation	Chainlink Automation	Trustless installment due-date checking; replaces centralised cron job
Proof of Reserve	Chainlink PoR	Cryptographically verifiable World-BankReserve balance
Event Indexing	The Graph (subgraph)	GraphQL query layer over LoanApplication and ReserveRatioSnapshot events

This appendix table summarizes assistant integration components and configuration aspects used in the prototype (UI, API proxying, and local model server). The purpose is to make the implementation reproducible and to clarify which parts are prototype conveniences versus production requirements.

Appendix D

Smart Contract Capabilities

The complete banking architecture is designed around fifteen modular contracts. The three-contract lending core is specified at pre-thesis stage; the remaining twelve modules are planned across Phases II and III of the final thesis phase.

Specified core contracts (Phase I target):

- **World Bank Reserve Contract:** Reserve custody, deposit handling, national bank registration, capital allocation to national banks, system pause and unpause, emergency withdrawal, and system statistics.
- **National Bank Contract:** Local bank registration, borrowing from the World Bank reserve, capital allocation to local banks, and network status queries.
- **Local Bank Contract:** Client registration, loan application processing, approval and rejection workflows, installment generation and payment processing, approver management, user account management, and a `freezeAccount(clientId)` function gated by `onlyApprover` modifier, used by the SAR workflow (Section 3.18.5) to suspend loan and payment operations for a flagged client pending compliance review.

Planned contracts – Banking product suite (Phase II):

- **SavingsVault Contract:** Standard savings account management, variable yield accrual, withdrawal processing, and deposit-to-lending pool integration.
- **FixedDeposit Contract:** Term deposit creation, lock period enforcement, agreed APY accrual, early withdrawal penalty calculation, and maturity release.
- **GroupLendingPool Contract:** Group formation, multi-signature consent recording, shared collateral management, per-member disbursement, mutual liability enforcement, and group credit history recording.
- **FXModule Contract:** Oracle-priced currency conversion, spread calculation, dual-currency lending denomination support, and conversion audit trail.
- **InsuranceFund Contract:** Captures 5% of all interest collected across the lending hierarchy, processes default coverage disbursement claims, and publishes its reserve balance to Chainlink Proof of Reserve for external verifiability.
- **CurrentAccount Contract:** Transactional account management, atomic peer transfers, recurring payment scheduling, and payroll deposit handling.

Planned contracts – Multi-entity and cross-tier operations (Phases II and III):

- **InterBankLendingPool Contract** (Section 3.14.1): one instance per tier (IBLP_NB, IBLP_LB). Short-tenor (1 / 7 / 30 day) same-tier borrowing with a utilization-kinked rate model bounded by the tier-above downward rate; default cascade through reserve buffer → InsuranceFund → parent-tier backstop.
- **UpwardDepositFacility Contract** (Section 3.14.2): role-restricted extension of SavingsVault accounting that lets a lower-tier bank deposit excess reserves into its par-

ent tier, earning a strictly lower yield than the downward borrowing rate; withdrawal gate enforces depositor reserve-ratio safety.

- **SyndicatedLoan Contract** (Section 3.14.3): syndicate proposal, subscription window, supermajority on-chain consent vote (75%-by-capital-share), atomic disbursement, pro-rata interest and recovery distribution; supports cross-tier syndication (mixed NB / LB co-funders).
- **TranchedPool Contract** (Section 3.14.4): senior / junior tranching with a subordination-ratio policy, waterfall interest and principal payments, first-loss absorption by the junior tranche; suitable for risk-segmented institutional and impact-aligned capital.
- **TreasurySwap Contract** (Section 3.14.5): oracle-priced atomic asset swap restricted to wallets holding a banking role; cross-tier asset exchange (e.g., NB ETH treasury → USDC) with a tighter spread than retail FX; post-swap reserve-ratio invariant enforced before state change.
- **NettingEngine Contract** (Future Work, Section 6): multilateral netting with partial settlement and challenge-period dispute resolution—not specified at implementation depth in this report.

Appendix C: Planned Testnet Deployment Manifest

The following table is a **planned deployment manifest template** for Polygon zkEVM Cardona testnet (Phase IV). No contracts are claimed as live-deployed at pre-thesis submission. Addresses will be recorded in the project repository under `/deployments/cardona/` after Phase IV deployment and can be verified on the Cardona block explorer at <https://explorer-ui.cardona.zkevm-rpc.com>.

Table D.1: Planned Cardona testnet deployment manifest (Phase IV template).

Contract	Network	Address / manifest path
WorldBankReserve	Polygon zkEVM Cardona	<code>deployments/cardona/WorldBankReserve.json</code>
NationalBank (Scaffold)	Polygon zkEVM Cardona	<code>deployments/cardona/NationalBank.json</code>
LocalBank (Scaffold)	Polygon zkEVM Cardona	<code>deployments/cardona/LocalBank.json</code>
LendingController	Polygon zkEVM Cardona	<code>deployments/cardona/LendingController.json</code>

Note: Exact hex addresses will be populated after Phase IV testnet deployment. The repository manifest is the authoritative record. This appendix documents the intended contract inventory only; it does not assert live deployment at pre-thesis stage.

Appendix D: WorldBankReserve Contract Interface

The following Solidity interface documents the public API of the WorldBankReserve contract—the formally specified Tier 1 contract (Phase I deployment target). Three design annotations follow the listing:

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 import "@openzeppelin/contracts/security/ReentrancyGuard.sol";
5 import "@openzeppelin/contracts/access/AccessControl.sol";
6
7 /// @title WorldBankReserve
8 /// @notice Tier 1 reserve contract: manages global capital allocation
9 /// to registered National Bank addresses.
10 contract WorldBankReserve is ReentrancyGuard, AccessControl {
11
12     bytes32 public constant NATIONAL_BANK_ROLE = keccak256("NATIONAL_BANK_ROLE");
13     bytes32 public constant AUDITOR_ROLE = keccak256("AUDITOR_ROLE");
14
15     uint256 public totalReserve;
16     uint256 public minimumReserveRatio; // e.g. 1000 = 10.00%
17     mapping(address => uint256) public allocatedTo; // NationalBank => amount
18
19     event CapitalAllocated(address indexed nationalBank, uint256 amount);
20     event RepaymentReceived(address indexed nationalBank, uint256 amount);
21     event ReserveRatioUpdated(uint256 newRatio);
22
23     /// @notice Allocate capital downward to a registered National Bank (CEI
24     pattern)
25     function allocateCapital(address nationalBank, uint256 amount)
26         external onlyRole(DEFAULT_ADMIN_ROLE) nonReentrant;
27
28     /// @notice Record repayment from a National Bank (CEI pattern)
29     function recordRepayment(uint256 amount)
30         external onlyRole(NATIONAL_BANK_ROLE) nonReentrant;
31
32     /// @notice Query current utilization ratio (scaled 1e4)
33     function utilizationRate() external view returns (uint256);
34
35     /// @notice Returns true if reserve ratio constraint is satisfied
36     function isReserveAdequate() external view returns (bool);
37
38     /// @notice Returns a four-value reserve summary for Proof of Reserve
39     /// verification and the Reserve Transparency Dashboard.
40     /// @return totalDeposited Gross deposits ever made to this reserve tier
41     /// @return totalLoaned Outstanding principal lent to National Banks
42     /// @return reserveRatio (totalDeposited - totalLoaned) * 1e4 /
43     /// totalDeposited (e.g. 5000 = 50.00%)
44     /// @return insuranceFundBalance Current balance of the InsuranceFund contract
45     function getReserveSummary() external view returns (
46         uint256 totalDeposited,
47         uint256 totalLoaned,
48         uint256 reserveRatio,
49         uint256 insuranceFundBalance
50     );
51 }
```

Listing D.1: WorldBankReserve.sol – Tier 1 public interface. Full implementation in project repository.

Design annotation 1 — nonReentrant on state-changing functions. The

`nonReentrant` modifier from OpenZeppelin's `ReentrancyGuard` sets a boolean lock before execution and clears it after, preventing recursive calls. It is applied to `allocateCapital` and `recordRepayment` because both involve ETH transfers that could trigger attacker-controlled fallback functions. The full Checks-Effects-Interactions analysis is in Section 3.9.

Design annotation 2 — AccessControl mapping to RBAC design. OpenZeppelin's `AccessControl` maps each on-chain role (e.g., `NATIONAL_BANK_ROLE`) to a `bytes32` `keccak256` hash stored in a role-to-address mapping. Role assignment is controlled by the `DEFAULT_ADMIN_ROLE` (the World Bank admin). This directly implements the four-tier RBAC hierarchy described in Section 3.1: only addresses granted `NATIONAL_BANK_ROLE` can call `recordRepayment`, and only the World Bank admin can call `allocateCapital`.

Design annotation 3 — Events enable off-chain indexing for the analytics layer. `CapitalAllocated`, `RepaymentReceived`, and `ReserveRatioUpdated` are emitted at every significant state transition. The off-chain Express.js event listener subscribes to these events and writes them to PostgreSQL, enabling the AI/ML monitoring layer to construct loan lifecycle timelines, compute utilization windows, and feed the Random Forest fraud detector without requiring costly repeated `SLOAD` calls.

Appendix E: Internal Industry Research Notes

This appendix catalogues supplementary industry-research documents prepared for the Crypto World Bank project. They informed Chapter 3 (actor taxonomy), Chapter 5 (CEX comparison and revenue analogues), and the no-native-token design constraint. All files reside in the project repository under Documentation/2nd Phase (Bokhtiar)/Improvements/.

Table D.2: Internal industry research corpus.

ID	Document	Thesis use
IR-1	Binance_FTX_Full_Report.md	CEX vs. CWB (Table 5.24); FTT/BNB token lesson; SAFU and Binance Earn analogues; PoR and commingling safeguards
IR-2	blockchain platforms research.csv	Five-exchange matrix (Binance, Bybit, Coinbase, OKX, Kraken) for Table 5.24
IR-3	Binance Business Analysis.md	Exchange revenue taxonomy for Table 5.25 (fees, listing, OTC)
IR-4	binance_software_engineering_architecture.md	Network taxonomy (Section 3.8); use-case and sequence-diagram conventions
IR-5	binance_architecture_research.m	Off-chain PostgreSQL/Redis patterns (background; not cited in body text)

These documents are project-internal research notes, not peer-reviewed publications. They are listed here for reproducibility and examiner traceability; formal citations in the body text use bibliography entries [128]–[130] for the three documents most directly referenced.

References

- [1] S. M. Werner, D. Perez, L. Gudgeon, A. Klages-Mundt, D. Harz, and W. J. Knottenbelt. 2022. SoK: Decentralized Finance (DeFi). *arXiv preprint arXiv:2101.08778*. <https://arxiv.org/abs/2101.08778>
- [2] M. Bastankhah, V. Nadkarni, C. Jin, S. Kulkarni, and P. Viswanath. 2024. Thinking Fast and Slow: Data-Driven Adaptive DeFi Borrow-Lending Protocol. In *Proc. 6th Conf. Advances in Financial Technologies (AFT)* (LIPIcs, Vol. 316). Article 27, 23 pages. <https://doi.org/10.4230/LIPIcs.AFT.2024.27>
- [3] G. Palaiokrassas, S. Scherrers, I. Ofeidis, and L. Tassiulas. 2024. Leveraging Machine Learning for Multichain DeFi Fraud Detection. In *Proc. IEEE Int. Conf. Blockchain and Cryptocurrency (ICBC)*. <https://doi.org/10.1109/ICBC59979.2024.10634350>
- [4] D. Adom, E. O. Acheampong, and M. Boateng. 2022. LIME and SHAP: A Comparison of Model-Agnostic Approaches to Explainability in Loan Approval Systems. *International Journal of Advanced Computer Science and Applications* 13, 11. https://thesai.org/Downloads/Volume13No11/Paper_90-LIME_and_SHAP_A_Comparison.pdf
- [5] B. W. Tan. 2023. Central Bank Digital Currency and Financial Inclusion. *IMF Working Paper WP/23/69*. <https://www.imf.org/en/Publications/WP/Issues/2023/03/27/Central-Bank-Digital-Currency-and-Financial-Inclusion-531273>
- [6] F. T. Liu, K. M. Ting, and Z.-H. Zhou. 2008. Isolation Forest. In *Proc. IEEE Int. Conf. Data Mining (ICDM)*, pages 413–422. <https://doi.org/10.1109/ICDM.2008.17>
- [7] N. Atzei, M. Bartoletti, and T. Cimoli. 2017. A Survey of Attacks on Ethereum Smart Contracts (SoK). In *Proc. Int. Conf. Principles of Security and Trust (POST)*, pages 164–186. https://doi.org/10.1007/978-3-662-54455-6_8
- [8] S. M. Lundberg and S.-I. Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 4765–4774. <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- [9] P. Bracke, A. Datta, C. Jung, and S. Sen. 2019. Machine Learning Explainability in Finance: An Application to Default Risk Analysis. *Bank of England Staff Working Paper No. 816*. <https://www.bankofengland.co.uk/working-paper/2019/machine-learning-explainability-in-finance-an-application-to-default-risk-analysis>
- [10] R. Beck, C. Müller-Bloch, and J. L. King. 2018. Governance in the Blockchain Economy: A Framework and Research Agenda. *Journal of the Association for Information Systems*, 19, 10, pages 1020–1034. <https://doi.org/10.17705/1jais.00518>
- [11] D. Mhlanga. 2022. Blockchain Technology for Financial Inclusion and Sustainable Development. In *Digital Financial Inclusion*. Springer. <https://doi.org/10.1007/978-3-031-05725-7>
- [12] OpenZeppelin. 2024. OpenZeppelin Contracts. <https://openzeppelin.com/contracts>
- [13] DefiLlama. 2024. Lending Protocols — DeFi TVL and Protocol Rankings. <https://defillama.com/protocols/Lending>
- [14] World Bank. 2021. The Global Findex Database 2021: Financial Inclusion, Digital Payments, and Resilience. <https://www.worldbank.org/en/publication/globalfindex>
- [15] Aave. 2024. Aave Protocol Documentation. <https://docs.aave.com/>

- [16] Compound. 2024. Compound Protocol Documentation. <https://docs.compound.finance/>
- [17] BCOLBD. 2025. Blockchain Olympiad Bangladesh: Guideline and Evaluation Scheme. <https://bcolbd.org/rules>
- [18] Galaxy Digital. 2024. The State of Crypto Lending and Borrowing. Galaxy Research. <https://www.galaxy.com/insights/research/the-state-of-crypto-lending>
- [19] World Bank. 2022. World Development Report 2022: Finance for an Equitable Recovery. <https://www.worldbank.org/en/publication/wdr2022>
- [20] International Finance Corporation (IFC). 2017. MSME Finance Gap: Assessment of the Shortfalls and Opportunities in Financing Micro, Small, and Medium Enterprises in Emerging Markets. World Bank. <https://openknowledge.worldbank.org/entities/publication/ff4c9839-21ac-5676-a23a-7cf6f745df0c>
- [21] Bank for International Settlements. 2024. DeFi Lending: Intermediation Without Information?. BIS Working Paper No. 1183. <https://www.bis.org/publ/work1183.pdf>
- [22] Deloitte. 2024. Global Banking Outlook 2024. Deloitte Insights. Retrieved June 9, 2026 from <https://www.deloitte.com/global/en/Industries/financial-services/research-global-banking-outlook.html>
- [23] Michigan State University Libraries. 2025. Literature Table and Synthesis for Systematic Reviews. MSU LibGuides. Retrieved June 9, 2026 from <https://libguides.lib.msu.edu/nursinglitreview/table>
- [24] Committee on Payments and Market Infrastructures and Bank for International Settlements. 2016. Correspondent Banking — Technical Report. CPMI Papers No. 147. <https://www.bis.org/cpmi/publ/d147.pdf>
- [25] Ripple Labs. 2025. RLUSD Stablecoin Product Overview. Ripple Documentation. <https://ripple.com/solutions/stablecoin/>
- [26] World Bank. 2024. Remittance Prices Worldwide: Making Markets More Transparent. Migration and Development Brief 40. <https://remittanceprices.worldbank.org/>
- [27] DefiLlama. 2026. Aave V3 TVL, Fees, Revenue & Income Statement. March. <https://defillama.com/protocol/aave-v3>
- [28] World Inequality Lab. 2026. Exorbitant Privilege. *World Inequality Report*. <https://wir2026.wid.world/insight/exorbitant-privilege/>
- [29] DefiLlama. 2026. Compound V3 TVL, Fees, Revenue & Income Statement. March. <https://defillama.com/protocol/compound-v3>
- [30] Fensory. 2026. Sky Protocol Projects \$611M Revenue in 2026 as USDS Supply Targets \$20.6 Billion. February. <https://www.fensory.com/intelligence/rwa/sky-protocol-tokenization-regatta-solana-february-2026>
- [31] Fensory. 2026. Maple Finance — Institutional DeFi Lending Protocol. <https://www.fensory.com/insights/protocols/maple-finance>
- [32] Fensory. 2026. DeFi Credit Platforms Hit \$2.4B Milestone. March. <https://www.fensory.com/intelligence/rwa/defi-private-credit-platforms-2-4-billion-loans-ethereum>
- [33] Financial Stability Board. 2020. Enhancing Cross-border Payments: Stage 3 Roadmap. <https://www.fsb.org/2020/10/enhancing-cross-border-payments-stage-3-roadmap/>
- [34] A. Beyer, B. Gasperini, and P. Theodoridis. 2025. Monetary Policy and Inequality: Distributional Effects of Asset Purchase Programs. *Journal of International Money and Finance* 157. <https://doi.org/10.1016/j.jimonfin.2024.103123>

- [35] Federal Reserve Board. 2025. Monetary Policy and the Distribution of Income: Evidence from U.S. Metropolitan Areas. FEDS Notes, March. <https://www.federalreserve.gov/econres/notes/feds-notes/monetary-policy-and-the-distribution-of-income-evidence-from-us-metropolitan-areas-20250331.html>
- [36] R. Cantillon. 1755. *Essai sur la Nature du Commerce en Général* (Essay on the Nature of Commerce in General). Posthumous edition.
- [37] GlobeNewswire. 2025. R3's Corda Leads Tokenized RWA Market with Over \$10 Billion in On-chain Assets. February. <https://www.globenewswire.com/news-release/2025/02/13/3025637/0/en/R3-s-Corda-leads-tokenized-RWA-market-with-over-10-billion-in-on-chain-assets-and-unrivalled-industry-adoption.html>
- [38] Centrifuge. 2025. Real-World Asset Tokenization: Key Trends from 2025. <https://centrifuge.io/blog/real-world-asset-tokenization-trends-2025>
- [39] World Bank. 2025. World Bank Group Tracks Project Funds with New Blockchain Tool. Press Release, September. <https://www.worldbank.org/en/news/press-release/2025/09/26/world-bank-group-tracks-project-funds-with-new-blockchain-tool>
- [40] JPMorgan. 2025. Kinexys Digital Payments: Real-Time Multicurrency Payments. <https://www.jpmorgan.com/onyx/coin-system.htm>
- [41] World Economic Forum. 2022. Why Decentralized Finance Is a Leapfrog Technology for the 1.1 Billion People Who Are Unbanked. September. <https://www.weforum.org/stories/2022/09/decentralized-finance-a-leapfrog-technology-for-the-unbanked/>
- [42] Opera Newsroom. 2026. 160M CELO Allocation Proposal to Grow Opera from Distribution Partner into Key Network Stakeholder. March. <https://press.opera.com/2026/03/19/opera-celo-partnership-2026/>
- [43] Morpho. 2026. Network Data. March. <https://data.morpho.org/>
- [44] Stellar. 2025. End of Year 2025 Report — Execution at Scale. <https://www.stellar.org/blog/foundation-news/2025-year-in-review>
- [45] Human Rights Foundation. 2025. Tracking CBDCs Before They Track You. <https://hrf.org/latest/tracking-cbdcs-before-they-track-you>
- [46] Y. Chen, S. Bin, and H. He. 2024. Design and Implementation of a Multi-Chain Lending Model in Blockchain. In *Proc. 3rd Int. Conf. Artificial Intelligence and Computer Information Technology (AICIT)*, pages 97–102.
- [47] S. Yang and W. Cui. 2023. An Evaluation System for DeFi Lending Protocols. *arXiv preprint arXiv:2303.01022*. <https://arxiv.org/abs/2303.01022>
- [48] J. Hartmann and O. Hasan. 2021. A Social-Capital Based Approach to Blockchain-Enabled Peer-to-Peer Lending. In *Proc. IEEE Int. Conf. Blockchain Computing and Applications (BCCA)*, pages 105–110. <https://hal.science/hal-03371872>
- [49] P. T. Hasan, H. K. T. Akhter, M. R. Tahsinur, A. B. Haque, A. K. M. N. Islam, and R. M. Rahman. 2022. Blockchain and Machine Learning for Fraud Detection: A Privacy-Preserving and Adaptive Incentive Based Approach. *IEEE Access*, vol. 10, pages 87115–87131. <https://ieeexplore.ieee.org/document/9857827>
- [50] Y. Wang, J. Liu, and Z. Chen. 2020. ContractWard: Automated Vulnerability Detection Models for Ethereum Smart Contracts. *IEEE Trans. Network Science and Engineering*, 8, 2, pages 1133–1144. <https://doi.org/10.1109/TNSE.2020.2968505>
- [51] C.-F. Liao, T.-H. Tsai, and C.-J. Chen. 2019. SoliAudit: Smart Contract Vulnerability Assessment Based on Machine Learning and Fuzz Testing. In *Proc. IEEE Int. Conf. Internet of Things (iThings)*, pages 458–465. <https://doi.org/10.1109/iThings/GreenCom/CPSCoM/SmartData.2019.00098>

- [52] S. So, M. Lee, J. Park, H. Lee, and H. Oh. 2020. VERISMArt: A Highly Precise Safety Verifier for Ethereum Smart Contracts. In *Proc. IEEE Symp. Security and Privacy (S&P)*, pages 1678–1694. <https://doi.org/10.1109/SP40000.2020.00032>
- [53] Bank for International Settlements et al. 2020. Central Bank Digital Currencies: Foundational Principles and Core Features. BIS and Central Banks Joint Report. <https://www.bis.org/publ/othp33.htm>
- [54] M. Asaduzzaman, F. Hasib, and Z. B. Hafiz. 2020. Towards Using Blockchain Technology for Microcredit Industry in Bangladesh. In *Proc. 23rd Int. Conf. Computer and Information Technology (ICCIT)*, pages 1–6. <https://doi.org/10.1109/ICCIT51783.2020.9392730>
- [55] H. Kim and D. Kim. 2024. Optimal Gas Fee Minimization in DeFi: Enhancing Efficiency and Security on the Ethereum Blockchain. *IEEE Access*, vol. 12, pages 173810–173823. <https://ieeexplore.ieee.org/document/10750187>
- [56] X. Yuan. 2024. SHAP-based Interpretable Models for Credit Default Assessment Using Machine Learning. In *Proc. 14th Int. Conf. Software Technology and Engineering (ICSTE)*, pages 213–217. <https://doi.org/10.1109/ICSTE63875.2024.00044>
- [57] S. Nakamoto. 2008. Bitcoin: A Peer-to-Peer Electronic Cash System. <https://bitcoin.org/bitcoin.pdf>
- [58] G. Wood. 2023. Ethereum: A Secure Decentralised Generalised Transaction Ledger. (Yellow Paper), Ethereum Foundation, 2014 (revised). <https://ethereum.github.io/yellowpaper/paper.pdf>
- [59] Aave. 2022. Aave Protocol V3 Technical Paper. https://github.com/aave/aave-v3-core/blob/master/techpaper/Aave_V3_Technical_Paper.pdf
- [60] T. Dao, T. Trinh, and V. Pham. 2025. Optimizing Credit Scoring Models for Decentralized Financial Applications. In *Information and Communication Technology*, Springer. https://doi.org/10.1007/978-981-96-4282-3_36
- [61] G.-L. Gücük, S. Leible, and J. Edinger. 2024. Blockchain-Based Microlending for Financial Inclusivity: A Literature Review of Its Privacy and Trust. In *Blockchain and Applications*. Springer. https://doi.org/10.1007/978-3-032-12801-0_22
- [62] A. Beniiche. 2020. A Study of Blockchain Oracles. *arXiv preprint arXiv:2004.07140*. <https://arxiv.org/pdf/2004.07140>
- [63] A. Pasdar, Y. C. Lee, and Z. Dong. 2023. Connect API with Blockchain: A Survey on Blockchain Oracle Implementation. *ACM Computing Surveys*, 55, 10. <https://doi.org/10.1145/3567582>
- [64] L. Gudgeon, S. M. Werner, D. Perez, and W. J. Knottenbelt. 2020. DeFi Protocols for Loanable Funds: Interest Rates, Liquidity and Market Efficiency. In *Proc. 2nd ACM Conf. Advances in Financial Technologies (AFT)*, pages 92–112. <https://doi.org/10.1145/3419614.3423254>
- [65] T. Mackinga, T. Nadahalli, and R. Wattenhofer. 2023. Attacks on Dynamic DeFi Interest Rate Curves. *arXiv preprint arXiv:2307.13139*.
- [66] K. W. Wu. 2024. Strengthening DeFi Security: A Static Analysis Approach to Flash Loan Vulnerabilities. *arXiv preprint arXiv:2411.01230*.
- [67] Y. Li et al. 2025. A Comprehensive Study of Exploitable Patterns in Smart Contracts: From Vulnerability to Defense. *arXiv preprint arXiv:2504.21480*. <https://arxiv.org/abs/2504.21480>
- [68] E. Albert, J. Correias, P. Gordillo, G. Román-Díez, and A. Rubio. 2020. GASOL: Gas Analysis and Optimization for Ethereum Smart Contracts. In *Proc. TACAS*, Springer. https://doi.org/10.1007/978-3-030-45237-7_7

- [69] F. Piper, K. Wolf, and J. Heiss. 2025. Privacy-Preserving On-chain Permissioning for KYC-Compliant Decentralized Applications. TU Berlin, *arXiv preprint arXiv:2510.05807*.
- [70] N. Decker. 2025. Zero-Knowledge Proofs: Cryptographic Model for Financial Compliance and Global Banking Security. *SSRN Working Paper No. 5170068*.
- [71] E. Toufaily and T. Zalan. 2024. How Can Blockchain-Based Lending Platforms Support Microcredit Activities in Developing Countries? An Empirical Validation of Its Opportunities and Challenges. *Technological Forecasting and Social Change*, vol. 203. <https://doi.org/10.1016/j.techfore.2024.123403>
- [72] S. Howlader and P. Halder. 2025. Fintech's Impact on Financial Inclusion Through Mobile Financial Services in Bangladesh. *Sage Publications*. <https://doi.org/10.1177/09763996251356998>
- [73] Grameen Bank. 2024. Grameen Bank at a Glance. Grameen Communications. <https://www.grameen-info.org/>
- [74] A. Carstens et al. 2021. Stablecoins: Risks, Potential and Regulation. *BIS Working Papers No. 905*, Bank for International Settlements. <https://www.bis.org/publ/work905.pdf>
- [75] M. Cen, J. He, A. S. Kyle, and D. Xiong. 2025. Stablecoin Devaluation Risk. *The European Journal of Finance*. <https://doi.org/10.1080/1351847X.2025.2505757>
- [76] J. A. Ocampo and K. Gallagher. 2024. The Role of Multilateral Development Banks and Development Assistance in the Provision of Global Public Goods. UNDP Background Paper. <https://hdr.undp.org/system/files/documents/background-paper-document/2024jaocampokdgonzaleztheroleofmultilateraldevlmtbanks.pdf>
- [77] Anonymous. 2025. Commit-Reveal²: Securing Randomness Beacons with Randomized Reveal Order in Smart Contracts. *arXiv preprint arXiv:2504.03936*. <https://arxiv.org/abs/2504.03936>
- [78] F. A. Aponte-Novoa, A. L. S. Orozco, R. Villanueva-Polanco, and P. Wightman. 2021. The 51% Attack on Blockchains: A Mining Behavior Study. *IEEE Access*, vol. 9, pages 140549–140564. <https://doi.org/10.1109/ACCESS.2021.3119291>
- [79] DL News Research. 2026. State of DeFi 2025. DL News, March. <https://www.dlnews.com/research/internal/state-of-defi-2025/>
- [80] H. Qu, K. M. Gogol, F. Grötschla, and C. J. Tessone. 2025. From Rules to Rewards: Reinforcement Learning for Interest Rate Adjustment in DeFi Lending. *arXiv preprint arXiv:2506.00505*. <https://arxiv.org/abs/2506.00505>
- [81] W3C. 2022. Decentralized Identifiers (DIDs) v1.0 — Core Architecture, Data Model, and Representations. W3C Recommendation, July ; see also W3C, “Verifiable Credentials Data Model v1.1,” W3C Recommendation, March. <https://www.w3.org/TR/did-core/>
- [82] Sygnum Bank. 2026. Institutional DeFi in 2025: From Niche to Mainstream Finance. Sygnum Research Report, February. <https://www.sygnum.com/research/>
- [83] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, vol. 372, n71. <https://doi.org/10.1136/bmj.n71>
- [84] P. Treleaven, R. Gendal Brown, and D. Yang. 2017. Blockchain Technology in Finance. *Computer*, 50, 9, pages 14–17. <https://doi.org/10.1109/MC.2017.3571047>
- [85] D. Cheng, X. Wang, Y. Zhang, and L. Zhang. 2022. Graph Neural Network for Fraud Detection via Spatial-Temporal Attention. *IEEE Trans. Knowledge and Data Engineering*, 34, 8, pages 3800–3813. <https://ieeexplore.ieee.org/document/9356317>
- [86] D. Wang, B. Wu, X. Yuan, L. Wu, Y. Zhou, and H. Cui. 2024. DeFiGuard: A Price Manipulation

- Detection Service in DeFi Using Graph Neural Networks. *arXiv preprint arXiv:2406.11157*. <https://arxiv.org/abs/2406.11157>
- [87] G. Palaiochrassas, S. Scherrers, E. Makri, and L. Tassioulas. 2024. Machine Learning in DeFi: Credit Risk Assessment and Liquidation Prediction. In *Proc. IEEE Int. Conf. Blockchain and Cryptocurrency (ICBC)*. <https://doi.org/10.1109/ICBC59979.2024.10634350>
- [88] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*.
- [89] Z. Chen, J. Chen, M. Sra, et al. 2025. Standard Benchmarks Fail — Auditing LLM Agents in Finance Must Prioritize Risk. *arXiv preprint arXiv:2502.15865*. <https://arxiv.org/abs/2502.15865>
- [90] Grameen Bank. 2024. Annual Report 2023. Grameen Bank, Dhaka, Bangladesh. <https://grameen.com/wp-content/uploads/2024/04/Annual-Report-2023-English.pdf>
- [91] W3C. 2022. Decentralized Identifiers (DIDs) v1.0: Core Architecture, Data Model, and Representations. W3C Recommendation (July 2022). <https://www.w3.org/TR/did-core/>. See also W3C. 2022. Verifiable Credentials Data Model v1.1. <https://www.w3.org/TR/vc-data-model/>
- [92] MicroSave Consulting. 2025. Shaping a Sustainable Microfinance Sector in Bangladesh. MSC Signature Project, September. https://www.microsave.net/signature_projects/shaping-a-sustainable-microfinance-sector-in-bangladesh-through-mscs-transformation-feasibility-study/
- [93] Bank for International Settlements. 2025. BIS Survey on Central Bank Digital Currency and Crypto-Assets. BIS Papers No. 147, July. <https://www.bis.org/publ/bppdf/bispap147.pdf>
- [94] Morgan Stanley. 2023. Key Milestone in Innovation Journey with OpenAI. Press Release, March. <https://www.morganstanley.com/press-releases/key-milestone-in-innovation-journey-with-openai>
- [95] McKinsey & Company. 2024. The Economic Potential of Generative AI: The Next Productivity Frontier (Banking Sector Update). McKinsey Global Institute, June. <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier>
- [96] G. Cheng et al. 2025. Uncovering the Vulnerability of Large Language Models in the Financial Domain via Risk Concealment. *arXiv preprint arXiv:2509.10546*. <https://arxiv.org/abs/2509.10546>
- [97] European Parliament and Council of the European Union. 2024. Regulation (EU) 2023/1114 of 31 May 2023 on Markets in Crypto-Assets (MiCA). *Official Journal of the European Union*, L 150/40, June 2023; fully applicable from 30 December. <https://eur-lex.europa.eu/eli/reg/2023/1114/oj>
- [98] European Banking Authority. 2024. Guidelines on the Minimum Content of the Governance Arrangements for Issuers of Asset-Referenced Tokens under MiCAR. EBA/GL//06, June. <https://www.eba.europa.eu/activities/single-rulebook/regulatory-activities/asset-referenced-and-e-money-tokens-micar/guidelines-internal-governance-arrangements-issuers-arts-under-micar>
- [99] United States Congress. 2025. Guiding and Establishing National Innovation for U.S. Stablecoins (GENIUS) Act. Pub. L. No. 119-27, July 18. <https://www.congress.gov/bill/119th-congress/senate-bill/919>
- [100] Bank for International Settlements Innovation Hub. 2024. Project mBridge: Connecting

- Economies through CBDC – Reaches Minimum Viable Product. BIS Innovation Hub Report, June. https://www.bis.org/about/bisih/topics/cbdc/mcbdc_bridge.htm
- [101] Bank for International Settlements and Central Banks of France, Japan, Korea, Mexico, Switzerland, the UK, and the US. 2024. Project Agora: Tokenization of Cross-Border Payments using Unified Ledgers. BIS Innovation Hub Working Paper, April. <https://www.bis.org/about/bisih/topics/fmis/agora.htm>
- [102] DL News. 2024. DeFi Lending Protocols: Market Overview and Analysis. <https://www.dlnews.com/>
- [103] Bangladesh Bank. 2025. FE Circular No. 06: Electronic Communication for Letters of Credit and Trade Finance Instruments. Foreign Exchange Policy Department, January. <https://www.bb.org.bd/mediaroom/circulars/fepd/jan172025fepd06.pdf>
- [104] H. Qu, K. M. Gogol, F. Grötschla, and C. J. Tessone. 2025. From Rules to Rewards: Reinforcement Learning for Interest Rate Adjustment in DeFi Lending. *arXiv preprint arXiv:2506.00505*. <https://arxiv.org/abs/2506.00505>
- [105] Bangladesh Bank. 2025. Financial Inclusion in Bangladesh: Progress and Gender-Disaggregated Outlook 2024–25. Special Publication SP-02, Financial Inclusion Department, June. <https://www.bb.org.bd/pub/special/SP2025-02.pdf>
- [106] BRAC. 2025. BRAC Microfinance Annual Performance Report 2025: Women-Centric Lending Outcomes. BRAC Research and Evaluation Division. <https://www.brac.net/program/microfinance/>
- [107] Ethereum Improvement Proposals. 2025. EIP-4626: Tokenized Vault Standard. Ethereum Foundation, April 2022. ; “EIP-7540: Asynchronous ERC-4626 Tokenized Vaults,” Ethereum Foundation, 2023. ; “EIP-3643: T-REX – Token for Regulated EXchanges,” Ethereum Foundation, 2021. See also: Spectral Finance, “On-Chain Credit Scoring for DeFi Lending,” 2023. ; RociFi Labs, “Undercollateralized DeFi Lending via On-Chain Credit Score,” 2023. ; Qwen Team (Alibaba Cloud), “Qwen3 Technical Report,” *arXiv preprint arXiv:2505.09388*. <https://eips.ethereum.org/EIPS/eip-4626>
- [108] Ethereum Improvement Proposals. 2022–2025. EIP-4626, EIP-7540, EIP-3643; Spectral Finance and RociFi documentation; Qwen Team. 2025. Qwen3 Technical Report (*arXiv:2505.09388*). <https://eips.ethereum.org/EIPS/eip-4626>
- [109] Elliptic. 2019. The Elliptic Data Set. Kaggle. <https://www.kaggle.com/datasets/elliptico/elliptic-data-set>
- [110] Y. Elmougy, S. Liu, and J. Liu. 2023. Demystifying Fraudulent Transactions and Illicit Nodes in the Bitcoin Network for Financial Forensics. In *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*. <https://github.com/git-disl/EllipticPlusPlus>
- [111] M. Weber, G. Domeniconi, J. Chen, D. K. I. Weidele, C. Bellei, T. Robinson, and C. E. Leiser-son. 2019. Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics. *arXiv preprint arXiv:1908.02591*, KDD Workshop on Anomaly Detection in Finance. <https://arxiv.org/abs/1908.02591>
- [112] T. Larabel. 2024. Laravel 11 Released. Laravel News. <https://laravel-news.com/laravel-11>
- [113] C. He, X. Zhou, D. Wang, H. Xu, W. Liu, and C. Miao. 2026. Harness Engineering for Language Agents: The Harness Layer as Control, Agency, and Runtime. *Preprints.org*. <https://www.preprints.org/manuscript/202603.1756>
- [114] M. Alizadeh, M. Gholampour, A. Imani, and J. Schulz. 2026. Simple Prompt Injection Attacks Can Leak Personal Data Observed by LLM Agents During Task Execution. *arXiv preprint arXiv:2506.01055*, University of Zurich / IPM, June.

- [115] OWASP Foundation. 2026. OWASP Top 10 for Large Language Model Applications and Agents, 2026 Edition. OWASP Foundation. <https://owasp.org/www-project-top-10-for-large-language-model-applications/>
- [116] Microsoft Security. 2026. Running OpenClaw Safely: Identity, Isolation, and Runtime Risk. Microsoft Security Blog, February . <https://www.microsoft.com/en-us/security/blog/2026/02/19/running-openclaw-safely-identity-isolation-runtime-risk/>
- [117] NVIDIA. 2026. NVIDIA and Microsoft Reinvent Windows PCs for the Age of Personal AI. NVIDIA Newsroom, GTC Taipei / COMPUTEX, May–June ; see also NVIDIA Blog, “RTX PCs and DGX Spark Supercomputers Run AI Agents Locally” (Nemotron on-premise inference for finance, healthcare, and legal workloads). <https://nvidianews.nvidia.com/news/nvidia-microsoft-windows-pcs-agents-rtx-spark>
- [118] E. Gamma, R. Helm, R. Johnson, and J. Vlissides. 1994. *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley.
- [119] T. Larabel. 2024. Laravel 11 Released. Laravel News. <https://laravel-news.com/laravel-11>
- [120] G. Aspris and J. Svec. 2025. Institutionalizing Risk Curation in Decentralized Credit. *arXiv preprint arXiv:2512.11976*. <https://arxiv.org/abs/2512.11976>
- [121] S. Jaffer. 1999. Microfinance and the Mechanics of Solidarity Lending. Harvard Law School, SSRN Working Paper No. 162548. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=162548
- [122] R. Sherratt. 2016. From Solidarity to Coercion: The Dynamics of Group Liability. *Oxford Economic Papers*, 68, 4, pages 955–974. <https://doi.org/10.1093/oep/gpw018>
- [123] Y. Chang, H. Kim, and S. Park. 2024. Anomalous Node Detection in Blockchain Networks Based on Graph Neural Networks. *Sensors*, 25, 1, art. 1. <https://doi.org/10.3390/s25010001>
- [124] B. Luo, Z. Zhang, Q. Wang, A. Ke, S. Lu, and B. He. 2024. AI-powered Fraud Detection in Decentralized Finance: A Project Life Cycle Perspective. *ACM Comput. Surv.*, 57, 4, art. 96. <https://doi.org/10.1145/3705296>
- [125] G. Caldarelli. 2025. Can Artificial Intelligence Solve the Blockchain Oracle Problem? Unpacking the Challenges and Possibilities. *Frontiers in Blockchain*, vol. 8, art. 1682623. <https://doi.org/10.3389/fbloc.2025.1682623>
- [126] H. Fazliani, A. Pasdar, and Y. C. Lee. 2025. Ensemble-Based Semi-Supervised Learning for Illicit Account Detection in Ethereum DeFi Transactions. *arXiv preprint arXiv:2412.02408*, rev. October. <https://arxiv.org/abs/2412.02408>
- [127] H. Alhasawi, A. Almutairi, and S. Alharbi. 2025. FedFraud: A Federated Approach to Scalable and Trustworthy Financial Fraud Detection. *Security and Privacy*, Wiley. <https://doi.org/10.1002/spy2.70012>
- [128] European Parliament and Council of the European Union. 2026. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (AI Act). *Official Journal of the European Union*, L 2024/1689, July 2024; high-risk credit-scoring provisions applicable from August. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>
- [129] M. Bartoletti, F. Fioravanti, G. Matricardi, R. Pettinau, and F. Sainas. 2024. Towards Benchmarking of Solidity Verification Tools. In *Proc. FMBC (OASICs)*, vol. 118, pages 6:1–6:15. <https://doi.org/10.4230/OASICs.FMBC.2024.6>
- [130] Crypto World Bank Project Team. 2026. Binance and FTX: Comparative Industry Analysis (Full Report). internal research note, Documentation/2nd Phase (Bokhtiar)/Improvements/, 2025– See Appendix D, entry IR-1.

- [131] Crypto World Bank Project Team. 2026. Blockchain Platforms Comparative Research (CSV). internal research dataset, Documentation/2nd Phase (Bokhtiar)/Improvements/, 2025– See Appendix D, entry IR-2.
- [132] Crypto World Bank Project Team. 2026. Binance Software Engineering Architecture and System Analysis. internal research note, Documentation/2nd Phase (Bokhtiar)/Improvements/, 2025– See Appendix D, entry IR-4.
- [133] Federal Reserve Bank of New York and Bank for International Settlements Innovation Hub. 2025. Project Pine: Central Bank Open Market Operations with Smart Contracts. BIS Innovation Hub / NY Fed Joint Report, May. <https://www.bis.org/publ/othp95.htm>
- [134] L. Dong, Y. Qiu, and F. Xu. 2023. Blockchain-Enabled Deep-Tier Supply Chain Finance. *Manufacturing & Service Operations Management*, 25, 6, pages 2021–2037. <https://doi.org/10.1287/msom.2022.1123>
- [135] U. Bindseil. 2020. Tiered CBDC and the Financial System. *ECB Working Paper Series*, no. 2351, January. <https://www.ecb.europa.eu/pub/pdf/scpwps/ecb.wp2351~c8c18bbd60.en.pdf>
- [136] J. Kiff, J. Alwazir, S. Davidovic, A. Farias, A. Khan, T. Khiaonarong, M. Malaika, H. Monroe, N. Sugimoto, H. Tourpe, and P. Zhou. 2020. A Survey of Research on Retail Central Bank Digital Currency. *IMF Working Paper WP/20/104*, June. <https://www.imf.org/en/Publications/WP/Issues/2020/06/26/A-Survey-of-Research-on-Retail-Central-Bank-Digital-Currency-49069>
- [137] R. Auer and R. Böhme. 2020. The Technology of Retail Central Bank Digital Currency. *BIS Quarterly Review*, March; see also BIS Working Paper No. 948. https://www.bis.org/publ/qtrpdf/r_qt2003j.htm
- [138] L. Ante. 2025. From Adoption to Continuance: Stablecoins in Cross-Border Remittances and the Role of Digital and Financial Literacy. *Telematics and Informatics*, vol. 97, art. 102230. <https://doi.org/10.1016/j.tele.2024.102230>
- [139] Y. Li, Y. Xiang, Q. Wang, T. H. Yuen, A. Deppeler, and J. Yu. 2026. SoK: Stablecoins in Retail Payments. *arXiv preprint arXiv:2601.00196*. <https://arxiv.org/abs/2601.00196>
- [140] W. Du, C. Huang, and D. Scharfstein. 2026. Competing Rails for Cross-Border Payments: Banks, Fintechs, and Stablecoins. Harvard Business School Working Paper, February. <https://www.hbs.edu/faculty/Pages/item.aspx?num=66992>
- [141] M. Fröhlich, J. A. Vega Vermehren, F. Alt, and A. Schmidt. 2022. Implementation and Evaluation of a Point-Of-Sale Payment System Using Bitcoin Lightning. In *Proc. NordiCHI*. <http://www.medien.ifi.lmu.de/pubdb/publications/pub/froehlich2022nordichi/froehlich2022nordichi.pdf>
- [142] K. Zhang, T.-M. Choi, S.-H. Chung, Y. Dai, and X. Wen. 2024. Blockchain Adoption in Retail Operations: Stablecoins and Traceability. *European Journal of Operational Research*, 315, 1, pages 147–160. <https://doi.org/10.1016/j.ejor.2023.11.026>
- [143] J. Lin, M. Liu, S. Li, and X. Wang. 2025. SecurePay: Enabling Secure and Fast Payment Processing for Platform Economy. In *Proc. IEEE/ACM Int. Symp. Quality of Service (IWQoS)*; *arXiv:2505.17466*. <https://arxiv.org/abs/2505.17466>
- [144] Committee on Payments and Market Infrastructures. 2023. Considerations for the Use of Stablecoin Arrangements in Cross-Border Payments. Bank for International Settlements, July. <https://www.bis.org/cpmi/publ/d220.pdf>
- [145] A. Kosse, J. Lisack, and N. Boakye-Agyeman. 2023. The Impact of Stablecoins on the International Monetary and Financial System. *BIS Papers*, no. 170, February. <https://www.bis.org/publ/bppdf/bispap170.pdf>

- [146] E. Cerutti, M. Firat, and H. Pérez-Saiz. 2025. Estimating the Impact of Digital Money on Cross-Border Flows: Scenario Analysis Covering the Intensive Margin. *IMF Fintech Notes* 2025/002, February. <https://www.imf.org/-/media/files/publications/ftn063/2025/english/ftnea2025002.pdf>
- [147] Financial Action Task Force. 2021. Updated Guidance for a Risk-Based Approach to Virtual Assets and Virtual Asset Service Providers. FATF, Paris, October. <https://www.fatf-gafi.org/en/publications/Fatfrecommendations/Va-Vasp-Guidance-R16-R21.html>
- [148] G. Cimaszewski, F. Da Dalt, T. Moser, and A. Perrig. 2025. SCION and Cross-Border Payments: Enhancing Security and Compliance in Distributed Ledger Networks. *SNB Working Papers* 15/2025, Swiss National Bank. https://www.snb.ch/en/publications/research/working-papers/2025/working_paper_2025_15
- [149] Bank for International Settlements Innovation Hub. 2024. Project Sela: An Accessible and Secure Retail CBDC Ecosystem. BIS. <https://www.bis.org/publ/othp74.htm>
- [150] S. Chaliasos, M. Charalambous, L. Zhou, R. Galanopoulou, A. Gervais, D. Mitropoulos, and B. Livshits. 2024. Smart Contract and DeFi Security: Insights from Tool Evaluations and Practitioner Surveys. In *Proc. IEEE/ACM 46th Int. Conf. Software Engineering (ICSE)*, pages 160:1–160:13. <https://doi.org/10.1145/3597503.3623302>
- [151] Z. Chen, S. M. Beillahi, P. Barahimi, C. Minwalla, H. Du, A. Veneris, and F. Long. 2026. Enforcing Control Flow Integrity on DeFi Smart Contracts. In *Proc. IEEE/ACM 48th Int. Conf. Software Engineering (ICSE)*. <https://doi.org/10.1145/3744916.3773266>
- [152] S. Ren, L. He, T. Tu, D. Wu, J. Liu, K. Ren, and C. Chen. 2025. LookAhead: Preventing DeFi Attacks via Unveiling Adversarial Contracts. *Proc. ACM Softw. Eng.*, 2, FSE, art. FSE083. <https://doi.org/10.1145/3729353>
- [153] ACM Transactions on the Web. 2026. Zero-Knowledge Proof Framework for Identity Verification and Interoperable Payments on the Decentralized Web. *ACM Trans. Web*. <https://doi.org/10.1145/3795774>
- [154] ACM Transactions on the Web. 2026. Web3-Based Identity and KYC Innovations for Next-Generation FinTech. *ACM Trans. Web*. <https://doi.org/10.1145/3771991>
- [155] Multiple Authors. 2025. Blockchain-Based Fraud Detection: A Comparative Systematic Literature Review of Federated Learning and Machine Learning Approaches. *Electronics*, 14, 24, art. 4952. <https://doi.org/10.3390/electronics14244952>
- [156] IEEE. 2025. Blockchain-Powered Platform for Microloans and Lending Solutions. In *IEEE Conf. Proc.* <https://ieeexplore.ieee.org/document/11042350/>
- [157] CAAW Workshop. 2025. Price Oracle Accuracy Across Blockchains: A Measurement and Analysis. In *Proc. Workshop on Crypto Asset Analytics (CAAW)*. https://caaw.io/2025/papers/CAAW25_paper_22.pdf
- [158] K. Qin, L. Zhou, B. Livshits, and A. Gervais. 2021. Attacking the DeFi Ecosystem with Flash Loans for Fun and Profit. In *Financial Cryptography and Data Security (FC)*, pages 3–32. https://doi.org/10.1007/978-3-662-63514-3_1
- [159] T. Mackinga, T. Nadahalli, and R. Wattenhofer. 2024. SoK: Attacks on DAOs. *arXiv preprint arXiv:2406.15071*. <https://arxiv.org/abs/2406.15071>
- [160] P. Daian, S. Goldfeder, T. Kell, Y. Li, X. Zhao, I. Bentov, L. Breidenbach, and A. Juels. 2020. Flash Boys 2.0: Frontrunning, Transaction Reordering, and Consensus Instability in Decentralized Exchanges. In *Proc. IEEE Symp. Security and Privacy (S&P)*, pages 910–927. <https://doi.org/10.1109/SP40000.2020.00040>
- [161] P. K. Ozili. 2024. Central Bank Digital Currency Adoption Challenges. *Journal of Central Banking Theory and Practice*, 13, 1, pages 5–24. <https://doi.org/10.2478/jcbtp-2024->

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